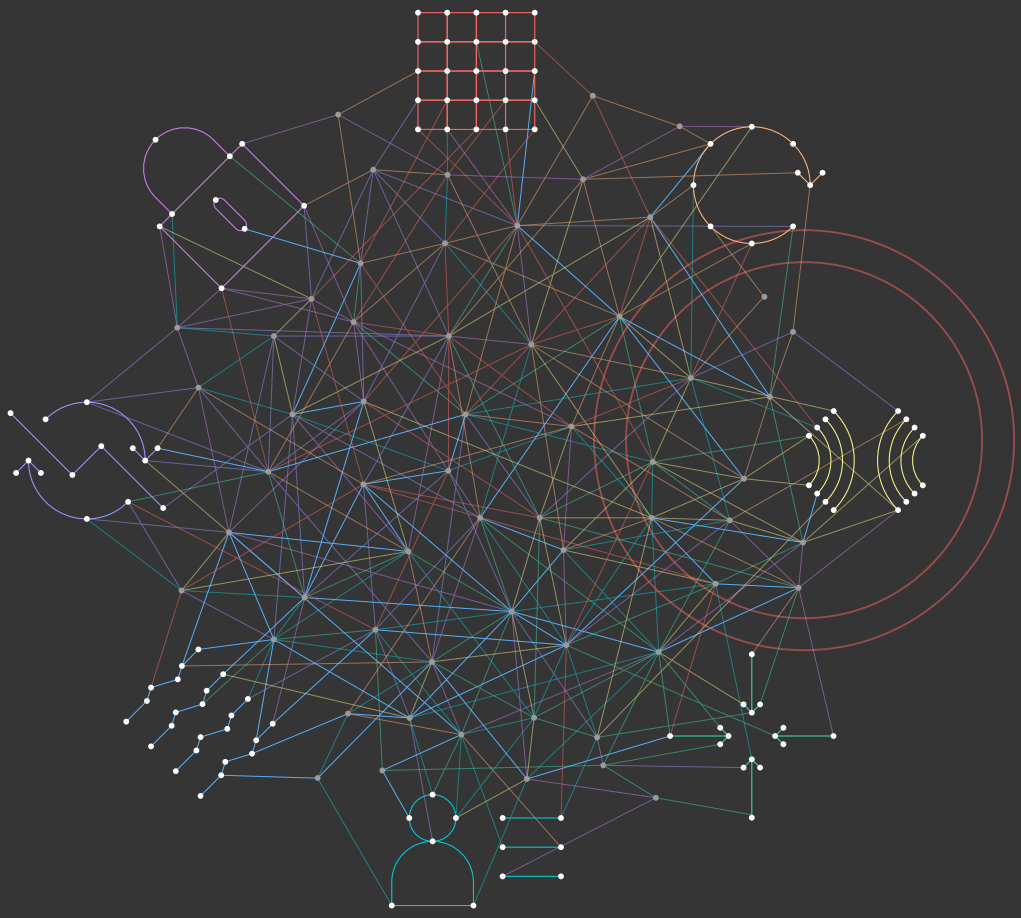
DISTRIBUTED SYSTEMS



MAARTEN VAN STEEN
ANDREW S. TANENBAUM

4TH EDITION

VERSION 01

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DISTRIBUTED SYSTEMS

Fourth edition

Version 4.01

(January 2023)

Maarten van Steen
Andrew S. Tanenbaum

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Published by Maarten van Steen
The first and second edition of this book were previously published by Pearson Education, Inc.

ISBN: 978-90-815406-3-6 (printed version)

ISBN: 978-90-815406-4-3 (digital version)

Edition: 4. Version: 01 (January 2023)

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To Mariëlle, Max, and Elke
– MvS
To Suzanne, Barbara, Marvin, Aron, Nathan, Olivia, and Mirte
– AST

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PREFACE

This is the fourth edition of "Distributed Systems." We have stayed close to the setup of the third edition, including examples of (part of) existing distributed systems close to where general principles are discussed. For example, we have included material on blockchain systems, and discuss their various components throughout the book. We have, again, used special boxed sections for material that can be skipped at first reading.

The text has been thoroughly reviewed, revised, and updated. In particular, all the Python code has been updated to Python3, while at the same time the channel package has been almost completely revised and simplified. The coding examples in the book leave out many details for readability, but the complete examples are available through the book's Website, hosted at `www.distributed-systems.net`. We have made sure that virtually all examples can be instantly executed through a simple script. However, it will be necessary to download and install special packages, such as Redis.

As before, the Website also contains slides in PDF and PPT, as well as the sources for producing slides using LaTeX with the Beamer class. All figures, now including those for tables and coding examples, are available in PDF and PNG formats.

Like the previous edition, the book can be (freely) downloaded, making it much easier to use hyperlinks where appropriate. At the same time, we are offering a printed version through Amazon.com, available at minimal costs.

The book being fully digital allows us to incorporate updates when needed. We plan to run updates on a yearly basis, while keeping previous versions digitally available, as well as the original printed version. Running frequent updates is not always the right thing to do from the perspective of teaching, but yearly updates and maintaining previous versions seems a good compromise. Updates generally consist of small corrections, the kind that usually pop up in errata lists. Next to that, the book now also contains an index section, as well as a glossary section. These are typically sections that generally take a huge time to compile, but are also sections that typically grow as the book is being used. Likewise, we may find ways to improve how hyperlinks have been incorporated. Such matters do not affect the main text, while potentially improving the usability of the digital version.

Acknowledgements

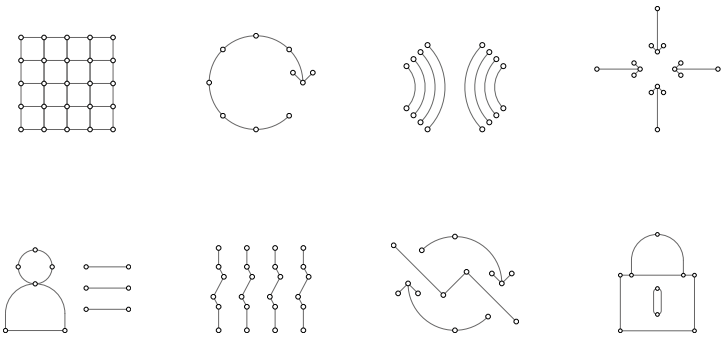
Many teachers and students have helped with spotting the inevitable errors in the third edition, and we owe them many thanks, but in any case Michael May, Linh Phan, Juan Abadie, and Christian Zirpins. For the fourth edition, a handful of colleagues were so kind to review parts of the material. In particular, we wish to thank Armin Stocker, Hermann de Meer, Pim Otte, Johan Pouwelse, Michael P. Anderson, Ivo Varenhorst, Aditya Pappu, and Alexander Iosup. A special thanks goes to Hein Meling of Stavanger University. Not only did he help us tremendously in understanding how Paxos works (already for the third edition), we have gratefully adopted, and adapted, his LaTeX style file for generating message-sequence charts. These figures are now much more consistent. We thank Max van Steen for designing the cover.

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01

INTRODUCTION



The pace at which computer systems change was, is, and continues to be overwhelming. From 1945, when the modern computer era began, until about 1985, computers were large and expensive. Moreover, lacking a way to connect them, these computers operated independently of one another.

Starting in the mid-1980s, however, two advances in technology began to change that situation. The first was the development of powerful microprocessors. Initially, these were 8-bit machines, but soon 16-, 32-, and 64-bit CPUs became common. With powerful multicore CPUs, we now are again facing the challenge of adapting and developing programs to exploit parallelism. In any case, the current generation of machines have the computing power of the mainframes deployed 30 or 40 years ago, but for 1/1000th of the price or less.

The second development was the invention of high-speed computer networks. **Local-area networks** or **LANs** allow thousands of machines within a building to be connected in such a way that small amounts of information can be transferred in a few microseconds or so. Larger amounts of data can be moved between machines at rates of billions of *bits per second* (**bps**). **Wide-area network** or **WANs** allow hundreds of millions of machines all over the earth to be connected at speeds varying from tens of thousands to hundreds of millions bps and more.

Parallel to the development of increasingly powerful and networked machines, we have also been able to witness miniaturization of computer systems, with perhaps the smartphone as the most impressive outcome. Packed with sensors, lots of memory, and a powerful multicore CPU, these devices are nothing less than full-fledged computers. Of course, they also have networking capabilities. Along the same lines, so-called **nano computers** have become readily available. These small, single-board computers, often the size of a credit card, can easily offer near-desktop performance. Well-known examples include Raspberry Pi and Arduino systems.

And the story continues. As digitalization of our society continues, we become increasingly aware of how many computers are actually being used, regularly embedded into other systems such as cars, airplanes, buildings, bridges, the power grid, and so on. This awareness is, unfortunately, increased when such systems suddenly turn out to be hackable. For example, in 2021, a fuel pipeline in the United States was effectively shut down by a ransomware attack. In this case, the computer system consisted of a mix of sensors, actuators, controllers, embedded computers, servers, etc., all brought together into a single system. What many of us do not realize, is that vital infrastructures, such as fuel pipelines, are monitored and controlled by **networked computer systems**. Along the same lines, it may be time to start realizing that a modern car is actually an autonomously operating, mobile networked computer. In this case, instead of the mobile computer being carried by a person, we need to deal with the mobile computer carrying people.

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The size of a networked computer system may vary from a handful of devices, to millions of computers. The interconnection network may be wired, wireless, or a combination of both. Moreover, these systems are often highly dynamic, in the sense that computers can join and leave, with the topology and performance of the underlying network almost continuously changing.

It is difficult to think of computer systems that are *not* networked. And as a matter of fact, most networked computer systems can be accessed from any place in the world because they are hooked up to the Internet. Studying to understand these systems can easily become exceedingly complex. In this chapter, we start with shedding some light on what needs to be understood to build up the bigger picture without getting lost.

1.1 From networked systems to distributed systems

Before we dive into various aspects of distributed systems, let us first consider what distribution, or decentralization, actually entails.

1.1.1 Distributed versus decentralized systems

When considering various sources, there are quite a few opinions on distributed versus decentralized systems. Often, the distinction is illustrated by three different organizations of networked computer systems, as shown in Figure 1.1, where each node represents a computer system and an edge a communication link between two nodes.

To what extent such distinctions are useful remains to be seen, especially when discussions open on the pros and cons of each organization. For example, it is often stated that centralized organizations do not scale well. Likewise, distributed organizations are said to be more robust against failures. As we shall see, none of these claims are generally true.

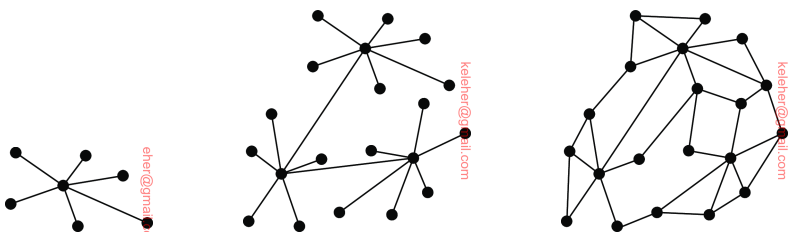


Figure 1.1: The organization of a (a) centralized, (b) decentralized, and (c) distributed system, according to various popular sources. We take a different approach, as figures such as these are really not that meaningful.

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We take a different approach. If we think of a networked computer system as a collection of computers connected in a network, we can ask ourselves how these computers even became connected to each other in the first place. There are roughly two views that one can take.

The first, **integrative view**, is that there was a need to connect existing (networked) computer systems to each other. Typically, this happens when services running on a system need to be made available to users and applications that were not thought of before. This may happen, for example, when integrating financial services with project management services, as is often the case within a single organization. In the scientific-research domain, we have seen efforts to connect a myriad of often expensive resources (special-purpose computers, supercomputers, very large database systems, etc.) into what came to be known as a grid computer.

The second, **expansive view** is that an existing system required an extension through additional computers. This view is the one most often related to the field of distributed systems. It entails expanding a system with computers to hold resources close to where those resources are needed. An expansion may also be driven by the need to improve dependability: if one computer fails, then there are others who can take over. An important type of expansion is when a service needs to be made available for remote users and applications, for example, by offering a Web interface or a smartphone application. This last example also shows that the distinction between an integrative and an expansive view is not a clear-cut.

In both cases, we see that the networked system runs services, where each service is implemented as a collection of processes and resources spread across multiple computers. The two views lead to a natural distinction between two types of networked computer systems:

- A **decentralized system** is a networked computer system in which processes and resources are **necessarily** spread across multiple computers.
- A **distributed system** is a networked computer system in which processes and resources are **sufficiently** spread across multiple computers.

Before we discuss why this distinction is important, let us look at a few examples of each type of system.

Decentralized systems are mainly related to the integrative view of networked computer systems. They come to being because we want to connect systems, yet may be hindered by administrative boundaries. For example, many applications in the artificial-intelligence domain require massive amounts of data for building reliable predictive models. Normally, data is brought to the high-performance computers that literally train models before they can be used. But when data needs to stay within the perimeter of an organization (and there can be many reasons why this is necessary), we need to bring the training to the data. The result is known as **federated learning**,

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and is implemented by a decentralized system, where the need for spreading processes and resources is dictated by administrative policies.

Another example of a decentralized system is that of **distributed ledger**, also known as a **blockchain**. In this case, we need to deal with the situation that participating parties do not trust each other enough to set up simple schemes for collaboration. Instead, what they do is essentially make transactions among each other fully public (and verifiable) by an extend-only ledger that keeps records of those transactions. The ledger itself is fully spread across the participants, and the participants are the ones who validate transactions (of others) before admitting them to the ledger. The result is a decentralized system in which processes and resources are, indeed, *necessarily* spread across multiple computers, in this case due to lack of trust.

As a last example of a decentralized system, consider systems that are naturally geographically dispersed. This occurs typically with systems in which an actual location needs to be monitored, for example, in the case of a power plant, a building, a specific natural environment, and so on. The system, controlling the monitors and where decisions are made, may easily be placed somewhere else than the location being monitored. One obvious example is monitoring and controlling of satellites, but also more mundane situations as monitoring and controlling traffic, trains, etc. In these examples, the necessity for spreading processes and resources comes from a spatial argument.

As we mentioned, distributed systems are mainly related to the expansive view of networked computer systems. A well-known example is making use of e-mail services, such as Google Mail. What often happens is that a user logs into the system through a Web interface to read and send mails. More often, however, is that users configure their personal computer (such as a laptop) to make use of a specific mail client. To that end, they need to configure a few settings, such as the incoming and outgoing server. In the case of Google Mail, these are imap.gmail.com and smtp.gmail.com, respectively. Logically, it seems as if these two servers will handle all your mail. However, with an estimate of close to 2 billion users as of 2022, it is unlikely that only two computers can handle all their e-mails (which was estimated to be more than 300 billion per year, that is, some 10,000 mails *per second*). Behind the scenes, of course, the entire Google Mail service has been implemented and spread across many computers, jointly forming a distributed system. That system has been set up to make sure that so many users can process their mails (i.e., ensures scalability), but also that the risk of losing mail because of failures, is minimal (i.e., the system ensures fault tolerance). To the user, however, the image of just two servers is kept up (i.e., the distribution itself is highly transparent to the user). The distributed system implementing an e-mail service, such as Google Mail, typically expands (or shrinks) as dictated by dependability requirements, in turn, dependent on the number of its users.

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An entirely different type of distributed system is formed by the collection of so-called **Content Delivery Networks**, or **CDNs** for short. A well-known example is Akamai with, in 2022, over 400,000 servers worldwide. We will discuss the principle working of CDNs later in Chapter 3. What it boils down to, is that the content of an actual Website, is copied and spread across various servers of the CDN. When visiting a Website, the user is transparently redirected to a nearby server that holds all or part of the content of that Website. The choice for which server to direct a user to may depend on many things, but surely when dealing with streaming content, a server is selected for which good performance in terms of latency and bandwidth can be guaranteed. The CDN dynamically ensures that the selected server will have the required content readily available, as well as update that content when needed, or remove it from the server when there are no or very few users to service there. Meanwhile, the user knows nothing about what is going on behind the scenes (which, again, is a form of distribution transparency). We also see in this example, that content is not copied to all servers, yet only to where it makes sense, that is, *sufficiently*, and for reasons of performance. CDNs also copy content to multiple servers to provide high levels of dependability.

As a final, much smaller distributed system, consider a setup based on a **Network-Attached Storage device**, also called a **NAS**. For domestic use, a typical NAS consists of 2–4 slots for internal hard disks. The NAS operates as a file server: it is accessible through a (generally wireless) network for any authorized device, and as such can offer services like shared storage, automated backups, streaming media, and so on. The NAS itself can best be seen as a single computer optimized for storing files, and offering the ability to easily share those files. The latter is important, and together with multiple users, we essentially have a setup of a distributed system. The users will be working with a set of files that are locally (i.e., from their laptop) easily accessible (in fact, seemingly integrated into the local file system), while also directly accessible by and for other users. Again, where and how the shared files are stored is hidden (i.e., the distribution is transparent). Assuming that sharing files is the goal, then we see that indeed a NAS can provide sufficient spreading of processes and resources.

Note 1.1 (More information: Are centralized solutions bad?)
There appears to be a stubborn misconception that centralized solutions cannot scale. Moreover, they are almost always associated with introducing a single point of failure. Both reasons are often seen to be enough to dismiss centralized solutions as being a good choice when designing distributed systems.

What many people forget is that a difference should be made between logical and physical designs. A logically centralized solution can be implemented in a highly scalable distributed manner. An excellent example is the **Domain Name System** (**DNS**), which we discuss extensively in Chapter 6. Logically, DNS is

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organized as a huge tree, where each path from the root to a leaf node represents a fully qualified name, such as `www.distributed-systems.net`. It would be a mistake to think that the root node is implemented by just a single server. In fact, the root node is implemented by 13 different root servers, each server, in turn, implemented as a large cluster computer. The physical organization of DNS also shows that the root is *not* a single point of failure. Being highly replicated, it would take serious efforts to bring that root down and so far, all attempts to do so have failed.

Centralized solutions are not bad just because they seem to be centralized. In fact, as we shall encounter many times throughout this book, (logically, and even physically) centralized solutions are often much better than distributed counterparts for the simple reason that there *is* a single point of failure. It makes them much easier to manage, for example, and certainly in comparison where there may be multiple points of failures. Moreover, that single point of failure can be hardened against many kinds of failures as well as many kinds of security attacks. When it comes to being a performance bottleneck, we will also see that many things can be done to ensure that even that cannot be held against centralization.

In this sense, let us not forget that centralized solutions have even proven to be extremely scalable and robust. They are called cloud-based solutions. Again, their implementations can make use of very sophisticated distributed solutions, yet even then, we shall see that even those solutions may sometimes need to rely on a small set of physical machines, if only to guarantee performance.

1.1.2 Why making the distinction is relevant

Why do we make this distinction between decentralized and distributed systems? It is important to realize that centralized solutions are generally much simpler, and also simpler along different criteria. Decentralization, that is, the act of spreading the implementation of a service across multiple computers because we believe it is *necessary*, is a decision that needs to be considered carefully. Indeed, distributed and decentralized solutions are inherently difficult:

- There are many, often unexpected, dependencies that hinder understanding the behavior of these systems.
- Distributed and decentralized systems suffer almost continuously from **partial failures**: some process or resource, somewhere at one of the participating computers, is not operating according to expectations. Discovering that failure may actually take some time, while also such failures are preferably masked (i.e., they go unnoticed for users and applications), including the recovery from failures.

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- Much related to partial failures is the fact that in many networked computer systems, participating nodes, processes, resources, and so on, come and go. This makes these systems highly dynamic, in turn requiring forms of automated management and maintenance, in turn increasing the complexity.

- The fact that distributed and decentralized systems are networked, used by many users and applications, and often cross multiple administrative boundaries, make them particularly vulnerable to security attacks. Therefore, understanding these systems and their behavior, requires that we understand how they can be, and are secured. Unfortunately, understanding security is not that easy.

Our distinction is one between *sufficiency* and *necessity* for spreading processes and resources across multiple computers. Throughout this book, we take the standpoint that decentralization can never be a goal in itself, and that it should focus on the *sufficiency* for spreading processes and resources across computers. In principle, the less spreading, the better. Yet at the same time, we need to realize that spreading is sometimes truly necessary, as illustrated by the examples of decentralized systems. From this point of sufficiency, the book is truly about distributed systems and where appropriate, we shall speak of decentralized systems.

Along the same lines, considering that distributed and decentralized systems are inherently complex, it is equally important to consider solutions that are as simple as possible. Therefore, we shall hardly discuss optimizations to solutions, firmly believing that the impact of their negative contribution to increased complexity outweighs the importance of their positive contribution to an increase of any type of performance.

1.1.3 Studying distributed systems

Considering that distributed systems are inherently difficult, it is important to take a systematic approach toward studying them. One of our major concerns is that there are so many explicit and implicit dependencies in distributed systems. For example, there is no such thing as a separate communication module, or a separate security module. Our approach is to take a look at distributed systems from a limited number, yet different perspectives. Each perspective is considered in a separate chapter.

- There are many ways in which distributed systems are organized. We start our discussion by taking the **architectural perspective**: what are common organizations, what are common styles? The architectural perspective will help in getting a first grip on how various components of existing systems interact and depend on each other.

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- Distributed systems are all about processes. The **process perspective** is all about understanding the different forms of processes that occur in distributed systems, be they threads, their virtualization of hardware processes, clients, servers, and so on. Processes form the software backbone of distributed systems, and their understanding is essential for understanding distributed systems.

- Obviously, with multiple computers at stake, communication between processes is essential. The **communication perspective** concerns the facilities that distributed systems provide to exchange data between processes. It essentially entails mimicking procedure calls across multiple computers, high-level message passing with a wealth of semantic options, and various sorts of communication between sets of processes.

- To make distributed systems work, what happens under the hood on top of which applications are executed, is that processes coordinate things. They jointly coordinate, for example, to compensate for the lack of global clock, for realizing mutual exclusive access to shared resources, and so on. The **coordination perspective** describes a number of fundamental coordination tasks that need to be carried out as part of most distributed systems.

- To access processes and resources, we need naming. In particular, we need naming schemes that, when used, will lead to the process, resources, or whatever other type of entity that is being named. As simple as this may seem, naming not only turns out to be crucial in distributed systems, there are also many ways in which naming is supported. The **naming perspective** focuses entirely on resolving a name to the access point of the named entity.

- A critical aspect of distributed systems is that they perform well in terms of efficiency and in terms of dependability. The key instrument for both aspects is replicating resources. The only problem with replication is that updates may happen, implying that all copies of a resource need to be updated as well. It is here, that keeping up the appearance of a nondistributed system becomes challenging. The **consistency and replication perspective** essentially concentrates on the trade-offs between consistency, replication, and performance.

- We already mentioned that distributed systems are subject to partial failures. The **perspective of fault tolerance** dives into the means for masking failures and their recovery. It has proven to be one of the toughest perspectives for understanding distributed systems, mainly because there are so many trade-offs to be made, and also because completely masking failures and their recovery is provably impossible.

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- As also mentioned, there is no such thing as a nonsecured distributed system. The **security perspective** focuses on how to ensure authorized access to resources. To that end, we need to discuss trust in distributed systems, along with authentication, namely verifying a claimed identity. The security perspective comes last, yet later in this chapter we shall discuss a few basic instruments that are needed to understand the role of security in the previous perspectives.

1.2 Design goals

Just because it is possible to build distributed systems does not necessarily mean that it is a good idea. In this section, we discuss four important goals that should be met to make building a distributed system worth the effort. A distributed system should make resources easily accessible; it should hide the fact that resources are distributed across a network; it should be open; and it should be scalable.

1.2.1 Resource sharing

An important goal of a distributed system is to make it easy for users (and applications) to access and share remote resources. Resources can be virtually anything, but typical examples include peripherals, storage facilities, data, files, services, and networks, to name just a few. There are many reasons for wanting to share resources. One obvious reason is that of economics. For example, it is cheaper to have a single high-end reliable storage facility be shared than having to buy and maintain storage for each user separately.

Connecting users and resources also makes it easier to collaborate and exchange information, as is illustrated by the success of the Internet with its simple protocols for exchanging files, mail, documents, audio, and video. The connectivity of the Internet has allowed geographically widely dispersed groups of people to work together by all kinds of **groupware**, that is, software for collaborative editing, teleconferencing, and so on, as is illustrated by multinational software-development companies that have outsourced much of their code production to Asia, but also the myriad of collaboration tools that became (more easily) available due to the COVID-19 pandemic.

Resource sharing in distributed systems is also illustrated by the success of file-sharing peer-to-peer networks like **BitTorrent**. These distributed systems make it simple for users to share files across the Internet. Peer-to-peer networks are often associated with distribution of media files such as audio and video. In other cases, the technology is used for distributing large amounts of data, as in the case of software updates, backup services, and data synchronization across multiple servers.

Seamless integration of resource-sharing facilities in a networked environment is also now commonplace. A group of users can simply place files into a

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special shared folder that is maintained by a third party somewhere on the Internet. Using special software, the shared folder is barely distinguishable from other folders on a user's computer. In effect, these services replace the use of a shared directory on a local distributed file system, making data available to users independent of the organization they belong to, and independent of where they are. The service is offered for different operating systems. Where exactly data are stored is completely hidden from the end user.

1.2.2 Distribution transparency

An important goal of a distributed system is to hide the fact that its processes and resources are physically distributed across multiple computers, possibly separated by large distances. In other words, it tries to make the distribution of processes and resources **transparent**, that is, invisible, to end users and applications. As we shall discuss more extensively in Chapter 2, achieving distribution transparency is realized through what is known as **middleware**, sketched in Figure 1.2 (see Gazis and Katsiri [2022] for a first introduction).

In essence, what applications get to see is the same interface everywhere, whereas behind that interface, where and how processes and resources are and how they are accessed is kept transparent. There are different types of transparency, which we discuss next.

Types of distribution transparency

The concept of transparency can be applied to several aspects of a distributed system, of which the most important ones are listed in Figure 1.3. We use the term *object* to mean either a process or a resource.

Access transparency deals with hiding differences in data representation and the way that objects can be accessed. At a basic level, we want to hide differences in machine architectures, but more important is that we reach

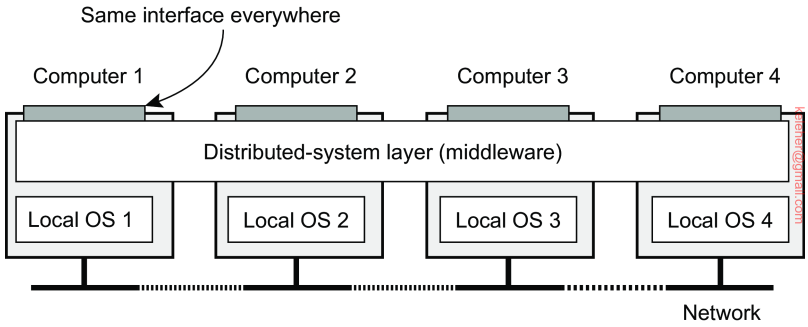


Figure 1.2: Realizing distribution transparency through a middleware layer.

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Transparency	Description
Access	Hide differences in data representation and how an object is accessed
Location	Hide where an object is located
Relocation	Hide that an object may be moved to another location while in use
Migration	Hide that an object may move to another location
Replication	Hide that an object is replicated
Concurrency	Hide that an object may be shared by several independent users
Failure	Hide the failure and recovery of an object

Figure 1.3: Different forms of transparency in a distributed system (see ISO [1995]). An object can be a resource or a process.

agreement on how data is to be represented by different machines and operating systems. For example, a distributed system may have computer systems that run different operating systems, each having their own file-naming conventions. Differences in naming conventions, differences in file operations, or differences in how low-level communication with other processes is to take place, are examples of access issues that should preferably be hidden from users and applications.

An important group of transparency types concerns the location of a process or resource. **Location transparency** refers to the fact that users cannot tell where an object is physically located in the system. Naming plays an important role in achieving location transparency. In particular, location transparency can often be achieved by assigning only logical names to resources, that is, names in which the location of a resource is not secretly encoded. An example of a such a name is the **uniform resource locator** (**URL**) `https://www.distributed-systems.net/`, which gives no clue about the actual location of the Web server where this book is offered. The URL also gives no clue whether files at that site have always been at their current location or were recently moved there. For example, the entire site may have been moved from one data center to another, yet users should not notice. The latter is an example of **relocation transparency**, which is becoming increasingly important in the context of **cloud computing**: the phenomenon by which services are provided by huge collections of remote servers. We return to cloud computing in subsequent chapters, and, in particular, in Chapter 2.

Where relocation transparency refers to *being* moved by the distributed system, **migration transparency** is offered by a distributed system when it supports the mobility of processes and resources initiated by users, without affecting ongoing communication and operations. A typical example is communication between mobile phones: regardless whether two people

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are actually moving, mobile phones will allow them to continue their conversation. Other examples that come to mind include online tracking and tracing of goods as they are being transported from one place to another, and teleconferencing (partly) using devices that are equipped with mobile Internet.

As we shall see, replication plays an important role in distributed systems. For example, resources may be replicated to increase availability or to improve performance by placing a copy close to the place where it is accessed. **Replication transparency** deals with hiding the fact that several copies of a resource exist, or that several processes are operating in some form of lockstep mode so that one can take over when another fails. To hide replication from users, it is necessary that all replicas have the same name. Consequently, a system that supports replication transparency should generally support location transparency as well, because it would otherwise be impossible to refer to replicas at different locations.

We already mentioned that an important goal of distributed systems is to allow sharing of resources. In many cases, sharing resources is done cooperatively, as in the case of communication channels. However, there are also many examples of competitive sharing of resources. For example, two independent users may each have stored their files on the same file server or may be accessing the same tables in a shared database. In such cases, it is important that each user does not notice that the other is making use of the same resource. This phenomenon is called **concurrency transparency**. An important issue is that concurrent access to a shared resource leaves that resource in a consistent state. Consistency can be achieved through locking mechanisms, by which users are, in turn, given exclusive access to the desired resource. A more refined mechanism is to make use of transactions, but these may be difficult to implement in a distributed system, notably when scalability is an issue.

Last, but certainly not least, it is important that a distributed system provides **failure transparency**. This means that a user or application does not notice that some piece of the system fails to work properly, and that the system subsequently (and automatically) recovers from that failure. Masking failures is one of the hardest issues in distributed systems and is even impossible when certain apparently realistic assumptions are made, as we will discuss in Chapter 8. The main difficulty in masking and transparently recovering from failures lies in the inability to distinguish between a dead process and a painfully slowly responding one. For example, when contacting a busy Web server, a browser will eventually time out and report that the Web page is unavailable. At that point, the user cannot tell whether the server is actually down or that the network is badly congested.

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Degree of distribution transparency

Although distribution transparency is generally considered preferable for any distributed system, there are situations in which blindly attempting to hide all distribution aspects from users is not a good idea. A simple example is requesting your electronic newspaper to appear in your mailbox before 7 AM local time, as usual, while you are currently at the other end of the world living in a different time zone. Your morning paper will not be the morning paper you are used to.

Likewise, a wide-area distributed system that connects a process in San Francisco to a process in Amsterdam cannot be expected to hide the fact that Mother Nature will not allow it to send a message from one process to the other in less than approximately 35 milliseconds. Practice shows that it actually takes several hundred milliseconds using a computer network. Signal transmission is not only limited by the speed of light, but also by limited processing capacities and delays in the intermediate switches.

There is also a trade-off between a high degree of transparency and the performance of a system. For example, many Internet applications repeatedly try to contact a server before finally giving up. Consequently, attempting to mask a transient server failure before trying another one may slow down the system as a whole. In such a case, it may have been better to give up earlier, or at least let the user cancel the attempts to make contact.

Another example is where we need to guarantee that several replicas, located on different continents, must be consistent all the time. In other words, if one copy is changed, that change should be propagated to all copies before allowing any other operation. A single update operation may now even take seconds to complete, something that cannot be hidden from users.

Finally, there are situations in which it is not at all obvious that hiding distribution is a good idea. As distributed systems are expanding to devices that people carry around and where the very notion of location and context awareness is becoming increasingly important, it may be best to actually *expose* distribution rather than trying to hide it. An obvious example is making use of location-based services, which can often be found on mobile phones, such as finding a nearest shop or any nearby friends.

There are other arguments against distribution transparency. Recognizing that full distribution transparency is simply impossible, we should ask ourselves whether it is even wise to *pretend* that we can achieve it. It may be much better to make distribution explicit so that the user and application developer are never tricked into believing that there is such a thing as transparency. The result will be that users will much better understand the (sometimes unexpected) behavior of a distributed system, and are thus much better prepared to deal with this behavior.

The conclusion is that aiming for distribution transparency may be a nice goal when designing and implementing distributed systems, but that

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it should be considered together with other issues such as performance and comprehensibility. The price for achieving full transparency may be surprisingly high.

Note 1.2 (Discussion: Against distribution transparency)
Several researchers have argued that hiding distribution will lead to only further complicating the development of distributed systems, exactly for the reason that full distribution transparency can never be achieved. A popular technique for achieving access transparency is to extend procedure calls to remote servers. However, Waldo et al. [1997] already pointed out that attempting to hide distribution by such remote procedure calls can lead to poorly understood semantics, for the simple reason that a procedure call *does* change when executed over a faulty communication link.

As an alternative, various researchers and practitioners are now arguing for less transparency, for example, by more explicitly using message-style communication, or more explicitly posting requests to, and getting results from remote machines, as is done on the Web when fetching pages. Such solutions will be discussed in detail in the next chapter.

A somewhat radical standpoint was taken by Wams [2012] by stating that partial failures preclude relying on the successful execution of a remote service. If such reliability cannot be guaranteed, it is then best to always perform only local executions, leading to the **copy-before-use** principle. According to this principle, data can be accessed only after they have been transferred to the machine of the process wanting that data. Moreover, modifying a data item should not be done. Instead, it can only be updated to a new version. It is not difficult to imagine that many other problems will surface. However, Wams shows that many existing applications can be retrofitted to this alternative approach without sacrificing functionality.

1.2.3 Openness

Another important goal of distributed systems is openness. An **open distributed system** is essentially a system that offers components that can easily be used by, or integrated into other systems. At the same time, an open distributed system itself will often consist of components that originate from elsewhere.

Interoperability, composability, and extensibility

To be open means that components should adhere to standard rules that describe the syntax and semantics of what those components have to offer (i.e., which service they provide). A general approach is to define services through **interfaces** using an **Interface Definition Language** (**IDL**). Interface definitions written in an IDL nearly always capture only the syntax of services. In other words, they specify precisely the names of the functions that are available

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together with types of the parameters, return values, possible exceptions that can be raised, and so on. The hard part is specifying precisely what those services do, that is, the semantics of interfaces. In practice, such specifications are given in an informal way by natural language.

If properly specified, an interface definition allows an arbitrary process that needs a certain interface, to talk to another process that provides that interface. It also allows two independent parties to build entirely different implementations of those interfaces, leading to two separate components that operate in exactly the same way.

Proper specifications are complete and neutral. Complete means that everything that is necessary to make an implementation has indeed been specified. However, many interface definitions are not at all complete, so that it is necessary for a developer to add implementation-specific details. Just as important is the fact that specifications do not prescribe what an implementation should look like; they should be neutral.

As pointed out in Blair and Stefani [1998], completeness and neutrality are important for interoperability and portability. **Interoperability** characterizes the extent by which two implementations of systems or components from different manufacturers can co-exist and work together by merely relying on each other's services as specified by a common standard. **Portability** characterizes to what extent an application developed for a distributed system A can be executed, without modification, on a different distributed system B that implements the same interfaces as A.

Another important goal for an open distributed system is that it should be easy to configure the system out of different components (possibly from different developers). Moreover, it should be easy to add new components or replace existing ones without affecting those components that stay in place. In other words, an open distributed system should also be **extensible**. For example, in an extensible system, it should be relatively easy to add parts that run on a different operating system, or even to replace an entire file system. Relatively simple examples of extensibility are plug-ins for Web browsers, but also those for Websites, such as the ones used for WordPress.

Note 1.3 (Discussion: Open systems in practice)
Of course, what we have just described is an ideal situation. Practice shows that many distributed systems are not as open as we would like, and that still a lot of effort is needed to put various bits and pieces together to make a distributed system. One way out of the lack of openness is to simply reveal all the gory details of a component and to provide developers with the actual source code. This approach is becoming increasingly popular, leading to so-called open-source projects, where large groups of people contribute to improving and debugging systems. Admittedly, this is as open as a system can get, but whether it is the best way is questionable.

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Separating policy from mechanism

To achieve flexibility in open distributed systems, it is crucial that the system be organized as a collection of relatively small and easily replaceable or adaptable components. This implies that we should provide definitions of not only the highest-level interfaces, that is, those seen by users and applications, but also definitions for interfaces to internal parts of the system and describe how those parts interact. This approach is relatively new. Many older and even contemporary systems are constructed using a monolithic approach in which components are only logically separated but implemented as one, huge program. This approach makes it hard to replace or adapt a component without affecting the entire system. Monolithic systems thus tend to be closed instead of open.

The need for changing a distributed system is often caused by a component that does not provide the optimal policy for a specific user or application. As an example, consider caching in Web browsers. There are many different parameters that need to be considered:

Storage: Where is data to be cached? Typically, there will be an in-memory cache next to storage on disk. In the latter case, the exact position in the local file system needs to be considered.

Exemption: When the cache fills up, which data is to be removed so that newly fetched pages can be stored?

Sharing: Does each browser make use of a private cache, or is a cache to be shared among browsers of different users?

Refreshing: When does a browser check if cached data is still up-to-date? Caches are most effective when a browser can return pages without having to contact the original Website. However, this bears the risk of returning stale data. Note also that refresh rates are highly dependent on which data is actually cached: whereas timetables for trains hardly change, this is not the case for Web pages showing current highway-traffic conditions, or worse yet, stock prices.

What we need is a separation between **policy** and **mechanism**. In the case of Web caching, for example, a browser should ideally provide facilities for only storing documents (i.e., a mechanism) and at the same time allow users to decide which documents are stored and for how long (i.e., a policy). In practice, this can be implemented by offering a rich set of parameters that the user can set (dynamically). When taking this a step further, a browser may even offer facilities for plugging in policies that a user has implemented as a separate component.

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Note 1.4 (Discussion: Is a strict separation really what we need?)
In theory, strictly separating policies from mechanisms seems to be the way to go. However, there is an important trade-off to consider: the stricter the separation, the more we need to make sure that we offer the appropriate collection of mechanisms. In practice, this means that a rich set of features is offered, in turn leading to many configuration parameters. As an example, the popular Firefox browser comes with a few hundred configuration parameters. Just imagine how the configuration space explodes when considering large distributed systems consisting of many components. In other words, strict separation of policies and mechanisms may lead to highly complex configuration problems.

One option to alleviate these problems is to provide reasonable defaults, and this is what often happens in practice. An alternative approach is one in which the system observes its own usage and dynamically changes parameter settings. This leads to what are known as **self-configurable systems**. Nevertheless, the fact alone that many mechanisms need to be offered to support a wide range of policies often makes coding distributed systems very complicated. Hard-coding policies into a distributed system may reduce complexity considerably, but at the price of less flexibility.

1.2.4 Dependability

As its name suggests, **dependability** refers to the degree that a computer system can be relied upon to operate as expected. In contrast to single-computer systems, dependability in distributed systems can be rather intricate due to **partial failures**: somewhere there is a component failing while the system as a whole still seems to be living up to expectations (up to a certain point or moment). Although single-computer systems can also suffer from failures that do not appear immediately, having a potentially large collection of networked computer systems complicates matters considerably. In fact, one should assume that at any time, there are always partial failures occurring. An important goal of distributed systems is to mask those failures, as well as mask the recovery from those failures. This masking is the essence of being able to tolerate faults, accordingly referred to as **fault tolerance**.

Basic concepts

Dependability is a term that covers several useful requirements for distributed systems, including the following [Kopetz and Verissimo, 1993]:

- Availability
- Reliability
- Safety
- Maintainability

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Availability is defined as the property that a system is ready to be used immediately. In general, it refers to the probability that the system is operating correctly at any given moment and is available to perform its functions on behalf of its users. In other words, a highly available system is one that will most likely be working at a given instant in time.

Reliability refers to the property that a system can run continuously without failure. In contrast to availability, reliability is defined in terms of a time interval instead of an instant in time. A highly reliable system is one that will most likely continue to work without interruption during a relatively long period of time. This is a subtle but important difference when compared to availability. If a system goes down on average for one, seemingly random millisecond every hour, it has an availability of more than 99.9999 percent, but is still unreliable. Similarly, a system that never crashes but is shut down for two specific weeks every August has high reliability but only 96 percent availability. The two are not the same.

Safety refers to the situation that when a system temporarily fails to operate correctly, no catastrophic event happens. For example, many process-control systems, such as those used for controlling nuclear power plants or sending people into space, are required to provide a high degree of safety. If such control systems temporarily fail for only a very brief moment, the effects could be disastrous. Many examples from the past (and probably many more yet to come) show how hard it is to build safe systems.

Finally, **maintainability** refers to how easily a failed system can be repaired. A highly maintainable system may also show a high degree of availability, especially if failures can be detected and repaired automatically. However, as we shall see, automatically recovering from failures is easier said than done.

Traditionally, fault-tolerance has been related to the following three metrics:

- **Mean Time To Failure** (***MTTF***): The average time until a component fails.
- **Mean Time To Repair** (***MTTR***): The average time needed to repair a component.
- **Mean Time Between Failures** (***MTBF***): Simply *MTTF* + *MTTR*.

Note that these metrics make sense only if we have an accurate notion of what a failure actually is. As we will encounter in Chapter 8, identifying the occurrence of a failure may actually not be so obvious.

Faults, errors, failures

A system is said to **fail** when it cannot meet its promises. In particular, if a distributed system is designed to provide its users with several services, the system has failed when one or more of those services cannot be (completely) provided. An **error** is a part of a system's state that may lead to a failure. For

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example, when transmitting packets across a network, it is to be expected that some packets have been damaged when they arrive at the receiver. Damaged in this context means that the receiver may incorrectly sense a bit value (e.g., reading a 1 instead of a 0), or may even be unable to detect that something has arrived.

The cause of an error is called a **fault**. Clearly, finding out what caused an error is important. For example, a wrong or bad transmission medium may easily cause packets to be damaged. In this case, it is relatively easy to remove the fault. However, transmission errors may also be caused by bad weather conditions, such as in wireless networks. Changing the weather to reduce or prevent errors is a bit trickier.

As another example, a crashed program is clearly a failure, which may have happened because the program entered a branch of code containing a programming bug (i.e., a programming error). The cause of that bug is typically a programmer. In other words, the programmer is the cause of the error (programming bug), in turn leading to a failure (a crashed program).

Building dependable systems closely relates to controlling faults. As explained by Avizienis et al. [2004], a distinction can be made between preventing, tolerating, removing, and forecasting faults. For our purposes, the most important issue is **fault tolerance**, meaning that a system can provide its services even in the presence of faults. For example, by applying error-correcting codes for transmitting packets, it is possible to tolerate, to a certain extent, relatively poor transmission lines and reducing the probability that an error (a damaged packet) may lead to a failure.

Faults are generally classified as transient, intermittent, or permanent. **Transient faults** occur once and then disappear. If the operation is repeated, the fault goes away. A bird flying through the beam of a microwave transmitter may cause lost bits on some network (not to mention a roasted bird). If the transmission times out and is retried, it will probably work the second time.

An **intermittent fault** occurs, then vanishes on its own accord, then reappears, and so on. A loose contact on a connector will often cause an intermittent fault. Intermittent faults cause a great deal of aggravation because they are difficult to diagnose. Typically, when the fault doctor shows up, the system works fine.

A **permanent fault** is one that continues to exist until the faulty component is replaced. Burnt-out chips, software bugs, and disk-head crashes are examples of permanent faults.

Dependable systems are also required to provide security, especially in terms of confidentiality and integrity. **Confidentiality** is the property that information is disclosed only to authorized parties, while **integrity** relates to ensuring that alterations to various assets can be made only in an authorized way. Indeed, can we speak of a dependable system when confidentiality and integrity are not in place? We return to security next.

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1.2.5 Security

A distributed system that is not secure, is not dependable. As mentioned, special attention is needed to ensure confidentiality and integrity, both of which are directly coupled to authorized disclosure and access of information and resources. In any computer system, **authorization** is done by checking whether an identified entity has proper access rights. In turn, this means that the system should know it is indeed dealing with the proper entity. For this reason, **authentication** is essential: verifying the correctness of a claimed identity. Equally important is the notion of **trust**. If a system can positively authenticate a person, what is that authentication worth if the person cannot be trusted? For this reason alone, proper authorization is important, as it may be used to limit any damage that a person, who could in hindsight not be trusted, can cause. For example, in financial systems, authorization may limit the amount of money a person is allowed to transfer between various accounts. We will discuss trust, authentication, and authorization at length in Chapter 9.

Key elements needed to understand security

An essential technique to making distributed systems secure is cryptography. This is not the place in this book to extensively discuss cryptography (which we also defer until Chapter 9), yet to understand how security fits into various perspectives in the following chapters, we informally introduce some of its basic elements.

Keeping matters simple, security in distributed systems is all about **encrypting** and **decrypting** data using **security keys**. The easiest way of considering a security key K is to see it as a function operating on some data data. We use the notation $K(\mathsf{data})$ to express the fact that the key K operates on data.

There are two ways of encrypting and decrypting data. In a **symmetric cryptosystem**, encryption and decryption takes place with a single key. Denoting by $E_K(\mathsf{data})$ the encryption of data using key $\mathsf{E_K}$, and likewise $D_K(\mathsf{data})$ for decryption with key $\mathsf{D_K}$, then in a symmetric cryptosystem, the same key is used for encryption and decryption, i.e.,

$$\textit{if } \mathsf{data} = D_K(E_K(\mathsf{data})) \textit{ then } \mathsf{D_K} = \mathsf{E_K}.$$

Note that in a symmetric cryptosystem, the key will need to be kept secret by all parties that are authorized to encrypt or decrypt data. In an **asymmetric cryptosystem**, the keys used for encryption and decryption are different. In particular, there is a **public key** PK that can be used by anyone, and a **secret key** SK that is, as its name suggests, to be kept secret. Asymmetric cryptosystems are also called **public-key systems**.

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Encryption and decryption in public-key systems can be used in two, fundamentally different ways. First, if Alice wants to encrypt data that can be decrypted only by Bob, she should use Bob's public key, $\mathsf{PK_B}$, leading to the encrypted data $PK_B(\mathsf{data})$. Only the holder of the associated secret key can decrypt this information, i.e., Bob, who will apply the operation $SK_B(PK_B(\mathsf{data}))$, which returns data.

A second, and widely applied use case, is that of realizing **digital signatures**. Suppose Alice makes some data available for which it is important that any party, but let us assume it is Bob, needs to know for sure that it comes from Alice. In that case, Alice can encrypt the data with her secret key $\mathsf{SK_A}$, leading to $SK_A(\mathsf{data})$. If it can be assured that the associated public key $\mathsf{PK_A}$ indeed belongs to Alice, then successfully decrypting $SK_A(\mathsf{data})$ to data, is proof that Alice knows about data: she is the only one holding the secret key $\mathsf{SK_A}$. Of course, we need to make the assumption that Alice is indeed the only one who holds $\mathsf{SK_A}$. We return to some of these assumptions in Chapter 9.

As it turns out, proving that an entity has seen, or knows about some data, returns frequently in secured distributed systems. Practical placement of digital signatures is generally more efficient by a **hash function**. A hash function H has the property that when operating on some data, i.e., $H(\mathsf{data})$, it returns a fixed-length string, regardless of the length of data. Any change of data to data^* will lead to a different hash value $H(\mathsf{data}^*)$. Moreover, given a hash value h, it is computationally impossible in practice, to discover the original data. What this all means, is that for placing a digital signature, Alice computes $\mathsf{sig} = SK_A(H(\mathsf{data}))$ as her signature, and tells Bob about data, H and sig. Bob, in turn, can then verify that signature by computing $PK_A(\mathsf{sig})$ and verifying that it matches the value $H(\mathsf{data})$.

Using cryptography in distributed systems

The application of cryptography in distributed systems comes in many forms. Besides its general use for encryption and digital signatures, cryptography forms the basis for realizing a **secure channel** between two communicating parties. Such channels basically let two parties know for sure that they are indeed communicating to the entities that they expected to communicate with. In other words, a communication channel that supports mutual authentication. A practical example of a secure channel is using https when accessing Websites. Now, many browsers demand that Websites support this protocol, and at the very least will warn the user when this is not the case. In general, using cryptography is necessary to realize authentication (and authorization) in distributed systems.

Cryptography is also used to realize secure distributed data structures. A well-known example is that of a **blockchain**, which is, literally, a chain of blocks. The basic idea is simple: hash the data in a block $\mathsf{B_i}$, and place that hash value as part of the data in its succeeding block $\mathsf{B_{i+1}}$. Any change in

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B_i (for example, as the result of an attack), will require that the attacker also changes the stored hash value in B_{i+1}. However, because the successor of B_{i+1} contains the hash computed over the data in B_{i+1}, and thus including the *original* hash value of B_i, the attacker will also have to change the new has value of B_i as stored in B_{i+1}. Yet changing that value, also means changing the hash value of B_{i+1}, and thus the value stored in B_{i+2}, in turn, requiring that a new hash value is to be computed for B_{i+2}, and so on. In other words, by securely linking blocks into a chain, any successful change to a block requires that *all* successive blocks be modified as well. These modifications should go unnoticed, which is virtually impossible.

Cryptography is also used for another important mechanism in distributed systems: delegating access rights. The basic idea is that Alice may want to delegate some rights to Bob, who, in turn, may want to pass some of those rights on to Chuck. Using appropriate means (which we discuss in Chapter 9, a service can securely check that Chuck has indeed been authorized to perform certain operations, without the need for that service to check with Alice whether the delegation is in place. Note that delegation is something we are now used to: many of us delegate access rights that we have as a user to specific applications, such as an e-mail client.

An upcoming distributed application of cryptography is so-called **multi-party computation**: the means for two or three parties to compute a value for which the data of those parties is needed, but without having to actually share that data. An often-used example is computing the number of votes without having to know who voted for whom.

We will see many more examples of security in distributed systems in the following chapters. With the brief explanations of the cryptographic basis, it should suffice to see how security is applied. We shall consistently use the notations as shown in Figure 1.4. Alternatively, security examples can be skipped until having studied Chapter 9.

Notation	Description
$K_{A,B}$	Secret key shared by A and B
PK_A	Public key of A
SK_A	Private (secret) key of A
$E_K(\text{data})$	Encryption of data using key E_K (or key K)
$D_K(\text{data})$	Decryption of (encrypted) data using key D_K (or key K)
$H(\text{data})$	The hash of data computed using function H

Figure 1.4: Notations for cryptosystems used in this book.

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1.2.6 Scalability

For many of us, worldwide connectivity through the Internet is as common as being able to send a package to anyone anywhere around the world. Moreover, where until recently, we were used to having relatively powerful desktop computers for office applications and storage, we are now witnessing that such applications and services are being placed in what has been coined "the cloud," in turn leading to an increase of much smaller networked devices such as tablet computers or even cloud-only laptops such as Google's Chromebook. With this in mind, scalability has become one of the most important design goals for developers of distributed systems.

Scalability dimensions

Scalability of a system can be measured along at least three different dimensions (see [Neuman, 1994]):

Size scalability: A system can be scalable regarding its size, meaning that we can easily add more users and resources to the system without any noticeable loss of performance.

Geographical scalability: A geographically scalable system is one in which the users and resources may lie far apart, but the fact that communication delays may be significant is hardly noticed.

Administrative scalability: An administratively scalable system is one that can still be easily managed even if it spans many independent administrative organizations.

Let us take a closer look at each of these three scalability dimensions.

Size scalability When a system needs to scale, very different types of problems need to be solved. Let us first consider scaling regarding size. If more users or resources need to be supported, we are often confronted with the limitations of centralized services, although often for very different reasons. For example, many services are centralized in the sense that they are implemented by a single **server** running on a specific machine in the distributed system. In a more modern setting, we may have a group of collaborating servers co-located on a cluster of tightly coupled machines physically placed at the same location. The problem with this scheme is obvious: the server, or group of servers, can simply become a bottleneck when it needs to process an increasing number of requests. To illustrate how this can happen, let us assume that a service is implemented on a single machine. In that case, there are essentially three root causes for becoming a bottleneck:

- The computational capacity, limited by the CPUs
- The storage capacity, including the I/O transfer rate

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- The network between the user and the centralized service

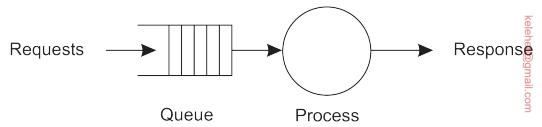
Let us first consider the computational capacity. Just imagine a service for computing optimal routes taking real-time traffic information into account. It is not difficult to imagine that this may be primarily a compute-bound service, requiring several (tens of) seconds to complete a request. If there is only a single machine available, then even a modern high-end system will eventually run into problems if the number of requests increases beyond a certain point.

Likewise, but for different reasons, we will run into problems when having a service that is mainly I/O bound. A typical example is a poorly designed centralized search engine. The problem with content-based search queries is that we essentially need to match a query against an entire data set. Even with advanced indexing techniques, we may still face the problem of having to process a huge amount of data exceeding the main-memory capacity of the machine running the service. As a consequence, much of the processing time will be determined by the relatively slow disk accesses and transfer of data between disk and main memory. Simply adding more or higher-speed disks will prove not to be a sustainable solution as the number of requests continues to increase.

Finally, the network between the user and the service may also be the cause of poor scalability. Just imagine a video-on-demand service that needs to stream high-quality video to multiple users. A video stream can easily require a bandwidth of 8 to 10 Mbps, meaning that if a service sets up point-to-point connections with its customers, it may soon hit the limits of the network capacity of its own outgoing transmission lines.

There are several solutions to attack size scalability, which we discuss below after having looked into geographical and administrative scalability.

Note 1.5 (Advanced: Analyzing size scalability)



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Figure 1.5: A simple model of a service as a queuing system.

Size scalability problems for centralized services can be formally analyzed using queuing theory and making a few simplifying assumptions. At a conceptual level, a centralized service can be modeled as the simple queuing system shown in Figure 1.5: requests are submitted to the service, where they are queued until further notice. As soon as the process can handle a next request, it fetches it from the queue, does its work, and produces a response. We largely follow Menasce and Almeida [2002] in explaining the performance of a centralized service.

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Often, we may assume that the queue has an infinite capacity, meaning that there is no restriction on the number of requests that can be accepted for further processing. Strictly speaking, this means that the arrival rate of requests is not influenced by what is currently in the queue or being processed. Assuming that the arrival rate of requests is λ requests per second, and that the processing capacity of the service is μ requests per second, one can compute that the fraction of time p_k that there are k requests in the system is equal to:

$$p_k = (1 - \frac{\lambda}{\mu})(\frac{\lambda}{\mu})^k$$

If we define the **utilization** U of a service as the fraction of time that it is busy, then clearly,

$$U = \sum_{k>0} p_k = 1 - p_0 = \frac{\lambda}{\mu} \Rightarrow p_k = (1-U)U^k$$

We can then compute the average number $\overline{N}$ of requests in the system as

$$\overline{N} = \sum_{k\geq 0} k \cdot p_k = \sum_{k\geq 0} k \cdot (1-U)U^k = (1-U)\sum_{k\geq 0} k \cdot U^k = \frac{(1-U)U}{(1-U)^2} = \frac{U}{1-U}.$$

What we are truly interested in, is the response time R: how long does it take before the service to process a request, including the time spent in the queue. To that end, we need the average throughput X. Considering that the service is "busy" when at least one request is being processed, and that this then happens with a throughput of μ requests per second, and during a fraction U of the total time, we have:

$$X = \underbrace{U \cdot \mu}_{\text{server at work}} + \underbrace{(1-U) \cdot 0}_{\text{server idle}} = \frac{\lambda}{\mu} \cdot \mu = \lambda$$

Using Little's formula [Trivedi, 2002], we can then derive the response time as

$$R = \frac{\overline{N}}{X} = \frac{S}{1-U} \Rightarrow \frac{R}{S} = \frac{1}{1-U}$$

where $S = \frac{1}{\mu}$, the actual service time. Note that if U is small, the response-to-service time ratio is close to 1, meaning that a request is virtually instantly processed, and at the maximum speed possible. However, as soon as the utilization comes closer to 1, we see that the response-to-server time ratio quickly increases to very high values, effectively meaning that the system is coming close to a grinding halt. This is where we see scalability problems emerge. From this simple model, we can see that the only solution is bringing down the service time S. We leave it as an exercise to the reader to explore how S may be decreased.

Geographical scalability Geographical scalability has its own problems. One of the main reasons why it is still difficult to scale existing distributed systems

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that were designed for local-area networks is that many of them are based on **synchronous communication**. In this form of communication, a party requesting a service, generally referred to as a **client**, blocks until a reply is sent back from the **server** implementing the service. More specifically, we often see a communication pattern consisting of many client-server interactions, as may be the case with database transactions. This approach generally works fine in LANs, where communication between two machines is often at worst a few hundred microseconds. However, in a wide-area system, we need to consider that interprocess communication may be hundreds of milliseconds, three orders of magnitude slower. Building applications using synchronous communication in wide-area systems requires a great deal of care (and not just a little patience), notably with a rich interaction pattern between client and server.

Another problem that hinders geographical scalability is that communication in wide-area networks is inherently much less reliable than in local-area networks. In addition, we generally also need to deal with limited bandwidth. The effect is that solutions developed for local-area networks cannot always be easily ported to a wide-area system. A typical example is streaming video. In a home network, even when having only wireless links, ensuring a stable, fast stream of high-quality video frames from a media server to a display is quite simple. Simply placing that same server far away and using a standard TCP connection to the display will surely fail: bandwidth limitations will instantly surface, but also maintaining the same level of reliability can easily cause headaches.

Yet another issue that pops up when components lie far apart is the fact that wide-area systems generally have only very limited facilities for multipoint communication. In contrast, local-area networks often support efficient broadcasting mechanisms. Such mechanisms have proven to be extremely useful for discovering components and services, which is essential from a management perspective. In wide-area systems, we need to develop separate services, such as naming and directory services, to which queries can be sent. These support services, in turn, need to be scalable as well and often no obvious solutions exist as we will encounter in later chapters.

Administrative scalability Finally, a difficult, and often open, question is how to scale a distributed system across multiple, independent administrative domains. A major problem that needs to be solved is that of conflicting policies regarding resource usage (and payment), management, and security. To illustrate, for many years, scientists have been looking for solutions to share their (often expensive) equipment in what is known as a **computational grid**. In these grids, a global decentralized system is constructed as a federation of local distributed systems, allowing a program running on a computer at an organization A to directly access resources at the organization B.

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Many components of a distributed system that reside within a single domain can often be trusted by users that operate within that same domain. In such cases, system administration may have tested and certified applications, and may have taken special measures to ensure that such components cannot be tampered with. In essence, the users trust their system administrators. However, this trust does not expand naturally across domain boundaries.

If a distributed system expands to another domain, two types of security measures need to be taken. First, the distributed system has to protect itself against malicious attacks from the new domain. For example, users from the new domain may have only read access to the file system in its original domain. Likewise, facilities such as expensive image setters or high-performance computers may not be made available to unauthorized users. Second, the new domain has to protect itself against malicious attacks from the distributed system. A typical example is that of downloading programs, such as in the case of federated learning. Basically, the new domain does not know what to expect from such foreign code. The problem, as we shall see in Chapter 9, is how to enforce those limitations.

As a counterexample of distributed systems spanning multiple administrative domains that apparently *do not* suffer from administrative scalability problems, consider modern file-sharing peer-to-peer networks. In these cases, end users simply install a program implementing distributed search and download functions and within minutes can start downloading files. Other examples include peer-to-peer applications for telephony over the Internet such as older Skype systems [Baset and Schulzrinne, 2006], and (again older) peer-assisted audio-streaming applications such as Spotify [Kreitz and Niemelä, 2010]. A more modern application (that has yet to prove itself in terms of scalability) are blockchains. What these decentralized systems have in common is that *end users*, and not administrative entities, collaborate to keep the system up and running. At best, underlying administrative organizations such as **Internet Service Providers** (**ISPs**) can police the network traffic that these peer-to-peer systems cause.

Scaling techniques

Having discussed some scalability problems brings us to the question of how those problems can generally be solved. In most cases, scalability problems in distributed systems appear as performance problems caused by limited capacity of servers and network. Simply improving their capacity (e.g., by increasing memory, upgrading CPUs, or replacing network modules) is often a solution, referred to as **scaling up**. When it comes to **scaling out**, that is, expanding the distributed system by essentially deploying more machines, there are basically only three techniques we can apply: hiding communication latencies, distribution of work, and replication (see also Neuman [1994]).

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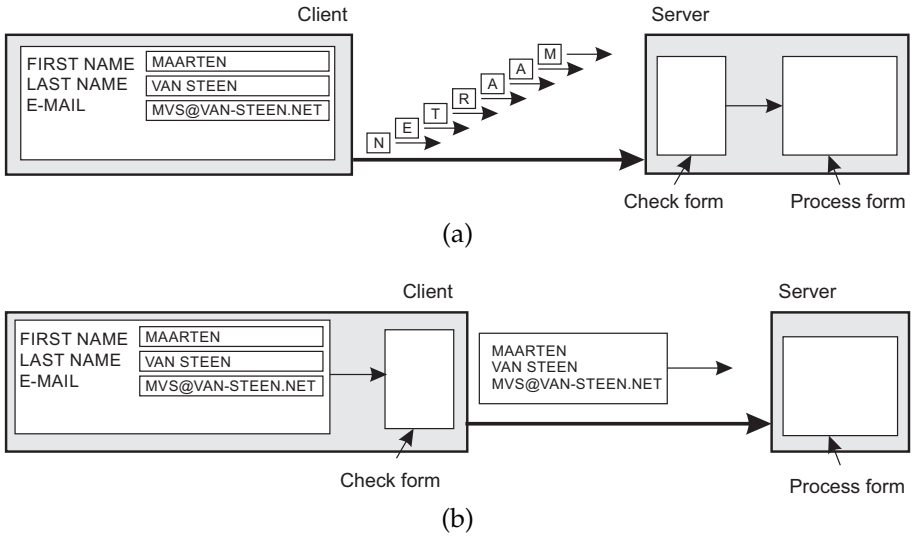


Figure 1.6: The difference between letting (a) a server or (b) a client check forms as they are being filled.

Hiding communication latencies Hiding communication latencies is applicable in the case of geographical scalability. The basic idea is simple: try to avoid waiting for responses to remote-service requests as much as possible. For example, when a service has been requested at a remote machine, an alternative to waiting for a reply from the server is to do other useful work at the requester's side. Essentially, this means constructing the requesting application in such a way that it uses only **asynchronous communication**. When a reply comes in, the application is interrupted and a special handler is called to complete the previously issued request. Asynchronous communication can often be used in batch-processing systems and parallel applications, in which independent tasks can be scheduled for execution while another task is waiting for communication to complete. Alternatively, a new thread of control can be started to perform the request. Although it blocks waiting for the reply, other threads in the process can continue.

However, there are many applications that cannot make effective use of asynchronous communication. For example, in interactive applications when a user sends a request, she will generally have nothing better to do than to wait for the answer. In such cases, a much better solution is to reduce the overall communication, for example, by moving part of the computation that is normally done at the server to the client process requesting the service. A typical case where this approach works is accessing databases using forms. Filling in forms can be done by sending a separate message for each field and waiting for an acknowledgment from the server, as shown in Figure 1.6(a). For example, the server may check for syntactic errors before accepting an entry.

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A much better solution is to ship the code for filling in the form, and possibly checking the entries, to the client, and have the client return a completed form, as shown in Figure 1.6(b). This approach of shipping code is widely supported by the Web through JavaScript.

Partitioning and distribution Another important scaling technique is partitioning and distribution, which involves taking a component or other resource, splitting it into smaller parts, and subsequently spreading those parts across the system. A good example of partitioning and distribution is the Internet Domain Name System (DNS). The DNS name space is hierarchically organized into a tree of **domains**, which are divided into nonoverlapping **zones**, as shown for the original DNS in Figure 1.7. The names in each zone are handled by a single name server. Without going into too many details now (we return to DNS extensively in Chapter 6), one can think of each path name being the name of a host on the Internet, and is thus associated with a network address of that host. Basically, resolving a name means returning the network address of the associated host. Consider, for example, the name flits.cs.vu.nl. To resolve this name, it is first passed to the server of zone Z1 (see Figure 1.7) which returns the address of the server for zone Z2, to which the rest of the name, flits.cs.vu, can be handed. The server for Z2 will return the address of the server for zone Z3, which is capable of handling the last part of the name, and will return the address of the associated host.

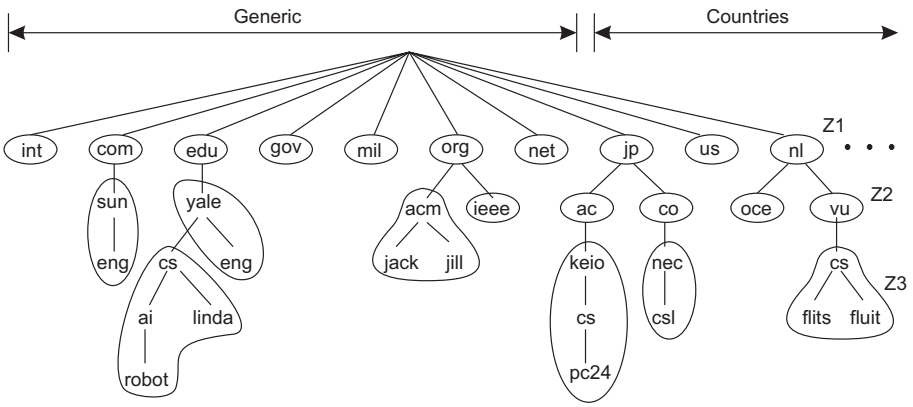


Figure 1.7: An example of dividing the (original) DNS name space into zones.

These examples illustrate how the **naming service** as provided by DNS, is distributed across several machines, thus avoiding that a single server has to deal with all requests for name resolution.

As another example, consider the World Wide Web. To most users, the Web appears to be an enormous document-based information system, in which each document has its own unique name in the form of a URL. Conceptually,

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it may even appear as if there is only a single server. However, the Web is physically partitioned and distributed across *a few hundreds of millions* of servers, each handling often a number of Websites, or parts of Websites. The name of the server handling a document is encoded into that document's URL. It is only because of this distribution of documents that the Web has been capable of scaling to its current size. Yet, note that finding out how many servers provide Web-based services is virtually impossible: A Website today is so much more than a few static Web documents.

Replication Considering that scalability problems often appear in the form of performance degradation, it is generally a good idea to actually **replicate** components or resources, etc., across a distributed system. Replication not only increases availability, but also helps to balance the load between components, leading to better performance. Moreover, in geographically widely dispersed systems, having a copy nearby can hide much of the communication latency problems mentioned before.

Caching is a special form of replication, although the distinction between the two is often hard to make or even artificial. As in the case of replication, caching results in making a copy of a resource, generally in the proximity of the client accessing that resource. However, in contrast to replication, caching is a decision made by the client of a resource and not by the owner of a resource.

There is one serious drawback to caching and replication that may adversely affect scalability. Because we now have multiple copies of a resource, modifying one copy makes that copy different from the others. Consequently, caching and replication leads to **consistency** problems.

To what extent inconsistencies can be tolerated depends on the usage of a resource. For example, many Web users find it acceptable that their browser returns a cached document of which the validity has not been checked for the last few minutes. However, there are also many cases in which strong consistency guarantees need to be met, such as in the case of electronic stock exchanges and auctions. The problem with strong consistency is that an update must be immediately propagated to all other copies. Moreover, if two updates happen concurrently, it is often also required that updates are processed in the same order everywhere, introducing a global ordering problem. To make things worse, combining consistency with desirable properties such as availability may simply be impossible, as we discuss in Chapter 8.

Replication therefore often requires some global synchronization mechanism. Unfortunately, such mechanisms are extremely hard or even impossible to implement in a scalable way, if alone because network latencies have a natural lower bound. Consequently, scaling by replication may introduce other, inherently nonscalable solutions. We return to replication and consistency extensively in Chapter 7.

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Discussion When considering these scaling techniques, one could argue that size scalability is the least problematic from a technical perspective. Often, increasing the capacity of a machine will save the day, although perhaps there is a high monetary cost to pay. Geographical scalability is a much tougher problem, as network latencies are naturally bound from below. As a consequence, we may be forced to copy data to locations close to where clients are, leading to problems of maintaining copies consistent. Practice shows that combining distribution, replication, and caching techniques with different forms of consistency generally leads to acceptable solutions. Finally, administrative scalability seems to be the most difficult problem to solve, partly because we need to deal with nontechnical issues, such as politics of organizations and human collaboration. The introduction and now widespread use of peer-to-peer technology has successfully demonstrated what can be achieved if end users are put in control [Lua et al., 2005; Oram, 2001]. However, peer-to-peer networks are obviously not the universal solution to all administrative scalability problems.

1.3 A simple classification of distributed systems

We have discussed distributed versus decentralized systems, yet it is also useful to classify distributed systems according to what they are being developed and used for. We make a distinction between systems that are developed for (high performance) computing, for general information processing, and those that are developed for pervasive computing, i.e., for the "Internet of Things." As with many classifications, the boundaries between these three types are not strict and combinations can easily be thought of.

1.3.1 High-performance distributed computing

An important class of distributed systems is the one used for high-performance computing tasks. Roughly speaking, one can make a distinction between two subgroups. In **cluster computing** the underlying hardware consists of a collection of similar compute nodes, interconnected by a high-speed network, often alongside a more common local-area network for controlling the nodes. In addition, each node generally runs the same operating system.

The situation becomes very different in the case of **grid computing**. This subgroup consists as decentralized systems that are often constructed as a federation of computer systems, where each system may fall under a different administrative domain, and may be very different when it comes to hardware, software, and deployed network technology.

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Note 1.6 (More information: Parallel processing)
High-performance computing more or less started with the introduction of **multiprocessor machines**. In this case, multiple CPUs are organized in such a way that they all have access to the same physical memory, as shown in Figure 1.8(a). In contrast, in a **multicomputer system** several computers are connected through a network and there is no sharing of main memory, as shown in Figure 1.8(b). The shared-memory model turned out to be highly convenient for improving the performance of programs, and it was relatively easy to program.

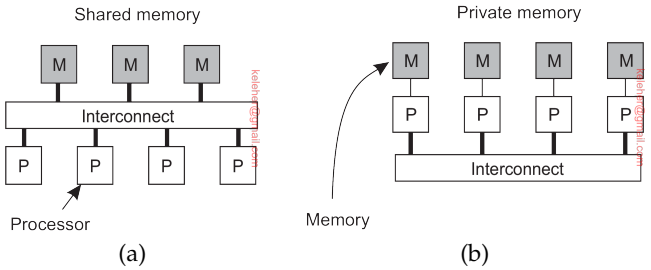


Figure 1.8: A comparison between (a) multiprocessor and (b) multicomputer architectures.

Its essence is that multiple threads of control are executing at the same time, while all threads have access to shared data. Access to that data is controlled through well-understood synchronization mechanisms like semaphores (see Ben-Ari [2006] or Herlihy et al. [2021] for more information on developing parallel programs). Unfortunately, the model does not easily scale: so far, machines have been developed in which only a few tens (and sometimes hundreds) of CPUs have efficient access to shared memory. To a certain extent, we are seeing the same limitations for multicore processors.

To overcome the limitations of shared-memory systems, high-performance computing moved to distributed-memory systems. This shift also meant that many programs had to make use of message passing instead of modifying shared data as a means of communication and synchronization between threads. Unfortunately, message-passing models have proven to be much more difficult and error-prone compared to the shared-memory programming models. For this reason, there has been significant research in attempting to build so-called **distributed shared-memory multicomputers**, or simply **DSM systems** [Amza et al., 1996].

In essence, a DSM system allows a processor to address a memory location at another computer as if it were local memory. This can be achieved using existing techniques available to the operating system, for example, by mapping all main-memory pages of the various processors into a single virtual address space. Whenever a processor A addresses a page located at another processor B, a page fault occurs at A allowing the operating system at A to fetch the content of the referenced page at B in the same way that it would normally fetch it locally from disk. At the same time, the processor B would be informed that the page is currently not accessible.

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Mimicking shared-memory systems using multicomputers eventually had to be abandoned because performance could never meet the expectations of programmers, who would rather resort to far more intricate, yet better (predictably) performing message-passing models. An important side effect of exploring the hardware-software boundaries of parallel processing is a thorough understanding of consistency models, to which we return extensively in Chapter 7.

Cluster computing

Cluster computing systems became popular when the price/performance ratio of personal computers and workstations improved. At a certain point, it became financially and technically attractive to build a supercomputer using off-the-shelf technology by simply hooking up a collection of relatively simple computers in a high-speed network. In virtually all cases, cluster computing is used for parallel programming, in which a single (compute intensive) program is run in parallel on multiple machines. The principle of this organization is shown in Figure 1.9.

This type of high-performance computing has evolved considerably. As discussed extensively by Gerofi et al. [2019], the developments of supercomputers organized as clusters have reached a point where we see clusters with more than 100,000 CPUs, with each CPU having 8 or 16 cores. There are multiple networks. Most important is a network formed by dedicated high-speed interconnects between the various nodes (in other words, there is often no such thing as a shared high-speed network for computations). A separate management network, as well as nodes, are used to monitor and control

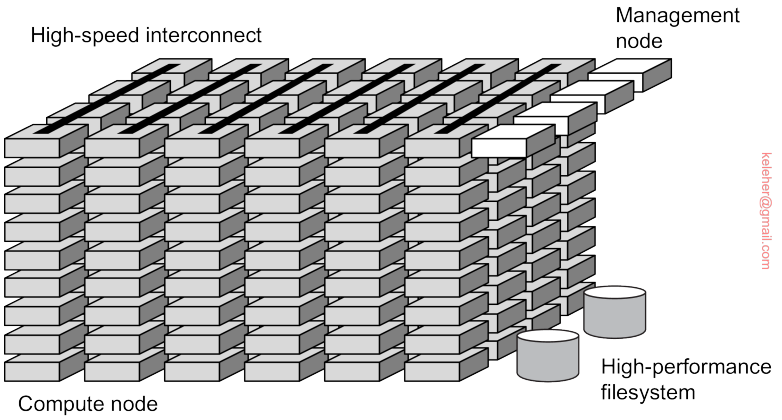


Figure 1.9: An example of a cluster computing system (adapted from [Gerofi et al., 2019].)

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the organization and performance of the system as a whole. In addition, a special high-performance file system or database is used, again with its own local, dedicated network. Figure 1.9 does not show additional equipment, notably high-speed I/O as well as networking facilities for remote access and communication.

A management node is generally responsible for collecting jobs from users, to subsequently distribute the associated tasks among the various compute nodes. In practice, several management nodes are used when dealing with very large clusters. As such, a management node actually runs the software needed for the execution of programs and management of the cluster, while the compute nodes are equipped with a standard operating system extended with typical functions for communication, storage, fault tolerance, and so on.

An interesting development, as explained in Gerofi et al. [2019], is the role of the operating system. There has been a clear trend to minimize the operating system to lightweight kernels, essentially ensuring the least possible overhead. A drawback is that such operating systems become highly specialized and fine-tuned toward the underlying hardware. This specialization affects compatibility, or openness. To compensate, we are now gradually seeing so-called multikernel approaches, in which a full-fledged operating system operates next to a lightweight kernel, thus achieving the best of two worlds. This combination is also necessary given increasingly more often, a high-performance compute node is required to run multiple, independent jobs simultaneously. At present, 95% of all high-performance computers run Linux-based systems; multikernel approaches are developed for multicore CPUs, with most cores running a lightweight kernel and the other running a regular Linux system. In this way, new developments such as containers (which we discuss in Chapter 3) can also be supported. The effects for computing performance still needs to be seen.

Grid computing

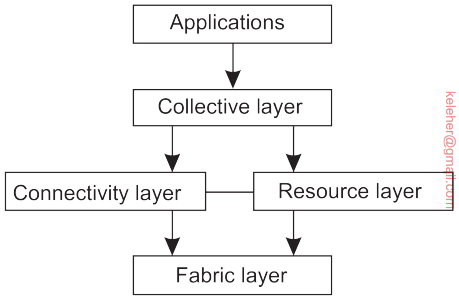
A characteristic feature of traditional cluster computing is its homogeneity. In most cases, the computers in a cluster are largely the same, have the same operating system, and are all connected through the same network. However, as we just discussed, there is a continuous trend toward more hybrid architectures in which nodes are specifically configured for certain tasks. This diversity is even more prevalent in **grid-computing systems**: no assumptions are made concerning similarity of hardware, operating systems, networks, administrative domains, security policies, etc. [Rajaraman, 2016].

A key issue in a grid-computing system is that resources from different organizations are brought together to allow the collaboration of a group of people from different institutions, indeed forming a federation of systems. Such a collaboration is realized in the form of a **virtual organization**. The processes belonging to the same virtual organization have access rights to the

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resources that are provided to that organization. Typically, resources consist of compute servers (including supercomputers, possibly implemented as cluster computers), storage facilities, and databases. In addition, special networked devices such as telescopes, sensors, etc., can be provided as well.

Given its nature, much of the software for realizing grid computing evolves around providing access to resources from different administrative domains, and to only those users and applications that belong to a specific virtual organization. For this reason, focus is often on architectural issues. An architecture initially proposed by Foster et al. [2001] is shown in Figure 1.10, which still forms the basis for many grid computing systems.



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Figure 1.10: A layered architecture for grid computing systems.

The architecture consists of four layers. The lowest **fabric layer** provides interfaces to local resources at a specific site. Note that these interfaces are tailored to allow sharing of resources within a virtual organization. Typically, they will provide functions for querying the state and capabilities of a resource, along with functions for actual resource management (e.g., locking resources).

The **connectivity layer** consists of communication protocols for supporting grid transactions that span the usage of multiple resources. For example, protocols are needed to transfer data between resources, or to simply access a resource from a remote location. In addition, the connectivity layer will contain security protocols to authenticate users and resources. Note that in many cases, human users are not authenticated; instead, programs acting on behalf of the users are authenticated. In this sense, delegating rights from a user to programs is an important function that needs to be supported in the connectivity layer. We return to delegation when discussing security in distributed systems in Chapter 9.

The **resource layer** is responsible for managing a single resource. It uses the functions provided by the connectivity layer and calls directly the interfaces made available by the fabric layer. For example, this layer will offer functions for obtaining configuration information on a specific resource, or, in general, to perform specific operations such as creating a process or reading data. The

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resource layer is thus seen to be responsible for access control, and hence will rely on the authentication performed as part of the connectivity layer.

The next layer in the hierarchy is the **collective layer**. It deals with handling access to multiple resources and typically consists of services for resource discovery, allocation and scheduling of tasks onto multiple resources, data replication, and so on. Unlike the connectivity and resource layer, each consisting of a relatively small, standard collection of protocols, the collective layer may consist of many protocols reflecting the broad spectrum of services it may offer to a virtual organization.

Finally, the **application layer** consists of the applications that operate within a virtual organization and which make use of the grid computing environment.

Typically, the collective, connectivity, and resource layer form the heart of what could be called a grid middleware layer. These layers jointly provide access to and management of resources that are potentially dispersed across multiple sites.

An important observation from a middleware perspective is that in grid computing, the notion of a site (or administrative unit) is common. This prevalence is emphasized by the gradual shift toward a **service-oriented architecture** in which sites offer access to the various layers through a collection of Web services [Joseph et al., 2004]. This, by now, has led to the definition of an alternative architecture known as the **Open Grid Services Architecture** (**OGSA**) [Foster et al., 2006]. OGSA is based upon the original ideas as formulated by Foster et al. [2001], yet having gone through a standardization process makes it complex, to say the least. OGSA implementations generally follow Web service standards.

1.3.2 Distributed information systems

Another important class of distributed systems is found in organizations that were confronted with a wealth of networked applications, but for which interoperability turned out to be a painful experience. Many of the existing middleware solutions are the result of working with an infrastructure in which it was easier to integrate applications into an enterprise-wide information system [Alonso et al., 2004; Bernstein, 1996; Hohpe and Woolf, 2004].

We can distinguish several levels at which integration can take place. Often, a networked application simply consists of a server running that application (often including a database) and making it available to remote programs, called **clients**. Such clients send a request to the server for executing a specific operation, after which a response is sent back. Integration at the lowest level allows clients to wrap several requests, possibly for different servers, into a single larger request and have it executed as a **distributed transaction**. The key idea is that all, or none of the requests are executed.

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As applications became more sophisticated and were gradually separated into independent components (notably distinguishing database components from processing components), it became clear that integration should also take place by letting applications communicate directly with each other. This has now led to an industry on **enterprise application integration** (**EAI**).

Distributed transaction processing

To clarify our discussion, we concentrate on database applications. In practice, operations on a database are carried out in the form of **transactions**. Programming using transactions requires special primitives that must either be supplied by the underlying distributed system or by the language runtime system. Typical examples of transaction primitives are shown in Figure 1.11. The exact list of primitives depends on what kinds of objects are being used in the transaction [Gray and Reuter, 1993; Bernstein and Newcomer, 2009]. In a mail system, there might be primitives to send, receive, and forward mail. In an accounting system, they might be quite different. READ and WRITE are typical examples, however. Ordinary statements, procedure calls, and so on, are also allowed inside a transaction. In particular, **remote procedure calls** (**RPC**), that is, procedure calls to remote servers, are often also encapsulated in a transaction, leading to what is known as a **transactional RPC**. We discuss RPCs extensively in Section 4.2.

Primitive	Description
BEGIN_TRANSACTION	Mark the start of a transaction
END_TRANSACTION	Terminate the transaction and try to commit
ABORT_TRANSACTION	Kill the transaction and restore the old values
READ	Read data from a file, a table, or otherwise
WRITE	Write data to a file, a table, or otherwise

Figure 1.11: Example primitives for transactions.

BEGIN_TRANSACTION and END_TRANSACTION are used to delimit the scope of a transaction. The operations between them form the body of the transaction. The characteristic feature of a transaction is either all of these operations are executed or none are executed. These may be system calls, library procedures, or bracketing statements in a language, depending on the implementation.

This all-or-nothing property of transactions is one of the four characteristic properties that transactions have. More specifically, transactions adhere to the so-called **ACID** properties:

- **Atomic**: To the outside world, the transaction happens indivisibly
- **Consistent**: The transaction does not violate system invariants

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- **Isolated**: Concurrent transactions do not interfere with each other
- **Durable**: Once a transaction commits, the changes are permanent

In distributed systems, transactions are often constructed as a number of subtransactions, jointly forming a **nested transaction** as shown in Figure 1.12. The top-level transaction may fork off children that run in parallel with one another, on different machines, to gain performance or simplify programming. Each of these children may also execute one or more subtransactions, or fork off its own children.

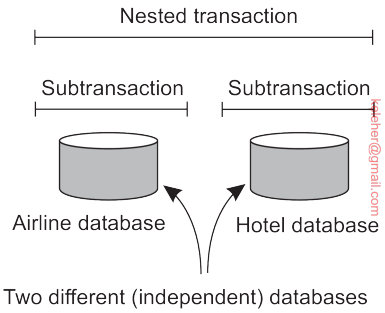


Figure 1.12: A nested transaction.

Subtransactions give rise to a subtle, but important, problem. Imagine that a transaction starts several subtransactions in parallel, and one of these commits, making its results visible to the parent transaction. After further computation, the parent aborts, restoring the entire system to the state it had before the top-level transaction started. Consequently, the results of the subtransaction that committed must nevertheless be undone. Thus, the permanence referred to above applies only to top-level transactions.

Since transactions can be nested arbitrarily deep, considerable administration is needed to get everything right. The semantics are clear, however. When any transaction or subtransaction starts, it is conceptually given a private copy of all data in the entire system for it to manipulate as it wishes. If it aborts, its private universe just vanishes, as if it had never existed. If it commits, its private universe replaces the parent's universe. Thus, if a subtransaction commits and then later a new subtransaction is started, the second one sees the results produced by the first one. Likewise, if an enclosing (higher level) transaction aborts, all its underlying subtransactions have to be aborted as well. And if several transactions are started concurrently, the result is as if they ran sequentially in some unspecified order.

Nested transactions are important in distributed systems, for they provide a natural way of distributing a transaction across multiple machines. They follow a *logical* division of the work of the original transaction. For example, a transaction for planning a trip by which three different flights need to be

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reserved can be logically split up into three subtransactions. Each of these subtransactions can be managed separately and independently.

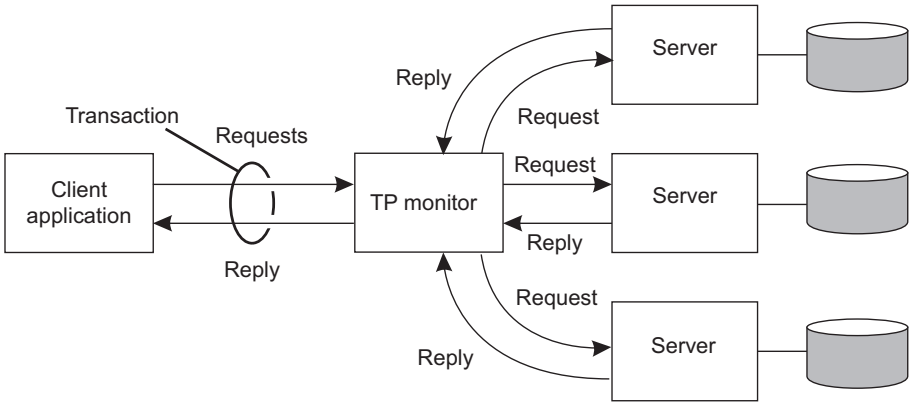


Figure 1.13: The role of a TP monitor in distributed systems.

Ever since the early days of enterprise middleware systems, the component that handles distributed (or nested) transactions belongs to the core for integrating applications at the server or database level. This component is called a **transaction-processing monitor** or **TP monitor** for short. Its main task is to allow an application to access multiple server/databases by offering it a transactional programming model, as shown in Figure 1.13. Essentially, the TP monitor coordinates the commitment of subtransactions following a standard protocol known as **distributed commit**, which we discuss in detail in Section 8.5.

An important observation is that applications wanting to coordinate several subtransactions into a single transaction do not have to implement this coordination themselves. By simply making use of a TP monitor, this coordination is done for them. This is precisely where middleware comes into play: it implements services that are useful for many applications, avoiding that such services have to be reimplemented over and over again by application developers.

Enterprise application integration

As mentioned, the more applications became decoupled from the databases they were built upon, the more evident it became that facilities were needed to integrate applications independently of their databases. In particular, application components should be able to communicate directly with each other and not merely by means of the request/reply behavior that was supported by transaction processing systems.

This need for interapplication communication led to many communication models. The main idea was that existing applications could directly exchange information, as shown in Figure 1.14.

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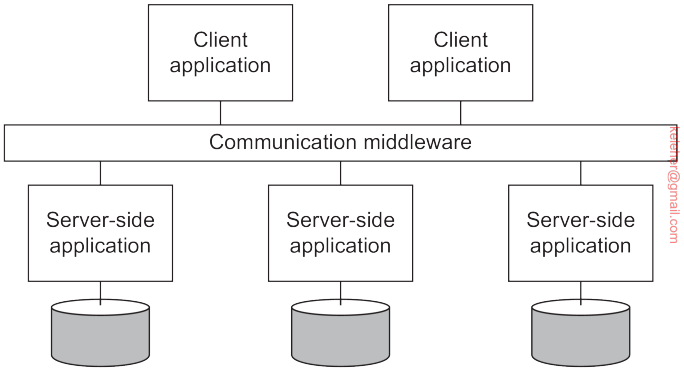


Figure 1.14: Middleware as a communication facilitator in enterprise application integration.

Several types of communication middleware exist. With **remote procedure calls** (**RPC**), an application component can effectively send a request to another application component by doing a local procedure call, which results in the request being packaged as a message and sent to the callee. Likewise, the result will be sent back and returned to the application as the result of the procedure call.

As the popularity of object technology increased, techniques were developed to allow calls to remote objects, leading to what is known as **remote method invocations** (**RMI**). An RMI is essentially the same as an RPC, except that it operates on objects instead of functions.

RPC and RMI have the disadvantage that the caller and callee both need to be up and running at the time of communication. In addition, they need to know exactly how to refer to each other. This tight coupling is often experienced as a serious drawback, and has led to what is known as **message-oriented middleware**, or simply **MOM**. In this case, applications send messages to logical contact points, often described by a subject. Likewise, applications can indicate their interest for a specific type of message, after which the communication middleware will take care that those messages are delivered to those applications. These so-called **publish-subscribe** systems form an important and expanding class of distributed systems.

Note 1.7 (More information: On integrating applications)
Supporting enterprise application integration is an important goal for many middleware products. In general, there are four ways to integrate applications [Hohpe and Woolf, 2004]:

File transfer: The essence of integration through file transfer, is that an application produces a file containing shared data that is subsequently read by

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other applications. The approach is technically simple, making it appealing. The drawback, however, is that there are numerous things that need to be agreed upon:

- File format and layout: text, binary, its structure, and so on. Nowadays, the **extended markup language** (**XML**) has become popular as its files are, in principle, self-describing.
- File management: where are they stored, how are they named, who is responsible for deleting files?
- Update propagation: When an application produces a file, there may be several applications that need to read that file to provide the view of a single coherent system. As a consequence, sometimes separate programs need to be implemented that notify applications of file updates.

Shared database: Many of the problems associated with integration through files are alleviated when using a shared database. All applications will have access to the same data, and often through a high-level database language such as SQL. Furthermore, it is easy to notify applications when changes occur, as triggers are often part of modern databases. There are, however, two major drawbacks. First, there is still a need to design a common data schema, which may be far from trivial if the set of applications that need to be integrated is not completely known in advance. Second, when there are many reads and updates, a shared database can easily become a performance bottleneck.

Remote procedure call: Integration through files or a database implicitly assumes that changes by one application can easily trigger other applications to act. However, practice shows that sometimes small changes should actually trigger many applications to take actions. In such cases, it is not really the change of data that is important, but the execution of a series of actions.

Series of actions are best captured through the execution of a procedure (which may, in turn, lead to all kinds of changes in shared data). To prevent that every application needs to know all the internals of those actions (as implemented by another application), standard encapsulation techniques should be used, as deployed with traditional procedure calls or object invocations. For such situations, an application can best offer a procedure to other applications in the form of a remote procedure call, or RPC. In essence, an RPC allows an application A to make use of the information available only to the application B, without giving A direct access to that information. There are many advantages and disadvantages to remote procedure calls, which are discussed in depth in Chapter 4.

Messaging: A main drawback of RPCs is that caller and callee need to be up and running at the same time in order for the call to succeed. However, in many scenarios, this simultaneous activity is often difficult or impossible

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to guarantee. In such cases, offering a messaging system carrying requests from the application A to perform an action at the application B, is what is needed. The messaging system ensures that eventually the request is delivered, and if needed, that a response is eventually returned as well. Obviously, messaging is not the panacea for application integration: it also introduces problems concerning data formatting and layout, it requires an application to know where to send a message to, there need to be scenarios for dealing with lost messages, and so on. Like RPCs, we will be discussing these issues extensively in Chapter 4.

What these four approaches tell us, is that application integration will generally not be simple. Middleware (in the form of a distributed system), however, can significantly help in integration by providing the right facilities such as support for RPCs or messaging. As said, enterprise application integration is an important target field for many middleware products.

1.3.3 Pervasive systems

The distributed systems discussed so far are largely characterized by their stability: nodes are fixed and have a more or less permanent and high-quality connection to a network. To a certain extent, this stability is realized through the various techniques for achieving distribution transparency. For example, there are many ways how we can create the illusion that only occasionally components may fail. Likewise, there are all kinds of means to hide the actual network location of a node, effectively allowing users and applications to believe that nodes stay put.

However, matters have changed since the introduction of mobile and embedded computing devices, leading to what are generally referred to as **pervasive systems**. As its name suggests, pervasive systems are intended to blend into our environment naturally. Many of their components are *necessarily* spread across multiple computers, making them arguably a type of decentralized system in our view. At the same time, most pervasive systems have many components that are *sufficiently* spread throughout the system, for example, to handle failures and such. In this sense, they are also arguably distributed systems. The seemingly strict separation between decentralized and distributed systems is thus seen to be less strict than one could initially imagine.

What makes them unique, in comparison to the computing and information systems described so far, is that the separation between users and system components is much more blurred. There is often no single dedicated interface, such as a screen/keyboard combination. Instead, a pervasive system is often equipped with many **sensors** that pick up various aspects of a user's behavior. Likewise, it may have a myriad of **actuators** to provide information and feedback, often even purposefully aiming to *steer* behavior.

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Many devices in pervasive systems are characterized by being small, battery-powered, mobile, and having only a wireless connection, although not all these characteristics apply to all devices. These are not necessarily restrictive characteristics, as is illustrated by smartphones [Roussos et al., 2005] and their role in what is now coined as the **Internet of Things** [Mattern and Floerkemeier, 2010; Stankovic, 2014]. Nevertheless, notably, the fact that we often need to deal with the intricacies of wireless and mobile communication, will require special solutions to make a pervasive system as transparent or unobtrusive as possible.

In the following, we make a distinction between three different types of pervasive systems, although there is considerable overlap between the three types: ubiquitous computing systems, mobile systems, and sensor networks. This distinction allows us to focus on different aspects of pervasive systems.

Ubiquitous computing systems

So far, we have been talking about pervasive systems to emphasize that its elements have spread through in many parts of our environment. In a ubiquitous computing system, we go one step further: the system is pervasive and continuously present. The latter means that a user will be continuously interacting with the system, often not even being aware that interaction is taking place. Poslad [2009] describes the core requirements for a **ubiquitous computing system** roughly as follows:

1. (**Distribution**) Devices are networked, distributed, and accessible transparently
2. (**Interaction**) Interaction between users and devices is highly unobtrusive
3. (**Context awareness**) The system is aware of a user's context to optimize interaction
4. (**Autonomy**) Devices operate autonomously without human intervention, and are thus highly self-managed
5. (**Intelligence**) The system as a whole can handle a wide range of dynamic actions and interactions

Let us consider these requirements from a distributed-systems perspective.

Ad. 1: Distribution As mentioned, a ubiquitous computing system is an example of a distributed system: the devices and other computers forming the nodes of a system are simply networked and work together to form the illusion of a single, coherent system. Distribution also comes naturally: there will be devices close to users (such as sensors and actuators), connected to computers hidden from view and perhaps even operating remotely in a

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cloud. Most, if not all, of the requirements regarding distribution transparency mentioned in Section 1.2.2, should therefore hold.

Ad. 2: Interaction When it comes to interaction with users, ubiquitous computing systems differ a lot in comparison to the systems we have been discussing so far. End users play a prominent role in the design of ubiquitous systems, meaning that special attention needs to be paid to how the interaction between users and core system takes place. For ubiquitous computing systems, much of the interaction by humans will be implicit, with an **implicit action** being defined as one "that is not primarily aimed to interact with a computerized system but which such a system understands as input" [Schmidt, 2000]. In other words, a user could be mostly unaware of the fact that input is being provided to a computer system. From a certain perspective, ubiquitous computing can be said to seemingly *hide* interfaces.

A simple example is where the settings of a car's driver's seat, steering wheel, and mirrors are fully personalized. If Bob takes a seat, the system will recognize that it is dealing with Bob and subsequently makes the appropriate adjustments. The same happens when Alice uses the car, while an unknown user will be steered toward making his or her own adjustments (to be remembered for later). This example already illustrates an important role of sensors in ubiquitous computing, namely as input devices that are used to identify a situation (a specific person apparently wanting to drive), whose input analysis leads to actions (making adjustments). In turn, the actions may lead to natural reactions, for example that Bob slightly changes the seat settings. The system will have to take all (implicit and explicit) actions by the user into account and react accordingly.

Ad. 3: Context awareness Reacting to the sensory input, but also the explicit input from users, is more easily said than done. What a ubiquitous computing system needs to do, is to take the **context** in which interactions take place into account. Context awareness also differentiates ubiquitous computing systems from the more traditional systems we have been discussing before, and is described by Dey and Abowd [2000] as "any information that can be used to characterize the situation of entities (i.e., whether a person, place, or object) that are considered relevant to the interaction between a user and an application, including the user and the application themselves." In practice, context is often characterized by location, identity, time, and activity: the *where*, *who*, *when*, and *what*. A system will need to have the necessary (sensory) input to determine one or several of these context types. As discussed by Alegre et al. [2016], developing context-aware systems is difficult, if only for the reason that the notion of context is difficult to grasp.

What is important, from a distributed-systems perspective, is that raw data as collected by various sensors is lifted to a level of abstraction that can be used by applications. A concrete example is detecting where a person is,

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for example in terms of GPS coordinates, and subsequently mapping that information to an actual location, such as the corner of a street, or a specific shop or other known facility. The question is where this processing of sensory input takes place: is all data collected at a central server connected to a database with detailed information on a city, or is it the user's smartphone where the mapping is done? Clearly, there are trade-offs to be considered.

Dey [2010] discusses more general approaches toward building context-aware applications. When it comes to combining flexibility and potential distribution, so-called **shared data spaces** in which processes are decoupled in time and space are attractive, yet as we shall see in later chapters, suffer from scalability problems. A survey on context-awareness and its relation to middleware and distributed systems is provided by Baldauf et al. [2007].

Ad. 4: Autonomy An important aspect of most ubiquitous computing systems is that explicit systems management has been reduced to a minimum. In a ubiquitous computing environment, there is simply no room for a systems administrator to keep everything up and running. As a consequence, the system as a whole should be able to act autonomously, and automatically react to changes. This requires a myriad of techniques, of which several will be discussed throughout this book. To give a few simple examples, think of the following:

Address allocation: In order for networked devices to communicate, they need an IP address. Addresses can be allocated automatically using protocols like the **Dynamic Host Configuration Protocol** (**DHCP**) [Droms, 1997] (which requires a server) or **Zeroconf** [Guttman, 2001].

Adding devices: It should be easy to add devices to an existing system. A step towards automatic configuration is realized by the **Universal Plug and Play protocol** (**UPnP**) [UPnP Forum, 2008]. Using UPnP, devices can discover each other and make sure that they can set up communication channels between them.

Automatic updates: Many devices in a ubiquitous computing system should be able to regularly check through the Internet if their software should be updated. If so, they can download new versions of their components and ideally continue where they left off.

Admittedly, these are simple examples, but the picture should be clear that manual intervention is to be kept to a minimum. We will be discussing many techniques related to self-management in detail throughout the book.

Ad. 5: Intelligence Finally, Poslad [2009] mentions that ubiquitous computing systems often use methods and techniques from the field of artificial

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intelligence. What this means, is that often a wide range of advanced algorithms and models need to be deployed to handle incomplete input, quickly react to a changing environment, handle unexpected events, and so on. The extent to which this can or should be done in a distributed fashion is crucial from the perspective of distributed systems. Unfortunately, distributed solutions for many problems in the field of artificial intelligence are yet to be found, meaning that there may be a natural tension between the first requirement of networked and distributed devices, and advanced distributed information processing.

Mobile computing systems

As mentioned, mobility often forms an important component of pervasive systems, and many, if not all aspects that we have just discussed also apply to **mobile computing**. There are several issues that set mobile computing aside to pervasive systems in general (see also Adelstein et al. [2005] and Tarkoma and Kangasharju [2009]).

First, the devices that form part of a (distributed) mobile system may vary widely. Typically, mobile computing is done with devices such as smartphones and tablet computers. However, entirely different types of devices are now using the Internet Protocol (IP) to communicate, placing mobile computing in a different perspective. Such devices include remote controls, pagers, active badges, car equipment, various GPS-enabled devices, and so on. A characteristic feature of all these devices is that they use wireless communication. Mobile implies wireless, so it seems (although there are exceptions to the rules).

Second, in mobile computing, the location of a device is assumed to change over time. A changing location has its effects on many issues. For example, if the location of a device changes regularly, so will perhaps the services that are locally available. As a consequence, we may need to pay special attention to dynamically discovering services, but also letting services announce their presence. In a similar vein, we often also want to know where a device actually is. This may mean that we need to know the actual geographical coordinates of a device such as in tracking and tracing applications, but it may also require that we can simply detect its network position (as in mobile IP [Perkins, 2010; Perkins et al., 2011].

Changing locations may also have a profound effect on communication. For some time, researchers in mobile computing have been concentrating on what are known as **mobile ad hoc networks**, also known as **MANETs**. The basic idea was that a group of local mobile computers would jointly set up a local, wireless network and to subsequently share resources and services. The idea never really became popular. Along the same lines, there are researchers who believe that end users are willing to share their local resources for another user's compute, storage, or communication requirements (see, e.g., Ferrer

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et al. [2019]). Practice has shown over and over again that voluntarily making resources available is not something users are willing to do, even if they have resources in abundance. The effect is that in the case of mobile computing, we generally see single mobile devices setting up connections to stationary servers. Changing locations then simply means that those connections need to be handed over by routers on the path from the mobile device to the server. Mobile computing is then brought back to its essence: a mobile device connected to a server (and nothing else). In practice, this means that mobile computing is all about mobile devices making use of cloud-based services, as sketched in Figure 1.15(a).

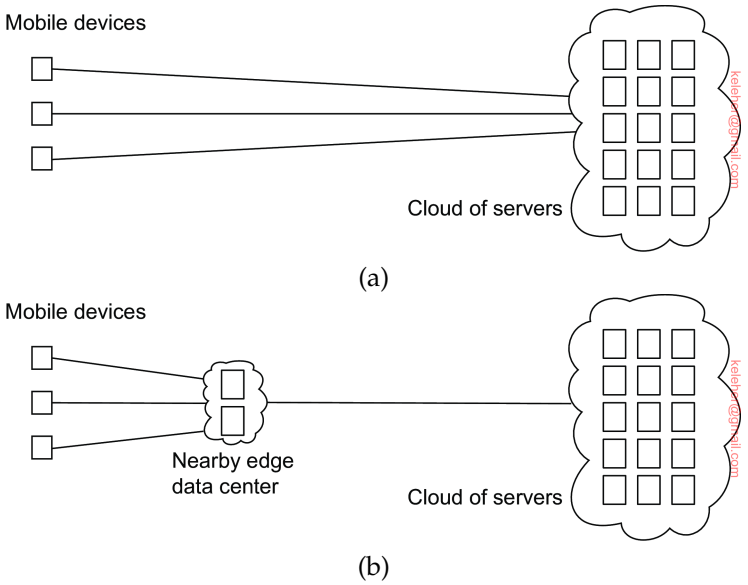


Figure 1.15: (a) Mobile Cloud Computing versus (b) Mobile Edge Computing.

Despite the conceptual simplicity of this model of mobile computing, the mere fact that so many devices make use of remote services has led to what is known as **Mobile Edge Computing**, or simply **MEC** [Abbas et al., 2018], in contrast to **Mobile Cloud Computing** (**MCC**). As we shall discuss further in Chapter 2, (mobile) edge computing, as sketched in Figure 1.15(b), is becoming increasingly important in those cases where latency, but also computational issues play a role for the mobile device. Typical example applications that require short latencies *and* computational resources include augmented reality, interactive gaming, real-time sports monitoring, and various health applications [Dimou et al., 2022]. In these examples, the combination of monitoring, analyses, and immediate feedback in general make it difficult to rely on servers that may be placed thousands of miles from the mobile devices.

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Sensor networks

Our last example of pervasive systems is **sensor networks**. These networks in many cases form part of the enabling technology for pervasiveness, and we see that many solutions for sensor networks return in pervasive applications. What makes sensor networks interesting from a distributed system's perspective is that they are more than just a collection of input devices. Instead, as we shall see, sensor nodes often collaborate to process the sensed data efficiently in an application-specific manner, making them very different from, for example, traditional computer networks. Akyildiz et al. [2002] and Akyildiz et al. [2005] provide an overview from a networking perspective. A more systems-oriented introduction to sensor networks is given by Zhao and Guibas [2004], but also [Hahmy, 2021] will show to be useful.

A sensor network generally consists of tens to hundreds or thousands of relatively small nodes, each equipped with one or more sensing devices. In addition, nodes can often act as actuators [Akyildiz and Kasimoglu, 2004], a typical example being the automatic activation of sprinklers when a fire has been detected. Many sensor networks use wireless communication, and the nodes are often battery powered. Their limited resources, restricted communication capabilities, and constrained power consumption demand that efficiency is high on the list of design criteria.

When zooming into an individual node, we see that, conceptually, they do not differ a lot from "normal" computers: above the hardware there is a software layer akin to what traditional operating systems offer, including low-level network access, access to sensors and actuators, memory management, and so on. Normally, support for specific services is included, such as localization, local storage (think of additional flash devices), and convenient communication facilities such as messaging and routing. However, similar to other networked computer systems, additional support is needed to effectively deploy sensor network *applications*. In distributed systems, this takes the form of middleware. For sensor networks, we can, in principle, follow a similar approach in those cases that sensor nodes are sufficiently powerful and that energy consumption is not a hindrance for running a more elaborate software stack. Various approaches are possible (see also [Zhang et al., 2021b]).

From a programming perspective, and extensively surveyed by Mottola and Picco [2011], it is important to take the scope of communication primitives into account. This scope can vary between addressing the physical neighborhood of a node, and providing primitives for systemwide communication. In addition, it may also be possible to address a specific group of nodes. Likewise, computations may be restricted to an individual node, a group of nodes, or affect all nodes. To illustrate, Welsh and Mainland [2004] use so-called **abstract regions** allowing a node to identify a neighborhood from where it can, for example, gather information:

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```
1    region  = k_nearest_region.create(8);
2    reading = get_sensor_reading();
3    region.putvar(reading_key, reading);
4    max_id  = region.reduce(OP_MAXID, reading_key);
```

In line 1, a node first creates a region of its eight nearest neighbors, after which it fetches a value from its sensor(s). This reading is subsequently written to the previously defined region to be defined using the key `reading_key`. In line 4, the node checks whose sensor reading in the defined region was the largest, which is returned in the variable `max_id`.

When considering that sensor networks produce data, one can also focus on the data-access model. This can be done directly by sending messages to and between nodes, or either moving code between nodes to locally access data. More advanced is to make remote data directly accessible, as if variables and such were available in a shared data space. Finally, and also quite popular, is to let the sensor network provide a view of a single database. Such a view is easy to understand when realizing that many sensor networks are deployed for measurement and surveillance applications [Bonnet et al., 2002]. In these cases, an operator would like to extract information from (a part of) the network by simply issuing queries such as "What is the northbound traffic load on highway 1 at Santa Cruz?" Such queries resemble those of traditional databases. In this case, the answer will probably need to be provided through collaboration of many sensors along highway 1, while leaving other sensors untouched.

To organize a sensor network as a distributed database, there are essentially two extremes, as shown in Figure 1.16. First, sensors do not cooperate but simply send their data to a centralized database located at the operator's site. The other extreme is to forward queries to relevant sensors and to let each compute an answer, requiring the operator to aggregate the responses.

Neither of these solutions is very attractive. The first one requires that sensors send all their measured data through the network, which may waste network resources and energy. The second solution may also be wasteful, as it discards the aggregation capabilities of sensors, which would allow much fewer data to be returned to the operator. What is needed are facilities for **in-network data processing**, similar to the previous example of abstract regions.

In-network processing can be done in numerous ways. One obvious way is to forward a query to all sensor nodes along a tree encompassing all nodes and to subsequently aggregate the results as they are propagated back to the root, where the initiator is located. Aggregation will take place where two or more branches of the tree come together. As simple as this scheme may sound, it introduces difficult questions:

- How do we (dynamically) set up an efficient tree in a sensor network?
- How does aggregation of results take place? Can it be controlled?

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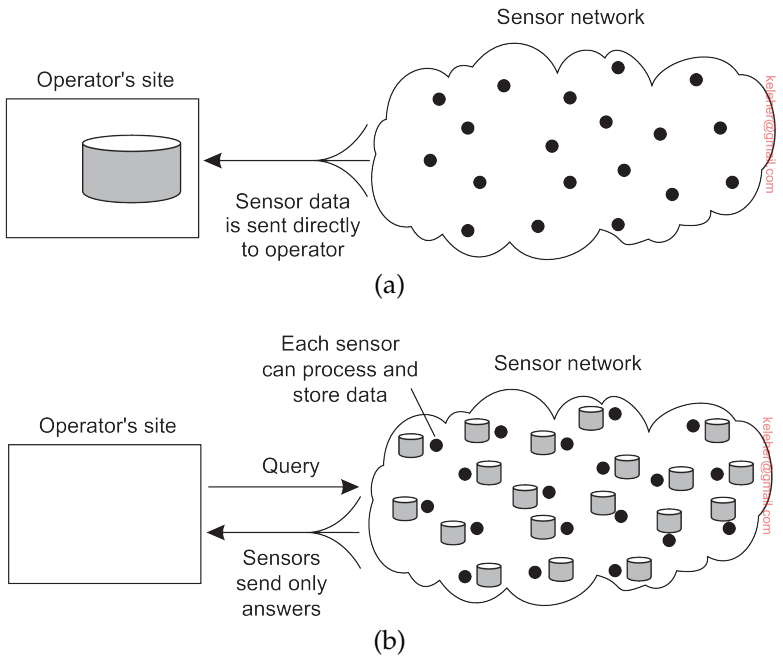


Figure 1.16: Organizing a sensor network database, while storing and processing data (a) only at the operator's site or (b) only at the sensors.

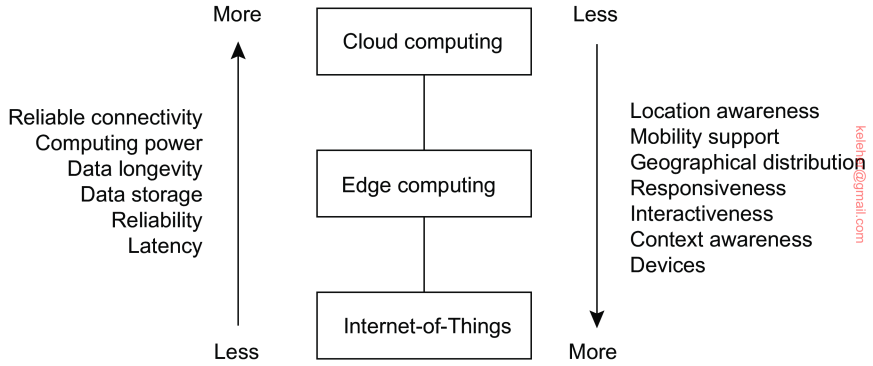
- What happens when network links fail?

These questions have been partly addressed in TinyDB, which implements a declarative (database) interface to wireless sensor networks [Madden et al., 2005]. In essence, TinyDB can use any tree-based routing algorithm. An intermediate node will collect and aggregate the results from its children, along with its own findings, and send that toward the root. To make matters efficient, queries span a period of time, allowing for careful scheduling of operations so that network resources and energy are optimally consumed.

However, when queries can be initiated from different points in the network, using single-rooted trees such as in TinyDB may not be efficient enough. As an alternative, sensor networks may be equipped with special nodes where results are forwarded to, as well as the queries related to those results. To give a simple example, queries and results related to temperature readings may be collected at a different location than those related to humidity measurements. This approach corresponds directly to the notion of publish-subscribe systems.

The interesting aspect of sensor networks, as discussed along these lines, is that we really need to concentrate on the organization of sensor *nodes*, and not the sensors themselves. Likewise, many sensor nodes will be equipped with **actuators**, i.e., devices that directly influence an environment. A typical

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Figure 1.17: A hierarchical view from clouds to devices (adapted from Yousefpour et al. [2019]).

actuator is one that controls the temperature in a room, or switches devices on or off. By viewing and organizing the network as a distributed system, an operator is provided with a higher level of abstraction to monitor and control a situation.

Cloud, edge, things

As may have become clear by now, distributed systems span a huge range of different networked computer systems. Many of such systems operate in a setting in which the various computers are connected through a local-area network. Yet with the growth of the Internet-of-Things and the connectivity with remote services offered through cloud-based systems, new organizations across wide-area networks are emerging. Figure 1.17 presents this more hierarchical approach.

Typically, higher up the hierarchy we see that typical qualities of distributed systems improve: they become more reliable, have more capacity, and, in general, perform better. Lower in the hierarchy, we see that location-related aspects are easier facilitated, as well as performance qualities related to latencies. At the same time, the lower parts show in an increase in the number of devices and computers, whereas higher up, the number of computers becomes less.

1.4 Pitfalls

It should be clear by now that developing a distributed system is a formidable task. As we will see many times throughout this book, there are so many issues to consider, while it seems that only complexity can be the result.

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Nevertheless, by following several design principles, distributed systems can be developed that strongly adhere to the goals we set out in this chapter.

Distributed systems differ from traditional software because components are dispersed across a network. Not taking this dispersion into account during design time is what makes so many systems needlessly complex and results in flaws that need to be patched later on. Peter Deutsch, at the time working at Sun Microsystems, formulated these flaws as the following false assumptions that many make when developing a distributed application for the first time:

- The network is reliable
- The network is secure
- The network is homogeneous
- The topology does not change
- Latency is zero
- Bandwidth is infinite
- Transport cost is zero
- There is one administrator

Note how these assumptions relate to properties that are unique to distributed systems: reliability, security, heterogeneity, and topology of the network; latency and bandwidth; transport costs; and finally administrative domains. When developing nondistributed applications, most of these issues will most likely not show up.

Most of the principles we discuss in this book relate immediately to these assumptions. In all cases, we will be discussing solutions to problems that are caused by the fact that one or more assumptions are false. For example, reliable networks simply do not exist and lead to the impossibility of achieving failure transparency. We devote an entire chapter to deal with the fact that networked communication is inherently insecure. We have already argued that distributed systems need to be open and take heterogeneity into account. Likewise, when discussing replication for solving scalability problems, we are essentially tackling latency and bandwidth problems. We will also touch upon management issues at various points throughout this book.

1.5 Summary

A distributed system is a collection of networked computer systems in which processes and resources are spread across different computers. We make a distinction between *sufficiently* and *necessarily* spread, where the latter relates to decentralized systems. This distinction is important to make, as spreading processes and resources cannot be considered to be a goal by itself. Instead,

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most choices for coming to a distributed system come from the need to improve the performance of a single computer system in terms of, for example, reliability, scalability, and efficiency. However, considering that most centralized systems are still much easier to manage and maintain, one should think twice before deciding to spread processes and resources. There are also cases when there is simply no choice, for example when connecting systems belonging to different organizations, or when computers simply operate from different locations (as in mobile computing).

Design goals for distributed systems include sharing resources and ensuring openness. Increasingly important is designing secure distributed systems. In addition, designers aim at hiding many of the intricacies related to the distribution of processes, data, and control. However, this distribution transparency not only comes at a performance price, in practical situations it can never be fully achieved. The fact that trade-offs need to be made between achieving various forms of distribution transparency is inherent to the design of distributed systems, and can easily complicate their understanding. One specific difficult design goal that does not always blend well with achieving distribution transparency is scalability. This is particularly true for geographical scalability, in which case hiding latencies and bandwidth restrictions can turn out to be difficult. Likewise, administrative scalability, by which a system is designed to span multiple administrative domains, may easily conflict with goals for achieving distribution transparency.

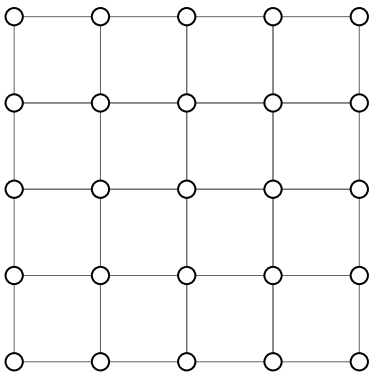
Different types of distributed systems exist which can be classified as being oriented toward supporting computations, information processing, and pervasiveness. Distributed computing systems are typically deployed for high-performance applications, often originating from the field of parallel computing. A field that emerged from parallel processing was initially grid computing with a strong focus on worldwide sharing of resources, in turn leading to what is now known as cloud computing. Cloud computing goes beyond high-performance computing and also supports distributed systems found in traditional office environments, where we see databases playing an important role. Typically, transaction processing systems are deployed in these environments. Finally, an emerging class of distributed systems is where components are small, the system is composed in an ad hoc fashion, but most of all is no longer managed through a system administrator. This last class is typically represented by pervasive computing environments, including mobile-computing systems as well as sensor-rich environments.

Matters are further complicated by the fact that many developers initially make assumptions about the underlying network that are fundamentally wrong. Later, when assumptions are dropped, it may turn out to be difficult to mask unwanted behavior. A typical example is assuming that network latency is not significant. Other pitfalls include assuming that the network is reliable, static, secure, and homogeneous.

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02

ARCHITECTURES



Distributed systems are often complex pieces of software, of which the components are by definition dispersed across multiple machines. To master their complexity, it is crucial that these systems be properly organized. There are different ways on how to view the organization of a distributed system, but an obvious one is to make a distinction between, on the one hand, the logical organization of the collection of software components, and on the other hand the actual physical realization.

The organization of distributed systems is mostly about the software components that constitute the system. These **software architectures** tell us how the various software components are to be organized and how they should interact. In this chapter, we will first pay attention to some commonly applied architectural styles toward organizing (distributed) computer systems.

An important goal of distributed systems is to separate applications from underlying platforms by providing a so-called middleware layer. Adopting such a layer is an important architectural decision, and its main purpose is to provide distribution transparency. However, trade-offs need to be made to achieve transparency, which has led to various techniques to adjust the middleware to the needs of the applications that make use of it. We discuss some of the more commonly applied techniques, as they affect the organization of the middleware itself.

The actual realization of a distributed system requires that we instantiate and place software components on real machines. There are many choices that can be made in doing so. The final instantiation of a software architecture is also referred to as a **system architecture**. In this chapter, we will look into traditional centralized architectures in which a single server implements most of the software components (and thus functionality), while remote clients can access that server using simple communication means. In addition, we consider decentralized peer-to-peer architectures in which all nodes more or less play equal roles. Many real-world distributed systems are often organized in a hybrid fashion, combining elements from centralized and decentralized architectures. We discuss several examples that illustrate the complexity of many real-world distributed systems.

2.1 Architectural styles

We start our discussion on architectures by first considering the logical organization of a distributed system into software components, also referred to as its **software architecture** [Bass et al., 2021; Richards and Ford, 2020]. Research on software architectures has matured considerably, and it is now commonly accepted that designing or adopting an architecture is crucial for the successful development of large software systems.

For our discussion, the notion of an **architectural style** is important. Such a style is formulated in terms of components, the way that components are

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connected to each other, the data exchanged between components, and finally, how these elements are jointly configured into a system. A **component** is a modular unit with well-defined required and provided **interfaces** that is *replaceable* within its environment [OMG, 2004]. That a component can be replaced, and, in particular, while a system continues to operate, is important. This is because often it is not an option to shut down a system for maintenance. At best, only parts of it may be put temporarily out of order. Replacing a component can be done only if its interfaces remain untouched. In practice, we see that replacing or updating a component means that a part of a system (such as a server), runs a regular update and switches to the refreshed components once their installation has finished. Special measures may need to be taken when a part of the distributed system does need to be restarted to let the updates take effect. Such measures may include having replicated standbys that take over while the partial restart is taking place.

A somewhat more difficult concept to grasp is that of a **connector**, which is generally described as a mechanism that mediates communication, coordination, or cooperation among components [Bass et al., 2021]. For example, a connector can be formed by the facilities for (remote) procedure calls, message passing, or streaming data. In other words, a connector allows for the flow of control and data between components.

Using components and connectors, we can come to various configurations, which, in turn, have been classified into architectural styles. Several styles have by now been identified, of which the most important ones for distributed systems are:

- Layered architectures
- Service-oriented architectures
- Publish-subscribe architectures

In the following, we discuss each of these styles separately. We note in advance that in most real-world distributed systems, many styles are combined. Notably, following an approach by which a system is subdivided into several (logical) layers is such a universal principle that it is generally combined with most other architectural styles.

2.1.1 Layered architectures

The basic idea for the layered style is simple: components are organized in a **layered fashion** where a component at layer L_j can make a **downcall** to a component at a lower-level layer L_i (with $i < j$) and generally expects a response. Only in exceptional cases will an **upcall** be made to a higher-level component. The three common cases are shown in Figure 2.1.

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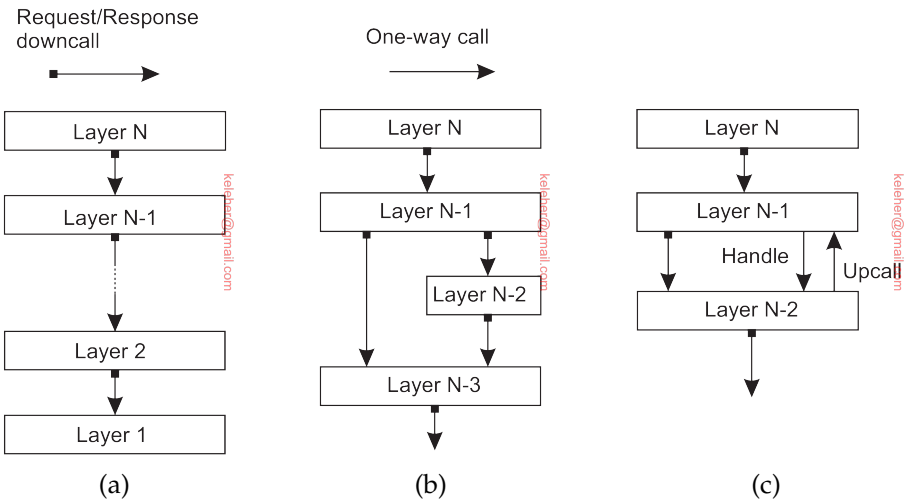


Figure 2.1: (a) Pure layered organization. (b) Mixed layered organization. (c) Layered organization with upcalls (adopted from [Krakowiak, 2009]).

Figure 2.1(a) shows a standard organization in which only downcalls to the next lower layer are made. This organization is commonly deployed in the case of network communication.

In many situations, we also encounter the organization shown in Figure 2.1(b). Consider, for example, an application A that makes use of a library L_{OS} to interface to an operating system. At the same time, the application uses a specialized mathematical library $\mathsf{L}_{\mathsf{math}}$ that has been implemented by also making use of L_{OS}. In this case, referring to Figure 2.1(b), A is implemented at layer $N-1$, $\mathsf{L}_{\mathsf{math}}$ at layer $N-2$, and L_{OS} which is common to both of them, at layer $N-3$.

Finally, a special situation is shown in Figure 2.1(c). In some cases, it is convenient to have a lower layer do an upcall to its next higher layer. A typical example is when an operating system signals the occurrence of an event, to which end it calls a user-defined operation for which an application had previously passed a reference (typically referred to as a **handle**).

Layered communication protocols

A well-known and ubiquitously applied layered architecture is that of **communication-protocol stacks**. We will concentrate here on the global picture only and defer a detailed discussion to Section 4.1.1.

In communication-protocol stacks, each layer implements one or several, **communication services** allowing data to be sent from a destination to one or several targets. To this end, each layer offers an **interface** specifying the

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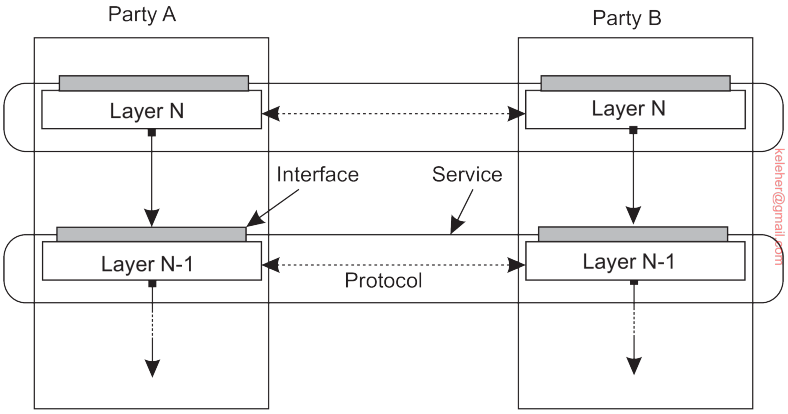


Figure 2.2: A layered communication-protocol stack, showing the difference between a service, its interface, and the protocol it deploys.

functions that can be called. In principle, the interface should completely hide the actual implementation of a service. Another important concept in the case of communication is that of a **(communication) protocol**, which describes the rules that parties will follow to exchange information. It is important to understand the difference between a service offered by a layer, the interface by which that service is made available, and the protocol that a layer implements to establish communication. This distinction is shown in Figure 2.2.

To make this distinction clear, consider a reliable, connection-oriented service, which is provided by many communication systems. In this case, a communicating party first needs to set up a connection to another party before the two can send and receive messages. Being reliable means that strong guarantees will be given that sent messages will indeed be delivered to the other side, even when there is a high risk that messages may be lost (as, for example, may be the case when using a wireless medium). In addition, such services generally also ensure that messages are delivered in the same order as that they were sent.

This kind of service is realized in the Internet by the **Transmission Control Protocol** (**TCP**). The protocol specifies which messages are to be exchanged for setting up or tearing down a connection, what needs to be done to preserve the ordering of transferred data, and what both parties need to do to detect and correct data that was lost during transmission. The service is made available in the form of a relatively simple programming interface, containing calls to set up a connection, send and receive messages, and to tear down the connection again. In fact, there are different interfaces available, often dependent on the operating system or programming language used. Likewise, there are many implementations of the protocol and its interfaces. (All the gory details can be found in [Stevens, 1994; Wright and Stevens, 1995].)

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Note 2.1 (Example: Two communicating parties)
To make this distinction between service, interface, and protocol more concrete, consider the following two communicating parties, also known as a **client** and a **server**, respectively, expressed in Python (note that some code has been removed for clarity).

```
1  from socket  import *
2
3  s = socket(AF_INET, SOCK_STREAM)
4  (conn, addr) = s.accept()  # returns new socket and addr. client
5  while True:                # forever
6    data = conn.recv(1024)   # receive data from client
7    if not data: break       # stop if client stopped
8    msg = data.decode()+"*"  # process the incoming data into a response
9    conn.send(msg.encode())  # return the response
10 conn.close()               # close the connection
```

(a) A simple server

```
1  from socket  import *
2
3  s = socket(AF_INET, SOCK_STREAM)
4  s.connect((HOST, PORT)) # connect to server (block until accepted)
5  msg = "Hello World"     # compose a message
6  s.send(msg.encode())    # send the message
7  data = s.recv(1024)     # receive the response
8  print(data.decode())    # print the result
9  s.close()               # close the connection
```

(b) A client

Figure 2.3: Two communicating parties.

In this example, a server is created that makes use of a **connection-oriented service** as offered by the `socket` library available in Python. This service allows two communicating parties to reliably send and receive data over a connection. The main functions available in its interface are:

- `socket()`: to create an object representing the connection
- `accept()`: a blocking call to wait for incoming connection requests; if successful, the call returns a new socket for a separate connection
- `connect()`: to set up a connection to a specified party
- `close()`: to tear down a connection
- `send()`, `recv()`: to send and receive data over a connection, respectively

The combination of constants `AF_INET` and `SOCK_STREAM` is used to specify that the TCP protocol should be used in the communication between the two parties. These two constants can be seen as part of the interface, whereas making use of TCP is part of the offered service. *How* TCP is implemented, or for that matter, any part of the communication service, is hidden completely from the applications.

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Finally, also note that these two programs implicitly adhere to an application-level protocol: apparently, if the client sends some data, the server will return it. Indeed, it operates as an echo server, where the server adds an asterisk to the data sent by the client.

Application layering

Let us now turn our attention to the logical layering of applications. Considering that a large class of distributed applications is targeted toward supporting users or applications access to databases, many people have advocated a distinction between three logical levels, essentially following a layered architectural style:

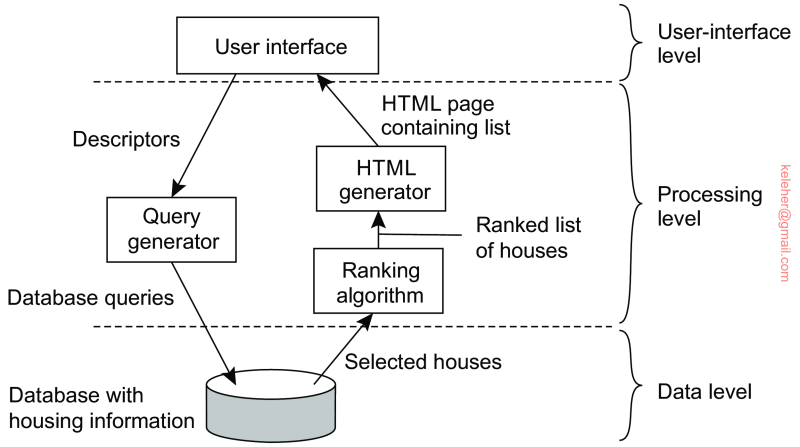
- The application-interface level
- The processing level
- The data level

In line with this layering, we see that applications can often be constructed from roughly three different pieces: a part that handles interaction with a user or some external application, a part that operates on a database or file system, and a middle part that generally contains the core functionality of the application. This middle part is logically placed at the processing level. In contrast to user interfaces and databases, there are not many aspects common to the processing level. Therefore, we shall give several examples to make this level clearer.

As a first example, consider a simple Internet search engine, for example one dedicated to buying houses. Such search engines appear as seemingly simple Websites through which someone can provide descriptors such as a city or region, a price range, the type of house, etc. The back end consists of a huge database of houses currently for sale. The processing layer does nothing else but transform the provided descriptors into a collection of database queries, retrieves the answers and post-processes these answers by, for example, ranking the output by relevance and subsequently generating an HTML page. Figure 2.4 shows this organization.

As another example, consider the organization of this book's Website, in particular, the interface that allows someone to get a personalized digital copy of the book in PDF. In this case, the interface consists of a WordPress-based Web server that merely collects the user's e-mail address (and some information on exactly which version of the book is being requested). This information is internally appended to a file requests.txt. The data layer is simple: it merely consists of a collection of LaTeX files and figures that jointly constitute the entire book.

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Figure 2.4: The simplified organization of an Internet search engine for housing, into three different layers.

Making a personalized copy consists of embedding the user's e-mail address into each of the figures. To this end, once every five minutes a separate process is started that takes the list of requests and one-by-one adds the requester's e-mail address into each bitmapped figure, generates a fresh copy of the book, stores the generated PDF in a special location (accessible through a unique URL), and sends the requester an e-mail that the copy is available for download. This process continues until all requests have been handled. In this example, we thus see that the processing layer outweighs the data layer or the application-interface layer in terms of computational efforts and actions to take.

2.1.2 Service-oriented architectures

Although the layered architectural style is popular, one of its major drawbacks is the often strong dependency between different layers. Good examples where these potential dependencies have been carefully considered are found in designing communication protocol stacks. Bad examples include applications that have essentially been designed and developed as compositions of existing components without much concern for the stability of interfaces or the components themselves, let alone the overlap of functionality between different components. (A compelling example is given by Kucharski [2020], who describes the dependency on a simple component that pads a given string with zeroes or spaces. The author withdrew the component from the NPM library, leaving thousands of programs affected.)

Such direct dependencies to specific components have led to an architectural style reflecting a more loose organization into a collection of separate,

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independent entities. Each entity encapsulates a service. Whether they are called **services**, **objects**, or **microservices**, they have in common that the service is executed as a separate process (or thread). Of course, running separate entities does not necessarily lower dependencies in comparison to a layered architectural style.

Object-based architectural style

Taking the object-based approach as an example, we have a logical organization as shown in Figure 2.5. In essence, each object corresponds to what we have defined as a component, and these components are connected through a procedure-call mechanism. In the case of distributed systems, a procedure call can also take place over a network, that is, the calling object need not be executed on the same machine as the called object. In fact, *where* exactly the called object is located can be transparent to the caller: the called object may equally well run as a separate process on the same machine.

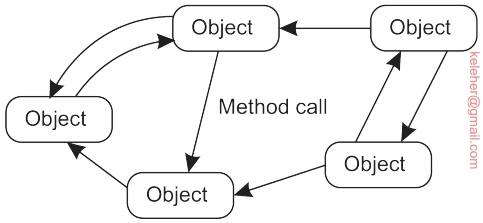


Figure 2.5: An object-based architectural style.

Object-based architectures are attractive because they provide a natural way of **encapsulating** data (called an object's **state**) and the operations that can be performed on that data (which are referred to as an object's **methods**) into a single entity. The **interface** offered by an object conceals implementation details, essentially meaning that we, in principle, can consider an object completely independent of its environment. As with components, this also means that if the interface is clearly defined and left otherwise untouched, an object should be replaceable with one having the same interface.

This separation between interfaces and the objects implementing these interfaces allows us to place an interface at one machine, while the object itself resides on another machine. This organization, which is shown in Figure 2.6 is commonly referred to as a **distributed object**, or every so often a **remote object**.

When a client **binds** to a distributed object, an implementation of the object's interface, called a **proxy**, is then loaded into the client's address space. A proxy is analogous to a so-called **client stub** in RPC systems. The only thing it does is pack method invocations into messages and unpack reply messages to return the result of the method invocation to the client. The actual object

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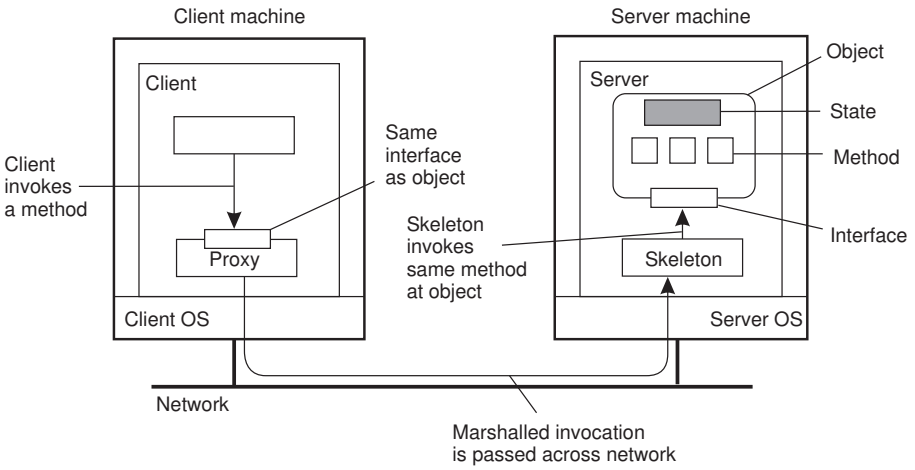


Figure 2.6: Common organization of a remote object with client-side proxy.

resides at a server machine, where it offers the same interface as it does on the client machine. Incoming invocation requests are first passed to a server stub, which unpacks them to make method invocations at the object's interface at the server. The server stub is also responsible for packing return values into a message, and forwarding these reply messages to the client-side proxy.

The server-side stub is often referred to as a **skeleton** as it provides the bare means for letting the server middleware access the user-defined objects. In practice, it often contains incomplete code in the form of a language-specific class that needs to be further specialized by the developer.

A characteristic, but somewhat counterintuitive, feature of most distributed objects is that their state is *not* distributed: it resides at a single machine. Only the interfaces implemented by the object are made available on other machines. Such objects are also referred to as **remote objects**. In a general distributed object, the state itself may be physically distributed across multiple machines, but this distribution is also hidden from clients behind the object's interfaces.

Microservice architectural style

One could argue that object-based architectures form the foundation of encapsulating services into independent units. **Encapsulation** is the keyword here: the service as a whole is realized as a self-contained entity, although it can possibly make use of other services. By clearly separating various services such that they can operate independently, we are paving the road toward **service-oriented architectures**, generally abbreviated as **SOA**s.

Stimulated by object-oriented designs and inspired by the Unix approach in which many, many small and mutually independent programs can be easily

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composed to form larger programs, software architects have been working on what are called **microservices**. Essential is that each microservice runs as a separate (network) process. The implementation of a microservice could be in the form of a remote object, but this is not a requirement. Furthermore, despite that people speak of *micro*services, there is no common agreement on what the size of such a service should be. Most important, however, is that a microservice truly represents a separate, independent service. In other words, modularization is key to designing microservices [Wolff, 2017].

Nevertheless, size does matter. By already stating that microservices run as separate networked processes, we are also given a choice *where* to place a microservice. As we shall see later in this chapter, in the advent of edge and fog infrastructures, discussions have started on the **orchestration** of deploying distributed applications across different layers. In other words, where do we place what.

Coarse-grained service composition

In a service-oriented architecture, a distributed application or system is essentially constructed as a composition of many services. A difference (although not strict) with microservices is that not all of these services may belong to the same administrative organization. We already came across this phenomenon when discussing cloud computing: it may very well be that an organization running its business application makes use of storage services offered by a cloud provider. These storage services are logically completely encapsulated into a single unit, of which an interface is made available to customers.

Of course, storage is a rather basic service, but more sophisticated situations easily come to mind. Consider, for example, a Web shop selling goods such as e-books. A simple implementation, following the application layering we discussed previously, may consist of an application for processing orders, which, in turn, operates on a local database containing the e-books. Order processing typically involves selecting items, registering and checking the delivery channel (perhaps by making use of e-mail), but also making sure that a payment takes place. The latter can be handled by a separate service, run by a different organization, to which a purchasing customer is redirected for the payment, after which the e-book organization is notified so that it can complete the transaction. This example also illustrates that where microservices are considered to be relatively small, a general service may be expected to be relatively large. In fact, it is not uncommon to implement a service as a collection of microservices.

In this way, we see that the problem of developing a distributed system is partly one of *service composition*, and making sure that those services operate in harmony. Indeed, this problem is completely analogous to the enterprise application integration issues discussed in Section 1.3.2. Crucial is, and remains, that each service offers a well-defined (programming) interface. In

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practice, this also means that each service offers *its own* interface, in turn, possibly making the composition of services far from trivial.

Resource-based architectures

As an increasing number of services became available over the Web and the development of distributed systems through service composition became more important, researchers started to rethink the architecture of mostly Web-based distributed systems. One of the problems with service composition is that connecting various components can easily turn into an integration nightmare.

As an alternative, one can also view a distributed system as a huge collection of resources that are individually managed by components. Resources may be added or removed by (remote) applications, and likewise can be retrieved or modified. This approach has now been widely adopted for the Web and is known as **Representational State Transfer** (**REST**) [Fielding, 2000]. There are four key characteristics of what are known as **RESTful architectures** [Pautasso et al., 2008]:

1. Resources are identified through a single naming scheme
2. All services offer the same interface, consisting of at most four operations, as shown in Figure 2.7
3. Messages sent to or from a service are fully self-described
4. After executing an operation at a service, that component forgets everything about the caller

The last property is also referred to as a **stateless execution**, a concept to which we return in Chapter 3.

Operation	Description
PUT	Modify a resource by transferring a new state
POST	Create a new resource
GET	Retrieve the state of a resource in some representation
DELETE	Delete a resource

Figure 2.7: The four operations available in RESTful architectures.

To illustrate how RESTful can work in practice, consider a cloud storage service, such as Amazon's **Simple Storage Service** (**Amazon S3**). Amazon S3, described in [Murty, 2008] and more recently in [Culkin and Zazon, 2022], supports two resources: *objects*, which are essentially the equivalent of files, and *buckets*, the equivalent of directories. There is no concept of placing buckets into buckets. An object named `ObjectName` contained in a bucket `BucketName` is referred to by the following **Uniform Resource Identifier** (**URI**):

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```
https://s3.amazonaws.com/BucketName/ObjectName
```

To create a bucket, or an object for that matter, an application would essentially send a `PUT` request with the URI of the bucket/object. In principle, the protocol that is used with the service is HTTP. In other words, it is just another HTTP request, which will subsequently be correctly interpreted by S3. If the bucket or object already exists, an HTTP error message is returned.

Similarly, to know which objects are contained in a bucket, an application would send a `GET` request with the URI of that bucket. S3 will return a list of object names, again as an ordinary HTTP response.

The RESTful architecture has become popular because of its simplicity. Pautasso et al. [2008] have compared RESTful services to service-specific interfaces, and, as to be expected, they both have their advantages and disadvantages. In particular, the simplicity of RESTful architectures can easily prohibit easy solutions to intricate communication schemes. One example is where distributed transactions are needed, which generally requires that services keep track of the state of execution. On the other hand, there are many examples in which RESTful architectures match perfectly a simple integration scheme of services, yet where the myriad of service interfaces will complicate matters.

Note 2.2 (Advanced: On interfaces)
Clearly, a service cannot be made easier or more difficult just because of the particular interface it offers. A service offers functionality, and at best the way that the service is accessed is determined by the interface. Indeed, one could argue that the discussion on RESTful versus service-specific interfaces is much about access transparency. To better appreciate why so many people are paying attention to this issue, let us zoom in on the **Amazon S3** service, which offers a REST interface as well as a more traditional interface (referred to as the SOAP interface). The fact that the latter has been deprecated out says a lot.

Bucket operations	Object operations
ListAllMyBuckets	PutObjectInline
CreateBucket	PutObject
DeleteBucket	CopyObject
ListBucket	GetObject
GetBucketAccessControlPolicy	GetObjectExtended
SetBucketAccessControlPolicy	DeleteObject
GetBucketLoggingStatus	GetObjectAccessControlPolicy
SetBucketLoggingStatus	SetObjectAccessControlPolicy

Figure 2.8: The operations in Amazon's S3 SOAP interface, by now deprecated.

The SOAP interface consists of approximately 16 operations, listed in Figure 2.8. However, if we were to access Amazon S3 using the Python boto3 library, we would have more than 100 operations available. In contrast, the REST interface

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offers only very few operations, essentially those listed in Figure 2.7. Where do these differences come from? The answer is, of course, in the parameter space. In the case of RESTful architectures, an application will need to provide all that it wants through the parameters it passes by one of the operations. In Amazon's SOAP interface, the number of parameters per operation is generally limited, and this is certainly the case if we were to use the Python `boto3` library.

Sticking to principles (so that we can avoid the intricacies of real code), suppose that we have an interface `bucket` that offers an operation `create`, requiring an input string such as `mybucket`, for creating a bucket with name "mybucket." Normally, the operation would be called roughly as follows:

```
import bucket
bucket.create("mybucket")
```

However, in a RESTful architecture, the call would need to be essentially encoded as a single string, such as

```
PUT "https://mybucket.s3.amazonsws.com/"
```

The difference is striking. For example, in the first case, many syntactical errors can often already be caught during compile time, whereas in the second case, checking needs to be deferred until runtime. Secondly, one can argue that specifying the semantics of an operation is much easier with specific interfaces than with ones that offer only generic operations. On the other hand, with generic operations, changes are much easier to accommodate, as they would generally involve changing the layout of strings that encode what is actually required.

2.1.3 Publish-subscribe architectures

As systems continue to grow and processes can more easily join or leave, it becomes important to have an architecture in which dependencies between processes become as loose as possible. A large class of distributed systems have adopted an architecture in which there is a strong separation between *processing* and *coordination*. The idea is to view a system as a collection of autonomously operating processes. In this model, **coordination** encompasses the communication and cooperation between processes. It forms the glue that binds the activities performed by processes into a whole [Gelernter and Carriero, 1992].

Cabri et al. [2000] provide a taxonomy of coordination models that can be applied equally to many types of distributed systems. Slightly adapting their terminology, we make a distinction between models along two different dimensions, temporal and referential, as shown in Figure 2.9.

When processes are temporally and referentially coupled, coordination takes place directly, referred to as **direct coordination**. The referential coupling generally appears in the form of explicit referencing in communication. For example, a process can communicate only if it knows the name or identifier of

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	Temporally coupled	Temporally decoupled
Referentially coupled	Direct	Mailbox
Referentially decoupled	Event-based	Shared data space

Figure 2.9: Examples of different forms of coordination.

the other processes it wants to exchange information with. Temporal coupling means that processes that are communicating will both have to be up and running. In real life, talking over cell phones (and assuming that a cell phone has only one owner), is an example of direct communication.

A different type of coordination occurs when processes are temporally decoupled, but referentially coupled, which we refer to as **mailbox coordination**. In this case, there is no need for two communicating processes to be active at the same time to let communication take place. Instead, communication takes place by putting messages in a (possibly shared) mailbox. Because it is necessary to explicitly address the mailbox that will hold the messages that are to be exchanged, there is a referential coupling.

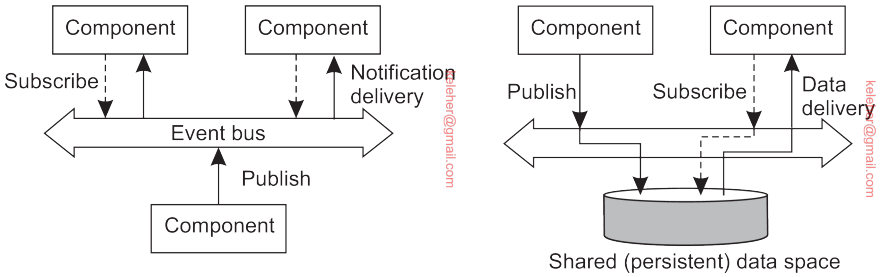
The combination of referentially decoupled and temporally coupled systems form the group of models for **event-based coordination**. In referentially decoupled systems, processes do not know each other explicitly. The only thing a process can do is **publish** a **notification** describing the occurrence of an event (e.g., that it wants to coordinate activities, or that it just produced some interesting results). Assuming that notifications come in all sorts and kinds, processes may **subscribe** to a specific kind of notification (see also [Mühl et al., 2006]). In an ideal event-based coordination model, a published notification will be delivered exactly to those processes that have subscribed to it. However, it is generally required that the subscriber is up-and-running at the time the notification was published.

A well-known coordination model is the combination of referentially and temporally decoupled processes, leading to what is known as a **shared data space**. The key idea is that processes communicate entirely through **tuples**, which are structured data records consisting of several fields, very similar to a row in a database table. Processes can put any type of tuple into the shared data space. To retrieve a tuple, a process provides a search pattern that is matched against the tuples. Any tuple that matches is returned.

Shared data spaces are thus seen to implement an associative search mechanism for tuples. When a process wants to extract a tuple from the data space, it specifies (some of) the values of the fields it is interested in. Any tuple that matches that specification is then removed from the data space and passed to the process.

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Shared data spaces are often combined with event-based coordination: a process subscribes to certain tuples by providing a search pattern; when a process inserts a tuple into the data space, matching subscribers are notified. In both cases, we are dealing with a **publish-subscribe** architecture, and indeed, the key characteristic feature is that processes have no explicit reference to each other. The difference between a pure event-based architectural style , and that of a shared data space, is shown in Figure 2.10. We have also shown an abstraction of the *mechanism* by which publishers and subscribers are matched, known as an **event bus**.



(a) (b)

Figure 2.10: The (a) event-based and (b) shared data-space architectural style.

Note 2.3 (Example: Linda tuple spaces)
To make matters a bit more concrete, we take a closer look at **Linda**, a programming model developed in the 1980s [Carriero and Gelernter, 1989]. The shared data space in Linda is known as a **tuple space**, which supports three operations:

- `in(t)`: remove a tuple that matches the template `t`
- `rd(t)`: obtain a copy of a tuple that matches the template `t`
- `out(t)`: add the tuple `t` to the tuple space

Note that if a process would call `out(t)` twice in a row, we would find that two copies of tuple `t` would have been stored. Formally, a tuple space is therefore always modeled as a *multiset*. Both `in` and `rd` are *blocking* operations: the caller will be blocked until a matching tuple is found, or has become available.

Consider a simple microblog application in which messages are tagged with the name of its poster and a topic, followed by a short string. Each message is modeled as a tuple `<string,string,string>` where the first string names the poster, the second string represents the topic, and the third one is the actual content. Assuming that we have created a shared data space called MicroBlog, Figure 2.11 shows how Alice and Bob can post messages to that space, and how

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Chuck can pick a (randomly selected) message. We have omitted some code for clarity. Note that neither Alice nor Bob knows who will read their postings.

```
1 blog = linda.universe._rd(("MicroBlog",linda.TupleSpace))[1]
2
3 blog._out(("bob","distsys","I am studying chap 2"))
4 blog._out(("bob","distsys","The linda example's pretty simple"))
5 blog._out(("bob","gtcn","Cool book!"))
```

(a) Bob's code for creating a microblog and posting three messages.

```
1 blog = linda.universe._rd(("MicroBlog",linda.TupleSpace))[1]
2
3 blog._out(("alice","gtcn","This graph theory stuff is not easy"))
4 blog._out(("alice","distsys","I like systems more than graphs"))
```

(b) Alice's code for creating a microblog and posting two messages.

```
1 blog = linda.universe._rd(("MicroBlog",linda.TupleSpace))[1]
2
3 t1 = blog._rd(("bob","distsys",str))
4 t2 = blog._rd(("alice","gtcn",str))
5 t3 = blog._rd(("bob","gtcn",str))
```

(c) Chuck reading a message from Bob's and Alice's microblog.

Figure 2.11: A simple example of using a shared data space.

In the first line of each code fragment, a process looks up the tuple space named "MicroBlog." Bob posts three messages: two on topic distsys, and one on gtcn. Alice posts two messages, one on each topic. Chuck, finally, reads three messages: one from Bob on distsys and one on gtcn, and one from Alice on gtcn.

Obviously, there is much room for improvement. For example, we should ensure that Alice cannot post messages under Bob's name. However, the important issue to note now, is that by providing only tags, a reader such as Chuck will be able to pick up messages without needing to directly reference the poster. In particular, Chuck could also read a randomly selected message on topic distsys through the statement

```
t = blog_rd((str,"distsys",str))
```

We leave it as an exercise to the reader to extend the code fragments such that a *next* message will be selected instead of a random one.

An important aspect of publish-subscribe systems is that communication takes place by describing the events that a subscriber is interested in. As a consequence, naming plays a crucial role. We return to naming later, but for now, the important issue is that often, data items are not explicitly identified by senders and receivers.

Let us first assume that events are described by a series of **attributes**. A notification describing an event is said to be **published** when it is made

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available for other processes to read. To that end, a **subscription** needs to be passed to the middleware, containing a description of the event that the subscriber is interested in. Such a description typically consists of some (*attribute, value*) pairs, which is common for so-called **topic-based publish-subscribe systems**.

As an alternative, in **content-based publish-subscribe systems**, a subscription may also consist of (*attribute, range*) pairs. In this case, the specified attribute is expected to take on values within a specified range. Descriptions can sometimes be given using all kinds of predicates formulated over the attributes, very similar in nature to SQL-like queries in the case of relational databases. Obviously, the more expressive a description is allowed to be, the more difficult it will be to test whether an event matches a description.

We are now confronted with a situation in which subscriptions need to be **matched** against notifications, as shown in Figure 2.12. Often, an event actually corresponds to data becoming available. In that case, when matching succeeds, there are two possible scenarios. In the first case, the middleware may decide to forward the published notification, along with the associated data, to its current set of subscribers, that is, processes with a matching subscription. As an alternative, the middleware can also forward only a notification, at which point subscribers can execute a `read` operation to retrieve the data item.

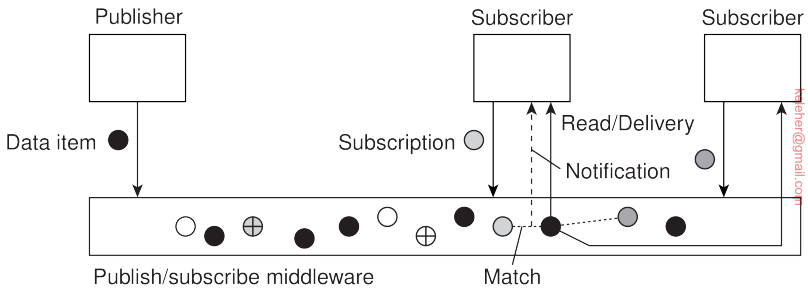


Figure 2.12: The principle of exchanging data items between publishers and subscribers.

In those cases, in which data associated with an event are immediately forwarded to subscribers, the middleware will generally not offer storage of data. Storage is either explicitly handled by a separate service, or is the responsibility of subscribers. In other words, we have a referentially decoupled, but temporally coupled system.

This situation is different when notifications are sent so that subscribers need to explicitly read the associated data. Necessarily, the middleware will have to store data items. In these situations, there are additional operations for data management. It is also possible to attach a **lease** to a data item such that when the lease expires that the data item is automatically deleted.

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Events can easily complicate the processing of subscriptions. To illustrate, consider a subscription such as "notify when room ZI.1060 is unoccupied and the door is unlocked." Typically, a distributed system supporting such subscriptions can be implemented by placing independent sensors for monitoring room occupancy (e.g., motion sensors) and those for registering the status of a door lock. Following the approach sketched so far, we would need to *compose* such primitive events into a publishable data item, to which processes can then subscribe. Event composition turns out to be a difficult task, notably when the primitive events are generated from sources dispersed across the distributed system.

Clearly, in publish-subscribe systems such as these, the crucial issue is the efficient and scalable implementation of matching subscriptions to notifications. From the outside, the publish-subscribe architecture provides lots of potential for building very large-scale distributed systems due to the strong decoupling of processes. On the other hand, devising scalable implementations without losing this independence is not a trivial exercise, notably in the case of content-based publish-subscribe systems. In this sense, although many claim that the publish-subscribe style offers the path toward scalable architectures, the fact is that implementations may easily form a bottleneck, certainly when security and privacy is at stake, as we will discuss in Chapter 9 and later in Chapter 5.

2.2 Middleware and distributed systems

To assist the development of distributed applications, distributed systems are often organized to have a separate layer of software that is logically placed on top of the respective operating systems of the computers that are part of the

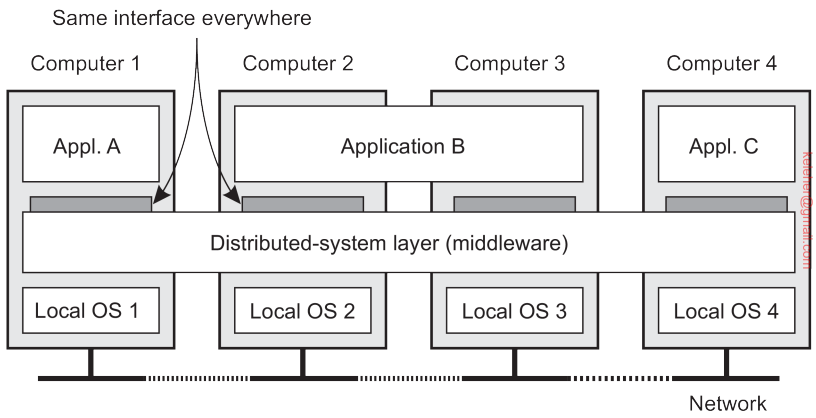


Figure 2.13: A distributed system organized in a middleware layer, which extends over multiple machines, offering each application the same interface.

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system. This organization is shown in Figure 2.13, leading to what is known as **middleware** [Bernstein, 1996].

Figure 2.13 shows four networked computers and three applications, of which application B is distributed across computers 2 and 3. Each application is offered the same interface. The distributed system provides the means for components of a single distributed application to communicate with each other, but also to let different applications communicate. At the same time, it hides, as best and reasonably as possible, the differences in hardware and operating systems from each application.

In a sense, middleware is the same to a distributed system as what an operating system is to a computer: a manager of resources offering its applications to efficiently share and deploy those resources across a network. Next to resource management, it offers services that can also be found in most operating systems, including:

- Facilities for interapplication communication.
- Security services.
- Accounting services.
- Masking of and recovery from failures.

The main difference with their operating-system equivalents, is that middleware services are offered in a networked environment. Note also that most services are useful to many applications. In this sense, middleware can also be viewed as a container of commonly used components and functions that now no longer have to be implemented by applications separately.

Note 2.4 (Historical note: The term middleware)
Although the term middleware became popular in the mid 1990s, it was most likely mentioned for the first time in a report on a NATO software engineering conference, edited by Peter Naur and Brian Randell in October 1968 [Naur and Randell, 1968]. Indeed, middleware was placed precisely between applications and service routines (the equivalent of operating systems).

2.2.1 Middleware organization

Let us now zoom into the actual organization of middleware. There are two important types of *design patterns* that are often applied to the organization of middleware: wrappers and interceptors. Each targets different problems, yet addresses the same goal for middleware: achieving openness (as we discussed in Section 1.2.3).

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Wrappers

When building a distributed system out of existing components, we immediately bump into a fundamental problem: the interfaces offered by the legacy component are most likely not suitable for all applications. In Section 1.3.2 we discussed how enterprise application integration could be established through middleware as a communication facilitator, but there we still implicitly assumed that, in the end, components could be accessed through their native interfaces.

A **wrapper** or **adapter** is a special component that offers an interface acceptable to a client application, of which the functions are transformed into those available at the component. In essence, it solves the problem of incompatible interfaces (see also Gamma et al. [1994]).

Although originally narrowly defined in the context of object-oriented programming, in the context of distributed systems wrappers are much more than simple interface transformers. For example, an **object adapter** is a component that allows applications to invoke remote objects, although those objects may have been implemented as a combination of library functions operating on the tables of a relational database.

As another example, reconsider Amazon's S3 storage service. As mentioned, there are two types of interfaces available, one adhering to a RESTful architecture, another following a more traditional approach. For the RESTful interface, clients will be using the HTTP protocol, essentially communicating with a traditional Web server which now acts as an adapter to the actual storage service, by partly dissecting incoming requests and subsequently handing them off to specialized servers internal to S3.

Wrappers have always played an important role in extending systems with existing components. Extensibility, which is crucial for achieving openness, used to be addressed by adding wrappers as needed. In other words, if an application A managed data that was needed by an application B, one approach would be to develop a wrapper specific for B so that it could have access to A's data. Clearly, this approach does not scale well: with N applications we would, in theory, need to develop $N \times (N-1) = \mathcal{O}(N^2)$ wrappers.

Again, facilitating a reduction of the number of wrappers is typically done through middleware. One way of doing this is implementing a so-called **broker**, which is logically a centralized component that handles all the accesses between different applications. An often-used type is a **message broker** of which we discuss the technicalities in Section 4.3.3. In the case of a message broker, applications simply send requests to the broker containing information on what they need. The broker, having knowledge of all relevant applications, contacts the appropriate applications, possibly combines and transforms the responses and returns the result to the initial application. In principle, because a broker offers a single interface to each application, we

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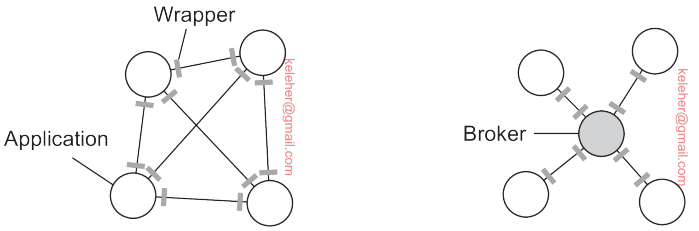


Figure 2.14: (a) Requiring each application to have a wrapper for each other application. (b) Reducing the number of wrappers by making use of a broker.

now need at most $2N = \mathcal{O}(N)$ wrappers instead of $\mathcal{O}(N^2)$. This situation is sketched in Figure 2.14.

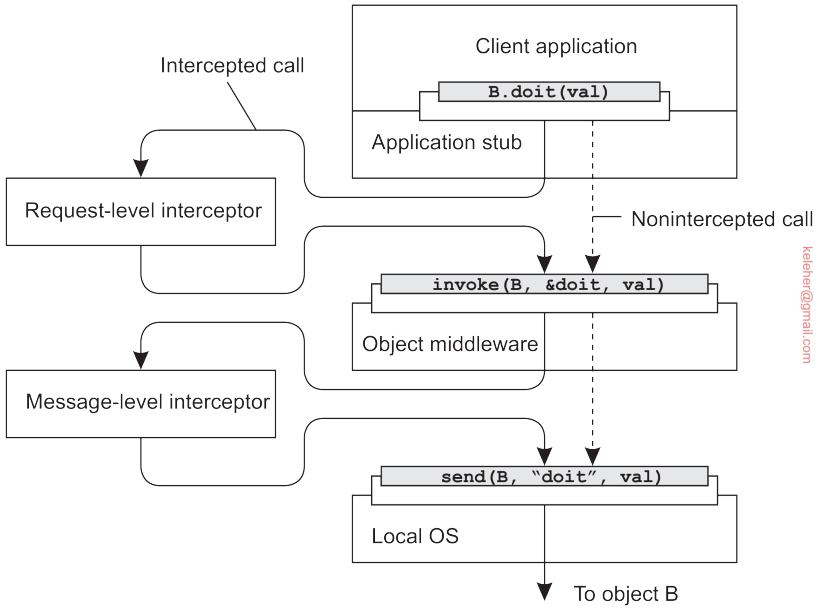
Interceptors

Conceptually, an **interceptor** is nothing but a software construct that will break the usual flow of control and allow other (application specific) code to be executed. Interceptors are a primary means for adapting middleware to the specific needs of an application. As such, they play an important role in making middleware open. To make interceptors generic may require a substantial implementation effort, as illustrated by Schmidt et al. [2000], and it is unclear whether in such cases generality should be preferred over restricted applicability and simplicity. Furthermore, often having only limited interception facilities will improve management of the software and the distributed system as a whole.

To make matters concrete, consider interception as supported in many object-based distributed systems. The basic idea is simple: an object A can call a method that belongs to an object B, while the latter resides on a different machine than A. As we explain in detail later in the book, such a remote-object invocation is carried out in three steps:

1. Object A is offered a local interface that is the same as the interface offered by object B. A calls the method available in that interface.
2. The call by A is transformed into a generic object invocation, made possible through a general object-invocation interface offered by the middleware at the machine where A resides.
3. Finally, the generic object invocation is transformed into a message that is sent through the transport-level network interface as offered by A's local operating system.

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Figure 2.15: Using interceptors to handle remote-object invocations.

This scheme is shown in Figure 2.15. After the first step, the call `B.doit(val)` is transformed into a generic call, such as `invoke(B,&doit,val)` with a reference to B's method and the parameters that go along with the call. Now imagine that object B is replicated. In that case, each replica should actually be invoked. This is a clear point where interception can help. What the **request-level interceptor** will do, is simply call `invoke(B,&doit,val)` for each of the replicas. The beauty of this all is that the object A need not be aware of the replication of B, but also the object middleware need not have special components that deal with this replicated call. Only the request-level interceptor, which may be *added* to the middleware, needs to know about B's replication.

In the end, a call to a remote object will have to be sent over the network. In practice, this means that the messaging interface as offered by the local operating system will need to be invoked. At that level, a **message-level interceptor** may assist in transferring the invocation to the target object. For example, imagine that the parameter `val` actually corresponds to a huge array of data. In that case, it may be wise to fragment the data into smaller parts to have it assembled again at the destination. Such a fragmentation may improve performance or reliability. Again, the middleware need not be aware of this fragmentation; the lower-level interceptor will transparently handle the rest of the communication with the local operating system.

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2.2.2 Modifiable middleware

What wrappers and interceptors offer are means to extend and adapt the middleware. The need for adaptation comes from the fact that the environment in which distributed applications are executed changes continuously. Changes include those resulting from mobility, a strong variance in the quality-of-service of networks, failing hardware, and battery drainage, among others. Rather than making applications responsible for reacting to changes, this task is placed in the middleware. Moreover, as the size of a distributed system increases, changing its parts can rarely be done by temporarily shutting it down. What is needed is being able to make changes on-the-fly.

These strong influences from the environment have brought many designers of middleware to consider the construction of *adaptive software*. We follow Parlavantzas and Coulson [2007] in speaking of **modifiable middleware** to express that middleware may not only need to be adaptive, but that we should be able to purposefully modify it without bringing it down. In this context, interceptors can be thought of offering a means to adapt the standard flow of control. Replacing software components at runtime is an example of modifying a system. And indeed, perhaps one of the most popular approaches toward modifiable middleware is that of dynamically constructing middleware from components.

Component-based design focuses on supporting modifiability through composition. A system may either be configured statically at design time, or dynamically at runtime. The latter requires support for late binding, a technique that has been successfully applied in programming language environments, but also for operating systems where modules can be loaded and unloaded at will. Automatically selecting the best implementation of a component during runtime is by now well understood [Yellin, 2003] but again, the process remains complex for distributed systems, especially when considering that replacement of one component requires to know exactly what the effect of that replacement on other components will be. Often, components are less independent as one may think.

The bottom line is that to accommodate dynamic changes to the software that makes up middleware, we need at least basic support to load and unload components at runtime. In addition, for each component explicit specifications of the interfaces it offers, as well the interfaces it requires, are needed. If state is maintained between calls to a component, then further special measures are needed. By-and-large, it should be clear that organizing middleware to be modifiable requires special attention.

2.3 Layered-system architectures

Let us now take a look at how many distributed systems are actually organized by considering where software components are placed. Deciding on software

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components, their interaction, and their placement leads to an instance of a software architecture, also known as a **system architecture** [Bass et al., 2021]. We start with discussing layered architectures. Other forms follow later.

Despite the lack of consensus on many distributed systems issues, there is one issue that many researchers and practitioners agree upon: thinking in terms of *clients* that request services from *servers* helps to understand and manage the complexity of distributed systems [Saltzer and Kaashoek, 2009]. In the following, we first consider a simple layered organization, followed by looking at multi-layered organizations.

2.3.1 Simple client-server architecture

In the basic client-server model, processes in a distributed system are divided into two (possibly overlapping) groups. A **server** is a process implementing a specific service, for example, a file system service or a database service. A **client** is a process that requests a service from a server by sending it a request and subsequently waiting for the server's reply. This client-server interaction, also known as **request-reply behavior** is shown in Figure 2.16.

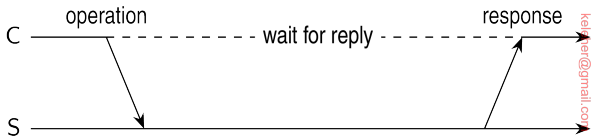


Figure 2.16: General interaction between a client C and a server S. C sends the operation oper and waits for the response from S.

Communication between a client and a server can be implemented by a simple connectionless protocol when the underlying network is fairly reliable, as in many local-area networks. In these cases, when a client requests a service, it simply packages a message for the server, identifying the service it wants, along with the necessary input data. The message is then sent to the server. The latter, in turn, will always wait for an incoming request, subsequently process it, and package the results in a reply message that is then sent to the client.

Using a connectionless protocol has the obvious advantage of being efficient. As long as messages do not get lost or corrupted, the request/reply protocol just sketched works fine. Unfortunately, making the protocol resistant to occasional transmission failures is not trivial. The only thing we can do is possibly let the client resend the request when no reply message comes in. The problem, however, is that the client cannot detect whether the original request message was lost, or that transmission of the reply failed. If the reply was lost, then resending a request may result in performing the operation twice. If the

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operation was something like "transfer $10,000 from my bank account," then clearly, it would have been better that we simply reported an error instead. On the other hand, if the operation was "tell me how much money I have left," it would be perfectly acceptable to resend the request. When an operation can be repeated multiple times without harm, it is said to be **idempotent**. Since some requests are idempotent and others are not, it should be clear that there is no single solution for dealing with lost messages. We defer a detailed discussion on handling transmission failures to Section 8.3.

As an alternative, many client-server systems use a reliable connection-oriented protocol. Although this solution is not entirely appropriate in a local-area network due to relatively low performance, it works perfectly fine in wide-area systems in which communication is inherently unreliable. For example, virtually all Internet application protocols are based on reliable TCP/IP connections. In this case, whenever a client requests a service, it first sets up a connection to the server before sending the request. The server generally uses that same connection to send the reply message, after which the connection is torn down. The trouble may be that setting up and tearing down a connection is relatively costly, especially when the request and reply messages are small.

The client-server model has been subject to many debates and controversies over the years. One of the main issues was how to draw a clear distinction between a client and a server. Not surprisingly, there is often no clear distinction. For example, a server for a distributed database may continuously act as a client because it is forwarding requests to different file servers responsible for implementing the database tables. In such a case, the database server itself only processes the queries.

2.3.2 Multitiered Architectures

The distinction into three logical levels, as discussed so far, suggests several possibilities for physically distributing a client-server application across several machines. The simplest organization is to have only two types of machines:

1. A client machine containing only the programs implementing (part of) the user-interface level
2. A server machine containing the rest, that is, the programs implementing the processing and data level

In this organization everything is handled by the server while the client is essentially no more than a dumb terminal, possibly with only a convenient graphical interface. There are, however, many other possibilities. As explained in Section 2.1.1, many distributed applications are divided into three layers: (1) a user-interface layer, (2) a processing layer, and (3) a data layer. One

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approach for organizing clients and servers is then to distribute these layers across different machines, as shown in Figure 2.17 (see also Umar [1997]). As a first step, we make a distinction between only two kinds of machines: client machines and server machines, leading to what is also referred to as a **(physically) two-tiered architecture**.

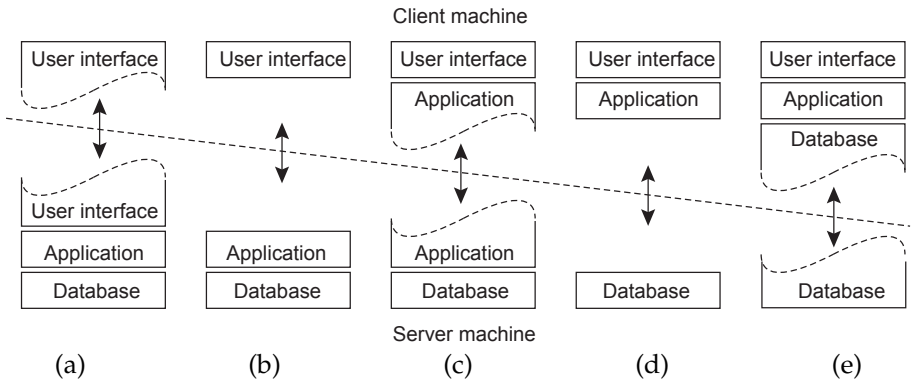


Figure 2.17: Client-server organizations in a two-tiered architecture.

One possible organization is to have only the terminal-dependent part of the user interface on the client machine, as shown in Figure 2.17(a), and give the applications remote control over the presentation of their data. An alternative is to place the entire user-interface software on the client side, as shown in Figure 2.17(b). In such cases, we essentially divide the application into a graphical front end, which communicates with the rest of the application (residing at the server) through an application-specific protocol. In this model, the front end (the client software) does no processing other than necessary for presenting the application's interface.

Continuing along this line of reasoning, we may also move part of the application to the front end, as shown in Figure 2.17(c). An example where this makes sense is where the application makes use of a form that needs to be filled in entirely before it can be processed. The front end can then check the correctness and consistency of the form, and where necessary interact with the user. Another example of the organization of Figure 2.17(c), is that of a word processor in which the basic editing functions execute on the client side where they operate on locally cached, or in-memory data, but where the advanced support tools such as checking the spelling and grammar execute on the server side.

In many client-server environments, the organizations shown in Figure 2.17(d) and Figure 2.17(e) are particularly popular. These organizations are used where the client machine is a PC or workstation, connected through a network to a distributed file system or database. Essentially, most of the

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application is running on the client machine, but all operations on files or database entries go to the server. For example, many banking applications run on an end-user's machine, where the user prepares transactions and such. Once finished, the application contacts the database on the bank's server and uploads the transactions for further processing. Figure 2.17(e) represents the situation where the client's local disk contains part of the data. For example, when browsing the Web, a client can gradually build a huge cache on local disk of most recent inspected Web pages.

Note 2.5 (More information: Is there something like the best organization?)
We note that there has been a strong trend to move away from the configurations shown in Figure 2.17(d) and Figure 2.17(e) in those cases, that client software is placed at end-user machines. Instead, most of the processing and data storage is handled at the server side. The reason for this is simple: although client machines do a lot, they are also more problematic to manage. Having more functionality on the client machine means that a wide range of end users will need to be able to handle that software. This implies that more effort needs to be spent on making software resilient to end-user behavior. In addition, client-side software is dependent on the client's underlying platform (i.e., operating system and resources), which can easily mean that multiple versions will need to be maintained. From a systems-management perspective, having what are called **fat clients** is not optimal. Instead, the **thin clients** as represented by the organizations shown in Figure 2.17(a)–(c) are much easier, perhaps at the cost of less sophisticated user interfaces and client-perceived performance.

Does this mean the end of fat clients? Not in the least. For one thing, there are many applications for which a fat-client organization is often still the best. We already mentioned office suites, but also many multimedia applications require that processing is done on the client's side. Moreover, when end users need to operate offline, we see that installing applications will be necessary. Second, with the advent of advanced Web browsing technology, it is now much easier to dynamically place and manage client-side software by simply uploading (the sometimes very sophisticated) scripts to the client. Combined with the fact that this type of client-side software runs in well-defined commonly deployed environments, and thus that platform dependency is much less of an issue, we see that the counter-argument of management complexity is often no longer valid. This has led to the deployment of **virtual desktop environments**, which we discuss further in Chapter 3.

Finally, note that moving away from fat clients does not imply that we no longer need distributed systems. On the contrary, what we continue to see is that server-side solutions are becoming increasingly more distributed as a single server is being replaced by multiple servers running on different machines. Cloud computing is a good example in this case: the complete server side is being executed in data centers, and generally on multiple servers.

When distinguishing only client and server machines as we did so far, we miss the point that a server may sometimes need to act as a client, as shown

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in Figure 2.18, leading to a **(physically) three-tiered architecture**.

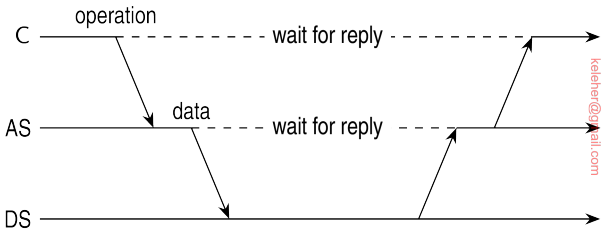


Figure 2.18: An example of an application server AS acting as client for a database server DS.

In this architecture, traditionally programs that form part of the processing layer are executed by a separate server, but may additionally be partly distributed across the client and server machines. A typical example of where a three-tiered architecture is used is in transaction processing. A separate process, called the transaction processing monitor, coordinates all transactions across possibly different data servers.

Another, but very different example where we often see a three-tiered architecture is in the organization of Websites. In this case, a Web server acts as an entry point to a site, passing requests to an application server where the actual processing takes place. This application server, in turn, interacts with a database server. We already came across such an organization when discussing this book's Website and the facilities for generating and downloading a personalized copy.

2.3.3 Example: The Network File System

Many distributed files systems are organized like client-server architectures, with Sun Microsystem's **Network File System** (**NFS**) being one of the most widely deployed ones for Unix systems [Callaghan, 2000; Haynes, 2015; Noveck and Lever, 2020].

The basic idea behind NFS is that each file server provides a standardized view of its local file system. In other words, it should not matter how that local file system is implemented; each NFS server supports the same model. This approach has been adopted for other distributed files systems as well. NFS comes with a communication protocol that allows clients to access the files stored on a server, thus allowing a heterogeneous collection of processes, possibly running on different operating systems and machines, to share a common file system.

The model underlying NFS and similar systems is that of a **remote file service**. In this model, clients are offered transparent access to a file system that is managed by a remote server. However, clients are normally unaware

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of the actual location of files. Instead, they are offered an interface to a file system that is similar to the interface offered by a conventional local file system. In particular, the client is offered only an interface containing various file operations, but the server is responsible for implementing those operations. This model is therefore also referred to as the **remote access model**. It is shown in Figure 2.19(a).

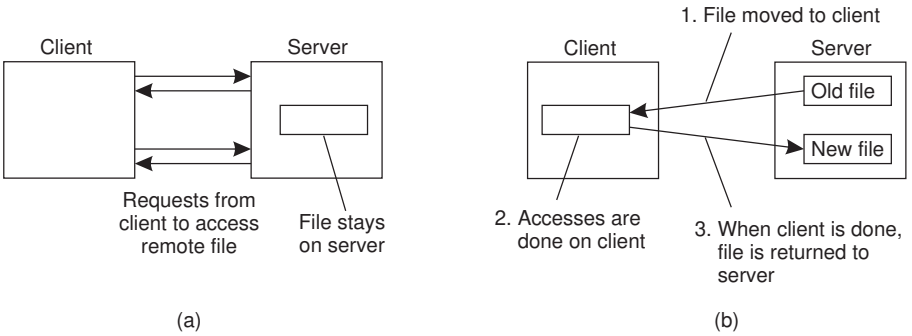


Figure 2.19: (a) The remote access model. (b) The upload/download model.

In contrast, in the **upload/download model** a client accesses a file locally after having downloaded it from the server, as shown in Figure 2.19(b) When the client is finished with the file, it is uploaded back to the server again so that it can be used by another client. The Internet's FTP service can be used this way when a client downloads a complete file, modifies it, and then puts it back.

NFS has been implemented for numerous operating systems, although the Unix versions are predominant. For virtually all modern Unix systems, NFS is generally implemented following the architecture shown in Figure 2.20.

A client accesses the file system using the system calls provided by its local operating system. However, the local Unix file system interface is replaced by an interface to the **Virtual File System** (**VFS**), which by now is a de facto standard for interfacing to different (distributed) file systems [Kleiman, 1986]. Virtually all modern operating systems provide VFS, and not doing so more or less forces developers to largely reimplement huge parts of an operating system when adopting a new file-system structure. With NFS, operations on the VFS interface are either passed to a local file system, or passed to a separate component known as the **NFS client**, which takes care of handling access to files stored at a remote server. In NFS, all client-server communication is done through so-called **remote procedure calls** (**RPC**s). As mentioned before, an RPC is essentially a standardized way to let a client on a machine A make an ordinary call to a procedure that is implemented on another machine B. We discuss RPCs extensively in Chapter 4. The NFS client implements the NFS file system operations as remote procedure calls to the server. Note that the operations offered by the VFS interface can be different from those offered by

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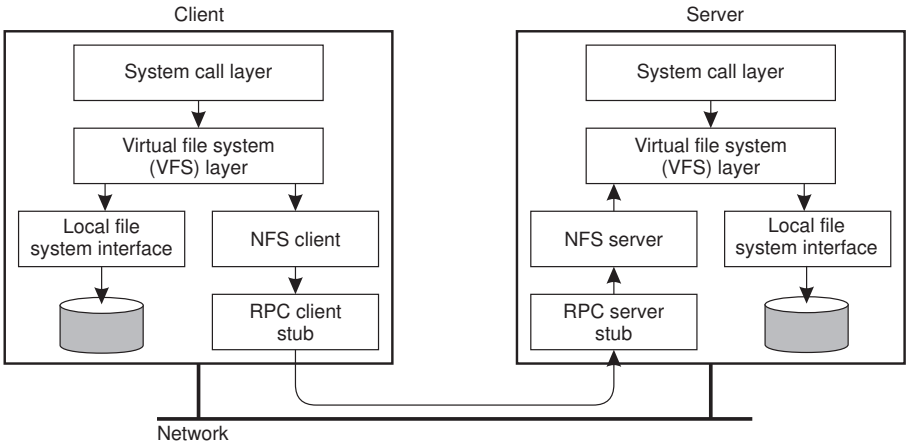


Figure 2.20: The basic NFS architecture for Unix systems.

the NFS client. The whole idea of the VFS is to hide the differences between various file systems.

On the server side, we see a similar organization. The **NFS server** is responsible for handling incoming client requests. The RPC component at the server converts incoming requests to regular VFS file operations that are subsequently passed to the VFS layer. Again, the VFS is responsible for implementing a local file system in which the actual files are stored.

An important advantage of this scheme is that NFS is largely independent of local file systems. It does not matter whether the operating system at the client or server uses a Unix file system, a Windows file system, or even an old MS-DOS file system. The only important issue is that these file systems are compliant with the file system model offered by NFS. For example, MS-DOS with its short file names cannot be used to implement an NFS server in a fully transparent way.

2.3.4 Example: The Web

The architecture of Web-based distributed systems is not fundamentally different from other distributed systems. However, it is interesting to see how the initial idea of supporting distributed documents has evolved since its inception in the 1990s. Documents turned from being purely static and passive to dynamically generated content. Furthermore, recently, many organizations have begun supporting services instead of just documents.

Simple Web-based systems

Many Web-based systems are still organized as relatively simple client-server architectures. The core of a Web site is formed by a process that has access to a

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local file system storing documents. The simplest way to refer to a document is by a reference called a **uniform resource locator** (**URL**). It specifies where a document is located by embedding the DNS name of its associated server along with a file name by which the server can look up the document in its local file system. Furthermore, a URL specifies the application-level protocol for transferring the document across the network.

A client interacts with Web servers through a **browser**, which is responsible for properly displaying a document. Furthermore, a browser accepts input from a user mostly by letting the user select a reference to another document, which it then subsequently fetches and displays. The communication between a browser and Web server is standardized: they both adhere to the **HyperText Transfer Protocol** (**HTTP**). This leads to the overall organization shown in Figure 2.21.

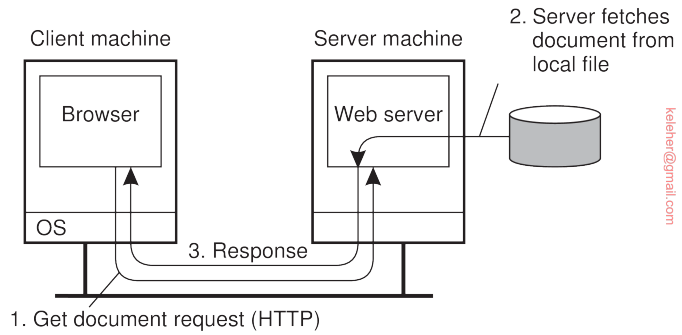


Figure 2.21: The overall organization of a traditional Web site.

Let us zoom in a bit into what a document actually is. Perhaps the simplest form is a standard text file. In that case, the server and browser have barely anything to do: the server copies the file from the local file system and transfers it to the browser. The latter, in turn, merely displays the content of the file ad verbatim without further ado.

More interesting are Web documents that have been marked up, which is usually done in the **HyperText Markup Language**, or simply **HTML**. In that case, the document includes various instructions expressing how its content should be displayed, similar to what one can expect from any decent word-processing system (although those instructions are normally hidden from the end user). For example, instructing text to be emphasized is done by the following markup:

```
<emph>Emphasize this text</emph>
```

There are many more of such markup instructions. The point is that the browser understands these instructions and will act accordingly when displaying the text.

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Documents can contain much more than just markup instructions. In particular, they can have complete programs embedded, of which **JavaScript** is the one most often deployed. In this case, the browser is warned that there is some code to execute as in:

```
<script type="text/javascript">....</script>
```

and as long as the browser has an appropriate embedded interpreter for the specified language, everything between "`<script>`" and "`</script>`" will be executed as any other program. The main benefit of including scripts is that it allows for much better interaction with the end user, including sending information back to the server. (The latter, by the way, has always been supported in HTML through **forms**.)

Much more can be said about Web documents, but this is not the place to do so. A good introduction on how to build Web-based applications can be found in [Sebesta, 2015].

Multitiered architectures

The Web started out as the relatively simple two-tiered client-server system shown in Figure 2.21. By now, this simple architecture has been extended to support much more sophisticated means of documents. In fact, one could justifiably argue that the term "document" is no longer appropriate. For one, most things that we get to see in our browser has been generated on the spot as the result of sending a request to a Web server. Content is stored in a database at the server's side, along with client-side scripts and such, to be composed on-the-fly into a document which is then subsequently sent to the client's browser. Documents have thus become completely dynamic.

One of the first enhancements to the basic architecture was support for simple user interaction by the **Common Gateway Interface** or simply **CGI**. CGI defines a standard way by which a Web server can execute a program taking user data as input. Usually, user data come from an HTML form; it specifies the program that is to be executed at the server side, along with parameter values that are filled in by the user. Once the form has been completed, the program's name and collected parameter values are sent to the server, as shown in Figure 2.22.

When the server sees the request, it starts the program named in the request and passes it the parameter values. At that point, the program simply does its work and generally returns the results in the form of a document that is sent back to the user's browser to be displayed.

CGI programs can be as sophisticated as a developer wants. For example, as shown in Figure 2.22 many programs operate on a database local to the Web server. After processing the data, the program generates an HTML document and returns that document to the server. The server will then pass the document to the client. An interesting observation is that to the server,

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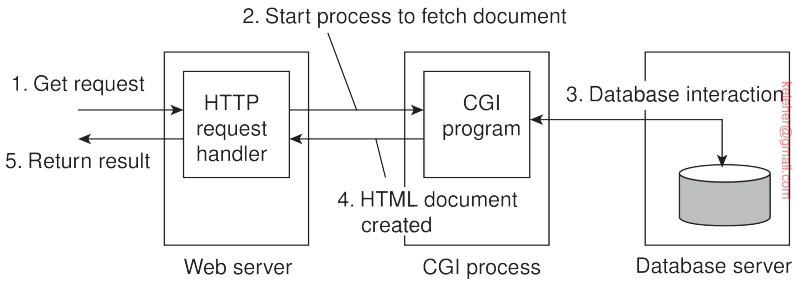


Figure 2.22: The principle of using server-side CGI programs.

it appears as if it is asking the CGI program to fetch a document. In other words, the server does nothing but delegate the fetching of a document to an external program.

The main task of a server used to be handling client requests by simply fetching documents. With CGI programs, fetching a document could be delegated in such a way that the server would remain unaware of whether a document had been generated on the fly, or actually read from the local file system. Note that we have just described a two-tiered organization of server-side software.

However, servers nowadays do much more than just fetching documents. One of the most important enhancements is that servers can also process a document before passing it to the client. In particular, a document may contain a **server-side script**, which is executed by the server when the document has been fetched locally. The result of executing a script is sent along with the rest of the document to the client. The script itself is not sent. In other words, using a server-side script changes a document by essentially replacing the script with the results of its execution. To make matters concrete, take a look at a simple example of dynamically generating a document. Assume a file is stored at the server with the following content:

```
<strong> <?php echo $_SERVER['REMOTE_ADDR']; ?> </strong>
```

The server will examine the file and subsequently process the PHP code (between "<?php" and "?>") replacing the code with the address of the requesting client. Much more sophisticated settings are possible, such as accessing a local database and subsequently fetching content from that database to be combined with other dynamically generated content.

2.4 Symmetrically distributed system architectures

Multitiered client-server architectures are a direct consequence of dividing distributed applications into a user interface, processing components, and

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data-management components. The different tiers correspond directly with the logical organization of applications. In many business environments, distributed processing is equivalent to organizing a client-server application as a multitiered architecture. We refer to this type of distribution as **vertical distribution**. The characteristic feature of vertical distribution is that it is achieved by placing logically different components on different machines. The term is related to the concept of *vertical fragmentation* as used in distributed relational databases, where it means that tables are split columnwise, and subsequently distributed across multiple machines [Özsu and Valduriez, 2020].

Again, from a systems-management perspective, having a vertical distribution can help: functions are logically and physically split across multiple machines, where each machine is tailored to a specific group of functions. However, vertical distribution is only one way of organizing client-server applications. In modern architectures, it is often the distribution of the clients and the servers that counts, which we refer to as **horizontal distribution**. In this type of distribution, a client or server may be physically split up into logically equivalent parts, but each part is operating on its own share of the complete data set, thus balancing the load. In this section, we will take a look at a class of modern system architectures that support horizontal distribution, known as **peer-to-peer systems**.

From a high-level perspective, the processes that constitute a peer-to-peer system are all equal. This means that the functions that need to be carried out are represented by every process that constitutes the distributed system. As a consequence, much of the interaction between processes is symmetric: each process will act as a client and a server at the same time (which is also referred to as acting as a **servant**).

Given this symmetric behavior, peer-to-peer architectures revolve around the question of how to organize the processes in an **overlay network** [Tarkoma, 2010]: a network in which the nodes are formed by the processes and the links represent the possible communication channels (which are often realized as TCP connections). A node may not be able to communicate directly with an arbitrary other node, but is required to send messages through the available communication channels. Two types of overlay networks exist: those that are structured and those that are not. These two types are surveyed extensively in Lua et al. [2005] along with numerous examples. Buford and Yu [2010] additionally includes an extensive list of various peer-to-peer systems. Aberer et al. [2005] provide a reference architecture that allows for a more formal comparison of the different types of peer-to-peer systems. A survey taken from the perspective of content distribution is provided by Androutsellis-Theotokis and Spinellis [2004]. Finally, Buford et al. [2009], Tarkoma [2010] and Vu et al. [2010] go beyond the level of surveys and form adequate textbooks for initial or further study.

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2.4.1 Structured peer-to-peer systems

As its name suggests, in a **structured peer-to-peer system** the nodes (i.e., processes) are organized in an overlay that adheres to a specific, deterministic topology: a ring, a binary tree, a grid, etc. This topology is used to efficiently look up data. Characteristic for structured peer-to-peer systems, is that they are generally based on using a so-called semantic-free index. What this means is that each data item that is to be maintained by the system, is uniquely associated with a key, and that this key is subsequently used as an index. To this end, it is common to use a hash function so that we get:

$$key(data\ item) = hash(data\ item's\ value).$$

The peer-to-peer system as a whole is now responsible for storing (*key*,*value*) pairs. To this end, each node is assigned an identifier from the same set of all possible hash values, and each node is made responsible for storing data associated with a specific subset of keys. In essence, the system is thus seen to implement a **distributed hash table**, generally abbreviated to a **DHT** [Balakrishnan et al., 2003].

Following this approach now reduces the essence of structured peer-to-peer systems to being able to look up a data item by its key. That is, the system provides an efficient implementation of a function *lookup* that maps a key to an *existing* node:

$$existing\ node = lookup(key).$$

This is where the topology of a structured peer-to-peer system plays a crucial role. Any node can be asked to look up a given key, which then boils down to efficiently *routing* that lookup request to the node responsible for storing the data associated with the given key.

To clarify these matters, let us consider a simple peer-to-peer system with a *fixed* number of nodes, organized into a hypercube. A **hypercube** is an n-dimensional cube. The hypercube shown in Figure 2.23 is four-dimensional. It can be thought of as two ordinary cubes, each with 8 vertices and 12 edges. To expand the hypercube to five dimensions, we would add another set of two interconnected cubes to the figure, connect the corresponding edges in the two halves, and so on.

For this (admittedly naive) system, each data item is associated with one of the 16 nodes. This can be achieved by hashing the value of a data item to a key $k \in \{0, 1, 2, \ldots, 2^4 - 1\}$. Now suppose that the node with identifier 0111 is requested to look up the data having key 14, corresponding to the binary value 1110. In this example, we assume that the node with identifier 1110 is responsible for storing all data items that have key 14. What node 0111 can simply do, is forward the request to a neighbor who is closer to node 1110. In this case, this is either node 0110 or node 1111. If it picks a node 0110, that

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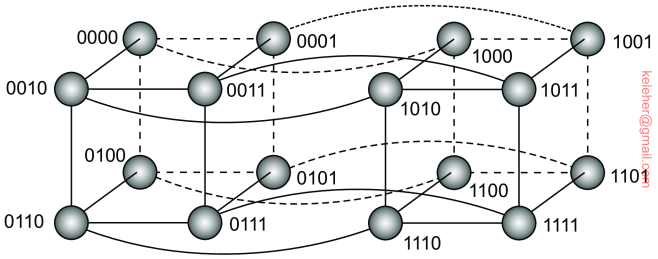


Figure 2.23: A simple peer-to-peer system organized as a four-dimensional hypercube.

node will then forward the request directly to a node 1110 from where the data can be retrieved.

Note 2.6 (Example: The Chord system)
The previous example illustrates two things: (1) the use of a hashing function to identify the node responsible for storing some data item, and (2) the routing along the topology of a peer-to-peer system when looking up a data item given its key. However, it is not a very realistic example, if only for the reason that we assumed that the total set of nodes is fixed. Let us therefore consider a more realistic example of a structured peer-to-peer system that is considered as belonging to the foundations for many other such systems.

In the **Chord system** [Stoica et al., 2003] the nodes are logically organized in a ring such that a data item with an m-bit key k is mapped to the node with the smallest (again, also m bit) identifier id $\geq k$. This node is referred to as the **successor** of key k and denoted as $succ(k)$. Keys and identifiers are typically 128 or 160 bits long. Figure 2.24 shows a much smaller Chord ring, where $m = 5$ and with nine nodes $\{1, 4, 9, 11, 14, 18, 20, 21, 28\}$. The successor of key 7 is equal to 9. Likewise, $succ(5) = 9$, but also $succ(9) = 9$. In Chord, each node maintains shortcuts to other nodes. A shortcut appears as a directed edge from one node to another. How these shortcuts are constructed is explained in Chapter 6. The construction is done in such a way that the length of the shortest path between any pair of nodes is of order $\mathcal{O}(\log N)$, where N is the total number of nodes.

To look up a key, a node will try to forward the request "as far as possible," but without passing it beyond the node responsible for that key. To clarify, suppose that in our example Chord ring, node 9 is asked to look up the node responsible for key 3 (which is node 4). Node 9 has four shortcuts: to nodes 11, 14, 18, and 28, respectively. As the node 28 is the farthest node 9 knows about and still preceding the one responsible for key 3, it will get the lookup request. Node 28 has three shortcuts: to nodes 1, 4, and 14, respectively. Note that node 28 has no knowledge about the existence of nodes between nodes 1 and 4. For this reason, the best what it can do is forward the request to the node 1. The latter knows that its successor in the ring is node 4, and thus that this is the node responsible for the key 3, to which it will subsequently forward the request.

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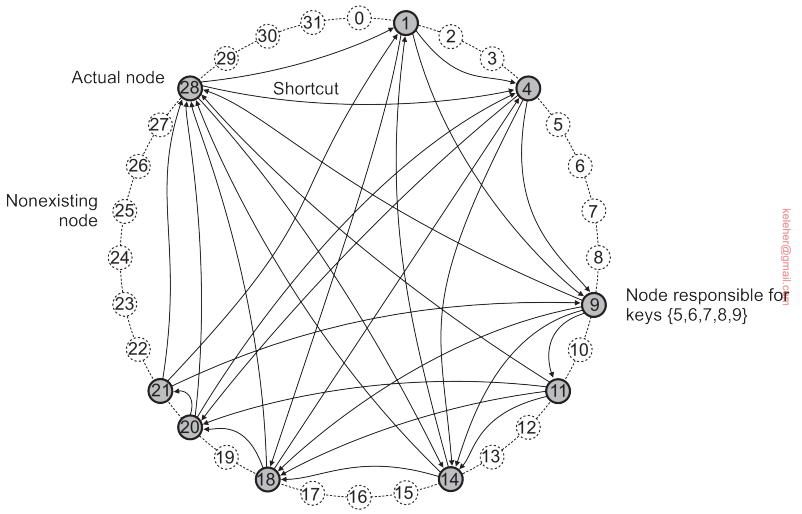


Figure 2.24: The organization of nodes and data items in Chord.

Now suppose that a node, with the unique identifier u, wants to join a Chord overlay. To that end, it contacts an arbitrary node and requests it to look up u, that is, return the value v= $succ(u)$. At that point, node u will simply need to insert itself between the predecessor of v and v itself, thus becoming the new predecessor of v. During this process, shortcuts from u to other nodes will be established, but also some existing ones previously directed toward v will now be adjusted to point to u (again, details are deferred until later chapters). Obviously, any data item with key k stored at v but for which $succ(k)$ is now equal to u is transferred from v to u. Leaving is just as simple: node u informs its departure to its predecessor and successor, and transfers its data items to $succ(u)$.

We return to Chord in more detail in Section 6.2.3 when discussing the resolution of random bit strings to network addresses.

2.4.2 Unstructured peer-to-peer systems

Structured peer-to-peer systems attempt to maintain a specific, deterministic overlay network. In contrast, in an **u**nstructured peer-to-peer system, each node maintains an ad hoc list of neighbors. The resulting overlay resembles what is known as a **random graph**: a graph in which an edge $\langle u, v \rangle$ between two nodes u and v exists only with a certain probability $\mathbb{P}[\langle u, v \rangle]$. Ideally, this probability is the same for all pairs of nodes, but in practice a wide range of distributions is observed.

In an unstructured peer-to-peer system, when a node joins, it often contacts a well-known node to obtain a starting list of other peers in the system. This list can then be used to find more peers, and perhaps ignore others, and so

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on. In practice, a node generally changes its local list almost continuously. For example, a node may discover that a neighbor is no longer responsive and that it needs to be replaced. There may be other reasons, which we will describe shortly.

Unlike structured peer-to-peer systems, looking up data cannot follow a predetermined route when lists of neighbors are constructed in an ad hoc fashion. Instead, in an unstructured peer-to-peer systems, we really need to resort to *searching* for data [Risson and Moors, 2006]. Let us look at two extremes and consider the case in which we are requested to search for specific data (e.g., identified by keywords).

Flooding: In the case of **flooding**, an issuing node u simply passes a request for a data item to all its neighbors. A request will be ignored when its receiving node, say v, had seen it before. Otherwise, v searches locally for the requested data item. If v has the required data, it can either respond directly to the issuing node u, or send it back to the original forwarder, who will then return it to *its* original forwarder, and so on. If v does not have the requested data, it forwards the request to all of its own neighbors.

Obviously, flooding can be expensive, for which reason a request often has an associated **time-to-live** or **TTL** value, giving the maximum number of hops a request is allowed to be forwarded. Choosing the right TTL value is crucial: too small means that a request will stay close to the issuer and may thus not reach a node having the data. Too large incurs high communication costs.

As an alternative to setting TTL values, a node can also start a search with an initial TTL value of 1, meaning that it will first query only its neighbors. If no, or not enough results are returned, the TTL is increased, and a new search is initiated.

Random walks: At the other end of the search spectrum, an issuing node u can simply try to find a data item by asking a randomly chosen neighbor, say v. If v does not have the data, it forwards the request to one of its randomly chosen neighbors, and so on. The result is known as a **random walk** [Gkantsidis et al., 2006; Lv et al., 2002]. Obviously, a random walk imposes much less network traffic, yet it may take much longer before a node is reached that has the requested data. To decrease the waiting time, an issuer can simply start n random walks simultaneously. Indeed, studies show that in this case, the time it takes before reaching a node that has the data drops approximately by a factor n. Lv et al. [2002] reports that relatively small values of n, such as 16 or 64, turn out to be effective.

A random walk also needs to be stopped. To this end, we can either again use a TTL, or alternatively, when a node receives a lookup request, check

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with the issuer whether forwarding the request to another randomly selected neighbor is still needed.

Note that neither method relies on a specific comparison technique to decide when requested data has been found. For structured peer-to-peer systems, we assumed the use of keys for comparison; for the two approaches just described, any comparison technique would suffice.

Between flooding and random walks lie **policy-based search methods**. For example, a node may decide to keep track of peers who responded positively, effectively turning them into preferred neighbors for succeeding queries. Likewise, we may want to restrict flooding to fewer neighbors, but in any case give preference to neighbors having many neighbors themselves.

Note 2.7 (Advanced: Flooding versus random walks)
When giving the matter some thought, it may come as a surprise that people have even considered a random walk as an alternative way to search. At first instance, it would seem like a technique resembling the search for a needle in a haystack. However, we need to realize that in practice we are dealing with *replicated* data, and even for minimal replication factors and different replication distributions, studies show that deploying random walks is not only effective, it can also be much more efficient in comparison to flooding.

To see why, we closely follow the model described in Lv et al. [2002] and Cohen and Shenker [2002]. Assume there are a total of N nodes and that each data item is replicated across r randomly chosen nodes. A search consists of repeatedly selecting a node at random until the item is found. If $\mathbb{P}[k]$ is the probability that the item is found after k attempts, we have

$$\mathbb{P}[k] = \frac{r}{N}(1 - \frac{r}{N})^{k-1}.$$

Let the average search size S be the expected number of nodes that need to be probed before finding the requested data item:

$$S = \sum_{k=1}^{N} k \cdot \mathbb{P}[k] = \sum_{k=1}^{N} k \cdot \frac{r}{N}(1 - \frac{r}{N})^{k-1} \approx N/r \text{ for } 1 \ll r \leq N.$$

By simply replicating every data item to each node, $S = 1$ and it is clear that a random walk will always outperform flooding even for TTL values of 1. More realistically, however, is to assume that r/N is relatively low, such as 0.1%, meaning that the average search size would be approximately 1000 nodes.

To compare this to flooding, assume that each node, on average, forwards a request to d randomly selected neighbors. After one step, the request will have arrived at d nodes, each of who will forward it to another $d - 1$ nodes (assuming that the node from where the request came is skipped), and so on. In other words, after k steps, and considering that a node can receive the request more than once, we will have reached (at most) the following number of nodes:

$$R(k) = d(d-1)^{k-1}$$

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Various studies show that $R(k)$ is a good estimate for the actual number of nodes reached, as long as we have only a few number of flooding steps. Of these nodes, we can expect a fraction of r/N to have the requested data item, meaning that when $\frac{r}{N} \cdot R(k) \geq 1$, we will most likely have found a node that has the data item.

To illustrate, let $r/N = 0.001 = 0.1\%$, which means that $S \approx 1000$. With flooding to, on average, $d = 10$ neighbors, we would require at least 4 flooding steps, reaching some 7290 nodes, which is considerably more than the 1000 nodes required when using a random walk. Only with $d = 33$ will we need to contact approximately also 1000 nodes in $k = 2$ flooding steps and having $r/N \cdot R(k) \geq 1$.

The obvious drawback of deploying random walks, is that it may take much longer before an answer is returned.

2.4.3 Hierarchically organized peer-to-peer networks

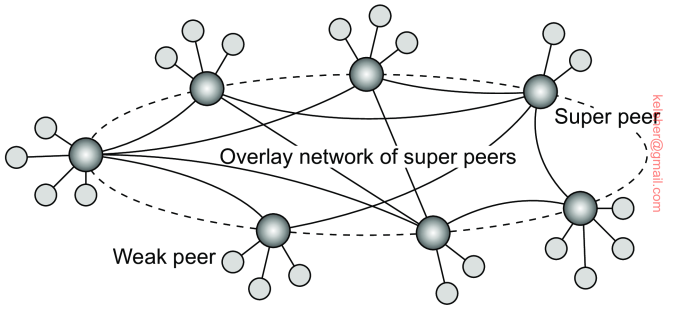
Notably in unstructured peer-to-peer systems, locating relevant data items can become problematic as the network grows. The reason for this scalability problem is simple: as there is no deterministic way of routing a lookup request to a specific data item, essentially the only technique a node can resort to is *searching* for the request by flooding or randomly walking through the network. As an alternative, many peer-to-peer systems have proposed to make use of special nodes that maintain an index of data items.

There are other situations in which abandoning the symmetric nature of peer-to-peer systems is sensible. Consider a collaboration of nodes that offer resources to each other. For example, in a collaborative **Content Delivery Network** (CDN), nodes may offer storage for hosting copies of Web documents allowing Web clients to access pages nearby, and thus to access them quickly. What is needed is a means to find out where documents can be stored best. In that case, making use of a **broker** that collects data on resource usage and availability for several nodes that are in each other's proximity allows selecting a node quickly with sufficient resources.

Nodes such as those maintaining an index or acting as a broker are generally referred to as **super peers**. As the name suggests, super peers are often also organized in a peer-to-peer network, leading to a hierarchical organization, as explained in Yang and Garcia-Molina [2003]. A simple example of such an organization is shown in Figure 2.25. In this organization, every regular peer, now referred to as a **weak peer**, is connected as a client to a super peer. All communication from and to a weak peer proceeds through that peer's associated super peer.

Often, the association between a weak peer and its super peer is fixed: whenever a weak peer joins the network, it attaches to one of the super peers and remains attached until it leaves the network. Obviously, it is expected that super peers are long-lived processes with high availability. To compensate for potential unstable behavior of a super peer, backup schemes can be deployed,

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Figure 2.25: A hierarchical organization of nodes into a super-peer network.

such as pairing every super peer with another one and requiring weak peers to attach to both.

Having a fixed association with a super peer may not always be the best solution. For example, in the case of file-sharing networks, it may be better for a weak peer to attach to a super peer that maintains an index of files that the weak peer is currently interested in. In that case, chances are bigger that when a weak peer is looking for a specific file, its super peer will know where to find it. Garbacki et al. [2010] describe a relatively simple scheme in which the association between weak peer and strong peer can change as weak peers discover better super peers to associate with. In particular, a super peer returning the result of a lookup operation is given preference over other super peers.

As we have seen, peer-to-peer networks offer a flexible means for nodes to join and leave the network. However, with super-peer networks a new problem is introduced, namely how to select the nodes that are eligible to become super peer. This problem is closely related to the **leader-election problem**, which we discuss in Section 5.4.

2.4.4 Example: BitTorrent

Let us consider the widely popular **BitTorrent** file-sharing system [Cohen, 2003] as an example of a (largely) unstructured peer-to-peer system. BitTorrent is a file-downloading system. Its principle working is shown in Figure 2.26. The basic idea is that when an end user is looking for a file, she downloads chunks of the file from other users until the downloaded chunks can be assembled, yielding the complete file. An important design goal was to ensure collaboration. In most file-sharing systems, a significant fraction of participants merely download files but otherwise contribute close to nothing [Adar and Huberman, 2000; Saroiu et al., 2003; Yang et al., 2005], a phenomenon referred to as **free riding**. To prevent this situation, in BitTorrent a file can

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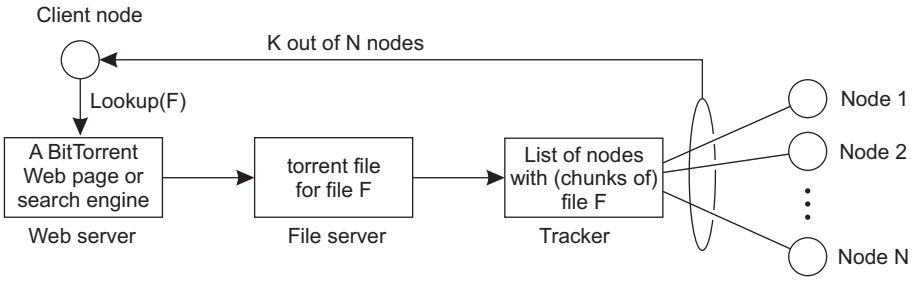


Figure 2.26: The principle working of BitTorrent [adapted with permission from Pouwelse et al. [2005].

be downloaded only when the downloading client is providing content to someone else.

To download a file, a user needs to access a global directory, which is generally just one of a few well-known Websites. Such a directory contains references to what are called torrent files. A **torrent file** contains the information that is needed to download a specific file. In particular, it contains a link to what is known as a **tracker**, which is a server that is keeping an accurate account of *active* nodes that have (chunks of) the requested file. An active node is one that is currently downloading the file as well. Obviously, there will be many trackers, although there will generally be only a single tracker per file (or collection of files).

Once the nodes have been identified from where chunks can be downloaded, the downloading node effectively becomes active. At that point, it will be forced to help others, for example by providing chunks of the file it is downloading that others do not yet have. This enforcement comes from a simple rule: if a node P notices that a node Q is downloading more than it is uploading, P can decide to decrease the rate at which it sends data to Q. This scheme works well, provided P has something to download from Q. For this reason, nodes are often supplied with references to many other nodes, putting them in a better position to trade data.

Clearly, BitTorrent combines centralized with decentralized solutions. As it turns out, the bottleneck of the system is easily formed by the trackers. In an alternative implementation of BitTorrent, a node also joins a separate structured peer-to-peer system (i.e., a DHT) to assist in tracking file downloads. In effect, a central tracker's load is now distributed across the participating nodes, with each node acting as a tracker for a relatively small set of torrent files. The original function of the tracker coordinating the collaborative downloading of a file is retained. However, we note that in many BitTorrent systems used today, the tracking functionality has actually been minimized to a one-time provisioning of peers currently involved in downloading the file. From that moment on, the newly participating peer will communicate only

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with those peers and no longer with the initial tracker. The initial tracker for the requested file is looked up in the DHT through a so-called **magnet link**. We return to DHT-based lookups in Section 6.2.3.

2.5 Hybrid system architectures

Real-world distributed systems are complex in the sense that they combine a myriad of architectures: centralized features are combined with peer-to-peer features are combined with hierarchical organizations, etc. The complexity is aggravated by the fact that many distributed systems cross organizational boundaries, leading to truly decentralized solutions in which even no single organization can take responsibility for a system's operation. In this section, we will take a closer look into these complex, hybrid system architectures.

2.5.1 Cloud computing

Organizations in charge of running data centers have been seeking ways for opening up their resources to customers. Eventually, this led to the concept of **utility computing** by which a customer could upload tasks to a data center and be charged on a per-resource basis. Utility computing formed the basis for what is now commonly referred to as **cloud computing**.

Following Vaquero et al. [2008], cloud computing is characterized by an easily usable and accessible pool of *virtualized* resources. Which and how resources are used can be configured dynamically, providing the basis for scalability: if more work needs to be done, a customer can simply acquire more resources. The link to utility computing is formed by the fact that cloud computing is generally based on a pay-per-use model in which guarantees are offered by customized **service-level agreements** (**SLAs**). Keeping it simple, clouds are organized into four layers, as shown in Figure 2.27.

Hardware: The lowest layer is formed by the means to manage the necessary hardware: processors, routers, but also power and cooling systems. It is generally implemented at data centers and contains the resources that customers normally never get to see directly.

Infrastructure: This is an important layer forming the backbone for most cloud computing platforms. It deploys virtualization techniques (discussed in Section 3.2) to provide customers an infrastructure consisting of virtual storage and computing resources. Indeed, nothing is what it seems: cloud computing evolves around allocating and managing virtual storage devices and virtual servers.

Platform: One could argue that the platform layer provides to a cloud-computing customer what an operating system provides to application developers, namely the means to easily develop and deploy applications

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that need to run in a cloud. In practice, an application developer is offered a vendor-specific API, which includes calls to uploading and executing a program in that vendor's cloud. In a sense, this is comparable to the Unix exec family of system calls, which take an executable file as a parameter and pass it to the operating system to be executed.

Furthermore, like operating systems, the platform layer provides higher-level abstractions for storage and such. For example, as we discussed, the **Amazon S3 storage system** [Murty, 2008; Culkin and Zazon, 2022] is offered to the application developer in the form of an API allowing (locally created) files to be organized and stored in **buckets**. By storing a file in a bucket, that file is automatically uploaded to the Amazon cloud.

Application: Actual applications run in this layer and are offered to users for further customization. Well-known examples include those found in office suites (text processors, spreadsheet applications, presentation applications, and so on). It is important to realize that these applications are again executed in the vendor's cloud. As before, they can be compared to the traditional suite of applications that are shipped when installing an operating system.

Cloud-computing providers offer these layers to their customers through various interfaces (including command-line tools, programming interfaces, and Web interfaces), leading to three different types of services:

- **Infrastructure-as-a-Service** (**IaaS**) covering the hardware and infrastructure layer.
- **Platform-as-a-Service** (**PaaS**) covering the platform layer.
- **Software-as-a-Service** (**SaaS**) in which their applications are covered.

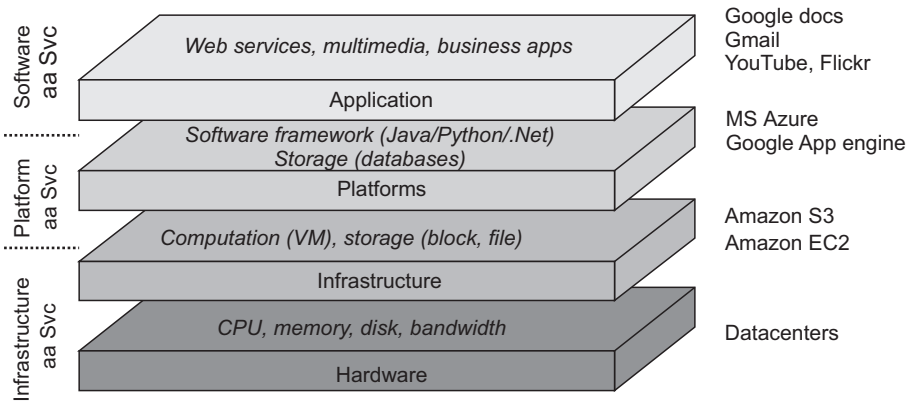


Figure 2.27: The organization of clouds (adapted from Zhang et al. [2010]).

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As of now, making use of clouds is relatively easy, and we discuss in later chapters more concrete examples of interfaces to cloud providers. As a consequence, cloud computing as a means for outsourcing local computing infrastructures has become a serious option for many enterprises.

From the perspective of a system architecture, which deals with configuring (micro)services across some infrastructure, one may argue that in the case of cloud computing, we are dealing with a highly advanced client-server architecture. However, let it be noted that the actual implementation of a server is generally completely hidden from the client: it is often unclear *where* the server actually is, and even whether the server is actually implemented in a fully distributed manner (which it often is). To further illustrate this point, the notion of a **Function-as-a-Service**, or simply **Faas**, allows a client to execute code without bothering even with starting a server to handle the code (see also Shahrad et al. [2019]).

2.5.2 The edge-cloud architecture

In the advent of increasingly more network-connected devices and the emergence of the **Internet-of-Things** (**IoT**) many became aware of the fact that we may need more than just cloud computing. **Edge computing** was born. There is a lot to say about edge computing, and much has already been said. And as is the case with so many topics in distributed systems, it simply takes a few years before things settle down a bit. In this section, we take a look at edge computing from an architectural perspective and will return to various elements throughout the book, often without even explicitly mentioning edge computing. An excellent overview of edge computing is given by Yousefpour et al. [2019] and the interested reader is referred to that paper to get a better grip on its nomenclature. We deliberately take a simplified and broad view of edge computing, using it as a general term for most of the things sitting between the devices that comprise the Internet-of-Things and the services typically offered through cloud computing. In this sense, we follow the discussion as presented by Horner [2021].

As its name suggests, edge computing deals with the placement of services "at the edge" of the network. This edge is often formed by the boundary between enterprise networks and the actual Internet, for example, as provided by an **Internet Service Provider** (**ISP**). For example, many universities reside on a campus consisting of various buildings, each having their own local network, in turn connected through a campuswide network. As part of the campus, there may be multiple on-premise services for storage, computing, security, lectures, and so on. On-premise means that the local IT department is responsible for hosting those services on servers directly hooked up to the campus network. Much of the traffic related to those services will never leave the campus network, and the network together with its servers and services form a typical **edge infrastructure**.

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At the same time, such servers may be connected to those of other universities and perhaps even making use of, again, other servers. In other words, instead of setting up connections between universities in a peer-to-peer fashion, we also see configurations in which various universities share services through a logically centralized infrastructure. Such an infrastructure may be situated "in the cloud," but it may equally well have been set up through a regional infrastructure using locally available data centers. As we move closer to cloud infrastructures, the term **fog computing** is often used. We thus see an overall picture emerge as the one shown in Figure 2.28, where the boundaries between cloud and edge are becoming blurred.

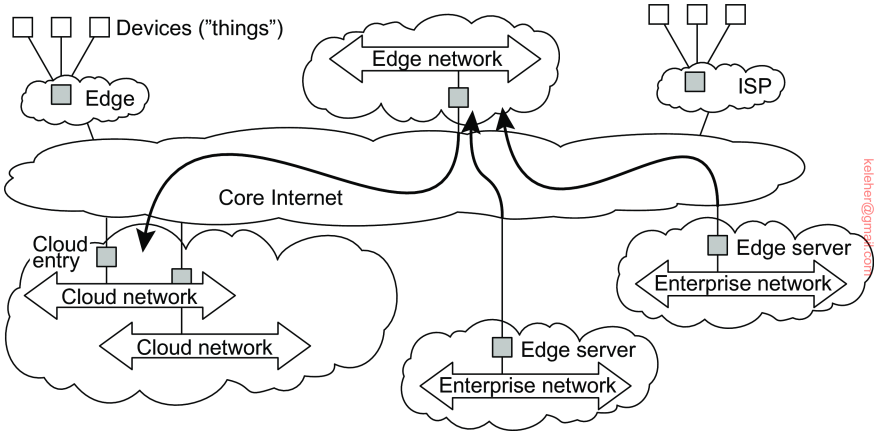


Figure 2.28: A collection of infrastructures involving edge devices, edge infrastructures and cloud infrastructures, and a possible setup between two enterprise edge infrastructures, an intermediate edge infrastructure, and a cloud infrastructure.

Many configurations for an edge infrastructure easily come to mind, ranging from infrastructures needed to keep track of your activities, layered video-streaming infrastructures, gaming infrastructures, etc. What all of these have in common is that there is some smart end device that one way or the other needs to (eventually) connect to a service hosted somewhere in the cloud. The question then pops up why an edge infrastructure is needed at all. Logically, it seems much simpler to just connect to the cloud service directly using existing and often excellent networking facilities. Let us take a critical look at a few arguments.

Latency and bandwidth What should have become clear from our examples is that edge infrastructures are considered to be close to the end devices. Closeness can be measured in terms of latency and often also bandwidth. Throughout the decades, bandwidth, or actually lack of bandwidth, has

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always been used as an argument for introducing solutions close to specific devices. However, if anything has become clear all this time, is that available bandwidth continues to increase, now reaching the point that one should seriously question how problematic it actually is, and whether installing and maintaining edge infrastructures for having insufficient bandwidth is a good reason. Nevertheless, there are situations in which closeness to end devices is actually needed to guarantee quality of service. The canonical example is formed by video services: the closer the video sources are, the better bandwidth guarantees can be given, reducing issues such as jitter.

More problematic is when Mother Nature gets in our way. This may easily happen when dealing with latency. It may take 100 ms to reach a cloud, rendering many interactive applications quite useless. One such important application is (semi-)autonomous driving. A car will need to continuously observe its environment through a myriad of sensors and react accordingly. Having to coordinate its movements through the cloud is not acceptable from a real-time aspect alone. This example also illustrates that cars may need to detect each other beyond the capabilities of their sensors, for example, when heading toward a junction with clear visibility. In a real-time system, cars may be able to provide their current position to a local edge infrastructure and reveal themselves to each other when approaching the junction.

Other examples in which latency plays a crucial role easily come to mind. Overcoming latency is one of the most compelling reasons for developing edge infrastructures.

Reliability Many argue that cloud connectivity is simply not reliable enough for many applications, for which reason edge infrastructures should be deployed. To what extent this is a valid argument remains to be seen. The fact is that for many networked applications, connectivity is generally good and reliable, if not excellent. Of course, there are situations in which relying on 24/7 reliability is not an option. This may be the case for hospitals, factories, and other critical settings in general. Yet, in those cases, measures have been traditionally already taken and to what extent edge computing brings in anything new is not always clear.

Security and privacy Finally, many argue that edge solutions enhance security and privacy. It all depends. One could argue that if a cloud solution is not secure, then there is no reason why an edge solution would be. An implicit assumption that many people make is that an edge infrastructure is owned by a specific organization and operates within the (protected) network boundaries of that organization. In that case, it often does indeed become simpler to protect data and operations, yet one should ask whether such protection is sufficient. As we shall discuss in Chapter 9, simply trying to protect assets by developing a secure wall around an organization does not help against

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insider attacks, whether they are intended or not. A same reasoning holds for privacy: if we cannot protect personal data in the cloud, then why would an edge infrastructure suffice for privacy? A thorough discussion on the role and position of privacy in the edge and edge devices is given by Hong [2017]. From that discussion, it is clear that there is still considerable work to be done.

However, there may be another reason related to security and privacy why edge infrastructures are needed. In many cases, organizations are simply not allowed, for whatever regulatory reasons, to place data in the cloud or have data be processed by a cloud service. For example, medical records may have to be kept on premise on certified servers and with strict audit procedures in place. In this case, an organization will have to resort to maintaining an edge infrastructure.

Introducing additional layers between end devices and cloud infrastructures opens a whole can of worms compared to the relatively simple situation of just having to deal with cloud computing. For the latter, one can argue that the cloud provider to a considerable extent decides where and how a service is actually implemented. In practice, we will be dealing with a data center in which the (micro)services that make up the entire service are distributed across multiple machines.

Matters become more intricate in the case of edge computing. In this case, the client organization will now have to make informed decisions on what to do where. Which services need to be placed on premise on a local edge infrastructure, and which can be moved to the cloud? To what extent does an edge infrastructure offer facilities for virtual resources, akin to the facilities offered in cloud computing? Moreover, where we may be able to assume that computational and storage resources are in abundance when dealing with a cloud, this is not necessarily the case for an edge infrastructure. In practice, the latter simply have less hardware resources available, but often also offer less flexibility in terms of available platforms.

By and large, allocating resources in the case of edge computing appears to be much more challenging in comparison to clouds. As summarized by Hong and Varghese [2019], we are dealing with limitations when it comes to resources, higher degrees of hardware heterogeneity, and much more dynamic workloads, which, when taken together, have led to a higher demand of **orchestration**. Moreover, where from a client's perspective the cloud appears to be hiding many of its internal intricacies, this is necessarily no longer the case, making it much more difficult to do the orchestration [Bittencourt et al., 2018]. Orchestration boils down to the following (see also Taleb et al. [2017]):

- **Resource allocation**: specific services require specific resources. The question is then to guarantee the availability of the resources required to perform a service. Typically, resources amount to CPU, storage, memory, and networking facilities.

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- **Service placement**: regardless the availability of resources, it is important to decide *when* and *where* to place a service. This is notably relevant for mobile applications, for in that case finding the edge infrastructure that is closest to that application may be crucial. A typical use case is that of video conferencing, for which the encoding is often not done on the mobile device, but at an edge infrastructure. In practice, one needs to decide at which edges the service should be installed. An extensive overview of service placement in the case of edge computing is provided by Salaht et al. [2020].
- **Edge selection**: related to service placement is deciding which edge infrastructure should be used when the service needs to be offered. It may seem logical to use the edge infrastructure closest to the end device, but all kinds of circumstances may ask for an alternative solution, for example the connectivity of that edge to the cloud provider.

Other issues play a role as well, but it should be clear by now that the edge-cloud architecture is much more demanding than one might initially think it to be. Moreover, the different perspectives on how the continuum between end devices and the cloud should be filled with edge components and solutions has still to converge [Antonini et al., 2019].

2.5.3 Blockchain architectures

An upcoming and much debated type of distributed system is that of so-called **blockchains**. Blockchain systems enable the registration of transactions, for which reason they are also referred to as **distributed ledgers**. The latter is actually more accurate, with blockchains forming one of different ways for implementing distributed ledgers.

The key issue in transaction systems is that a transaction is validated, effectuated, and subsequently stored for various auditing purposes. For example, Alice may decide to create a transaction stating that she transfers $10 to Bob's account. Normally, she would go to a bank, where she would have to sign the transaction to prove that she really wants it to be carried out. Whether this all happens physically or digitally does not really matter. The bank will check whether she has enough credit, whether Bob is eligible for receiving the money, and assuming everything is fine, will subsequently transfer the money. A record of the transaction is kept for all kinds of auditing purposes. Note that transactions in blockchains systems are taken very broad. Besides monetary transactions, systems have been developed for registering identification documents, registering resource usage and allocation, electronic voting, and sharing health records, to name a few.

The bank operates as a **trusted third party**. An important design assumption for blockchains is that participating parties can, in principle, not be trusted. This also excludes having a trusted third party that handles all

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transactions. We will return to trust in Section 9.4, and concentrate on the implications lack of trust has for the architecture of a system.

In the case of blockchains, we assume there is a (potentially very large) set of participants who jointly register transactions among them in a publicly available ledger. In this way, any participant can see what has happened and also verify the validity of a transaction. For example, in a blockchain system for digital coins, each having a unique and unforgeable ID, any participant should be able to check whether a coin has already been spent by checking all transactions that have taken place since the beginning.

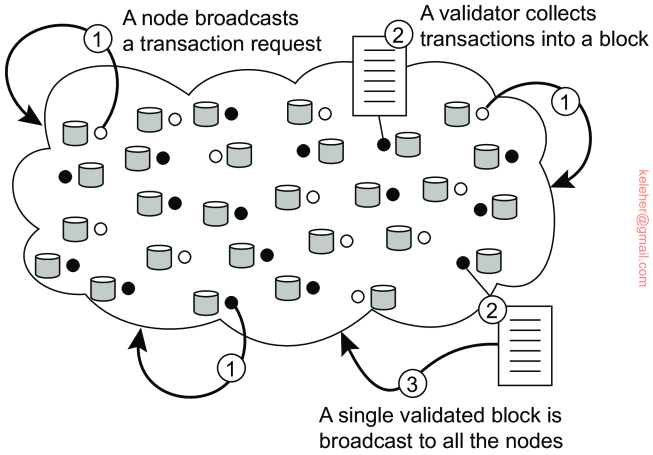


Figure 2.29: The principle operation of a blockchain.

To this end, when Alice wants to transfer $10 to Bob, she essentially tells all the participants in the blockchain system about this intent, thus allowing volunteers to validate the intended transaction. This is shown as Step 1 in Figure 2.29. To avoid having to check every transaction separately one-by-one as they are submitted, a validator groups several transactions into a block to increase efficiency, shown as Step 2 in Figure 2.29. If everything goes well, i.e., the transactions in the block are considered to be valid, the validator securely protects the block against any modifications, and appends the now immutable block to the chain of other blocks with validated transactions. It does so by broadcasting the validated block to all participants, shown as Step 3 in Figure 2.29.

An important observation is that there is logically only a single chain of blocks with validated transactions. Each block is immutable, in the sense that if an adversary decides to modify any transaction from any block in that chain, the modification can never go unnoticed. Securely protecting blocks of transactions against modifications, but also securely appending a block to an existing list, are well-understood techniques, which we will explain further

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in Section 9.4.3. The immutability of a block makes it an ideal fit for massive replication: it will never be changed anyway, so someone may just as well store it locally to make verification as simple as possible. Effectively, the logically single chain of immutable blocks may be physically massively replicated across the Internet among all participating parties. This is precisely what happens: each block is broadcast to every participating node in a blockchain, as we just explained.

What differentiates so many blockchain systems from each other is deciding on which node(s) may actually carry out validation tasks. In other words, we need to figure out who is allowed to append a block of validated transactions to the existing chain. Appending such a block means that there is global consensus on fulfilled transactions. It is therefore important that we also reach consensus on which validator can move ahead. All others will have to do their validation over again, as their transactions may be affected by the newly appended ones and thus may need to be revisited.

Deciding on which validator can move ahead requires **(distributed) consensus**. In principle, there are three options:

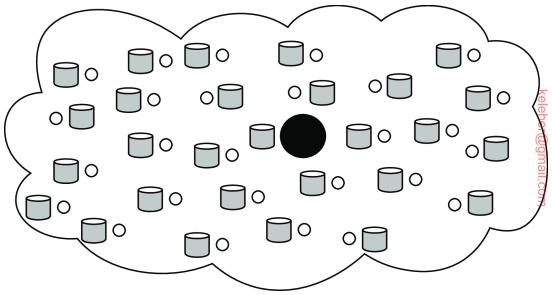
1. A centralized solution, in which a trusted third party validates transactions as before.
2. A distributed solution, in which a small, preselected group of processes takes over the role of a trusted third party.
3. A fully decentralized solution, in which, in principle, all participating nodes in the blockchain jointly reach consensus without any (distributed) third party.

These three options are shown in Figure 2.30. As mentioned, each node participating in the blockchain is assumed to have a full copy locally available.

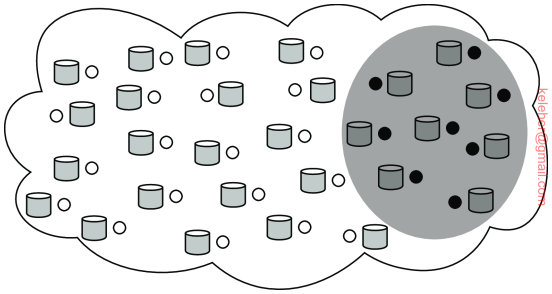
Obviously, a centralized architecture for a blockchain does not fit its design goals, which state that there is essentially no place for a trusted third party. The distributed architecture is an interesting one. In this case, there is a relatively small group of nodes that are *permissioned* to validate transactions. For blockchains, it is important to realize that none of these permissioned nodes are assumed to be trusted, yet they are assumed to run a consensus protocol that can withstand malicious behavior. Specifically, if there are n permissioned nodes, then it is assumed that at most $k \leq (n-1)/3$ will fail and perhaps act maliciously. One problem with such so-called **permissioned blockchains** is that n is quite limited, in practice, to less than a few tens of nodes.

Finally, in so-called **permissionless blockchains** all nodes collectively participate to validate transactions. In practice, this means that all nodes who want to validate transactions are engaged in a process called **leader election**. The process elected as leader appends a block to the current chain (and is

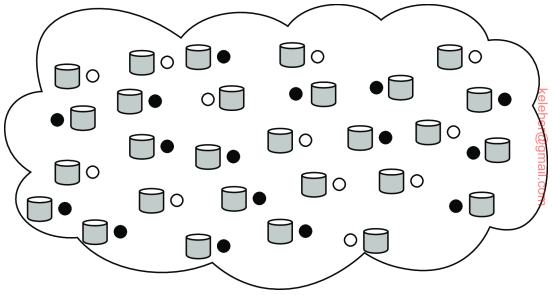
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(a) Centralized



(b) Distributed (permissioned)



(c) Decentralized (permissionless)

Figure 2.30: The three different organizations of blockchains: (a) centralized, (b) distributed, (c) fully decentralized. Filled nodes represent validators; other nodes are participants not engaged in validation.

often rewarded for that). In practice, not all participating nodes will want to act as validator, if only because the leader-election algorithm is costly in terms of resources. We return to leader elections in Section 5.4.

The architecture of a blockchain system is thus seen to be quite complex. In a permissioned system, we have a few tens of nodes for validating transactions. None of these nodes needs to be individually trusted beforehand, yet it may be argued that they form a centralized, fault-tolerant distributed group. Trust

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is needed in so far that we need to assume that not too many of those nodes act maliciously or collude against any decisions they should make. On the other hand, permissionless blockchains may be viewed as being fully decentralized, but here we see that special measures are needed to guarantee some form of fairness among willing validators. In fact, through the dynamics of permissionless blockchains, we often see that, in practice, only a relatively few number of nodes can carry out validation tasks, effectively also leading to a more centralized system. An overview of the various settings in blockchains from the perspective of architectures is given by Xu et al. [2017].

2.6 Summary

Distributed systems can be organized in many ways. We can make a distinction between software architecture and system architecture. The latter considers where the components that constitute a distributed system are placed across the various machines. The former is more concerned about the logical organization of the software: how do components interact, in what ways can they be structured, how can they be made independent, and so on?

A keyword when talking about architectures is architectural style. A style reflects the basic principle that is followed in organizing the interaction between the software components comprising a distributed system. Important styles include layering, service-oriented styles, and styles in which handling events are prominent, exemplified by are known as publish-subscribe styles.

There are many organizations of distributed systems. An important class is where machines are divided into clients and servers. A client sends a request to a server, who will then produce a result that is returned to the client. The client-server architecture reflects the traditional way of modularizing software, in which a module calls the functions available in another module. By placing different components on different machines, we obtain a natural physical distribution of functions across a collection of machines.

Client-server architectures are often highly centralized. In decentralized architectures, we often see an equal role played by the processes that constitute a distributed system, also known as peer-to-peer systems. In peer-to-peer systems, the processes are organized into an overlay network, which is a logical network in which every process has a local list of other peers that it can communicate with. The overlay network can be structured, in which case deterministic schemes can be deployed for routing messages between processes. In unstructured networks, the list of peers is more or less random, implying that search algorithms need to be deployed for locating data or other processes.

In hybrid architectures, elements from centralized and decentralized organizations are combined. A typical example is that of cloud computing, which logically follows a client-server architecture, but where the server is generally

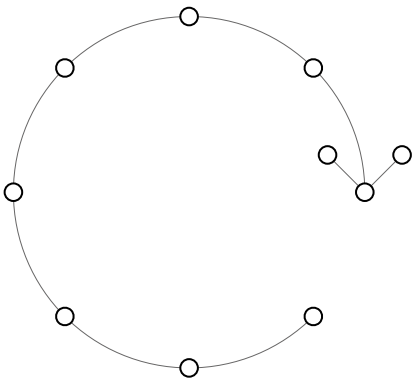
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completely distributed across a data center. In the last decade, we have seen a strong emergence of what is known as edge computing. Edge infrastructures form several steps between end devices and clouds and are demanding from the view point of organizing and configuring distributed systems. Finally, as an example in which decentralization plays a prominent role, the increasing popular blockchain architecture illustrates yet another class of hybrid system architectures.

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03

PROCESSES



In this chapter, we take a closer look at how the different types of processes play a crucial role in distributed systems. The concept of a process originates from the field of operating systems, where it is generally defined as a program in execution. From an operating-system perspective, the management and scheduling of processes are perhaps the most important issues to deal with. However, when it comes to distributed systems, other issues turn out to be equally or more important.

We start with extensively discussing threads and their role in distributed systems. As it turns out, threads play a crucial role in obtaining performance in multicore and multiprocessor environments, but also help in structuring clients and servers. There are many cases where we see threads being replaced by processes and using the underlying operating system for guaranteeing protection and facilitating communication. Nevertheless, when performance is at stake, threads continue to play an important role.

Since a few years, the concept of virtualization has regained much popularity. Virtualization allows an application, and possibly also its complete environment including the operating system, to run concurrently with other applications, but highly independent of the underlying hardware and platforms, leading to a high degree of portability. Moreover, virtualization helps in isolating failures caused by errors or security problems. It is an important concept for distributed systems, and we pay attention to it in a separate section.

Client-server organizations are important in distributed systems. In this chapter, we take a closer look at typical organizations of both clients and servers. We also pay attention to general design issues for servers, including those typically used in object-based distributed systems. A widely used Web server is Apache, to which we pay separate attention. The organization of server clusters remains important, especially when they need to collaboratively provide the illusion of a single system. we will discuss examples of how to achieve this perspective, including wide-area servers like PlanetLab.

An important issue, especially in wide-area distributed systems, is moving processes between different machines. Process migration or more specifically, code migration, can help in achieving scalability, but can also help to configure clients and servers dynamically. What is actually meant by code migration and what its implications are is also discussed in this chapter.

3.1 Threads

Although processes form a building block in distributed systems, practice indicates that the granularity of processes as provided by the operating systems on which distributed systems are built is not sufficient. Instead, it turns out that having a finer granularity in the form of multiple threads of control per process makes it much easier to build distributed applications

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and to get better performance. In this section, we take a closer look at the role of threads in distributed systems and explain why they are so important. More on threads and how they can be used to build applications can be found in [Lewis and Berg, 1998; Stevens, 1999; Robbins and Robbins, 2003]. Herlihy and Shavit [2008] is highly recommended to learn more about multithreaded concurrent programming in general.

3.1.1 Introduction to threads

To understand the role of threads in distributed systems, it is important to understand what a process is, and how processes and threads relate. To execute a program, an operating system creates a number of **virtual processors**, each one for running a different program. To keep track of these virtual processors, the operating system has a **process table**, containing entries to store CPU register values, memory maps, open files, accounting information, privileges, etc. Jointly, these entries form a **process context**.

A process context can be viewed as the software analog of the hardware's **processor context**. The latter consists of the minimal information that is automatically stored by the hardware to handle an interrupt, and to later return to where the CPU left off. The processor context contains at least the program counter, but sometimes also other register values such as the stack pointer.

A **process** is often defined as a program in execution, that is, a program that is currently being executed on one of the operating system's virtual processors. An important issue is that the operating system takes great care to ensure that independent processes cannot maliciously or inadvertently affect the correctness of each other's behavior. In other words, the fact that multiple processes may be concurrently sharing the same CPU and other hardware resources is made transparent. Usually, the operating system requires hardware support to enforce this separation.

This concurrency transparency comes at a price. For example, each time a process is created, the operating system must create a complete independent address space. Allocation can mean initializing memory segments by, for example, zeroing a data segment, copying the associated program into a text segment, and setting up a stack for temporary data. Likewise, switching the CPU between two processes may require some effort as well. Apart from saving the data as currently stored in various registers (including the program counter and stack pointer), the operating system will also have to modify registers of the **Memory Management Unit** (**MMU**) and invalidate address translation caches, such as in the **Translation Lookaside Buffer** (**TLB**). In addition, if the operating system supports more processes than it can simultaneously hold in main memory, it may have to swap processes between main memory and disk before the actual switch can take place.

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Like a process, a thread executes its own piece of code, independently of other threads. However, in contrast to processes, no attempt is made to achieve a high degree of concurrency transparency if this would result in performance degradation. Therefore, a thread system generally maintains only the minimum information to allow a CPU to be shared by several threads. In particular, a **thread context** often consists of nothing more than the processor context, along with some other information for thread management. For example, a thread system may keep track of the fact that a thread is currently blocked on a mutex variable, so as not to select it for execution. Information that is not strictly necessary to manage multiple threads is generally ignored. For this reason, protecting data against inappropriate access by threads within a single process is left entirely to application developers. We thus see that a processor context is contained in a thread context, and that, in turn, a thread context is contained in a process context.

There are two important implications of deploying threads, as we just sketched. First, the performance of a multithreaded application need hardly ever be worse than that of its single-threaded counterpart. In fact, often, multithreading even leads to a performance gain. Second, because threads are not automatically protected against each other the way processes are, development of multithreaded applications requires additional intellectual effort. Proper design and keeping things simple, as usual, help a lot. Unfortunately, current practice does not demonstrate that this principle is equally well understood.

Thread usage in nondistributed systems

Before discussing the role of threads in distributed systems, let us first consider their usage in traditional, nondistributed systems. There are several benefits to multithreaded processes that have increased the popularity of using thread systems.

The most important benefit comes from the fact that in a single-threaded process, whenever a blocking system call is executed, the process as a whole is blocked. To illustrate, consider an application such as a spreadsheet program, and assume that a user continuously and interactively wants to change values. An important property of a spreadsheet program is that it maintains the functional dependencies between different cells, often from different spreadsheets. Therefore, whenever a cell is modified, all dependent cells are automatically updated. When a user changes the value in a single cell, such a modification can trigger a large series of computations. If there is only a single thread of control, computation cannot proceed while the program is waiting for input. Likewise, it may be difficult to provide input while dependencies are being calculated. The easy solution is to have at least two threads of control: one for handling interaction with the user and one for updating the spreadsheet. Meanwhile, a third thread could be used for backing up the spreadsheet to disk while the other two are doing their work.

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Another advantage of multithreading is that it becomes possible to exploit parallelism when executing the program on a multiprocessor or multicore system. In that case, each thread is assigned to a different CPU or core, while shared data are stored in shared main memory. When properly designed, such parallelism can be transparent: the process will run equally well on a uniprocessor system, albeit slower. Multithreading for parallelism is becoming increasingly important with the availability of relatively cheap multiprocessor and multicore computers. Such computer systems are typically used for running servers in client-server applications, but are by now also extensively used in devices such as smartphones.

Multithreading is also useful in the context of large applications. Such applications are often developed as a collection of cooperating programs, each to be executed by a separate process. This approach is typical for a Unix environment. Cooperation between programs is implemented through **interprocess communication** (**IPC**) mechanisms. For Unix systems, these mechanisms typically include (named) pipes, message queues, and shared memory segments (see also Stevens and Rago [2005]). The major drawback of all IPC mechanisms is that communication often requires relatively extensive context switching, shown at three different points in Figure 3.1.

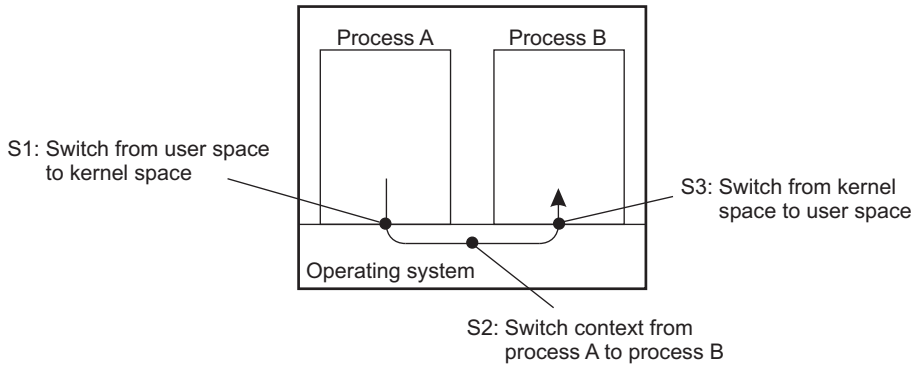


Figure 3.1: Context switching as the result of IPC.

Because IPC requires kernel intervention, a process will generally first have to switch from user mode to kernel mode, shown as *S*1 in Figure 3.1. This requires changing the memory map in the MMU, as well as flushing the TLB. Within the kernel, a process context switch takes place (*S*2 in the figure), after which the other party can be activated by switching from kernel mode to user mode again (*S*3 in Figure 3.1). The latter switch again requires changing the MMU map and flushing the TLB.

Instead of using processes, an application can also be constructed such that different parts are executed by separate threads. Communication between those parts is entirely dealt with by using shared data. Thread switching can

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sometimes be done entirely in user space, although in other implementations, the kernel is aware of threads and schedules them. The effect can be a dramatic improvement in performance.

Finally, there is also a pure software engineering reason to use threads: many applications are simply easier to structure as a collection of cooperating threads. Think of applications that need to perform several (more or less independent) tasks, like our spreadsheet example discussed previously.

Note 3.1 (Advanced: The cost of a context switch)
There have been many studies on measuring the performance effects of context switches. As in so many cases with measuring computer systems, finding the ground truth is not easy. Tsafrir [2007] notes that handling clock ticks has become more or less ubiquitous in operating systems, making it an excellent candidate to measure overheads. A clock handler is activated once every T milliseconds by a clock interrupt. Common values for T range between 0.5 and 20 milliseconds, corresponding to interrupt frequencies of 2000 Hz and 50 Hz, respectively. The handler typically assists in realizing various timing and CPU usage services, sends alarm signals, and assists in preempting running tasks for fair CPU sharing. By simply varying the frequency by which the hardware generates an interrupt, one can easily get an impression of the incurred overhead.

To measure the performance effects of an interrupt, a distinction is made between **direct overhead** and **indirect overhead**. The direct overhead consists of the time it takes to do the actual context switch, along with the time it takes for the handler to do its work and subsequently switching back to the interrupted task. The indirect overhead is everything else, and is mainly caused by cache perturbations (to which we will return shortly). For various Intel processors, Tsafrir [2007] found that the time to switch context is in the order of 0.5–1 microsecond, and that the handler itself takes in the order of 0.5–7 microseconds to do its work, depending strongly on the implementation.

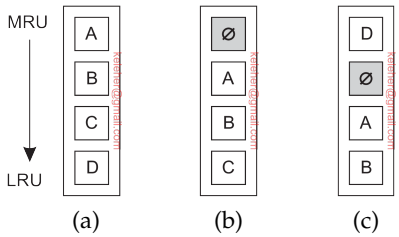


Figure 3.2: The organization of the cache when dealing with interrupts: (a) before the context switch, (b) after the context switch, and (c) after accessing block D. (Adapted from Liu and Solihin [2010].)

However, it turns out that the direct overhead is not really that influential. In a complimentary study, Liu and Solihin [2010] make clear that context switching can greatly perturbate the cache, resulting in a loss of performance in comparison to the situation before an interrupt occurred. In fact, for the simple case of clock

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interrupts, Tsafrir [2007] measured an indirect overhead of approximately 80%. To understand what is going on, consider the data organizations as sketched in Figure 3.2. Assume the cache is organized such that a least-recently used block of data is removed from the cache when room is needed for a fresh data block.

Figure 3.2(a) shows the situation before the interrupt occurs. After the interrupt has been handled, block D may have been evicted from the cache, leaving a hole as shown in Figure 3.2(b). Accessing block D will copy it back into the cache, possibly evicting block C, and so on. In other words, even a simple interrupt may cause a considerable, and relatively long-lasting reorganization of the cache, in turn, affecting the overall performance of an application.

```
1  from multiprocessing import Process
2  from time import *
3  from random import *
4
5  def sleeper(name):
6      t = gmtime()
7      s = randint(1,20)
8      txt = str(t.tm_min)+':'+str(t.tm_sec)+' '+name+' is going to sleep for '+str(s)+' seconds'
9      print(txt)
10     sleep(s)
11     t = gmtime()
12     txt = str(t.tm_min)+':'+str(t.tm_sec)+' '+name+' has woken up'
13     print(txt)
14
15 if __name__ == '__main__':
16     p = Process(target=sleeper, args=('eve',))
17     q = Process(target=sleeper, args=('bob',))
18     p.start(); q.start()
19     p.join(); q.join()
```

(a)

```
40:23 eve is going to sleep for 14 seconds
40:23 bob is going to sleep for 4 seconds
40:27 bob has woken up
40:37 eve has woken up
```

(b)

Figure 3.3: (a) A simple example in which two processes are started, and (b) the output after a run.

A simple example in Python

To make matters more concrete, let us look at a simple example in Python, also to illustrate the differences between processes and threads. Consider the code shown in Figure 3.3(a), which shows how we start to separate processes

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in Python using the `multiprocessing` package. The core of the example is formed by the function `sleeping` which simply puts the calling process to sleep for a randomly chosen number of seconds.

```
1  from multiprocessing import Process
2  from threading import Thread
3
4  shared_x = random.randint(10,99)
5
6  def sleeping(name):
7      global shared_x
8      s = randint(1,20)
9      sleep(s)
10     shared_x = shared_x + 1
11
12 def sleeper(name):
13     sleeplist = list()
14     for i in range(3):
15         subsleeper = Thread(target=sleeping, args=(name+' '+str(i),))
16         sleeplist.append(subsleeper)
17
18     for s in sleeplist: s.start()
19     for s in sleeplist: s.join()
```

(a)

```
eve sees shared x being 71
53:21 eve 0 is going to sleep for 20 seconds
bob sees shared x being 84
53:21 eve 1 is going to sleep for 15 seconds
53:21 eve 2 is going to sleep for 3 seconds
53:21 bob 0 is going to sleep for 8 seconds
53:21 bob 1 is going to sleep for 16 seconds
53:21 bob 2 is going to sleep for 8 seconds
53:24 eve 2 has woken up, seeing shared x being 72
53:29 bob 0 has woken up, seeing shared x being 85
53:29 bob 2 has woken up, seeing shared x being 86
53:36 eve 1 has woken up, seeing shared x being 73
53:37 bob 1 has woken up, seeing shared x being 87
bob sees shared x being 87
53:41 eve 0 has woken up, seeing shared x being 74
eve sees shared x being 74
```

(b)

Figure 3.4: (a) A multithreading example in which two processes are started, each with three threads, and (b) the output after a run.

To create two processes, we call the operation `Process` in lines 16 and 17, respectively, to subsequently start each of them. The `join` operation tells the

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main process to wait until the newly created processes have finished. A possible output is shown in Figure 3.3(b), indicating the time in "minutes:seconds" format when each process outputs some text.

The differences between threads and processes can be observed when we extend our example as shown in Figure 3.4 (where we have left out many statements for recording time and printing text). In this case, we again start two processes, named `eve` and `bob` (the code is the same as lines 15–19 in Figure 3.3 and has been omitted for clarity). Each process subsequently starts three threads, each, in turn, executing the function `sleeping`. The main difference is that there is now a shared variable `shared_x`. (To keep matters simple, we incorrectly assume that the assignment in line 10 is atomic. We explain atomic operations in detail in Chapter 5). What the output in Figure 3.4 shows is that this is a variable shared among the threads in a single process, but *not* shared between the two processes `eve` and `bob`. In other words, each *process* has its own instance of `shared_x`.

The output also shows that the `sleep` operation works at the thread level as well as the process level. In this case, each thread is suspended for a few seconds, yet the process hosting that thread is not blocked when `sleep` is called. Instead, another thread is scheduled (who subsequently also calls `sleep`). How this has been implemented is transparent to the programmer. Likewise, it is also transparent to what extent different threads are executed on different cores, if available, of the used CPU. If we had started 1000 threads per process, we would still see accurate timing. However, if we would replace the call to `sleep` with a busy waiting loop, such as in:

```
c = s * 26000000
for i in range(c):
    x = x + 1.0
```

we would see that the thread execution may be completely serialized *per process*, whereas each process would be assigned to a separate core and thus run in parallel. What exactly is done, depends on the underlying operating system and the used Python runtime system. The standard implementations of Python assign all threads within a single process to just one core.

Thread implementation

Threads are often provided in the form of a thread package. Such a package contains operations to create and destroy threads, as well as operations on synchronization variables such as mutexes and condition variables. There are basically two approaches to implement a thread package. The first approach is to construct a thread library that is executed entirely in user space. The second approach is to have the kernel be aware of threads and schedule them.

A user-level thread library has several advantages. First, it is cheap to create and destroy threads. Because all thread administration is kept in the

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user's address space, the price of creating a thread is primarily determined by the cost for allocating memory to set up a thread stack. Analogously, destroying a thread mainly involves freeing memory for the stack, which is no longer used. Both operations are cheap.

A second advantage of user-level threads is that switching thread context can often be done in just a few instructions. Basically, only the values of the CPU registers need to be stored and subsequently reloaded with the previously stored values of the thread to which it is being switched. There is no need to change memory maps, flush the TLB, do CPU accounting, and so on. Switching thread context is done when two threads need to synchronize, for example, when entering a section of shared data. However, as discussed in Note 3.1, much of the overhead of context switching is caused by perturbing memory caches.

A major drawback of user-level threads comes from deploying the **many-to-one threading model**: multiple threads are mapped to a single schedulable entity. We already saw this with our simple multithreaded Python example in Figure 3.4. As a consequence, the invocation of a blocking system call will immediately block the entire process to which the thread belongs, and thus also all the other threads in that process. As we explained, threads are particularly useful to structure large applications into parts that could be logically executed at the same time. In that case, blocking on I/O should not prevent other parts to be executed in the meantime. For such applications, user-level threads are of no help.

These problems can be mostly circumvented by implementing threads in the operating system's kernel, leading to what is known as the **one-to-one threading model** in which every thread is a schedulable entity. The price to pay is that every thread operation (creation, deletion, synchronization, etc.), will have to be carried out by the kernel, requiring a system call. Switching thread contexts may now become as expensive as switching process contexts. However, because the performance of context switching is generally dictated by ineffective use of memory caches, and not by the distinction between the many-to-one or one-to-one threading model, many operating systems now offer the latter model, if only for its simplicity.

Note 3.2 (Advanced: Many-to-many threading model)
An alternative to the two threading extremes is a hybrid form of user-level and kernel-level threads, a so-called **many-to-many threading model**. In the following, we simplify our terminology and speak of user threads and kernel threads. A kernel thread runs in the context of a single process, and there can be several kernel threads per process. In addition to managing kernel threads, a runtime system also offers a user-level thread package, offering applications the usual operations for creating and destroying threads. In addition, the package provides facilities for thread synchronization, such as mutexes and condition variables. The

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important issue is that the thread package is implemented entirely in user space: all operations on threads are carried out without intervention of the kernel.

The thread package can be shared by multiple kernel threads, as shown in Figure 3.5. This means that each kernel thread can be running its own (user level) thread. Multithreaded applications are constructed by creating user and kernel threads, and subsequently assigning each user thread to a kernel thread.

The combination of user threads and kernel threads works as follows. The thread package has a single routine to schedule the next thread. When creating a kernel thread (which is done through a system call), the kernel thread is given its own stack, and is instructed to execute the scheduling routine searching for a user thread to execute. If there are several kernel threads, then each of them executes the scheduler. The thread table, which is used to keep track of the current set of threads, is thus shared by the kernel threads. Protecting this table to guarantee mutually exclusive access is done through mutexes that are implemented entirely in user space. In other words, synchronization between kernel threads does not require any kernel support. A thread table is often implemented as a ready queue.

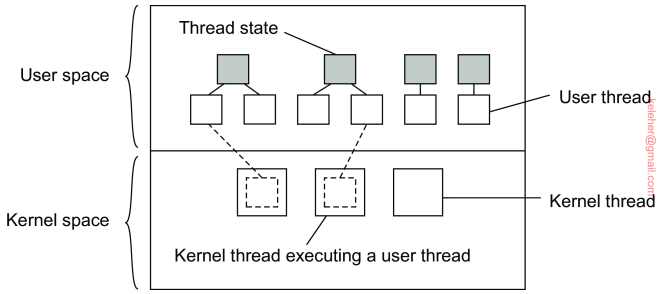


Figure 3.5: Combining kernel-level and user-level threads.

When a kernel thread finds a runnable user thread, it switches context to that thread. Meanwhile, other kernel threads may be looking for other runnable user threads as well. If a user thread needs to block on a mutex or condition variable, it does the necessary administration and eventually calls the scheduling routine. When another runnable user thread has been found, a context switch is made to that thread. The beauty of all this is that the kernel thread executing the user thread need not be informed: the context switch is implemented completely in user space and appears to the kernel thread as normal program code.

Now let us see what happens when a user thread does a blocking system call. In that case, execution changes from user mode to kernel mode, but still continues in the context of the current kernel thread. At the point where the current kernel thread can no longer continue, the operating system may decide to switch context to another kernel thread, which also implies that a context switch is made back to user mode. The selected kernel thread will simply continue where it had previously left off.

There are several advantages to using kernel threads with a user-level thread package. First, creating, destroying, and synchronizing threads is relatively cheap

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and involves no kernel intervention at all. Second, provided that a process has enough kernel threads, a blocking system call will not suspend the entire process. Third, there is no need for an application to know about the kernel threads. All it sees are user threads. Fourth, kernel threads can be easily used in multiprocessing environments by executing different kernel threads on different CPUs or different cores. This multiprocessing can be hidden entirely from the application.

The approach just sketched is actually the general form of combining user and kernel threads. This approach has been implemented in the Go programming language [Donovan and Kernighan, 2015]. The `libfibre` runtime system described and evaluated by Karsten and Barghi [2020] is also exemplary for the many-to-many threading model. A slightly different approach can be found in Arachne [Qin et al., 2018]. Arachne hides the kernel threads from applications, but instead assumes that an application can get full insight in the cores that have been assigned. It assigns one kernel thread per allocated core. An important consequence is that Arachne does not provide support for blocking I/O calls.

As a final note, it is important to realize that using threads is one way of organizing simultaneous and concurrent executions within an application. In practice, we often see that applications are constructed as a collection of concurrent *processes*, jointly making use of the interprocess facilities offered by an operating system (see also [Robbins and Robbins, 2003; Stevens, 1999]). A good example of this approach is the organization of the Apache Web server, which, by default, starts with a handful of processes for handling incoming requests. Each process forms a single-threaded instantiation of the server, yet is capable of communicating with other instances through standard means.

As argued by Srinivasan [2010], using processes instead of threads has the important advantage of separating the data space: each process works on its own part of data and is protected from interference from others through the operating system. The advantage of this separation should not be underestimated: thread programming is considered to be notoriously difficult because the developer is fully responsible for managing concurrent access to shared data. Using processes, data spaces, in the end, are protected by hardware support. If a process attempts to access data outside its allocated memory, the hardware will raise an exception, which is then further processed by the operating system. No such support is available for threads concurrently operating within the same process.

3.1.2 Threads in distributed systems

An important property of threads is that they can provide a convenient means of allowing blocking system calls without blocking the entire process in which the thread is running (assuming we do not have a **many-to-one threading model**). This property makes threads particularly attractive to use in distributed systems, as it makes it much easier to express communication

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in the form of maintaining multiple logical connections at the same time. We illustrate this point by taking a closer look at multithreaded clients and servers, respectively.

Multithreaded clients

To establish a high degree of distribution transparency, distributed systems that operate in wide-area networks may need to conceal long interprocess message propagation times. The round-trip delay in a wide-area network can easily be in the order of hundreds of milliseconds, or sometimes even seconds.

The usual way to hide communication latencies is to initiate communication and immediately proceed with something else. A typical example where this happens is in Web browsers. Often, a Web document consists of an HTML file containing plain text along with a collection of images, icons, etc. To fetch each element of a Web document, the browser has to set up a TCP/IP connection, read the incoming data, and pass it to a display component. Setting up a connection as well as reading incoming data are inherently blocking operations. When dealing with long-haul communication, we also have the disadvantage that the time for each operation to complete may be long.

A Web browser often starts with fetching the HTML page and subsequently displays it. To hide communication latencies as much as possible, some browsers start displaying data while it is still coming in. While the text is made available to the user, including the facilities for scrolling and such, the browser continues with fetching other files that make up the page, such as the images. The latter are displayed as they are brought in. The user need thus not wait until all the components of the entire page are fetched before the page is made available.

In effect, it is seen that the Web browser is doing several tasks simultaneously. As it turns out, developing the browser as a multithreaded client simplifies matters considerably. As soon as the main HTML file has been fetched, separate threads can be activated to take care of fetching the other parts. Each thread sets up a separate connection to the server and pulls in the data. Setting up a connection and reading data from the server can be programmed using the standard (blocking) system calls, assuming that a blocking call does not suspend the entire process. As is also illustrated in [Stevens, 1998], the code for each thread is the same and, above all, simple. Meanwhile, the user notices only delays in the display of images and such, but can otherwise browse through the document.

There is another important benefit to using multithreaded Web browsers, in which several connections can be opened simultaneously. In the previous example, several connections were set up to the same server. If that server is heavily loaded, or just plain slow, no real performance improvements will be noticed compared to pulling in the files that make up the page strictly one after the other.

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However, often, Web servers have been replicated across multiple machines, where each server provides the same set of Web documents. The replicated servers are located at the same site, and are known under the same name. When a request for a Web page comes in, the request is forwarded to one of the servers, often using a round-robin strategy or some other load-balancing technique. When using a multithreaded client, connections may be set up to different replicas, allowing data to be transferred in parallel, effectively establishing that the entire Web document is fully displayed in a much shorter time than with a nonreplicated server. This approach is possible only if the client can handle truly parallel streams of incoming data. Threads are ideal for this purpose.

Note 3.3 (Advanced: Exploiting client-side threads for performance)
Although there are obvious opportunities for using threads to reach high performance, it is interesting to see whether multithreading is effectively exploited. In a study to see to what extent multiple threads put a multicore processor to work, Blake et al. [2010] looked at the execution of various applications on modern architectures. Browsers, like many other client-side applications, are interactive innately, for which reason the expected processor idle time may be quite high. To properly measure to what extent a multicore processor is being used, Blake et al. used a metric known as **thread-level parallelism** (**TLP**). Let c_i denote the fraction of time that exactly i threads are being executed simultaneously. Thread-level parallelism is then defined as:

$$TLP = \frac{\sum_{i=1}^{N} i \cdot c_i}{1 - c_0}$$

where N is the maximum number of threads that (can) execute at the same time. In their study, a typical Web browser at that time had a TLP value between 1.5 and 2.5, meaning that to effectively exploit parallelism, the client machine should have two or three cores, or likewise, 2–3 processors.

These results are interesting when considering that modern Web browsers create hundreds of threads, and that tens of threads are active at the same time (note that an active thread is not necessarily running; it may be blocked waiting for an I/O request to complete). We thus see that multithreading is used to organize an application, but that this multithreading is not leading to dramatic performance improvements through hardware exploitation. That browsers can be effectively designed for exploiting parallelism is shown, for example, by Meyerovich and Bodik [2010]. By adapting existing algorithms, the authors manage to establish several-fold speedups.

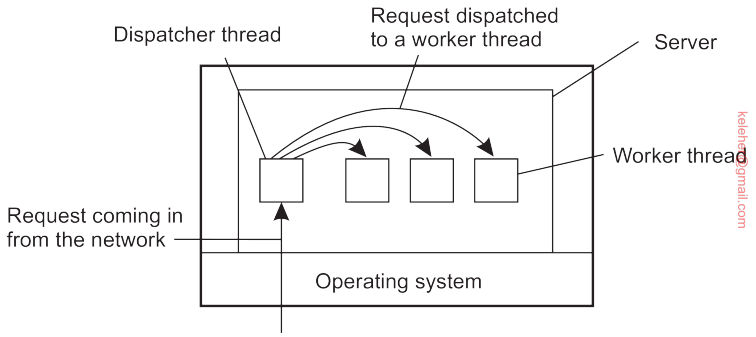
Multithreaded servers

Although there are important benefits to multithreaded clients, the main use of multithreading in distributed systems is found at the server side. Practice

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shows that multithreading not only simplifies server code considerably, but also makes it much easier to develop servers that exploit parallelism to attain high performance, even on uniprocessor systems. However, with modern multicore processors, multithreading for parallelism is an obvious path to follow.

To understand the benefits of threads for writing server code, consider the organization of a file server that occasionally has to block waiting for the disk. The file server normally waits for an incoming request for a file operation, subsequently carries out the request, and then sends back the reply. One possible, and particularly popular, organization is shown in Figure 3.6. Here, one thread, the **dispatcher**, reads incoming requests for a file operation. The requests are sent by clients to a well-known end point for this server. After examining the request, the server chooses an idle (i.e., blocked) **worker thread** and hands it the request.



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Figure 3.6: A multithreaded server organized in a dispatcher/worker model.

The worker proceeds by performing a blocking read on the *local* file system, which may cause the thread to be suspended until the data are fetched from disk. If the thread is suspended, another thread is selected to be executed. For example, the dispatcher may be selected to acquire more work. Alternatively, another worker thread can be selected that is now ready to run.

Now consider how the file server might have been written without threads. One possibility is to have it operate as a single thread. The main loop of the file server gets a request, examines it, and carries it out to completion before getting the next one. While waiting for the disk, the server is idle and does not process any other requests. Consequently, requests from other clients cannot be handled. In addition, if the file server is running on a dedicated machine, as is commonly the case, the CPU is simply idle while the file server is waiting for the disk. The net result is that many fewer requests per time unit can be processed. Thus threads gain considerable performance, but each thread is programmed sequentially, in the usual way.

So far, we have seen two possible designs: a multithreaded file server

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and a single-threaded file server. A third alternative is to run the server as a big single-threaded finite-state machine. When a request comes in, the one and only thread examines it. If it can be satisfied from the in-memory cache, fine, but if not, the thread must access the disk. However, instead of issuing a blocking disk operation, the thread schedules an asynchronous (i.e., nonblocking) disk operation for which it will be later interrupted by the operating system. To make this work, the thread will record the status of the request (namely, that it has a pending disk operation), and continues to see if there were any other incoming requests that require its attention.

Once a pending disk operation has been completed, the operating system will notify the thread, who will then, in due time, look up the status of the associated request and continue processing it. Eventually, a response will be sent to the originating client, again using a nonblocking call to send a message over the network.

In this design, the "sequential process" model that we had in the first two cases is lost. Every time the thread needs to do a blocking operation, it needs to record exactly where it was in processing the request, possibly also storing additional state. Once that has been done, it can start the operation and continue with other work. Other work means processing newly arrived requests, or post-processing requests for which a previously started operation has completed. Of course, if there is no work to be done, the thread may indeed block. In effect, we are simulating the behavior of multiple threads and their respective stacks the hard way. The process is being operated as a finite-state machine that gets an event and then reacts to it, depending on what is in it.

Model	Characteristics
Multithreading	Parallelism, blocking system calls
Single-threaded process	No parallelism, blocking system calls
Finite-state machine	Parallelism, nonblocking system calls

Figure 3.7: Three ways to construct a server.

It should now be clear what threads have to offer. They make it possible to retain the idea of sequential processes that make blocking system calls and still achieve parallelism. Blocking system calls make programming easier as they appear as just normal procedure calls. In addition, multiple threads allow for parallelism and thus performance improvement. The single-threaded server retains the ease and simplicity of blocking system calls, but may severely hinder performance in terms of number of requests that can be handled per time unit. The finite-state machine approach achieves high performance through parallelism, but uses nonblocking calls, which are generally hard to program and thus to maintain. These models are summarized in Figure 3.7.

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Again, note that instead of using threads, we can also use multiple processes to organize a server (leading to the situation that we actually have a multiprocess server). The advantage is that the operating system can offer more protection against accidental access to shared data. However, if processes need to communicate a lot, we may see a noticeable adverse effect on performance in comparison to using threads.

3.2 Virtualization

Threads and processes can be seen as a way to do more things at the same time. In effect, they allow us to build (pieces of) programs that appear to be executed simultaneously. On a single-processor (single core) computer, this simultaneous execution is, of course, an illusion. As there is only a single CPU, only an instruction from a single thread or process will be executed at a time. By rapidly switching between threads and processes, the illusion of parallelism is created.

This separation between having a single CPU and being able to pretend there are more can be extended to other resources as well, leading to what is known as **resource virtualization**. This virtualization has been applied for many decades, but has received renewed interest as (distributed) computer systems have become more commonplace and complex, leading to the situation that application software is mostly always outliving its underlying systems software and hardware.

3.2.1 Principle of virtualization

In practice, every (distributed) computer system offers a programming interface to higher-level software, as shown in Figure 3.8(a). There are many types of interfaces, ranging from the basic instruction set as offered by a CPU to the vast collection of application programming interfaces that are shipped with many current middleware systems. In its essence, **virtualization** deals with extending or replacing an existing interface to mimic the behavior of another system, as shown in Figure 3.8(b). We will come to discuss technical details on virtualization shortly, but let us first concentrate on why virtualization is important.

Virtualization and distributed systems

One of the most important reasons for introducing virtualization, back in the 1970s, was to allow legacy software to run on expensive mainframe hardware. The software not only included various applications, but in fact also the operating systems they were developed for. This approach toward supporting legacy software has been successfully applied on the IBM 370 mainframes (and

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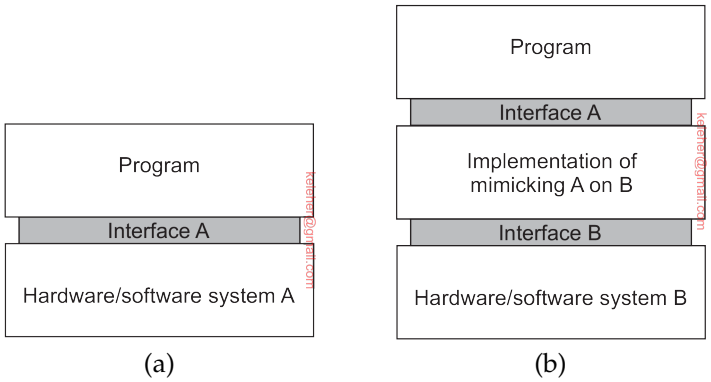


Figure 3.8: (a) General organization between a program, interface, and system. (b) General organization of virtualizing system A on top of B.

their successors) that offered a virtual machine to which different operating systems had been ported.

As hardware became cheaper, computers became more powerful, and the number of different operating system flavors was reducing, virtualization became less of an issue. However, matters have changed again since the late 1990s. First, while hardware and low-level systems software change reasonably fast, software at higher levels of abstraction (e.g., middleware and applications), are often much more stable. In other words, we are facing the situation that legacy software cannot be maintained in the same pace as the platforms it relies on. Virtualization can help here by porting the legacy interfaces to the new platforms, and thus immediately opening up the latter for large classes of existing programs.

Equally important is the fact that networking has become completely pervasive. It is hard to imagine that a modern computer is not connected to a network. In practice, this connectivity requires that system administrators maintain a large and heterogeneous collection of server computers, each one running very different applications, which can be accessed by clients. At the same time, the various resources should be easily accessible to these applications. Virtualization can help a lot: the diversity of platforms and machines can be reduced by essentially letting each application run on its own virtual machine, possibly including the related libraries *and* operating system, which, in turn, run on a common platform.

This last type of virtualization provides a high degree of portability and flexibility. For example, in order to realize content delivery networks that can easily support replication of dynamic content, Awadallah and Rosenblum [2002] have argued that management becomes much easier if edge servers would support virtualization, allowing a complete site, including its environ-

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ment, to be dynamically copied. These arguments are still valid, and indeed, portability is perhaps the most important reason why virtualization plays such a key role in many distributed systems.

Finally, an important reason for virtualization is that it provides an additional means of isolating code, which is particularly relevant in the case of cloud computing. At the same time, virtualization also introduces new security threats.

Note 3.4 (Discussion: Stable software?)
Although there is indeed a lot of legacy software that can benefit from stable interfaces to rapidly changing underlying hardware, it is a mistake to believe that the software for widely available services hardly changes. With the increasing shift toward server-side computing in the form of **Software-as-a-Service** (SaaS), much software can be maintained for a relatively homogeneous platform, owned entirely by the organization offering the associated service. As a consequence, maintaining software products can be much easier, as there is much lesser need to distribute changes to potentially millions of customers. In fact, changes may rapidly succeed each other following changes in available hardware and platform, but without any client actually noticing downtimes [Barroso et al., 2018].

Types of virtualization

There are many ways in which virtualization can be realized. An overview of these various approaches is described by Smith and Nair [2005a]. A more recent account is described by Bugnion et al. [2017], which provides many technical details on the realization of various forms of virtualization. To understand the differences in virtualization, it is important to realize that computer systems generally offer four different types of interfaces, at three different levels:

1. An interface between the hardware and software, referred to as the **instruction set architecture** (**ISA**), forming the set of machine instructions. This set is divided into two subsets:
   - Privileged instructions, which are allowed to be executed only by the operating system.
   - General instructions, which can be executed by any program.
2. An interface consisting of **system calls** as offered by an operating system.
3. An interface consisting of library calls, generally forming what is known as an **application programming interface** (**API**). Often, the aforementioned system calls are hidden by an API.

These different types are shown in Figure 3.9. The essence of virtualization is to mimic the behavior of these interfaces.

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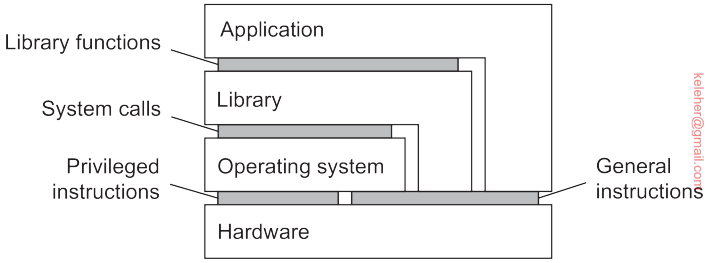


Figure 3.9: Various interfaces offered by computer systems.

Virtualization can take place in two different ways. First, we can build a runtime system that essentially provides an abstract instruction set that is to be used for executing applications. Instructions can be interpreted (as is the case for the Java runtime environment), but could also be emulated, as is done for running Windows applications on Unix platforms. Note that in the latter case, the emulator will also have to mimic the behavior of system calls, which has proven to be generally far from trivial. This type of virtualization, shown in Figure 3.10(a), leads to what Smith and Nair [2005a] call a **process virtual machine**, stressing that virtualization is only for a single process.

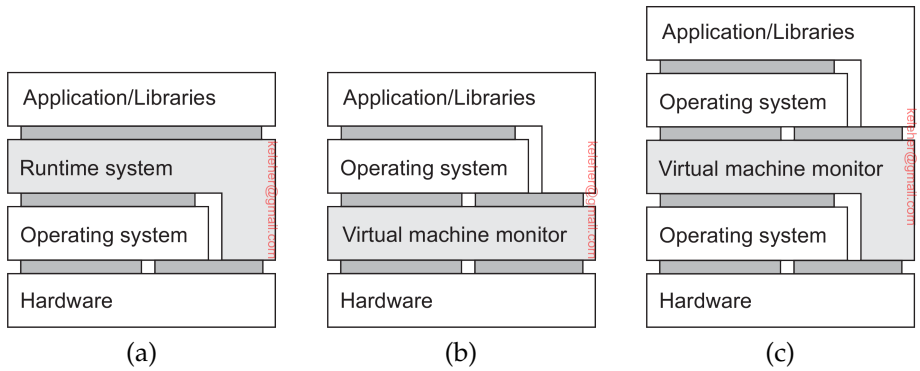


Figure 3.10: (a) A process virtual machine. (b) A native virtual machine monitor. (c) A hosted virtual machine monitor.

An alternative approach toward virtualization, shown in Figure 3.10(b), is to provide a system that is implemented as a layer shielding the original hardware, but offering the complete instruction set of that same (or other hardware) as an interface. This leads to what is known as a **native virtual machine monitor**. It is called native because it is implemented directly on top of the underlying hardware. Note that the interface offered by a virtual machine monitor can be offered *simultaneously* to different programs. As

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a result, it is now possible to have multiple, and different **guest operating systems** run independently and concurrently on the same platform.

A native virtual machine monitor will have to provide and regulate access to various resources, like external storage and networks. Like any operating system, this implies that it will have to implement device drivers for those resources. Rather than doing all this effort anew, a **hosted virtual machine monitor** will run on top of a trusted **host operating system** as shown in Figure 3.10(c). In this case, the virtual machine monitor can make use of existing facilities provided by that host operating system. It will generally have to be given special privileges instead of running as a user-level application. Using a hosted virtual machine monitor is highly popular in modern distributed systems such as data centers and clouds.

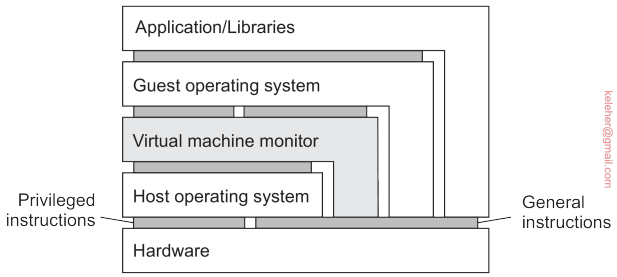
As argued by Rosenblum and Garfinkel [2005], virtual machines are important in the context of reliability and security for (distributed) systems. As they allow for the isolation of a complete application and its environment, a failure caused by an error or security attack need no longer affect a complete machine. In addition, as we also mentioned before, portability is greatly improved as virtual machines provide a further decoupling between hardware and software, allowing a complete environment to be moved from one machine to another. We return to migration in Section 3.5.

Note 3.5 (Advanced: On the performance of virtual machines)
Virtual machines perform surprisingly well. In fact, many studies show that modern virtual machines perform close to running applications directly on the host operating system. Let us take a closer look at what is going under the hood of virtual machines. A detailed and comprehensive account of virtual machines is provided by Smith and Nair [2005b].

Part of the answer to performance issues is shown in Figure 3.11, which forms an extension of Figure 3.10(c): a large part of the code constituting a virtual machine monitor, guest operating system, and application is running natively on the underlying hardware. In particular, all general (i.e., unprivileged) machine instructions are directly executed by the underlying machine.

This approach is not new and is founded on research by Popek and Goldberg [1974] who formalized the requirements for the efficient execution of virtual machines. In a nutshell, Popek and Goldberg assumed that the underlying machine provided at least two modes of operation (system and user mode), that a subset of the instructions could be executed only in system mode, and that memory addressing was relative (i.e., a physical address was obtained by adding a relative address to an offset found in a relocation register). A distinction was further made between two types of instructions. A **privileged instruction** is an instruction that is characterized by the fact that if and only if executed in user mode, it causes a **trap** to the operating system. **Nonprivileged instructions** are all other instructions.

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Figure 3.11: Applications, guest operating system, virtual machine monitor, and host operating system on a single hardware platform.

Given these formal assumptions, Popek and Goldberg defined two classes of special instructions. A **control-sensitive instruction** is one that may affect the configuration of a machine. A typical example is an instruction that affects the memory layout, for example, by changing the memory offset as stored in a relocation register. Another example is instructions that affect the interrupt table, containing pointers to interrupt handlers.

A **behavior-sensitive instruction** is one whose effect is partially determined by the context in which it is executed. For example, Intel x86 processors have instructions that may, or may not, affect certain registers depending on whether that instruction is executed in system mode or user mode. An example given in [Smith and Nair, 2005b] is that of the `POPF` instruction, which may set an interrupt-enabled flag, but only when executed in system mode.

We now have the following important result:

> *For any conventional computer, a virtual machine monitor may be constructed if the set of sensitive instructions for that computer is a subset of the set of privileged instructions.*

What this says is that as long as sensitive instructions are caught when executed in user mode, we can safely run all nonsensitive instructions natively on the underlying hardware. This also means that when designing instruction sets, if we take care that the above requirement is met, we will not be unnecessarily obstructing efficient virtualization of that instruction set.

Unfortunately, not all instruction sets have privileged-only sensitive instructions, including perhaps the most popular one, namely the Intel x86 instruction set. As it turns out, this set has 17 sensitive instructions that are not privileged [Robin and Irvine, 2000]. In other words, each of these instructions can be executed in user mode without causing a trap to the operating system, yet affect the way that the operating system is managing its resources. In these cases, there are essentially two solutions.

The first solution is to emulate all instructions. Of course, this would have a serious adverse effect on performance. To circumvent problems, an approach implemented in VMWare [Sugerman et al., 2001], is to scan the executable and to insert code around the nonprivileged sensitive instructions to divert control

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to the virtual machine monitor. There, appropriate emulation will take place, for example, by considering the context in which the instruction was to be executed. The effect is that full virtualization can take place, meaning that execution can take place without changing the guest operating system, nor the application itself.

An alternative solution is to apply **paravirtualization**, which requires the guest operating system to be modified. In particular, the guest operating system is modified such that all side effects of running nonprivileged sensitive instructions in user mode, which would normally be executed in system mode, are dealt with. For example, code can be rewritten such that these instructions simply no longer occur, or if they do, that their semantics are the same regardless whether being executed in user or system mode. Paravirtualization has been adopted by Xen [Barham et al., 2003; Chisnall, 2007].

3.2.2 Containers

Virtual machines offer a means to run applications relying on a specific operating environment, including its instruction set and operating system, to run independently across different platforms. As we have seen, this may require significant efforts to ensure portability and performance. However, often we see that applications are relatively stable when it comes to the used instruction set and operating system, yet do rely on specific libraries and other support software. In these cases, what we really want is to allow different applications to run side-by-side, yet each uses its own environment of support software without even noticing that there be other applications with a different environment. This is where **containers** come into the game.[1]

A container can be thought of a collection of binaries (also called images) that jointly constitute the software environment for running applications. The easiest way to think of a container is what a user would get to see when logging into, for example, a Unix system: it will consist of several standard directories containing executable programs, libraries, documentation, etc. A naive implementation of a container would be to copy an entire environment for a specific use case and install it as a subdirectory of, say, the root file system. Using a command such as `chroot`, the user would then be diverted to that subdirectory and run various applications, with this subdirectory now acting as the root. An application would see exactly the libraries and other dependencies it needed, while applications in other containers would have their own view on what the operating system is offering. In this sense, a container effectively virtualizes the software environment for an application.

However, this naive implementation is certainly not enough from a virtualization perspective. For one, applications and processes operating in different

[1] The material in this section has been inspired by Julia Evans's wonderful material at wizardzines.com.

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containers need to be isolated from each other. Likewise, simply copying an entire environment is not very efficient, certainly not because we may expect that many libraries and such are the same across different containers. Finally, it is important that an operating system hosting containers has some control over the usage of its own resources. All of these aspects are handled in Unix environments (i.e., notably Linux) through three important mechanisms:

- **Namespaces**, by which a collection of processes associated with a container is given their own view of identifiers and such, independent of other containers.
- **Union file system**, which allows to, literally, combine several file systems into a layered fashion with only the highest layer allowing for `write` operations (and the one being part of a container).
- **Control groups**, or simply **cgroups**, by which resource restrictions can be imposed upon a collection of processes.

Let us look a bit deeper into each of these mechanisms. **Namespaces** are necessary for giving a process running inside a container the illusion that it is on its own. As such, namespaces are important for isolating containers from each other. Perhaps the one most illustrative is setting the PID namespace. As every machine has only a single `init` process (with PID equal to 1), every container should see its own "init" process. This is established through the Unix `unshare` command:

```
unshare --pid --fork --mount-proc bash
```

will bring the calling process into a new shell in which the command `ps -ef` yields:

```
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 06:27 pts/0    00:00:00 bash
root           2       1  0 06:27 pts/0    00:00:00 ps -ef
```

Indeed, we see that there now seems to be a new collection of processes with just one having PID equal to 1. All other processes have become invisible when working from this new shell.

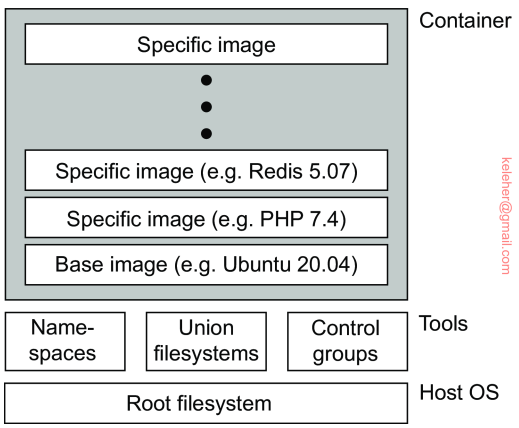
Another important mechanism is efficient sharing of existing file systems. Many containers will be based on a common instance of an operating system, say Ubuntu 20.4. Instead of copying that entire environment and installing it as a subdirectory as explained above, we can use it as a base layer and stack other parts on top of it. For example, we may decide to replace the entire collection of subdirectories that constitute PHP7.4 for an older version by simply stacking those directories on top the ones for version 7.4. The result is that a PHP application will be using the older version. Note that this approach is very similar to mounting a file system at a directory `dir`. Anything that was

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contained in `dir` will no longer be visible until the file system is unmounted again. Indeed, taking the union of file systems is done through successive calls to the `mount` system call, with each layer being mounted in read-only mode. Only the top layer can be written to, and will need to be explicitly saved when a running container finishes.

Finally, to control what a container can actually use, Unix systems offer **cgroups**. In essence, when creating a control group, the collection of processes running in that group may be restricted to the amount of main memory that they can use, the priority when it comes to using the CPU, etc. In this way, the hosting operating system can prevent that a single container is using too much of its resources, preventing perhaps other containers to do their work.

There are many other things related to isolate containers and properly restrict what processes can do within a container. In the end, a container can be thought of an archive of files that are placed somewhere in a filesystem, together with a specific stack of common, shared, existing read-only subdirectories. Processes running inside the context of a container are presented with a view that their context is the only one (through the use of namespaces) and they have certain abilities when it comes tot using resources (through cgroups, but also restrictions when it comes to, for example, system calls). This view is summarized in Figure 3.12. Besides the material from Julia Evans at `wizardzines.com`, the interested reader is referred to [Pahl et al., 2019] for an overview of container technologies.



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Figure 3.12: The organization of a container within a hosting environment.

Note 3.6 (Example: PlanetLab)
There are many examples of container technologies, yet at this point it is interesting to look at a specific case where the technology was used for developing a wide-area cluster of computers, even before it became popular in the context of cloud

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computing. **PlanetLab** was a collaborative distributed system in which different organizations each donated one or more computers, adding up to a total of hundreds of nodes. Together, these computers formed a 1-tier server cluster, where access, processing, and storage could all take place on each node individually. Management of PlanetLab was by necessity almost entirely distributed. The project closed down in 2020.

General organization In PlanetLab, a participating organization donated one or more nodes (i.e., computers) that were subsequently shared among all PlanetLab users. Each node was organized as shown in Figure 3.13. There are two important components [Bavier et al., 2004; Peterson et al., 2006]. The first one is the virtual machine monitor (VMM), which is an enhanced Linux operating system, essentially one capable of supporting containers along the lines sketched above. The enhancements mainly comprise adjustments for supporting the second component, namely (Linux) **Vservers**. A Vserver is essentially a container in execution.

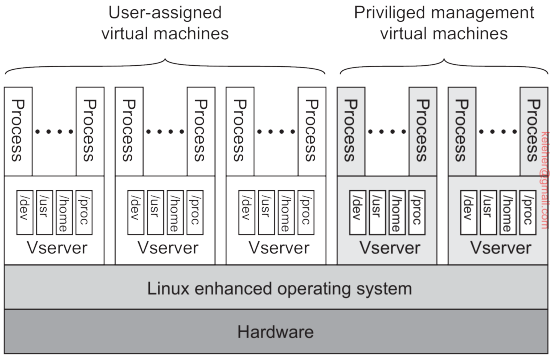


Figure 3.13: The basic organization of a PlanetLab node.

The Linux VMM ensured that Vservers were separated: processes in different Vservers are executed concurrently and independently, each making use only of the software packages and programs available in their own environment. The isolation between processes in different Vservers is strict. For example, two processes in different Vservers could have the same user ID, but this did not imply that they would stem from the same user. This separation considerably eased supporting users from different organizations that wanted to use PlanetLab as, for example, a testbed to experiment with entirely different distributed systems and applications. Note that this separation is precisely the one that is realized through the `unshare` command.

To support such experimentation, PlanetLab used **slices**, each slice being a set of Vservers, each Vserver running on a different node, as illustrated in Figure 3.14. A slice can thus be thought of as a virtual server cluster, implemented by a collection of containers connected through a wide-area network.

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Central to managing PlanetLab resources was the **node manager**. Each node had such a manager, implemented by a separate Vserver, whose only task was to create other Vservers on the node it managed and to control resource allocation. To create a new slice, each node would also run a **slice creation service** (**SCS**), which, in turn, could contact the node manager requesting it to create a Vserver and to allocate resources. The node manager itself could not be contacted directly over a network, allowing it to concentrate only on local resource management. In turn, the SCS would not accept slice-creation requests from just anybody. Only specific **slice authorities** were eligible for requesting the creation of a slice. Each slice authority would have access rights to a collection of nodes. The simplest model was that there is only a single, centralized slice authority that is allowed to request slice creation on all nodes. In practice, we saw that this slice authority was the one used to get a user up-and-running on PlanetLab.

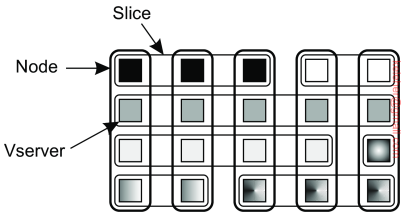


Figure 3.14: The principle of a PlanetLab slice, showing sets of associated Vservers across different nodes.

Keeping track of resources was done by a resource specification, or rspec for short. An rspec specified a time interval during which certain resources had been allocated. Resources include disk space, file descriptors, inbound and outbound network bandwidth, transport-level end points, main memory, and CPU usage. An rspec was identified through a globally unique 128-bit identifier known as a resource capability (rcap). Given an rcap, the node manager could look up the associated rspec in a local table.

Resources were bound to slices. In other words, to make use of resources, it was necessary to create a slice. Each slice was associated with a **service provider**, which can best be seen as an entity having an account on PlanetLab. Every slice could then be identified by a (principal_id, slice_tag) pair, where the principal_id identified the provider and slice_tag being an identifier chosen by the provider.

Vservers Let us now turn our attention to PlanetLab's Vservers, which have been described and evaluated by Soltesz et al. [2007]. A Vserver was organized as a container. The primary task of a Vserver was therefore to merely support a group of processes and keep that group isolated from processes running under the jurisdiction of another Vserver.

An important advantage of the container-based approach toward virtualization, in comparison to running separate guest operating systems, is that resource allocation could generally be much simpler. In particular, it was possible to overbook resources by allowing for dynamic resource allocation, just as is done

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with allocating resources to normal processes. Normally, when using a guest operating system, the guest will have to be allocated a fixed number of resources in advance (notably main memory). When considering that the nodes provided by participating PlanetLab organizations were required to have only a few GByte of main memory, it is not hard to imagine that memory would be a scarce resource. It was therefore necessary to dynamically allocate memory to allow tens of containers to be running at the same time on a single node. Vservers were ideal for this type of resource management; operating systems are much harder to support in such cases. Of course, this could not prevent a Vserver from using too much memory on a busy node. The PlanetLab policy in that case was simple: the Vserver, hogging memory when swap space was almost filled, was reset.

PlanetLab status Although the official PlanetLab closed down in 2020, a similar system is running worldwide as EdgeNet. Not surprisingly, where Vservers were containers *avant-la-lettre*, that technology has been replaced in EdgeNet by modern container technology, namely Docker in combination with Kubernetes.

3.2.3 Comparing virtual machines and containers

Ever since containers became popular, mainly caused by their introduction through Docker, an oftentimes heated debate has been going on what is better: virtual machines or containers? This debate has been hindered through the improper use of terminology, such as *lightweight* containers versus *heavyweight* virtual machines, immediately leading to often unsubstantiated performance statements (suggesting that heavyweight means slow). However, life is not so simple and understanding the technology of virtual machines as well as those of containers can help in making better judgments on when to use which technology. In this section, let us take a closer look at one specific, important aspect: performance. We return to portability later when discussing code migration.

Measuring the performance of any system requires looking at a multitude of criteria. Obvious ones include CPU and memory usage. Likewise, various I/O measurements are needed to get a good insight into how well a system is performing, in particular accessing disks and network I/O. On top of this, we need to ask ourselves how measurements are carried out. In other words, which workloads or benchmarks are used to evaluate and compare the performing systems.

A systematic study on comparing Linux containers (LXC) against Linux virtual machines (KVM) was conducted by Sharma et al. [2016]. If we just look at a baseline comparison in which an application is running either within a container or on top of a virtual machine, differences can be observed in favor of containers, yet these differences are not that big, except when it comes to I/O. In that case, we see that virtual machines perform significantly less than

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containers. This should not come as a surprise, as notably with traditional I/O, the operating system plays a crucial role: it has to execute many privileged instructions.

Nevertheless, even the obvious may come with surprises. In a more recent study on comparing different container technologies with virtual-machine approaches, van Rijn and Rellermeyer [2021] demonstrate that the differences between the two may often be close to negligible, even when looking at disk and network I/O. There are several reasons for this small difference, but one is that the host operating system caches results in main memory. In other words, when actually performing I/O operations, many subsequent operations are performed on in-memory data instead of data that is stored on disk. This makes benchmarking more difficult, yet also reflects realistic application-driven behavior. Furthermore, when running actual application-level benchmarks, such as those available for mysql, differences may exist, but can be small. Nevertheless, depending on the actual I/O behavior, the overall conclusion is that virtual machines do impose more overhead in comparison to containers.

Of course, it is more natural that several applications run side-by-side. The question then is to what extent the performance of one application influences that of another. In effect, this aspect boils down to the question how well scheduling resources among competitors actually is. In this case, the general trend is that containerization has more difficulty isolating independent applications and that scheduling for CPU usage as well as for disk performance is handled better through virtual machines.

What van Rijn and Rellermeyer [2021] and other recent studies show is that over the years many improvements have been made and that there is no real need for virtualization techniques to perform significantly slower in comparison to running applications directly on top of the hosting operating system.

3.2.4 Application of virtual machines to distributed systems

From the perspective of distributed systems, the most important application of virtualization lies in cloud computing. As we already mentioned in Section 1.3.1, cloud providers offer roughly three different types of services:

- **Infrastructure-as-a-Service** (**IaaS**) covering the basic infrastructure
- **Platform-as-a-Service** (**PaaS**) covering system-level services
- **Software-as-a-Service** (**SaaS**) containing actual applications

Virtualization plays a key role in IaaS. Instead of renting out a physical machine, a cloud provider will rent out a virtual machine (monitor) that may, or may not, be sharing a physical machine with other customers. The beauty of virtualization is that it allows for almost complete isolation between

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customers, who will indeed have the illusion that they have just rented a dedicated physical machine. Isolation is, however, never complete, if only for the fact that the actual physical resources are shared, in turn leading to observable lower performance.

To make matters concrete, let us consider the **Amazon Elastic Compute Cloud**, or simply **EC2**. EC2 allows one to create an environment consisting of several networked virtual servers, thus jointly forming the basis of a distributed system. To make life easy, there is a (large) number of pre-configured *machine images* available, referred to as **Amazon Machine Image**s, or simply **AMI**s. An AMI is an installable software package consisting of an operating-system kernel along with several services. An example of a simple, basic AMI is a **LAMP** image, consisting of a Linux kernel, the Apache Web server, a MySQL database system, and PHP libraries. More elaborate images containing additional software are also available, as well as images based on other Unix kernels or Windows. In this sense, an AMI is essentially the same as a boot disk (although there are a few important differences, to which we return shortly).

An EC2 customer needs to select an AMI, possibly after adapting or configuring one. An AMI can then be *launched*, resulting in what is called an **EC2 instance**: the actual virtual machine that can be used to host a customer's applications. An important issue is that a customer will hardly ever know exactly where an instance is actually being executed. Obviously, it is running on a single physical machine, but where that machine is located remains hidden. The closest one can get to pinpoint the location where an instance should run is by selecting one of a few regions provided by Amazon (US, South America, Europe, Asia).

To communicate, each instance obtains two IP addresses: a private one that can be used for internal communication between different instances, making use of EC2's internal networking facilities, and a public IP address allowing any Internet clients to contact an instance. The public address is mapped to the private one using standard **network-address translation** (**NAT**) technology. A simple way to manage an instance is to make use of an SSH connection, for which Amazon provides the means for generating the appropriate keys.

The EC2 environment in which an instance is executed provides different levels of the following services:

- **CPU**: allows selecting the number and type of core, including GPUs
- **Memory**: defines how much main memory is allocated to an instance
- **Storage**: defines how much local storage is allocated
- **Platform**: distinguishes between 32-bit or 64-bit architectures
- **Networking**: sets the bandwidth capacity that can be used

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In addition, extra resources can be requested, such as an additional networking interface. The local storage that comes with an instance is *transient*: when the instance stops, all the data stored locally is lost. To prevent data loss, a customer will need to explicitly save data to a persistent store, for example, by making use of Amazon's Simple Storage Service (S3). An alternative is to attach a storage device that is mapped to Amazon's **Elastic Block Store** (**EBS**). Again, this is yet another service, but one that can be used in the form of a virtual block device that is simply mounted as one would mount an additional hard disk. When an instance is stopped, all data that was stored on EBS will persist. And just as one would expect, an EBS device can be (re)mounted to any other instance as well.

It should be clear by now that, without having gone into any significant level of detail, the IaaS as offered by EC2 allows a customer to create a (potentially large) number of virtual machines, each configured with resources as needed, and capable of exchanging messages through an IP network. In addition, these virtual machines can be accessed from anywhere over the Internet (provided a client has the proper credentials). As such, Amazon EC2, like many other IaaS providers, offers the means to configure a complete distributed system consisting of networked virtual servers and running customer-supplied distributed applications. At the same time, those customers will not need to maintain any physical machine, which by itself is often already a huge gain, as we will encounter at several occasions throughout this text. One can indeed argue that virtualization lies at the core of modern cloud computing.

3.3 Clients

In the previous chapters we discussed the client-server model, the roles of clients and servers, and the ways they interact. Let us now take a closer look at the anatomy of clients and servers, respectively. We start in this section with a discussion of clients. Servers are discussed in the next section.

3.3.1 Networked user interfaces

A major task of client machines is to provide the means for users to interact with remote servers. There are roughly two ways in which this interaction can be supported. First, for each remote service, the client machine will have a separate counterpart that can contact the service over the network. A typical example is a calendar running on a user's smartphone that needs to synchronize with a remote, possibly shared calendar. In this case, an application-level protocol will handle the synchronization, as shown in Figure 3.15(a).

A second solution is to provide direct access to remote services by offering only a convenient user interface. Effectively, this means that the client machine is used only as a terminal with no need for local storage, leading

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to an application-neutral solution as shown in Figure 3.15(b). In the case of networked user interfaces, everything is processed and stored at the server. This **thin-client approach** has received much attention with the increase in Internet connectivity and the use of mobile devices. Thin-client solutions are also popular as they ease the task of system management.

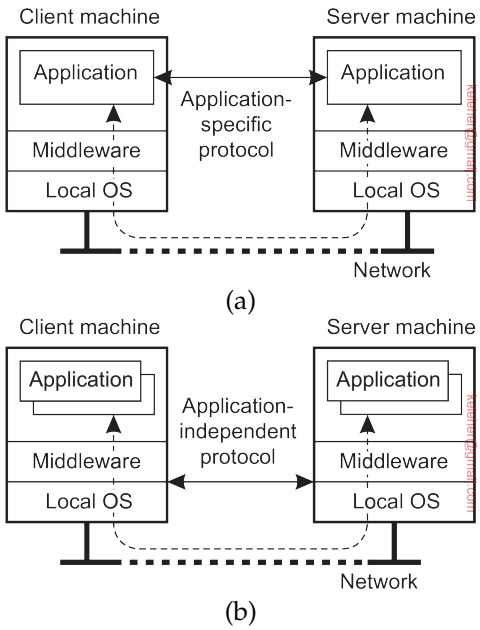


Figure 3.15: (a) A networked application with its own protocol. (b) A general solution to allow access to remote applications.

Example: The X window system

Perhaps one of the oldest and still widely used networked user interfaces is the **X Window System**. The X Window System, generally referred to simply as **X**, is used to control bit-mapped terminals, which include a monitor, keyboard, and a pointing device such as a mouse. Next to supporting traditional terminals as can be found with desktop computers and workstations, X also supports modern devices such a touchscreens on tablets and smartphones. In a sense, X can be viewed as that part of an operating system that controls the terminal. The heart of the system is formed by what we shall call the **X kernel**. It contains all the terminal-specific device drivers, and as such, is generally highly hardware dependent.

The X kernel offers a relatively low-level interface for controlling the screen, but also for capturing events from the keyboard and mouse. This

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interface is made available to applications as a library called Xlib. Its organization is shown in Figure 3.16. Note that Xlib is hardly ever used directly by applications, which instead deploy toolkits implemented on top of Xlib.

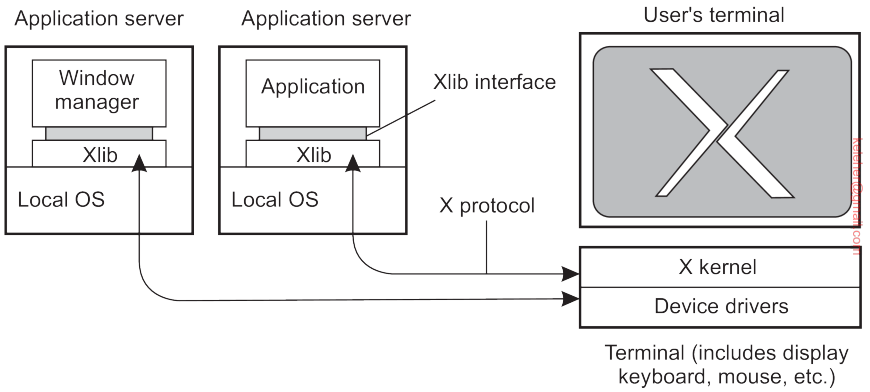


Figure 3.16: The basic organization of the X Window System.

The interesting aspect of X is that the X kernel and the X applications need not necessarily reside on the same machine. In particular, X provides the **X protocol**, which is an application-level communication protocol by which an instance of Xlib can exchange data and events with an X kernel. For example, Xlib can send requests to the X kernel for creating or killing a window, setting colors, and defining the type of cursor to display, among many other requests. In turn, the X kernel will react to local events such as keyboard and mouse input by sending event packets back to Xlib.

Several applications can communicate at the same time with the X kernel. There is one specific application that is given special rights, known as the **window manager**. This application can dictate the "look and feel" of the display as it appears to the user. For example, the window manager can prescribe how each window is decorated with extra buttons, how windows are to be placed on the display, and so on. Other applications will have to adhere to these rules. In practice, this means that much of the interaction between an application and an X terminal is redirected through a window manager.

It is interesting to note how the X window system actually fits into client-server computing. From what we have described so far, it should be clear that the X kernel receives requests to manipulate the display. It gets these requests from (possibly remote) applications. In this sense, the X kernel acts as a server, while the applications play the role of clients. This terminology has been adopted by X, and although strictly speaking it is correct, it can easily lead to confusion.

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Thin-client network computing

Obviously, applications manipulate a display using the specific display commands as offered by X. These commands are generally sent over the network where they are subsequently executed by the X kernel. By its nature, applications written for X should preferably separate application logic from user-interface commands. Unfortunately, this is often not the case. As reported by Lai and Nieh [2002] it turns out that much of the application logic and user interaction are tightly coupled, meaning that an application will send many requests to the X kernel for which it will expect a response before being able to make a next step. This synchronous behavior may adversely affect performance when operating over a wide-area network with long latencies.

There are several solutions to this problem. One is to re-engineer the implementation of the X protocol, as is done with NX [Pinzari, 2003]. An important part of this work concentrates on bandwidth reduction by reducing the size of X messages. To this end, messages are considered to consist of a fixed part, which is treated as an identifier, and a variable part. Often, multiple messages will have the same identifier, in which case they will often contain similar data. This property can be used to send only the differences between messages having the same identifier. By having the sender and receiver maintain identifiers, decoding at the receiver can be readily applied. Bandwidth reductions up to a factor 1000 have been reported, which allows X to also run through low-bandwidth links of only 9600 kbps.

As an alternative to using X, researchers and practitioners have also sought to let an application *completely* control the remote display, that is, up to the pixel level. This approach is also referred to as controlling a **remote desktop**. Changes in the bitmap are then sent over the network to the display, where they are immediately transferred to the local frame buffer. A well-known example of this approach is **Virtual Network Computing** (**VNC**) [Richardson et al., 1998], which has been around ever since the late 1990s. Obviously, letting the application control the display requires sophisticated encoding techniques to prevent bandwidth availability to become a problem. For example, consider displaying a video stream at 30 frames per second on a simple 320×240 screen. If each pixel is encoded by 24 bits, then without an efficient encoding scheme, we would need a bandwidth of approximately 53 Mbps. In practice, various encoding techniques are used, yet choosing the best one is generally application dependent.

3.3.2 Virtual desktop environment

As cloud computing further matured, and notably the number of cloud applications was growing, it became opportune to actually turn the cloud into a **virtual desktop environment** for end users. The only thing needed was the client-side software to access that desktop environment. One of the first

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providers of this model was Google with the introduction of **Chrome OS**. The basic idea is simple: let the browser provide the local desktop interface. Next to the browser, multiple stand-alone applications can be available of which each, in principle, eventually operates with a cloud-based counterpart. In this sense, the desktop is actually akin to what is offered by modern smartphones. With an increasing trend to transform applications to browser extensions, we can see that the browser is taking over the role of an operating system's user interface.

The anatomy of a Web browser

To get a better appreciation of what is going on, let us take a closer look at the anatomy of the Chrome browser, which has been reported to be the most widely used among all browsers.[2] The Chrome browser is an intricate piece of software consisting of over 25 million lines of code, comparable to the size of the Linux kernel. Where in the beginning the Web consisted of merely simple HTML pages, we are now dealing with not only a highly advanced markup language, but also an array of tools for interaction and client-side scripting languages. A high-level overview of how the Chrome browser works is shown in Figure 3.17.

At the core, we see the resource loader, responsible for fetching content from a Web server. It generally starts with fetching an HTML file and subsequently, in parallel, sets up connections for other material referenced in that page. This is typically done through separate threads. Note that the resource loader will need to partially parse the original HTML file to discover which other content it needs to fetch. Most of the parsing, however, is done by a separate component that constructs what is known as a **Document Object Model**, or **DOM** for short. In essence, a DOM is a tree representation of the HTML file.

Where the DOM can be said to represent structure, actual styling information is provided by a separate document. For example, it may state that every paragraph should be represented in a specific color and in a specific font. This information is parsed separately and essentially added to the DOM, leading to a render tree. This tree contains all the information for the next step: determining exactly where the various elements in the tree will be displayed. In particular, geometric regions need to be computed for different parts of the DOM, but also where lines will be broken (and considering language-specific issues such as reading text from left to right, or *vice versa* as is the case with Hebrew or Arabic. Line breaking, in turn, requires exact computations for which the styling component will need to take font characteristics into account. It is not difficult to see that the process of determining the layout can become

[2] See gs.statcounter.com.

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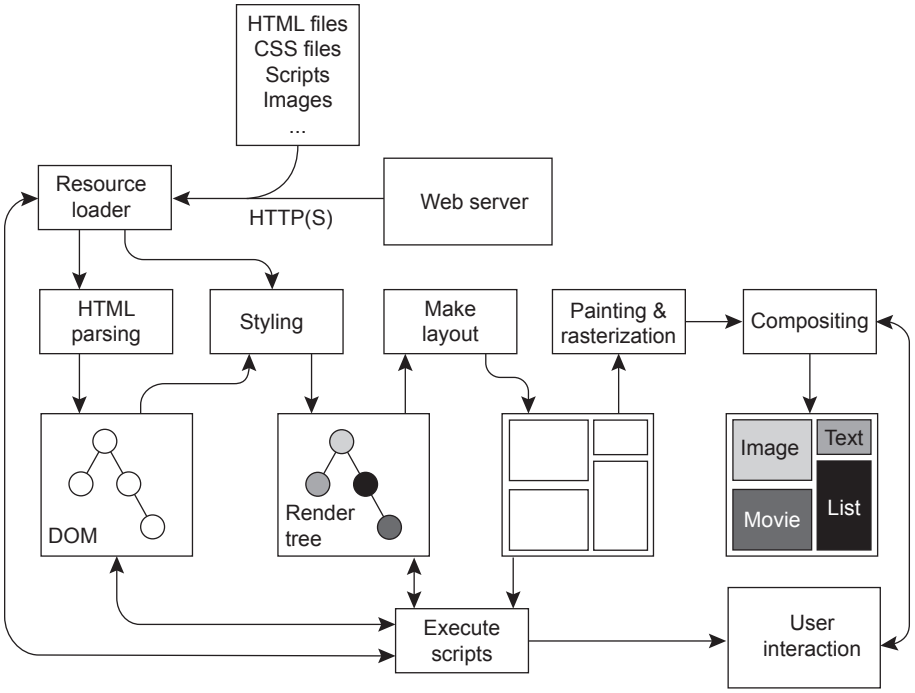


Figure 3.17: An overview of how the Chrome browser renders a Web page (adapted from Pourghassemi [2021]).

quite complex when, for example, floating images or multiple columns need to be rendered.

Once the layout has been determined, the painting component, takes the layout instructions and constructs a program consisting of paint operations, like `drawRectangle(x,y,height,width)`. There are many of such operations, which are subsequently executed by a rasterization process which fills in all the details for every pixel on the screen, notably, of course, its color. Rasterization also takes the embedded images into account (and based on the information provided by the painting process) ensures that each pixel on an embedded image gets properly displayed. What we have not shown explicitly in Figure 3.17, is that the DOM is actually decomposed into several layers, and that each layer is eventually rasterized separately. As a result, the layers need to be composed into a final image that can be displayed. Meanwhile, the compositor can interact with the user, who may be scrolling through a page.

Last, yet certainly not least, every browser now has a separate component for handling scripts, that is, executable code. A popular scripting language for client-side Web applications is **JavaScript**, but increasingly more often we also see **WebAssembly** codes [Sletten, 2022]. The latter offer performance that

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can be close to executing native code, certainly in combination with actual compilations. The generated code is always restricted to the permissions the browser has, making it (relatively) safe. The power of being able to run code inside a browser cannot be underestimated. It forms the core of giving the illusion to an end user that she is indeed running an application on the local desktop. We return to this phenomenon below.

Although we have skipped many details, it should be clear that actually rendering a Web page is a highly complex endeavor. Furthermore, it is not difficult to imagine that many of the elements just described, such as rasterizing layers, can be done in parallel and perhaps even on special processors, such a GPU. This is precisely what a modern Web browser does: spawning threads and processes, not only to provide more structure and organization to this immense complex software, but also to make sure that the rendering as a whole is done efficiently.

In fact, modern Web browsers also use processes to protect various parts of their code from each other. For example, the whole rendering in Chrome is done by a separate process (i.e., the components HTML parsing, styling, making the layout, painting, and compositing are taken together), while rasterization is separated to make use of a possibly available GPU. The rasterization process communicates with the main process through message passing. Moreover, every browser tab gets its own renderer process. To improve security, each renderer is running in a separate **sandbox**: a protection mechanism that precludes direct communication with the underlying operating system. We return to sandboxes when discussing mobile code.

Browsers and applications

As we already indicated, having a browser is often sufficient for offering a virtual desktop environment. Important is the fact that browsers can execute scripts as the client-side part of an otherwise remote application. Such scripts can largely handle everything that is needed to provide the illusion that a user is working on a local machine. This is precisely what Chrome OS offers. However, things become a bit more complicated when local resources are truly needed for offering a desktop environment. The first type of resources that come to mind are related to media: camera, microphone, speakers, etc. Many Web-based applications will ask the user for permission to use those resources.

It is often also possible to run applications natively on the computer hosting the client-side desktop. In essence, this is what happens with modern smartphones. Native applications (also referred to as mobile apps) operate as any other locally executing application. Their main advantage over running Web-based applications is that, in principle, a user can work offline. However, by simply examining how many mobile apps are useful when there is no

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network connectivity already illustrates that working offline is not assumed to be the default.

A development that is gradually emerging is that of using **Progressive Web apps** (**PWA**). These applications use the browser as their hosting environment, yet appear as an ordinary mobile app. In a nutshell, what a PWA does is move a lot of the server-side content that is not dependent on (high quality) network connectivity, to the client where it is subsequently cached. The effect is that many of these apps, which can run in the browser or even appear as a mobile app (i.e., the browser user interface itself is hidden) can be executed much faster and comparable to that of their mobile-app counterpart. Furthermore, because much less needs to be communicated with the server, many PWAs can operate well even when the quality of the network is at stake.

By and large, we see that virtual desk environments are moving extremely thin clients to ones that are hosts to much more functionality, and for which communication over the Internet has been highly optimized. As we mentioned before, a major advantage over the traditional fat clients that have mainly locally installed applications is that management of browser-based apps is much easier: the server can simply upload new parts to the client as needed.

3.3.3 Client-side software for distribution transparency

Client software comprises more than just user interfaces. Often, parts of the processing and data level in a client-server application are executed on the client side as well. A special class is formed by embedded client software, such as for automatic teller machines (ATMs), cash registers, barcode readers, TV set-top boxes, etc. In these cases, the user interface is a relatively small part of the client software, in contrast to the local processing and communication facilities.

Besides the user interface and other application-related software, client software comprises components for achieving **distribution transparency**. Ideally, a client should not be aware that it is communicating with remote processes. In contrast, distribution is often less transparent to servers for reasons of performance and correctness.

Access transparency is generally handled through the generation of a **client stub** from an interface definition of what the server has to offer. The stub provides the same interface as the one available at the server, but hides the possible differences in machine architectures, as well as the actual communication. The client stub transforms local calls to messages that are sent to the server, and *vice versa* transforms messages from the server to return values as one would expect when calling an ordinary procedure.

There are different ways to handle **location**, **migration**, and **relocation transparency** . Using a convenient naming system is crucial. Often, cooperation with client-side software is also important. For example, when a client is already bound to a server, the client can be directly informed when the server

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changes location. In this case, the client's middleware can hide the server's current network location from the user, and also transparently rebind to the server if necessary. At worst, the client's application may notice a temporary loss of performance.

Similarly, many distributed systems implement **replication transparency** employing client-side solutions. For example, imagine a distributed system with replicated servers, Such replication can be achieved by forwarding a request to each replica, as shown in Figure 3.18. Client-side software can transparently collect all responses and pass a single response to the client application.

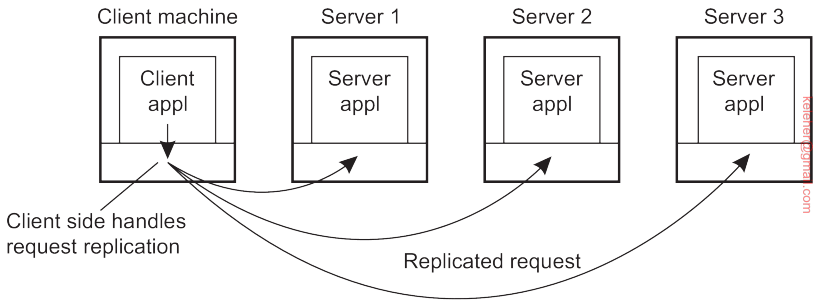


Figure 3.18: Transparent replication of a server using a client-side solution.

Regarding **failure transparency**, masking communication failures with a server is typically done through client middleware. For example, client middleware can be configured to repeatedly attempt to connect to a server, or perhaps try another server after several attempts. There are even situations in which the client middleware returns data it had cached during a previous session, as is sometimes done by Web browsers that fail to connect to a server.

Finally, **concurrency transparency** can be handled through special intermediate servers, notably transaction monitors, and requires less support from client software.

3.4 Servers

Let us now take a closer look at the organization of servers. In the following pages, we first concentrate on a number of general design issues for servers, followed by a discussion on server clusters.

3.4.1 General design issues

A server is a process implementing a specific service on behalf of a collection of clients. In essence, each server is organized in the same way: it waits for an

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incoming request from a client and subsequently ensures that the request is taken care of, after which it waits for the next incoming request.

Concurrent versus iterative servers

There are several ways to organize servers. In the case of an **iterative server**, the server itself handles the request and, if necessary, returns a response to the requesting client. A **concurrent server** does not handle the request itself, but passes it to a separate thread or another process, after which it immediately waits for the next incoming request. A multithreaded server is an example of a concurrent server. An alternative implementation of a concurrent server is to fork a new process for each new incoming request. This approach is followed in many Unix systems. The thread or process that handles the request is responsible for returning a response to the requesting client.

Contacting a server: end points

Another issue is where clients contact a server. In all cases, clients send requests to an **end point**, also called a **port**, at the machine where the server is running. Each server listens to a specific end point. How do clients know the end point of a service? One approach is to globally assign end points for well-known services. For example, servers that handle Internet FTP requests always listen to TCP port 21. Likewise, an HTTP server for the World Wide Web will always listen to TCP port 80. These end points have been assigned by the **Internet Assigned Numbers Authority** (**IANA**), and are documented in [Reynolds and Postel, 1994]. With assigned end points, the client needs to find only the network address of the machine where the server is running. Name services can be used for that purpose.

There are many services that do not require a preassigned end point. For example, a time-of-day server may use an end point that is dynamically assigned to it by its local operating system. In that case, a client will first have to look up the end point. One solution is to have a special daemon running on each machine that runs servers. The daemon keeps track of the current end point of each service implemented by a co-located server. The daemon itself listens to a well-known end point. A client will first contact the daemon, request the end point, and then contact the specific server, as shown in Figure 3.19(a).

It is common to associate an end point with a specific service. However, actually implementing each service by means of a separate server may be a waste of resources. For example, in a typical Unix system, it is common to have many servers running simultaneously, with most of them passively waiting until a client request comes in. Instead of having to keep track of so many passive processes, it can be more efficient to have a single **superserver** listening to each end point associated with a specific service, as shown in

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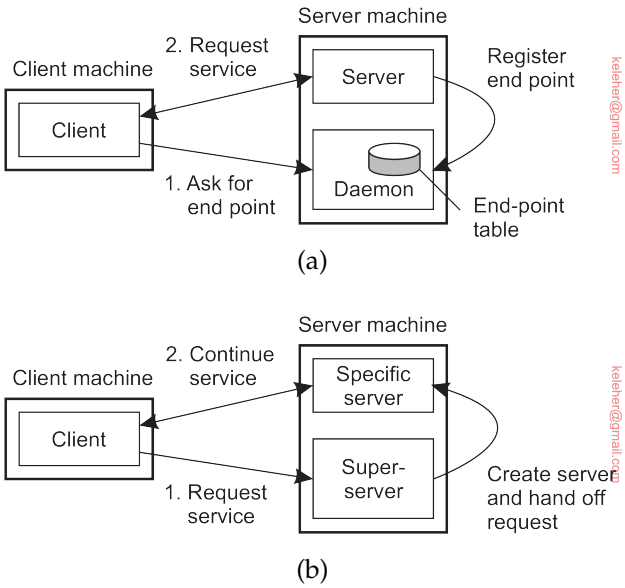


Figure 3.19: (a) Client-to-server binding using a daemon. (b) Client-to-server binding using a superserver.

Figure 3.19(b). For example, the inetd daemon in Unix listens to a number of well-known ports for Internet services. When a request comes in, the daemon forks a process to handle it. That process will exit when finished.

Interrupting a server

Another issue that needs to be considered when designing a server is whether and how a server can be interrupted. For example, consider a user who has just decided to upload a huge file to an FTP server. Then, suddenly realizing that it is the wrong file, she wants to interrupt the server to cancel further data transmission. There are several ways to do this. One approach that works only too well in the current Internet (and is sometimes the only alternative) is for the user to abruptly exit the client application (which will automatically break the connection to the server), immediately restart it, and pretend nothing happened. The server will eventually tear down the old connection, thinking the client has probably crashed.

A much better approach for handling communication interrupts is to develop the client and server such that it is possible to send **out-of-band data**, which is data that is to be processed by the server before any other data from that client. One solution is to let the server listen to a separate control end point to which the client sends out-of-band data, while at the same time listening (with a lower priority) to the end point through which the normal

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data passes. Another solution is to send out-of-band data across the same connection through which the client is sending the original request. In TCP, for example, it is possible to transmit urgent data. When urgent data are received at the server, the latter is interrupted (e.g., through a signal in Unix systems), after which it can inspect the data and handle them accordingly.

Stateless versus stateful servers

A final, important design issue, is whether the server is stateless. A **stateless server** does not keep information on the state of its clients, and can change its own state without having to inform any client [Birman, 2012]. A Web server, for example, is stateless. It merely responds to incoming HTTP requests, which can be either for uploading a file to the server or (most often) for fetching a file. When the request has been processed, the Web server forgets the client completely. Likewise, the collection of files that a Web server manages (possibly in cooperation with a file server), can be changed without clients having to be informed.

Note that in many stateless designs, the server actually does maintain information on its clients, but crucial is the fact that if this information is lost, it will not lead to a disruption of the service offered by the server. For example, a Web server generally logs all client requests. This information is useful, for example, to decide whether certain documents should be replicated, and where they should be replicated to. Clearly, there is no penalty, other than perhaps in the form of suboptimal performance if the log is lost.

A particular form of a stateless design is where the server maintains what is known as **soft state**. In this case, the server promises to maintain state on behalf of the client, but only for a limited time. After that time has expired, the server falls back to default behavior, thereby discarding any information it kept on account of the associated client. An example of this type of state is a server promising to keep a client informed about updates, but only for a limited time. Thereafter, the client is required to poll the server for updates. Soft-state approaches originate from protocol design in computer networks, but can be equally applied to server design [Clark, 1989; Lui et al., 2004].

In contrast, a **stateful server** generally maintains persistent information on its clients. This means that the information needs to be explicitly deleted by the server. A typical example is a file server that allows a client to keep a local copy of a file, even for performing update operations. Such a server would maintain a table containing *(client, file)* entries. Such a table allows the server to keep track of which client currently has the update permissions on which file, and thus possibly, also the most recent version of that file.

This approach can improve the performance of read and write operations as perceived by the client. Performance improvement over stateless servers is often an important benefit of stateful designs. However, the example also illustrates the major drawback of stateful servers. If the server crashes, it has

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to recover its table of *(client, file)* entries, or otherwise it cannot guarantee that it has processed the most recent updates on a file. In general, a stateful server needs to recover its entire state as it was just before the crash. Enabling recovery can introduce considerable complexity, as we discuss in Chapter 8. In a stateless design, no special measures need to be taken at all for a crashed server to recover. It simply starts running again, and waits for client requests to come in.

Ling et al. [2004] argue that one should actually make a distinction between (temporary) **session state** and **permanent state**. The example above is typical for session state: it is associated with a series of operations by a single user and should be maintained for some time, but not indefinitely. As it turns out, session state is often maintained in three-tiered client-server architectures, where the application server actually needs to access a database server through a series of queries before being able to respond to the requesting client. The issue here is that no real harm is done if session state is lost, provided that the client can simply re-issue the original request. This observation allows for simpler and less reliable storage of state.

What remains for permanent state is typically information maintained in databases, such as customer information, keys associated with purchased software, etc. However, for most distributed systems, maintaining session state already implies a stateful design requiring special measures when failures do happen and making explicit assumptions about the durability of state stored at the server. We will return to these matters when discussing fault tolerance.

When designing a server, the choice for a stateless or stateful design should not affect the services provided by the server. For example, if files have to be opened before they can be read from, or written to, then a stateless server should one way or the other mimic this behavior. A common solution is that the server responds to a read or write request by first opening the referred file, then does the actual read or write operation, and immediately closes the file again.

In other cases, a server may want to keep a record of a client's behavior so that it can more effectively respond to its requests. For example, Web servers sometimes offer the possibility to immediately direct a client to its favorite pages. This approach is possible only if the server has history information on that client. When the server cannot maintain state, a common solution is then to let the client send along additional information on its previous accesses. In the case of the Web, this information is transparently stored by the client's browser in what is called a **cookie**, which is a small piece of data containing client-specific information that is of interest to the server. Cookies are never executed by a browser; they are merely stored and sent to the server when accessed a next time.

The first time a client accesses a server, the latter sends a cookie along with the requested Web pages back to the browser, after which the browser safely

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tucks the cookie away. Each subsequent time the client accesses the server, its cookie for that server is sent along with the request.

3.4.2 Object servers

Let us take a look at the general organization of object servers needed for distributed objects. The important difference between a general object server and other (more traditional) servers is that an object server by itself does not provide a specific service. Specific services are implemented by the objects that reside in the server. Essentially, the server provides only the means to invoke local objects, based on requests from remote clients. As a consequence, it is relatively easy to change services by simply adding and removing objects.

An object server thus acts as a place where objects live. An object consists of two parts: data representing its state and the code for executing its methods. Whether these parts are separated, or whether method implementations are shared by multiple objects, depends on the object server. Furthermore, there are differences in the way an object server invokes its objects. For example, in a multithreaded server, each object may be assigned a separate thread, or a separate thread may be used for each invocation request. These and other issues are discussed next.

For an object to be invoked, the object server needs to know which code to execute, on which data it should operate, whether it should start a separate thread to take care of the invocation, and so on. A simple approach is to assume that all objects look alike and that there is only one way to invoke an object. Unfortunately, such an approach is generally inflexible and often unnecessarily constrains developers of distributed objects.

A much better approach is for a server to support different policies. Consider, for example, a **transient object**: an object that exists only as long as its server exists, but possibly for a shorter period of time. An in-memory, read-only copy of a file could typically be implemented as a transient object. Likewise, a calculator could also be implemented as a transient object. A reasonable policy is to create a transient object at the first invocation request and to destroy it as soon as no clients are bound to it anymore.

The advantage of this approach is that a transient object will need a server's resources only as long as the object is really needed. The drawback is that an invocation may take some time to complete because the object needs to be created first. Therefore, an alternative policy is sometimes to create all transient objects at the time the server is initialized, at the cost of consuming resources even when no client is making use of the object.

Similarly, a server could follow the policy that each of its objects is placed in a memory segment of its own. In other words, objects share neither code nor data. Such a policy may be necessary when an object implementation does not separate code and data, or when objects need to be separated for security reasons. In the latter case, the server will need to provide special

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measures, or require support from the underlying operating system, to ensure that segment boundaries are not violated.

The alternative approach is to let objects at least share their code. For example, a database containing objects that belong to the same class can be efficiently implemented by loading the class implementation only once into the server. When a request for an object invocation comes in, the server need only fetch that object's state and execute the requested method.

Likewise, there are many policies regarding threading. The simplest approach is to implement the server with only a single thread of control. Alternatively, the server may have several threads, one for each of its objects. Whenever an invocation request comes in for an object, the server passes the request to the thread responsible for that object. If the thread is currently busy, the request is temporarily queued.

The advantage of this approach is that objects are automatically protected against concurrent access: all invocations are serialized through the single thread associated with the object. Neat and simple. Of course, it is also possible to use a separate thread for each invocation request, requiring that objects should have already been protected against concurrent access. Independent of using a thread per object or thread per method is the choice of whether threads are created on demand or the server maintains a pool of threads. Generally, there is no single best policy. Which one to use depends on whether threads are available, how much performance matters, etc.

Decisions on how to invoke an object are commonly referred to as **activation policies**, to emphasize that often the object itself must first be brought into the server's address space (i.e., activated) before it can actually be invoked. What is needed then is a mechanism to group objects per policy. Such a mechanism is sometimes called an **object adapter**, or alternatively an **object wrapper**. An object adapter can best be thought of as software implementing a specific activation policy. The main issue, however, is that object adapters come as generic components to assist developers of distributed objects, and which need only to be configured for a specific policy.

An object adapter has one or more objects under its control. Because a server should be capable of simultaneously supporting objects that require different activation policies, several object adapters may reside in the same server. When an invocation request is delivered to the server, the request is first dispatched to the appropriate object adapter, as shown in Figure 3.20.

An important observation is that object adapters are unaware of the specific interfaces of the objects they control. Otherwise, they could never be generic. The only issue that is important to an object adapter is that it can extract an object reference from an invocation request, and subsequently dispatch the request to the referenced object, but now following a specific activation policy. As is also illustrated in Figure 3.20, rather than passing the request directly to the object, an adapter hands an invocation request to the server-side stub

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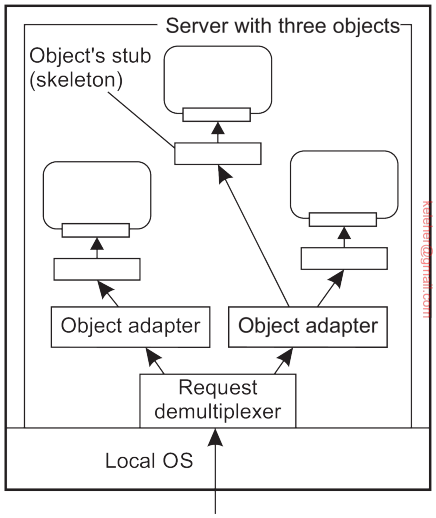


Figure 3.20: An object server supporting different activation policies.

of that object. The stub, also called a skeleton, is normally generated from the interface definitions of the object, unmarshals the request and invokes the appropriate method.

An object adapter can support different activation policies by simply configuring it at runtime. For example, in CORBA-compliant systems [OMG, 2001], it is possible to specify whether an object should continue to exist after its associated adapter has stopped. Likewise, an adapter can be configured to generate object identifiers, or to let the application provide one. An adapter can also be configured to operate in single-threaded or multithreaded mode.

Note that although in Figure 3.20 we have spoken about objects, we have said nothing about what these objects actually are. In particular, it should be stressed that as part of the implementation of such an object the server may (indirectly) access databases or call special library routines. The implementation details are hidden for the object adapter, who communicates only with a skeleton. As such, the actual implementation may have nothing to do with what we often see with language-level (i.e., compile-time) objects. For this reason, a different terminology is generally adopted. A **servant** is the general term for a piece of code that forms the implementation of an object.

Example: The Ice runtime system

Let us briefly consider the Ice distributed-object system, which has been partly developed in response to the intricacies of commercial object-based distributed systems [Henning, 2004]. An object server in Ice is nothing but an ordinary process that simply starts with initializing the Ice runtime system. The basis

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of the runtime environment is formed by what is called a *communicator*. A communicator is a component that manages several basic resources, of which the most important one is formed by a pool of threads. Likewise, it will have associated dynamically allocated memory, and so on. In addition, a communicator provides the means for configuring the environment. For example, it is possible to specify maximum message lengths, maximum invocation retries, and so on.

Normally, an object server would have only a single communicator. However, when different applications need to be fully separated and protected from each other, a separate communicator (with possibly a different configuration) can be created within the same process. At the very least, such an approach would separate the different thread pools so that if one application has consumed all its threads, then this would not affect the other application.

A communicator is used to create an object adapter, such as shown in Figure 3.21. In this example, we start with creating and initializing the runtime environment, which returns a communicator. Using the communicator, an object adapter is created. In this case, it is instructed to listen for incoming TCP connections on port 11000. Note that the adapter is created in the context of the just created communicator. We are now in the position to create objects and add those to the adapter. In this case, we create two objects, `object1` and `object2`, respectively. Both are added to the adapter, yet under different names. What we see here, is that the adapter will be listening for incoming requests on a single port, yet will use an object's name to invoke the proper object. Once the objects have been added, the adapter is *activated*, meaning that, under the hood, a thread is activated that will start listening for incoming requests.

On the client side, we see that two printer objects are created, one for each of their server-side counterparts. And, indeed, after starting the server and then the client, the output on the server's side will match:

```
Object1 says: Hello World from printer1!
Object2 says: Hello World from printer2!
```

Note that if we had associated `printer2` in line 9 of the client with `base1`, the output would have been:

```
Object1 says: Hello World from printer1!
Object1 says: Hello World from printer2!
```

In other words, `printer2` would have been associated to `object1` instead of `object2`.

This code does not yet show much differentiation in activation policies. Policies can be changed by modifying the *properties* of an adapter. One family of properties is related to maintaining an adapter-specific set of threads that are used for handling incoming requests. For example, one can specify that

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```
import sys, Ice
import Demo

class PrinterI(Demo.Printer):
    def __init__(self, t):
        self.t = t

    def printString(self, s, current=None):
        print(self.t, s)

communicator = Ice.initialize(sys.argv)

adapter = communicator.createObjectAdapterWithEndpoints("SimpleAdapter", "default -p 11000")
object1 = PrinterI("Object1 says:")
object2 = PrinterI("Object2 says:")
adapter.add(object1, communicator.stringToIdentity("SimplePrinter1"))
adapter.add(object2, communicator.stringToIdentity("SimplePrinter2"))
adapter.activate()

communicator.waitForShutdown()
```

(a)

```
import sys, Ice
import Demo

communicator = Ice.initialize(sys.argv)

base1 = communicator.stringToProxy("SimplePrinter1:default -p 11000")
base2 = communicator.stringToProxy("SimplePrinter2:default -p 11000")
printer1 = Demo.PrinterPrx.checkedCast(base1)
printer2 = Demo.PrinterPrx.checkedCast(base2)
if (not printer1) or (not printer2):
    raise RuntimeError("Invalid proxy")

printer1.printString("Hello World from printer1!")
printer2.printString("Hello World from printer2!")

communicator.waitForShutdown()
```

(b)

Figure 3.21: (a) An example of creating a simple object server in Ice, and (b) a corresponding client (slightly adapted from [ZeroC, 2022]).

there should always be only one thread, effectively serializing all accesses to objects that have been added to the adapter.

In the example above, an object is created as part of the application, after which it is added to an adapter. Effectively, this means that an adapter may need to support many objects at the same time, leading to potential scalability problems. An alternative solution is to dynamically load objects into memory when they are needed. To do this, Ice provides support for special objects known as *locators*. A locator can be viewed as a special type of servant; it is

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called when the adapter receives an incoming request for an object that has not been explicitly added. In that case, the request is forwarded to the locator, whose job is to further handle the request.

To make matters more concrete, suppose a locator is handed a request for an object of which the locator knows that its state is stored in a relational database system. Of course, there is no magic here: the locator has been programmed explicitly to handle such requests. In this case, the object's identifier may correspond to the key of a record in which that state is stored. The locator will then simply do a lookup on that key, fetch the state, and will then be able to further process the request.

There can be more than one locator added to an adapter. In that case, the adapter would keep track of which object identifiers would belong to the same locator. Using multiple locators allows supporting many objects by a single adapter. Of course, objects (or rather their state) would need to be loaded at runtime, but this dynamic behavior would possibly make the server itself relatively simple. More examples and detailed information on Ice can be found in [ZeroC, 2022].

3.4.3 Example: The Apache Web server

An interesting example of a server that balances the separation between policies and mechanisms is the **Apache Web server**. It is also an extremely popular server, estimated to be used to host approximately 25% of all Websites. Apache is a complex piece of software, and with the numerous enhancements to the types of documents that are now offered on the Web, it is important that the server be highly configurable and extensible, and at the same time largely independent of specific platforms.

Making the server platform independent is realized by essentially providing its own basic runtime environment, which is then subsequently implemented for different operating systems. This runtime environment, known as the **Apache Portable Runtime** (**APR**), is a library that provides a platform-independent interface for file handling, networking, locking, threads, and so on. When extending Apache, portability is largely guaranteed provided that only calls to the APR are made and none to platform-specific libraries.

From a certain perspective, Apache can be considered as a completely general server tailored to produce a response to an incoming request. Of course, there are all kinds of hidden dependencies and assumptions by which Apache turns out to be primarily suited for handling requests for Web documents. For example, as we mentioned, Web browsers and servers use HTTP as their communication protocol. HTTP is virtually always implemented on top of TCP, for which reason the core of Apache assumes that all incoming requests adhere to a TCP-based connection-oriented way of communication. Requests based on UDP cannot be handled without modifying the Apache core.

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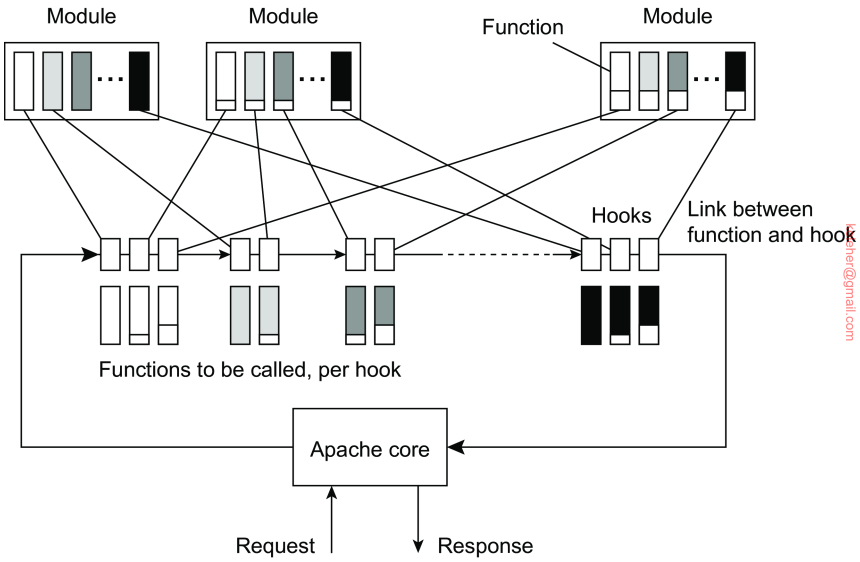


Figure 3.22: The general organization of the Apache Web server.

Otherwise, the Apache core makes few assumptions on how incoming requests should be handled. Its overall organization is shown in Figure 3.22. Fundamental to this organization are **modules**. A module consists of one, or several, functions that should be called for properly handling a request. This raises several questions:

1. How do we ensure that a function is called in the first place?
2. How do we ensure that a function is called at the right moment?
3. How do we prevent that a function is called for a request it was not supposed to handle?

The last question is actually the simplest: each function will be called (along with the request), yet each function will return the value `DECLINED` if the request was not meant for it.

Getting a function to be called at all is handled through a **hook**, which is a placeholder for a function. There is a hook for each function, and each module provides the Apache core with a list of hooks to its functions. By statically or dynamically linking modules to the Apache core, we establish a connection between a hook and its associated function.

For example, there is a hook to translate a URL to a local file name. Such a translation will almost certainly need to be done when processing a request. Likewise, there is a hook for writing information to a log, a hook for checking a client's identification, a hook for checking access rights, and a hook for

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checking which MIME type the request is related to (e.g., to make sure that the request can be properly handled). As shown in Figure 3.22, the hooks are processed in a specific order. It is here that we explicitly see that Apache enforces a specific flow of control concerning the processing of requests.

Often, functions can be processed independently of each other: they operate in perfect isolation on a request. However, this may not be generally the case, for which Apache distinguishes a number of phases. When hooking up a function to Apache, we need to specify whether that function should be called in the beginning, the middle, or the end of the total request-processing flow. If fine-grained control is necessary, one can also specify before or after which other module a function should be called. However, it is easily seen that trying to develop functions that operate in isolation (and are, in effect, stateless) contributes to a modular design.

Much more on the Apache Web server can be found in the (by now somewhat outdated) book by Laurie and Laurie [2002], as well as the developer's documentation provided by Apache itself.

3.4.4 Server clusters

In Chapter 1, we briefly discussed cluster computing as one of the many appearances of distributed systems. We now take a closer look at the organization of server clusters, along with the salient design issues. We first consider common server clusters that are organized in local-area networks. A special group is formed by wide-area server clusters, which we subsequently discuss.

Local-area clusters

Simply put, a server cluster is nothing else but a collection of machines connected through a network, where each machine runs one or more servers. The server clusters that we consider here, are the ones in which the machines are connected through a local-area network, often offering high bandwidth and low latency.

General organization Often, a server cluster is logically organized into three tiers, as shown in Figure 3.23. The first tier consists of a (logical) switch through which client requests are routed. Such a switch can vary widely. For example, transport-layer switches accept incoming TCP connection requests and pass requests on to one of the servers in the cluster. A completely different example is a Web server that accepts incoming HTTP requests, but that partly passes requests to application servers for further processing only to later collect results from those servers and return an HTTP response.

As in any multitiered client-server architecture, many server clusters also contain servers dedicated to application processing. In cluster computing, these are typically servers running on high-performance hardware dedicated

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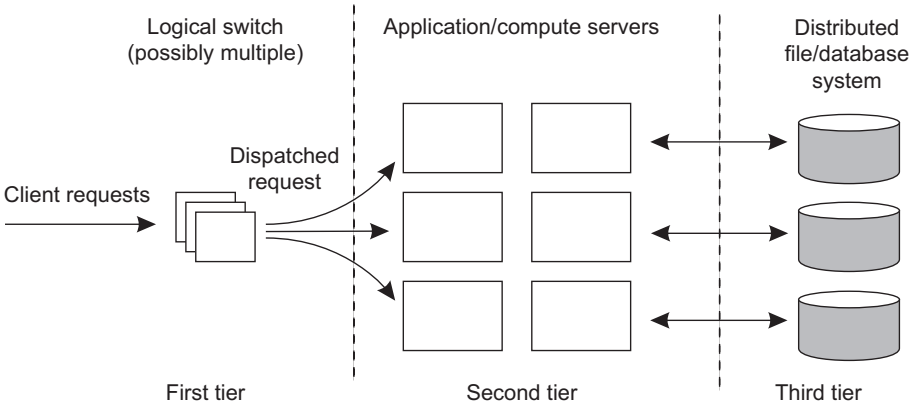


Figure 3.23: The general organization of a three-tiered server cluster.

to delivering compute power. However, in the case of enterprise server clusters, it may be the case that applications need only run on relatively low-end machines, as the required compute power is not the bottleneck, but access to storage is.

This brings us to the third tier, which consists of data-processing servers, notably file and database servers. Again, depending on the usage of the server cluster, these servers may be running on specialized machines, configured for high-speed disk access and having large server-side data caches.

Of course, not all server clusters will follow this strict separation. It is frequently the case that each machine is equipped with its own local storage, often integrating application and data processing in a single server, leading to a two-tiered architecture. For example, when dealing with streaming media using a server cluster, it is common to deploy a two-tiered system architecture, where each machine acts as a dedicated media server [Steinmetz and Nahrstedt, 2004].

When a server cluster offers multiple services, different machines may run different application servers. As a consequence, the switch will have to be able to distinguish services, or otherwise it cannot forward requests to the proper machines. As a consequence, we may find that certain machines are temporarily idle, while others are receiving an overload of requests. What would be useful is to temporarily migrate services to idle machines. A solution is to use virtual machines, allowing a relatively easy migration of services.

Request dispatching Let us now take a closer look at the first tier, consisting of the switch, also known as the **front end**. An important design goal for server clusters is to hide the fact that there are multiple servers. In other words, client applications running on remote machines should have no need to know anything about the internal organization of the cluster. This access

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transparency is invariably offered through a single access point, in turn implemented through some kind of hardware switch such as a dedicated machine.

The switch forms the entry point for the server cluster, offering a single network address. For scalability and availability, a server cluster may have multiple access points, where each access point is then realized by a separate dedicated machine. We consider only the case of a single access point.

In practice, we see two types of switches. In the case of **transport-layer switches**, the switch accepts incoming TCP connection requests, and hands off such connections to one of the servers. The client sets up a TCP connection such that all requests and responses pass through the switch. The switch, in turn, will set up a TCP connection with a selected server and pass client requests to that server, and also accept server responses (which it will pass on to the client). In effect, the switch sits in the middle of a TCP connection between the client and a selected server, rewriting the source and destination addresses when passing TCP segments. This approach is a form of **network-address translation** (NAT) [Srisuresh and Holdrege, 1999].

As an alternative, **application-layer switches** are used. As its name suggest, an application-layer switch operates by inspecting the content of requests instead of just looking at information available in TCP. For example, the switch can inspect the actual URL in case of a Web server. This distinction allows for developing server clusters in which dedicated machines can be configured for handling, for example, video or other media, next to those requiring access to specific databases, etc. In general, the more a switch knows about what is being requested, the better it can decide on which server to handle the request. An obvious drawback of application-layer switches is that they may be slower than transport-layer switches. On the other hand, using the right software and hardware, costs can be kept acceptable low. This is demonstrated, for example, by the nginx server, which has been designed to handle thousands of simultaneous connections and is used for many sites as what is also called a **reverse proxy** [DeJonghe, 2022]. Note that the Apache Web server can also be configured as an application-layer switch.

Note 3.7 (Example: TCP handoff)
In the days that request dispatching was relatively expensive, a simple and elegant way of ensuring that performance criteria could still be met, was by implementing **TCP handoff**. The idea is a switch can actually hand off the connection to a selected server such that all responses are directly communicated to the client without passing through the switch [Hunt et al., 1997; Pai et al., 1998]. The principle working is shown in Figure 3.24.

When the switch receives a TCP connection request, it first identifies the best server for handling that request, and forwards the request packet to that server. The server, in turn, will send an acknowledgment back to the requesting client, but

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inserting the switch's IP address as the source field of the header of the IP packet carrying the TCP segment. Note that this address rewriting is necessary for the client to continue executing the TCP protocol: it is expecting an answer back from the switch, not from some arbitrary server it has never heard of before. Clearly, a TCP-handoff implementation requires operating-system level modifications. TCP handoff is especially effective when responses are much larger than requests, as in the case of Web servers.

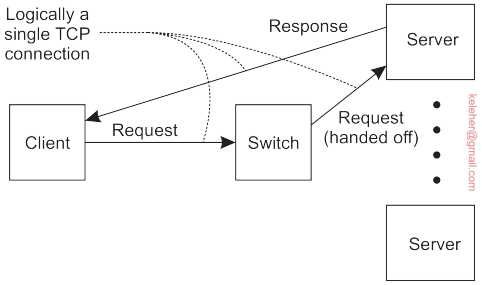


Figure 3.24: The principle of TCP handoff.

It can already be seen that the switch can play an important role in distributing the load among the various servers. By deciding where to forward a request to, the switch also decides which server is to handle further processing of the request. The simplest load-balancing policy that the switch can follow is round robin: each time it picks the next server from its list to forward a request to. Of course, the switch will have to keep track to which server it handed off a TCP connection, at least until that connection is torn down. As it turns out, maintaining this state and handing off subsequent TCP segments belonging to the same TCP connection, may actually slow down the switch.

Wide-area clusters

We came across wide-area clusters in our discussion on PlanetLab (see Note 3.6): an architecture in which participants contribute some (relatively simple) machines for hosting containers that are subsequently "sliced" across as many machines as a client needs. Thereafter, the client is on her own. Much more sophisticated and commercially deployed wide-area clusters exist today.

A straightforward version is seen with cloud providers like Amazon and Google, who manage several data centers placed at different locations worldwide. As such, they can offer an end user the ability to build a wide-area distributed system consisting of a potentially large collection of networked virtual machines, scattered across the Internet. An important reason for wanting such distributed systems is to provide locality: offering data and services that are close to clients. An example where such locality is important is streaming media: the closer a video server is located to a client, the easier

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it becomes to provide high-quality streams. This approach is followed by so-called cloud-based **Content Delivery Networks**, or simply CDNs, which we will discuss shortly.

An alternative is to have a single organization place servers across the Internet, effectively negotiating with local ISPs to decide how to make use of their facilities. This is the approach followed by the Akamai CDN [Dilley et al., 2002; Nygren et al., 2010], in 2022 having some 400,000 servers spread across 1350 ISPs and more than 135 countries.

The general organization of a CDN As CDNs form an important group of distributed systems that make use of wide-area clusters, let us take a closer look at how they are generally organized. We take a simplified view of the Akamai organization, shown in Figure 3.25. More information on CDNs can be found in [Passarella, 2012; Stocker et al., 2017] and [Zolfaghari et al., 2021]. The basic idea is that there is an **origin server** that hosts a Website with all its documents. As we will explain in Chapter 6, each document of such a site is referred to by a URL containing the **domain name** of the origin server, such as `www.example.com`. To access a document, the domain name needs to be resolved to a network address, which is done by the **Domain Name System**, which we will discuss in detail also in Chapter 6. To have the content of the origin server be hosted by a CDN, Akamai first makes sure that the domain name `www.example.com` is resolved to something like `www.example.com.akamai.net`, which, in turn, will refer to an edge server that is part of the Akamai CDN. To be able to still access the origin server, its domain name will have to be changed to, say, `org-www.example.com`. The principle is shown in Figure 3.25.

The client first looks up the regular domain name, but is redirected to the

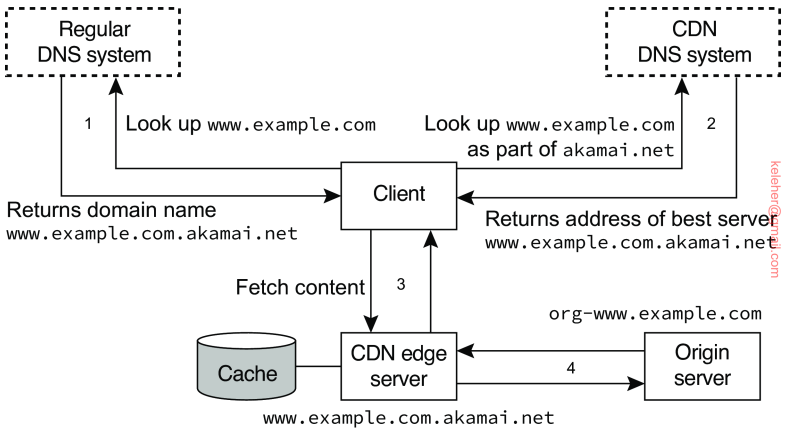


Figure 3.25: A simplified version of the working of the Akamai CDN.

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akamai.net resolvers (step 1). The Akamai name resolvers will look up the best edge server to serve the client, and return its network address (step 2). This allows the client to contact the edge server (step 3), who is aware of the new name of the origin server. If the requested content is not in the edge server's cache, it fetches documents from the origin server (step 4), caches those documents and returns the requested ones to the client. More details can be found in [Su et al., 2006].

Request dispatching This example already shows the importance of client-request redirection. In principle, by properly redirecting clients, a CDN can stay in control when it comes to client-perceived performance, but also considering global system performance by, for example, avoiding that requests are sent to heavily loaded servers. These so-called **adaptive redirection policies** can be applied when information on the system's current behavior is provided to the processes that take redirection decisions.

An important issue is whether request redirection is transparent to the client or not. In essence, there are only three redirection techniques: TCP handoff, DNS redirection, and HTTP redirection. We discussed TCP handoff in Note 3.7. This technique is applicable only for server clusters and does not scale to wide-area networks.

DNS redirection is a transparent mechanism by which the client can be kept completely unaware of where documents are located. Akamai's two-level redirection is one example of this technique. We can also directly deploy DNS to return one of several addresses. Note, however, that DNS redirection can be applied only to an entire site (or, more specifically, only at the level of a domain name): the name of individual documents does not fit into the DNS name space.

Unfortunately, DNS redirection is not perfect for two reasons. First, a client generally contacts the Domain Name System through a local DNS server that then acts as a proxy for that client. As a consequence, not the client's IP address, but that of the local DNS server, is used to identify the location of the client. Mao et al. [2002] have shown that there may be a huge additional communication cost, as the local DNS server is often not *that* local. Furthermore, even when clients are returned several options for selecting an edge server, simply picking the first on the list, which is what normally happens, may not be the best choice [Goel et al., 2015].

Secondly, depending on the scheme that is used for resolving a domain name, it may even be the case that the address of the local DNS server is not even being used. Instead, it may happen that the DNS server that is deciding on which IP address to return, may be fooled by the fact that the requester is yet another DNS server acting as an intermediate between the original client and the deciding DNS server (we will explain the details of this scheme in Chapter 6. In those cases, locality awareness has been completely lost.

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Despite that DNS-based redirection may not always be very accurate, it is widely deployed if only for the fact that it is relatively easy to implement and also transparent to the client. In addition, there is no need to rely on location-aware client-side software.

HTTP redirection, finally, is a nontransparent mechanism. When a client requests a specific document, it may be given an alternative URL as part of an HTTP response message, to which it is then redirected. An important observation is that this URL is visible to the client's browser. In fact, the user may decide to bookmark the referral URL, potentially rendering the redirection policy useless.

3.5 Code migration

So far, we have been mainly concerned with distributed systems in which communication is limited to passing data. However, there are situations in which passing programs, sometimes even while they are being executed, simplifies the design of a distributed system. In this section, we take a detailed look at what code migration actually is.

3.5.1 Reasons for migrating code

Traditionally, code migration in distributed systems took place in the form of **process migration** in which an entire process was moved from one node to another [Milojicic et al., 2000]. Moving a running process to a different machine is a costly and intricate task, and there had better be a good reason for doing so. Let us first consider why one would even want to migrate code from one machine to another.

Performance By far the most important reason for code migration has always been performance. The basic idea is that overall system performance can be improved if processes are moved from heavily loaded to lightly loaded machines. Load is often expressed in terms of the CPU queue length or CPU utilization, but other performance indicators are used as well. When completing their survey, Milojicic et al. had already come to the conclusion that process migration was no longer a viable option for improving distributed systems.

Instead of offloading machines, we can now witness that code is moved to make sure that a machine is *sufficiently* loaded. In particular, migrating complete virtual machines with their suite of applications to lightly loaded machines to minimize the total number of nodes being used is common practice in optimizing energy usage in data centers. Interestingly enough, although migrating virtual machines may require more resources, the task itself is far less intricate than migrating a process, as we discuss below.

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In general, load-distribution algorithms by which decisions are made concerning the allocation and redistribution of tasks regarding a set of machines, play an important role in compute-intensive systems. However, in many modern distributed systems, optimizing computing capacity is less an issue than, for example, trying to minimize communication. Moreover, due to the heterogeneity of the underlying platforms and computer networks, performance improvement through code migration is often based on qualitative reasoning instead of mathematical models.

Consider, as an example, a client-server system in which the server manages a huge database. If a client application needs to perform many database operations involving large quantities of data, it may be better to ship part of the client application to the server and send only the results across the network. Otherwise, the network may be swamped with the transfer of data from the server to the client. In this case, code migration is assuming that it generally makes sense to process data close to where those data reside.

This same reason can be used for migrating parts of the server to the client. For example, in many interactive database applications, clients need to fill in forms that are subsequently translated into a series of database operations. Processing the form at the client side, and sending only the completed form to the server, can sometimes avoid that a relatively large number of small messages need to cross the network. The result is that the client perceives better performance, while at the same time the server spends less time on form processing and communication. In the case of smartphones, moving code to be executed at the handheld instead of the server may be the only viable solution to obtain acceptable performance, *both* for the client and the server (see Kumar et al. [2013] for a survey on offloading computations).

Support for code migration can also help improve performance by exploiting parallelism, but without the usual intricacies related to parallel programming. A typical example is searching for information in the Web. It is relatively simple to implement a search query in the form of a small mobile program, called a **mobile agent**, that moves from site to site. By making several copies of such a program, and sending each off to different sites, we may be able to achieve a linear speed-up compared to using just a single program instance. However, Carzaniga et al. [2007] conclude that mobile agents have never become successful because they did not really offer an obvious advantage over other technologies.

Privacy and security Another reason for moving code to where the data is, has to do with security. Recently, the canonical example for this approach is found in **federated learning**. Federated learning comes from the need to train machine-learning models, notably all kinds of variants of artificial neural networks. To keep it simple, a neural network consists of a collection of nodes, organized as a series of several layers, and having weighted links between

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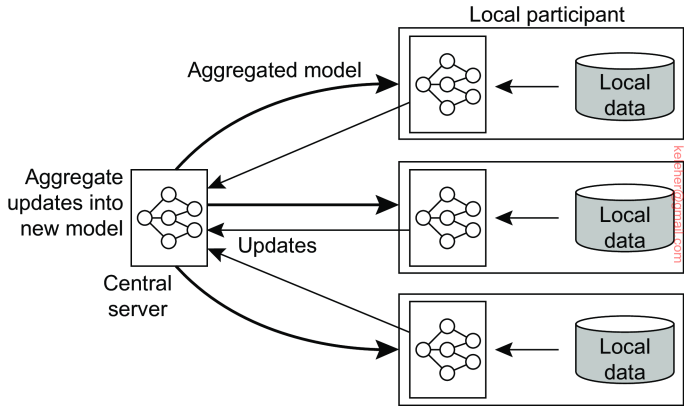


Figure 3.26: The principle of federated learning.

nodes of successive layers. Each node puts data on its outgoing links, which is the result of a (relatively simple) computation that it gets from data on its incoming links. Links have an adjustable weight. During a training phase, the weights of the links are gradually computed by feeding the network with data items for which it is known what the network should produce as a result. By systematically adjusting the weights to minimize the difference between what a network produces and what it should have produced, we eventually establish a final model. That model can then be used on unknown data items, for which we then take the produced result as a given.

A traditional approach is to collect the training data at a centralized location, often using specialized high-performance computers to construct an acceptable model. Obviously, this could mean that sensitive data, such as personal photo's and such, needed to be handed out to an organization that someone may not trust.

In such cases, a better approach is to bring the (partially) trained model to where the data is, and continue training with that local data. This will lead to an updated model (i.e., the weights have been further adjusted because of using new data). If several entities are involved in training with local data, a server can simply collect the different models, aggregate the results, and return an updated model to the local participants who can then continue with training. This iterative process, sketched in Figure 3.26 stops when the aggregated model is deemed sufficiently trained and fit for actual use.

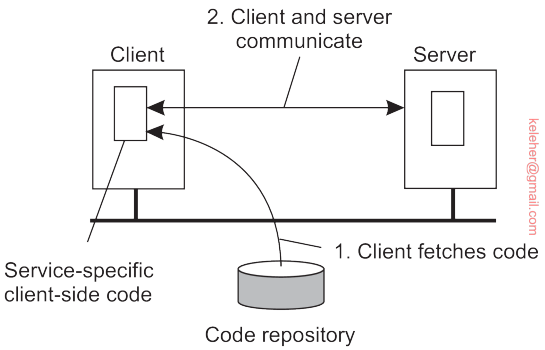
It must be said that federated learning is easier from a code-migration perspective than, e.g., migrating processes. In general, the code involved in a neural network is relatively straightforward, which makes it easier to have it adopted by a local participant. This has been a high-level view on federated learning, of which a good introduction is given by Zhang et al. [2021a]. As described in Lyu et al. [2020], simply doing local training may not be sufficient from a privacy and security perspective.

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Flexibility There are other reasons for supporting code migration as well. An important one is that of flexibility. The traditional approach to building distributed applications is to partition the application into different parts, and decide in advance where each part should be executed. This approach, for example, has lead to different multitiered client-server applications discussed in Chapter 2.

However, if code can move between different machines, it becomes possible to dynamically configure distributed systems. For example, suppose a server implements a standardized interface to a file system. To allow remote clients to access the file system, the server makes use of a proprietary protocol. Normally, the client-side implementation of the file system interface, which is based on that protocol, would need to be linked with the client application. This approach requires that the software be readily available to the client at the time the client application is being developed.

An alternative is to let the server provide the client's implementation no sooner than is strictly necessary, that is, when the client binds to the server. At that point, the client dynamically downloads the implementation, goes through the necessary initialization steps, and subsequently invokes the server. This principle is shown in Figure 3.27 (we note that the code repository is generally located as part of the server). This model of dynamically moving code from a remote site does require that the protocol for downloading and initializing code is standardized. Furthermore, it is necessary that the downloaded code can be executed on the client's machine. Typically, scripts that run in a virtual machine embedded in, for example, a Web browser, will do the trick. Arguably, this form of code migration has been key to the success of the dynamic Web. These and other solutions are discussed below and in later chapters.



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Figure 3.27: The principle of dynamically configuring a client to communicate with a server.

The important advantage of this model of dynamically downloading client-side software is that clients need not have all the software preinstalled to talk

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to servers. Instead, the software can be moved in as necessary, and likewise, discarded when no longer needed. Another advantage is that as long as interfaces are standardized, we can change the client-server protocol and its implementation as often as we like. Changes will not affect existing client applications that rely on the server.

There are, of course, also disadvantages. The most serious one, which we discuss in Chapter 9, has to do with security. Blindly trusting that the downloaded code implements only the advertised interface while accessing your unprotected hard disk and does not send the juiciest parts to heaven-knows-who may not always be such a good idea. Fortunately, it is well understood how to protect the client against malicious, downloaded code.

Note 3.8 (More information: Moving away from thin-client computing?)
By now, there is much more insight and expertise concerning transparent and safe dynamic migration of code to clients. As a result, the trend that we described in Note 2.5 of moving toward thin-client computing because managing client-side software often turned out to be cumbersome, has been partly reverted. By dynamically migrating client-side software, yet keeping the management of that software entirely at the server side (or rather, at its owner), having "richer" client-side software has become practically feasible. The most important domain where we see this happening is through applications running on smartphones. In fact, as soon as, for example, a bug is discovered, and update can take place. It is for this reason why many banks consider it safer to use a mobile app instead of a fixed piece of client-side software.

3.5.2 Models for code migration

Although code migration suggests that we move only code between machines, the term actually covers a much richer area. Traditionally, communication in distributed systems is concerned with exchanging data between processes. Code migration in the broadest sense deals with moving programs between machines, with the intention to have those programs be executed at the target. In some cases, as in process migration, the execution status of a program, pending signals, and other parts of the environment must be moved as well.

To get a better understanding of the different models for code migration, we use a framework proposed by Fuggetta et al. [1998]. In this framework, a process consists of three segments. The **code segment** is the part that contains the set of instructions that make up the program that is being executed. The **resource segment** contains references to external resources needed by the process, such as files, printers, devices, other processes, and so on. Finally, an **execution segment** is used to store the current execution state of a process, consisting of private data, the stack, and, of course, the program counter.

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A further distinction can be made between sender-initiated and receiver-initiated migration. In **sender-initiated** migration, migration is initiated at the machine where the code currently resides or is being executed. Typically, sender-initiated migration is done when uploading programs to a compute server. Another example is sending a query, or batch of queries, to a remote database server. In **receiver-initiated** migration, the initiative for code migration is taken by the target machine. Java applets are an example of this approach.

Receiver-initiated migration is simpler than sender-initiated migration. Often, code migration occurs between a client and a server, where the client takes the initiative for migration. Securely uploading code to a server, as is done in sender-initiated migration, often requires that the client has previously been registered and authenticated at that server. In other words, the server is required to know all its clients, the reason being is that the client will presumably want access to the server's resources such as its disk. Protecting such resources is essential. In contrast, downloading code, as in the receiver-initiated case, can often be done anonymously. Moreover, the server is generally not interested in the client's resources. Instead, code migration to the client is done only for improving client-side performance. To that end, only a limited number of resources need to be protected, such as memory and network connections.

This brings us to four different paradigms for code mobility, as shown in Figure 3.28. Following Fuggetta et al. [1998], we make a distinction between simple **client-server computing**, **remote evaluation**, **code-on-demand**, and **mobile agents**. Figure 3.28 shows the situation at respectively the client and the server, before and after execution of the mobile code.

In the case of client-server computing, the code, execution state, and resource segment are all located at the server, and after execution, only the execution state at the server is generally modified. This state modification is denoted using an asterisk. With the sender-initiated **remote evaluation**, the client migrates code to the server where that code is executed and leading to a modification of the execution state at the server. **Code-on-demand** is a receiver-initiated scheme by which the client obtains code from the server, with its execution modifying the client-side execution state and operating on the client's resources. Finally, **mobile agents** typically follow a sender-initiated approach, moving code as well as execution state from the client to the server, operating on both the client's and the server's resources. Running a mobile agent will generally lead to modification of the associated execution state.

The bare minimum for code migration is to provide only **weak mobility**. In this model, it is possible to transfer only the code segment, along with perhaps some initialization data. A characteristic feature of weak mobility is that a transferred program is always started anew. This is what happens, for example, with Java applets, which start from the same initial state. In

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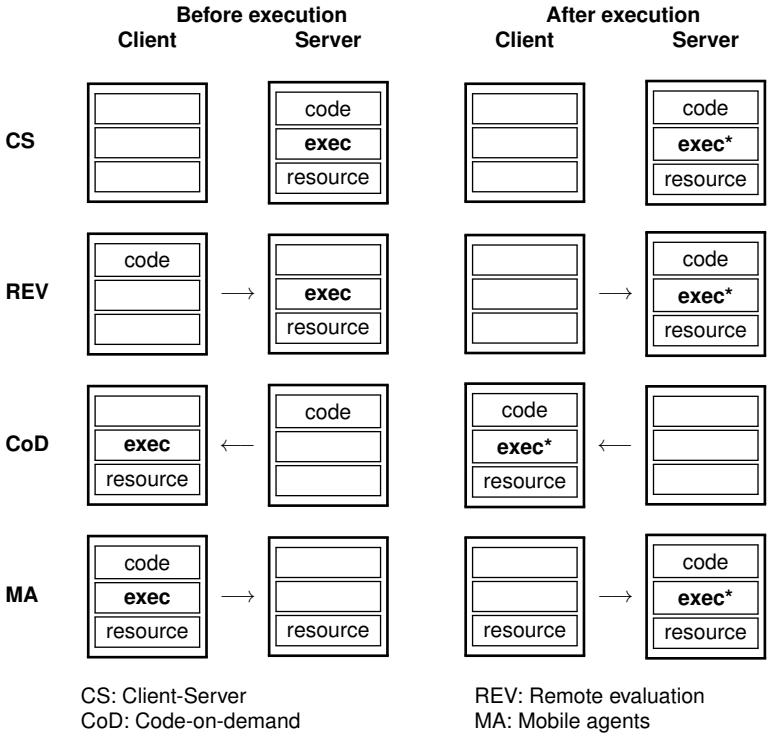


Figure 3.28: Four different paradigms for code mobility.

other words, no history from where the migrated code left off at a previous location is maintained by the underlying middleware. If such history needs to be preserved, it will have to be encoded as part of the mobile application itself. The benefit of weak mobility is its simplicity, as it requires only that the target machine can execute the code segment. In essence, this boils down to making the code portable. We return to these matters when discussing migration in heterogeneous systems.

In contrast to weak mobility, in systems that support **strong mobility** the execution segment can be transferred as well. The characteristic feature of strong mobility is that a running process can be stopped, subsequently moved to another machine, and then resume execution exactly where it left off. Clearly, strong mobility is much more general than weak mobility, but also much more difficult to implement. In particular, when migrating a process, the execution segment generally also contains data that is highly dependent on a specific *implementation* of the underlying operating system. For example, it may rely on information normally found in the operating system's process table. As a consequence, migrating to a different operating system, even one that belongs to the same family as the source, may cause plenty of headaches.

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In the case of weak mobility, it also makes a difference if the migrated code is executed by the target process, or whether a separate process is started. For example, Java applets are simply downloaded by a Web browser and are executed in the browser's address space. The benefit of this approach is that there is no need to start a separate process, thereby avoiding interprocess communication at the target machine. The main drawback, obviously, is that the target process needs to be protected against malicious or inadvertent code executions, which may be reason enough to isolate the migrated code in a separate process.

Instead of moving a running process, also referred to as process migration, strong mobility can also be supported by remote cloning. In contrast to process migration, cloning yields an exact copy of the original process, but now running on a different machine. The cloned process is executed in parallel to the original process. In Unix systems, remote cloning takes place by forking off a child process and letting that child continue on a remote machine. The benefit of cloning is that the model closely resembles the one that is already used in many applications. The only difference is that the cloned process is executed on a different machine. In this sense, migration by cloning is a simple way to improve distribution transparency.

3.5.3 Migration in heterogeneous systems

So far, we have tacitly assumed that the migrated code can be easily executed at the target machine. This assumption is in order when dealing with homogeneous systems. In general, however, distributed systems are constructed on a heterogeneous collection of platforms, each having their own operating system and machine architecture.

The problems coming from heterogeneity are in many respects the same as those of portability. Not surprisingly, solutions are also very similar. For example, at the end of the 1970s, a simple solution to alleviate many of the problems of porting Pascal to different machines was to generate machine-independent intermediate code for an abstract virtual machine [Barron, 1981]. That machine, of course, would need to be implemented on many platforms, but it would then allow Pascal programs to be run anywhere. Although this simple idea was widely used for some years, it never really caught on as the general solution to portability problems for other languages, notably C.

About 25 years later, code migration in heterogeneous systems is being tackled by scripting languages and highly portable languages such as Java and Python. In essence, these solutions adopt the same approach as was done for porting Pascal. All such solutions have in common that they rely on a (process) virtual machine that either directly interprets source code (as in the case of scripting languages), or otherwise interprets intermediate code generated by a compiler (as in Java). Being in the right place at the right time is also important for language developers.

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Further developments have weakened the dependency on programming languages. In particular, solutions have been proposed to migrate not only processes, but to migrate entire computing environments. The basic idea is to compartmentalize the overall environment and to provide processes in the same part their own view on their computing environment. That compartmentalization takes place in the form of virtual machine monitors running an operating system and a suite of applications.

With virtual machine migration, it becomes possible to decouple a computing environment from the underlying system and actually migrate it to another machine (see Medina and Garcia [2014] or Zhang et al. [2018] for overviews on migration mechanisms for virtual machines). A major advantage of this approach is that processes can remain ignorant of the migration itself: they need not be interrupted in their execution, nor should they experience any problems with used resources. The latter are either migrating along with a process, or the way that a process accesses a resource is left unaffected (at least, for that process).

As an example, Clark et al. [2005] concentrated on real-time migration of a virtualized operating system, typically something that would be convenient in a cluster of servers where a tight coupling is achieved through a single, shared local-area network. Under these circumstances, migration involves two major problems: migrating the entire memory image and migrating bindings to local resources.

As to the first problem, there are, in principle, three ways to handle migration (which can be combined):

1. Pushing memory pages to the new machine and resending the ones that are later modified during the migration process.
2. Stopping the current virtual machine; migrate memory, and start the new virtual machine.
3. Letting the new virtual machine pull in new pages as needed, that is, let processes start on the new virtual machine immediately and copy memory pages on demand.

The second option may lead to unacceptable downtime if the migrating virtual machine is running a live service, that is, one that offers continuous service. On the other hand, a pure on-demand approach as represented by the third option may extensively prolong the migration period, but may also lead to poor performance because it takes a long time before the working set of the migrated processes has been moved to the new machine.

As an alternative, Clark et al. [2005] propose to use a pre-copy approach which combines the first option, along with a brief stop-and-copy phase as represented by the second option. As it turns out, this combination can lead to very low service downtimes.

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Concerning local resources, matters are simplified when dealing only with a cluster server. First, because there is a single network, the only thing that needs to be done is to announce the new network-to-MAC address binding, so that clients can contact the migrated processes at the correct network interface. Finally, if it can be assumed that storage is provided as a separate tier (like we showed in Figure 3.23), then migrating binding to files is similarly simple, as it effectively means reestablishing network connections.

Matters become more intricate when we need to migrate a virtual machine to another data center. Although the transfer of memory can be largely done as before, we do need to actually transfer files to the target data center. Somewhat problematic is also the network connectivity: one way or the other clients need to be able to continue contacting the virtual machine while, and after, its migration to the new destination. Zhang et al. [2018] mention various solutions. In essence, many of these either boil down to extending the local network to the target using techniques such as tunneling, or using techniques that support transparent reassigning of network addresses (meaning that a client uses dynamic rebinding to actual network addresses).

Note 3.9 (Advanced: On the performance of live virtual machine migration)
One potential problem with virtual-machine migration is that it may take considerable time. This by itself need not be bad, as long as the services that are running on the migrating virtual machine can continue to operate. An approach used in practice was briefly described above. First, memory pages are copied to the target machine, possibly sending updates of pages that were modified while copying took place (remember that copying lots of memory may take tens of seconds, even across a high-speed local network). Second, when most pages have been faithfully copied, the current machine is stopped, the remaining dirty pages are copied to the target, where the now exact copy can be started where the original left off.

The downtime in which the remaining dirty pages need to be copied depends on the applications running on the virtual machine. Clark et al. [2005] report downtimes for specific configurations between 60 milliseconds and less than 4 seconds. Voorsluys et al. [2009] come to similar values. However, what may be more interesting is to observe what the response time is of the service running on the virtual machine while the latter is being migrated. The model in this case is that the service continues to operate on the original machine until full migration has completed. However, we cannot ignore that migration itself is a resource-intensive operation, requiring considerable processing capacity as well as network bandwidth.

Voorsluys et al. [2009] have observed that a complete migration may actually take tens of seconds, leading to a ten- to twentyfold increase in response time. In addition, we need to realize that during the migration, a service will be completely unavailable (i.e., unresponsive) for perhaps 4 seconds. The good news is that the response time goes up significantly only after the downtime to complete the migration, as shown in Figure 3.29.

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Often, virtual machines are migrated to optimize the usage of actual machines. However, it may also be desirable to clone a virtual machine, for example because the workload for the current machine is becoming too high. Such cloning is very similar to using multiple processes in concurrent servers, by which a dispatcher process creates worker processes to handle incoming requests. This scheme was explained in Figure 3.6 when discussing multithreaded servers.

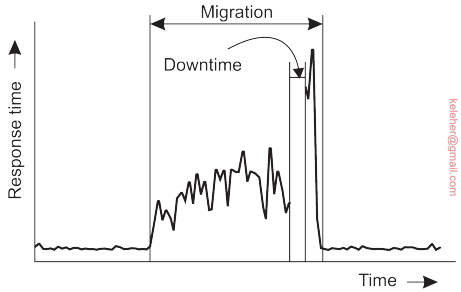


Figure 3.29: The effect on the response time of a service while migrating its underlying virtual machine. Adapted from Voorsluys et al. [2009].

When cloning for this type of performance, it often makes more sense *not* to first copy memory pages, but, in fact, start with as few pages as possible as the service running on the cloned machine will essentially start anew. Note that this behavior is very similar to the usual parent-child behavior we see when forking a Unix process. Namely, the child will start with loading its own executable, thereby effectively cleaning the memory it inherited from its parent. This analogy inspired Lagar-Cavilla et al. [2009] to develop an analogous mechanism for *forking* a virtual machine. However, unlike the mechanism used traditionally for migrating virtual machines, their **VM fork** copies pages primarily on demand. The result is an extremely efficient cloning mechanism.

It is thus seen that there is no single best way to place copies of a virtual machine on different physical machines: it very much depends on how and why a virtual machine is being deployed.

3.6 Summary

Processes play a fundamental role in distributed systems, as they form a basis for communication between different machines. An important issue is how processes are internally organized and, in particular, whether or not they support multiple threads of control. Threads in distributed systems are particularly useful to continue using the CPU when a blocking I/O operation is performed. In this way, it becomes possible to build highly efficient servers that run multiple threads in parallel, of which several may be blocking to wait until disk I/O or network communication completes. In general, threads are

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preferred over the use of processes when performance is at stake.

Virtualization has since long been an important field in computer science, but in the advent of cloud computing has regained tremendous attention. Popular virtualization schemes allow users to run a suite of applications on top of their favorite operating system and configure complete virtual distributed systems in the cloud. Impressively enough, performance remains close to running applications on the host operating system, unless that system is shared with other virtual machines or when the virtual machine is I/O bound. The flexible application of virtual machines has led to different types of services for cloud computing, including infrastructures, platforms, and software — all running in virtual environments. A special form of virtualization is that of containerization, which boils down to providing an application with its own environment (notably libraries and specific supporting programs) while sharing the same operating system. By sharing the same operating system, containers generally tend to perform better in the case of I/O-bound applications, yet it is fair to say that differences in performance between virtual machines and containers is diminishing.

Organizing a distributed application in terms of clients and servers has proven to be useful. Client processes generally implement user interfaces, which may range from simple displays to advanced interfaces that can handle compound documents. Client software is furthermore aimed at achieving distribution transparency by hiding details concerning the communication with servers, where those servers are currently located, and whether servers are replicated. In addition, client software is partly responsible for hiding failures and recovery from failures. An interesting phenomenon is the increasing popularity of virtual desktop environments, by which an entire desktop more or less runs in the cloud.

Servers are often more intricate than clients, but are nevertheless subject to only a relatively few design issues. For example, servers can either be iterative or concurrent, implement one or more services, and can be stateless or stateful. Other design issues deal with addressing services and mechanisms to interrupt a server after a service request has been issued and is possibly already being processed.

Special attention needs to be paid when organizing servers into a cluster. A common objective is to hide the internals of a cluster from the outside world. This means that the organization of the cluster should be shielded from applications. To this end, most clusters use a single entry point that can hand off messages to servers in the cluster. A challenging problem is to transparently replace this single entry point by a fully distributed solution.

Advanced object servers have been developed for hosting remote objects. An object server provides many services to basic objects, including facilities for storing objects, or to ensure serialization of incoming requests. Another important role is providing the illusion to the outside world that a collection

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of data and procedures operating on that data correspond to the concept of an object. This role is implemented by object adapters. Object-based systems have come to a point where we can build entire frameworks that can be extended for supporting specific applications. Java has proven to provide a powerful means for setting up more generic services, exemplified by the highly popular Enterprise Java Beans concept and its implementation.

An exemplary server for Web-based systems is the one from Apache. Again, the Apache server can be seen as a general solution for handling a myriad of HTTP-based queries. By offering the right hooks, we essentially obtain a flexibly configurable Web server. Apache has served as an example not only for traditional Websites, but also for setting up clusters of collaborative Web servers, even across wide-area networks.

An important development is that of content delivery networks in wide-area networks, which facilitate accessing data and other resources close to clients. An essential component is ensuring that resources from an origin server are copied to the proper edge servers transparently. To this end, the combination of client-request redirection techniques and advanced nearby caching is essential.

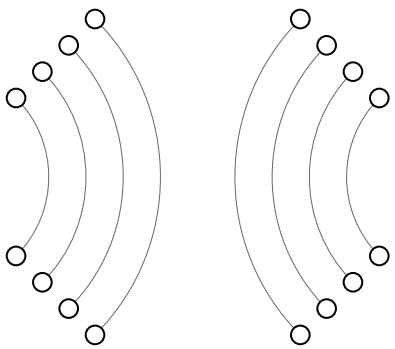
Another important topic for distributed systems is the migration of code between different machines. Two important reasons to support code migration are increasing performance and flexibility. When communication is expensive, we can sometimes reduce communication by shipping computations from the server to the client, and let the client do as much local processing as possible. Flexibility is increased if a client can dynamically download software needed to communicate with a specific server. The downloaded software can be specifically targeted to that server, without forcing the client to have it preinstalled. Lately, we see that also privacy and security have become reasons for migrating code, as is illustrated by federated learning.

Code migration brings along problems related to usage of local resources for which it is required that either resources are migrated as well, new bindings to local resources at the target machine are established, or for which systemwide network references are used. Another problem is that code migration requires that we take heterogeneity into account. Current practice indicates that the best solution to handle heterogeneity is to use virtual machines. These can take either the form of process virtual machines as in the case of, for example, Java, or through using virtual machine monitors that effectively allow the migration of a collection of processes along with their underlying operating system.

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04

COMMUNICATION



Interprocess communication is at the heart of all distributed systems. It makes no sense to study distributed systems without carefully examining the ways that processes on different machines can exchange information. Communication in distributed systems has traditionally always been based on low-level message passing as offered by the underlying network. Expressing communication through message passing is more difficult than using primitives based on shared memory, as available for nondistributed platforms. Modern distributed systems often consist of thousands or even millions of processes scattered across a network with unreliable communication, such as the Internet. Unless the primitive communication facilities of computer networks are replaced by something else, development of large-scale distributed applications is extremely difficult.

In this chapter, we start by discussing the rules that communicating processes must adhere to, known as protocols, and concentrate on structuring those protocols in the form of layers. We then look at two widely used models for communication: Remote Procedure Call (RPC), and Message-Oriented Middleware (MOM). We also discuss the general problem of sending data to multiple receivers, called multicasting.

Our first model for communication in distributed systems is the remote procedure call (RPC). An RPC aims at hiding most of the intricacies of message passing, and is ideal for client-server applications. However, realizing RPCs transparently is easier said than done. We look at several important details that cannot be ignored, while diving into actually code to illustrate to what extent distribution transparency can be realized such that performance is still acceptable.

In many distributed applications, communication does not follow the rather strict pattern of client-server interaction. In those cases, it turns out that thinking in terms of messages is more appropriate. The low-level communication facilities of computer networks are in many ways not suitable, again due to their lack of distribution transparency. An alternative is to use a high-level message-queuing model, in which communication proceeds much the same as in e-mail systems. Message-oriented communication is a subject important enough to warrant a section of its own. We look at numerous aspects, including application-level routing.

Finally, since our understanding of setting up multicast facilities has improved, novel and elegant solutions for data dissemination have emerged. We pay separate attention to this subject in the last section of this chapter, discussing traditional deterministic means of multicasting, as well as probabilistic approaches as used in flooding and gossiping. The latter have been receiving much attention over the past years due to their elegance, reliability, and simplicity.

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4.1 Foundations

Before we start our discussion on communication in distributed systems, we first recapitulate some fundamental issues related to communication. In the next section, we briefly discuss network communication protocols, as these form the basis for any distributed system. Thereafter, we take a different approach by classifying the different types of communication that usually occur in distributed systems.

4.1.1 Layered Protocols

Due to the absence of shared memory, all communication in distributed systems is based on sending and receiving (low level) messages. When process P wants to communicate with process Q, it first builds a message in its own address space. Then it executes a system call that causes the operating system to send the message over the network to Q. Although this basic idea sounds simple enough, to prevent chaos, P and Q have to agree on the meaning of the bits being sent.

The OSI reference model

To make it easier to deal with the numerous levels and issues involved in communication, the International Standards Organization (ISO) developed a reference model that clearly identifies the various levels involved, gives them standard names, and points out which level should do which job. This model is called the **Open Systems Interconnection Reference Model** [Day and Zimmerman, 1983] usually abbreviated as **ISO OSI** or sometimes just the **OSI model**. It should be emphasized that the protocols that were developed as part of the OSI model were never widely used and are essentially dead. However, the underlying model itself has proved to be quite useful for understanding computer networks. Although we do not intend to give a full description of this model and all of its implications here, a brief introduction will be helpful. For more details, see [Tanenbaum et al., 2021].

The OSI model is designed to allow open systems to communicate. An open system is one that is prepared to communicate with any other open system by using standard rules that govern the format, contents, and meaning of the messages sent and received. These rules are formalized in what are called **communication protocols**. To allow a group of computers to communicate over a network, they must all agree on the protocols to be used. A protocol is said to provide a **communication service**. There are two types of such services. In the case of a **connection-oriented service**, before exchanging data the sender and receiver first explicitly establish a connection, and possibly negotiate specific parameters of the protocol they will use. When they are done, they release (terminate) the connection. The telephone is a

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typical connection-oriented communication service. With a **connectionless services**, no setup in advance is needed. The sender just transmits the first message when it is ready. Dropping a letter in a mailbox is an example of making use of connectionless communication service. With computers, both connection-oriented and connectionless communication are common.

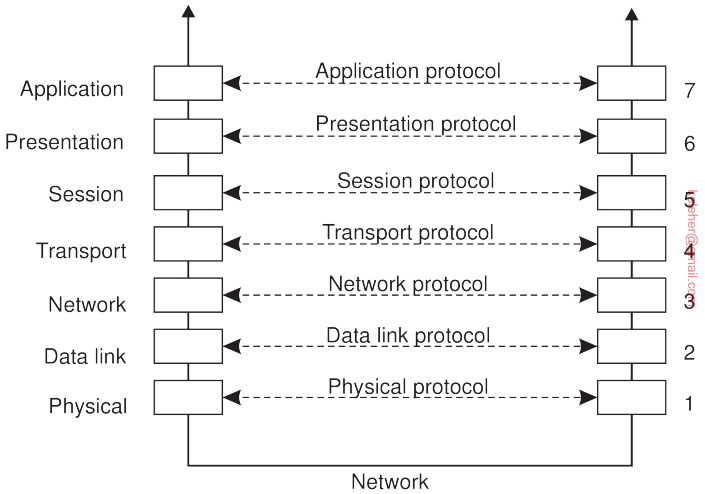


Figure 4.1: Layers, interfaces, and protocols in the OSI model.

In the OSI model, communication is divided into seven levels or layers, as shown in Figure 4.1. Each layer offers one or more specific communication services to the layer above it. In this way, the problem of getting a message from A to B can be divided into manageable pieces, each of which can be solved independently of the others. Each layer provides an **interface** to the one above it. The interface consists of a set of operations that together define the service the layer is prepared to offer. The seven OSI layers are:

Physical layer Deals with standardizing how two computers are connected and how 0s and 1s are represented.

Data link layer Provides the means to detect and possibly correct transmission errors, as well as protocols to keep a sender and receiver in the same pace.

Network layer Contains the protocols for routing a message through a computer network, as well as protocols for handling congestion.

Transport layer Mainly contains protocols for directly supporting applications, such as those that establish reliable communication, or support real-time streaming of data.

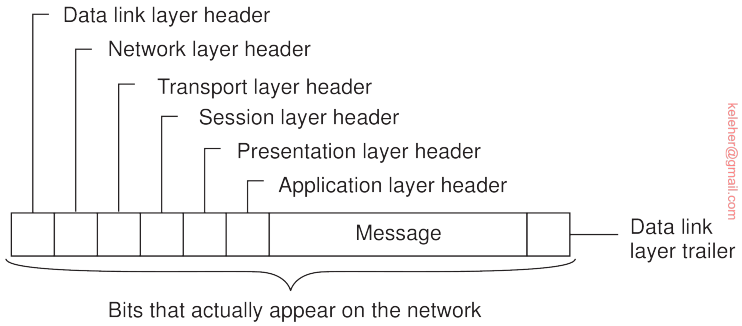
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Session layer Provides support for sessions between applications.

Presentation layer Prescribes how data is represented in a way that is independent of the hosts on which communicating applications are running.

Application layer Essentially, everything else: e-mail protocols, Web access protocols, file-transfer protocols, and so on.

When a process P wants to communicate with some remote process Q, it builds a message and passes that message to the application layer as offered to it through an interface. This interface will typically appear in the form of a library procedure. The application layer software then adds a *header* to the front of the message and passes the resulting message across the layer 6/7 interface to the presentation layer. The presentation layer, in turn, adds its own header and passes the result down to the session layer, and so on. Some layers add not only a header to the front, but also a trailer to the end. When it hits the bottom, the physical layer actually transmits the message (which by now might look as shown in Figure 4.2) by putting it onto the physical transmission medium.



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Figure 4.2: A typical message as it appears on the network.

When the message arrives at the remote machine hosting Q, it is passed upward, with each layer stripping off and examining its own header. Finally, the message arrives at the receiver, process Q, which may reply to it using the reverse path. The information in the layer-n header is used for the layer-n protocol.

In the OSI model, there are not two layers, but seven, as we saw in Figure 4.1. The collection of protocols used in a particular system is called a **protocol suite** or **protocol stack**. It is important to distinguish a *reference model* from its actual *protocols*. As said, the OSI protocols were never popular, in contrast to protocols developed for the Internet, such as TCP and IP.

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Note 4.1 (More information: Protocols in the OSI model)
Let us briefly examine each of the OSI layers in turn, starting at the bottom. Instead of giving examples of OSI protocols, where appropriate, we will point out some Internet protocols used in each layer.

Lower-level protocols The three lowest layers of the OSI protocol suite implement the basic functions that encompass a computer network.

The **physical layer** is concerned with transmitting the 0s and 1s. How many volts to use for 0 and 1, how many bits per second can be sent, and whether transmission can take place in both directions simultaneously are key issues in the physical layer. In addition, the size and shape of the network connector (plug), as well as the number of pins and meaning of each, are of concern here.

The physical layer protocol deals with standardizing the electrical, optical, mechanical, and signaling interfaces so that when one machine sends a 0 bit it is actually received as a 0 bit and not a 1 bit. Many physical layer standards have been developed (for different media), for example, the USB standard for serial communication lines.

The physical layer just sends bits. As long as no errors occur, all is well. However, real communication networks are subject to errors, so some mechanism is needed to detect and correct them. This mechanism is the main task of the **data link layer**. What it does is to group the bits into units, also called **frames**, and see that each frame is correctly received.

The data link layer does its work by putting a special bit pattern on the start and end of each frame to mark them, as well as computing a **checksum** by adding up all the bytes in the frame in a certain way. The data link layer appends the checksum to the frame. When the frame arrives, the receiver recomputes the checksum from the data and compares the result to the checksum following the frame. If the two agree, the frame is considered correct and is accepted. If they disagree, the receiver asks the sender to retransmit it. Frames are assigned sequence numbers (in the header), so everyone can tell which is which.

On a LAN, there is usually no need for the sender to locate the receiver. It just puts the message out on the network and the receiver takes it off. A wide-area network, however, consists of numerous machines, each with some number of lines to other machines, rather like a large-scale map showing major cities and roads connecting them. For a message to get from the sender to the receiver it may have to make a number of hops, at each one choosing an outgoing line to use. The question of how to choose the best path is called **routing**, and is essentially the primary task of the **network layer**.

The problem is complicated by the fact that the shortest route is not always the best route. What really matters is the amount of delay on a given route, which, in turn, is related to the amount of traffic and the number of messages queued up for transmission over the various lines. The delay can thus change over the course of time. Some routing algorithms try to adapt to changing loads, whereas others are content to make decisions based on long-term averages.

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At present, the most widely used network protocol is the connectionless **IP** (**Internet Protocol**), which is part of the Internet protocol suite. An IP **packet** (the technical term for a message in the network layer) can be sent without any setup. Each IP packet is routed to its destination independent of all others. No internal path is selected and remembered.

Transport protocols The **transport layer** forms the last part of what could be called a basic network protocol stack, in the sense that it implements all those services that are not provided at the interface of the network layer, but which are reasonably needed to build network applications. In other words, the transport layer turns the underlying network into something that an application developer can use.

Packets can be lost on the way from the sender to the receiver. Although some applications can handle their own error recovery, others prefer a reliable connection. The job of the transport layer is to provide this service. The idea is that the application layer should be able to deliver a message to the transport layer with the expectation that it will be delivered without loss.

Upon receiving a message from the application layer, the transport layer breaks it into pieces small enough for transmission, assigns each one a sequence number, and then sends them all. The discussion in the transport layer header concerns which packets have been sent, which have been received, how many more the receiver has room to accept, which should be retransmitted, and similar topics.

Reliable transport connections (which by definition are connection-oriented) can be built on top of connection-oriented or connectionless network services. In the former case all the packets will arrive in the correct sequence (if they arrive at all), but in the latter case it is possible for one packet to take a different route and arrive earlier than the packet sent before it. It is up to the transport layer software to put everything back to maintain the illusion that a transport connection is like a big tube–you put messages into it, and they come out undamaged and in the same order in which they went in. Providing this end-to-end communication behavior is an important aspect of the transport layer.

The Internet transport protocol is called **TCP** (**Transmission Control Protocol**) and is described in detail by Comer [2013]. The combination TCP/IP is now used as a de facto standard for network communication. The Internet protocol suite also supports a connectionless transport protocol called **UDP** (**Universal Datagram Protocol**), which is essentially just IP with some minor additions. User programs that do not need a connection-oriented protocol normally use UDP.

Additional transport protocols are regularly proposed. For example, to support real-time data transfer, the **Real-time Transport Protocol** (**RTP**) has been defined. RTP is a framework protocol in the sense that it specifies packet formats for real-time data without providing the actual mechanisms for guaranteeing data delivery. In addition, it specifies a protocol for monitoring and controlling data transfer of RTP packets [Schulzrinne et al., 2003]. Likewise, the **Streaming Control Transmission Protocol** (**SCTP**) has been proposed as an alternative to

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TCP [Stewart, 2007]. The main difference between SCTP and TCP is that SCTP groups data into messages, whereas TCP merely moves bytes between processes. Doing so may simplify application development.

Higher-level protocols Above the transport layer, OSI distinguishes three additional layers. In practice, only the **application layer** is ever used. In fact, in the Internet protocol suite, everything above the transport layer is grouped together. In the face of middleware systems, we shall see that neither the OSI nor the Internet approach is really appropriate.

The **session layer** is essentially an enhanced version of the transport layer. It provides dialog control, to keep track of which party is currently talking, and it provides synchronization facilities. The latter are useful to allow users to insert checkpoints into long transfers, so that in the event of a crash, it is necessary to go back only to the last checkpoint, rather than all the way back to the beginning. In practice, few applications are interested in the session layer, and it is rarely supported. It is not even present in the Internet protocol suite. However, in the context of developing middleware solutions, the concept of a session and its related protocols has turned out to be quite relevant, notably when defining higher-level communication protocols.

Unlike the lower layers, which are concerned with getting the bits from the sender to the receiver reliably and efficiently, the **presentation layer** is concerned with the meaning of the bits. Most messages do not consist of random bit strings, but more structured information such as people's names, addresses, amounts of money, and so on. In the presentation layer, it is possible to define records containing fields like these and then have the sender notify the receiver that a message contains a particular record in a certain format. This makes it easier for machines with different internal representations to communicate with each other.

The OSI application layer was originally intended to contain a collection of standard network applications such as those for electronic mail, file transfer, and terminal emulation. By now, it has become the container for all applications and protocols that in one way or the other do not fit into one of the underlying layers. From the perspective of the OSI reference model, virtually all distributed systems are just applications.

What is missing in this model is a clear distinction between applications, application-specific protocols, and general-purpose protocols. For example, the Internet **File Transfer Protocol** (**FTP**) [Postel and Reynolds, 1985; Horowitz and Lunt, 1997] defines a protocol for transferring files between a client and server machine. The protocol should not be confused with the `ftp` program, which is an end-user application for transferring files and which also (not entirely coincidentally) happens to implement the Internet FTP.

Another example of a typical application-specific protocol is the **HyperText Transfer Protocol** (**HTTP**) [Fielding and Reschke, 2014] which is designed to remotely manage and handle the transfer of Web pages. The protocol is implemented by applications such as Web browsers and Web servers. However, HTTP is now also used by systems that are not intrinsically tied to the Web. For example,

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Java's object-invocation mechanism can use HTTP to request the invocation of remote objects that are protected by a firewall.

There are also many general-purpose protocols that are useful to many applications, but which cannot be qualified as transport protocols. Often, such protocols fall into the category of middleware protocols.

Middleware protocols

Middleware is an application that logically lives (mostly) in the OSI application layer, but which contains many general-purpose protocols that warrant their own layers, independent of other, more specific applications. Let us briefly look at some examples.

The **Domain Name System** (**DNS**) [Liu and Albitz, 2006] is a distributed service that is used to look up a network address associated with a name, such as the address of a so-called **domain name** like www.distributed-systems.net. In terms of the OSI reference model, DNS is an application and therefore is logically placed in the application layer. However, it should be quite obvious that DNS is offering a general-purpose, application-independent service. Arguably, it forms part of the middleware.

As another example, there are various ways to establish **authentication**, that is, provide proof of a claimed identity. Authentication protocols are not closely tied to any specific application, but instead, can be integrated into a middleware system as a general service. Likewise, **authorization protocols** by which authenticated users and processes are granted access only to those resources for which they have authorization, tend to have a general, application-independent nature. Being labeled as applications in the OSI reference model, these are clear examples that belong in the middleware.

Distributed commit protocols establish that in a group of processes, possibly spread out across several machines, either all processes carry out a particular operation, or that the operation is not carried out at all. This phenomenon is also referred to as **atomicity** and is widely applied in transactions. As it turns out, commit protocols can present an interface independently of specific applications, thus providing a general-purpose transaction service. In such a form, they typically belong to the middleware and not to the OSI application layer.

As a last example, consider a **distributed locking protocol**, by which a resource can be protected against simultaneous access by a collection of processes that are distributed across multiple machines. It is not hard to imagine that such protocols can be designed in an application-independent fashion, and accessible through a relatively simple, again application-independent interface. As such, they generally belong in the middleware.

These protocol examples are not directly tied to communication, yet there

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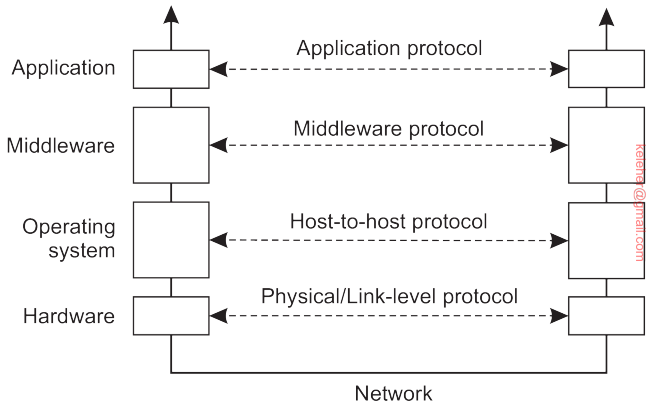


Figure 4.3: An adapted reference model for networked communication.

are also many middleware communication protocols. For example, with a so-called **remote procedure call**, a process is offered a facility to *locally* call a procedure that is effectively implemented on a *remote* machine. This communication service belongs to one of the oldest types of middleware services and is used for realizing access transparency. In a similar vein, there are high-level communication services for setting and synchronizing streams for transferring real-time data, such as needed for multimedia applications. As a last example, some middleware systems offer reliable multicast services that scale to thousands of receivers spread across a wide-area network.

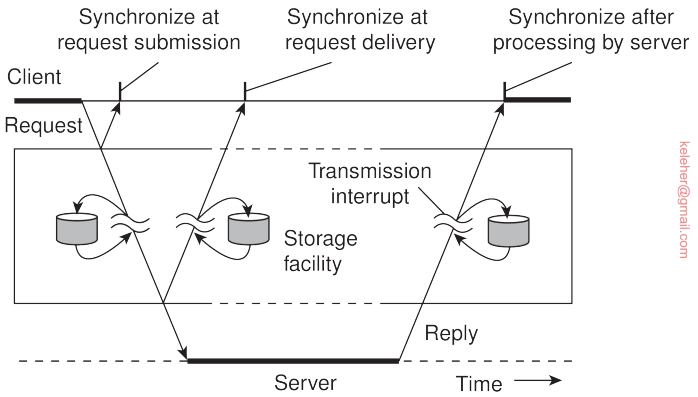
Taking this approach to layering leads to the adapted and simplified reference model for communication, as shown in Figure 4.3. Compared to the OSI model, the session and presentation layer have been replaced by a single middleware layer that contains application-independent protocols. These protocols do not belong in the lower layers we just discussed. Network and transport services have been grouped into communication services as normally offered by an operating system, which, in turn, manages the specific lowest-level hardware used to establish communication.

4.1.2 Types of Communication

In the remainder of this chapter, we concentrate on high-level middleware communication services. Before doing so, there are other general criteria for distinguishing (middleware) communication. To understand the various alternatives in communication that middleware can offer to applications, we view the middleware as an additional service in client-server computing, as shown in Figure 4.4. Consider, for example, an e-mail system. In principle, the core of the mail delivery system can be seen as a middleware communication service. Each host runs a user agent allowing users to compose, send, and receive e-mail. A sending user agent passes such mail to the mail delivery

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system, expecting it, in turn, to eventually deliver the mail to the intended recipient. Likewise, the user agent at the receiver's side connects to the mail delivery system to see whether any mail has come in. If so, the messages are transferred to the user agent so that they can be read by the user.



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Figure 4.4: Viewing middleware as an intermediate (distributed) service in application-level communication.

An e-mail system is a typical example in which communication is persistent. With **persistent communication**, a message that has been submitted for transmission is stored by the communication middleware as long as it takes to deliver it to the receiver. In this case, the middleware will store the message at one or several of the storage facilities shown in Figure 4.4. As a consequence, it is not necessary for the sending application to continue execution after submitting the message. Likewise, the receiving application need not be executing when the message is submitted.

In contrast, with **transient communication**, a message is stored by the communication system only as long as the sending and receiving application are executing. More precisely, in terms of Figure 4.4, if the middleware cannot deliver a message due to a transmission interrupt, or because the recipient is currently not active, it will simply be discarded. Typically, all transport-level communication services offer only transient communication. In this case, the communication system consists of traditional store-and-forward routers. If a router cannot deliver a message to the next one or the destination host, it will simply drop the message.

Besides being persistent or transient, communication can also be asynchronous or synchronous. The characteristic feature of **asynchronous communication** is that a sender continues immediately after it has submitted its message for transmission. This means that the message is (temporarily) stored immediately by the middleware upon submission. With **synchronous communication**, the sender is blocked until its request is known to be accepted. There are essentially three points where synchronization can take place. First,

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the sender may be blocked until the middleware notifies that it will take over transmission of the request. Second, the sender may synchronize until its request has been delivered to the intended recipient. Third, synchronization may take place by letting the sender wait until its request has been fully processed, that is, up to the time that the recipient returns a response.

Various combinations of persistence and synchronization occur in practice. Popular ones are persistence in combination with synchronization at request submission, which is a common scheme for many message-queuing systems, which we discuss later in this chapter. Likewise, transient communication with synchronization after the request has been fully processed is also widely used. This scheme corresponds with remote procedure calls, which we discuss in the following section.

4.2 Remote procedure call

Many distributed systems have been based on explicit message exchange between processes. However, the operations send and receive do not conceal communication at all, which is important to achieve access transparency in distributed systems. This problem has long been known, but little was done about it until researchers in the 1980s [Birrell and Nelson, 1984] introduced an entirely different way of handling communication. Although the idea is refreshingly simple (once someone has thought of it), the implications are often subtle. In this section we will examine the concept, its implementation, its strengths, and its weaknesses.

In a nutshell, the proposal was to allow programs to call procedures located on other machines. When a process on a machine A calls a procedure on a machine B, the calling process on A is suspended, and execution of the called procedure takes place on B. Information can be transported from the caller to the callee in the parameters and can come back in the procedure result. No message passing at all is visible to the programmer. This method is known as **remote procedure call**, or often just **RPC**.

While the basic idea sounds simple and elegant, subtle problems exist. To start with because the calling and called procedures run on different machines, they execute in different address spaces, which causes complications. Parameters and results also have to be passed, which can be complicated, especially if the machines are not identical. Finally, either or both machines can crash, and each of the possible failures causes different problems. Still, most of these can be dealt with, and RPC is a widely used technique that underlies many distributed systems.

4.2.1 Basic RPC operation

The idea behind RPC is to make a remote procedure call look as much as possible as a local one. In other words, we want RPC to be transparent—the

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calling procedure should not be aware that the called procedure is executing on a different machine or vice versa. Suppose that a program has access to a database that allows it to append data to a stored list, after which it returns a reference to the modified list. The operation is made available to a program by a routine append:

```
newlist = append(data, dbList)
```

In a traditional (single processor) system, append is extracted from a library by the linker and inserted into the object program. In principle, it can be a short procedure, which could be implemented by a few file operations for accessing the database.

Even though append eventually does only a few basic file operations, it is called in the usual way, by pushing its parameters onto the stack. The programmer does not know the implementation details of append, and this is, of course, how it is supposed to be.

Note 4.2 (More information: Conventional procedure calls)
To understand how RPC works and some of its pitfalls, it may help to first understand how a conventional (i.e., single machine) procedure call works. Consider the operation `newlist = append(data, dbList);`

We assume that the purpose of this call is to take a *globally* defined list object, referred here to as `dbList`, and append a simple data element to it, represented by the variable `data`. An important observation is that in various programming languages such as C, `dbList` is implemented as a *reference* to a list object (i.e., a pointer), whereas `data` may be represented directly by its value (which we assume to be the case here). When calling append, both the representations of `data` and `dbList` are pushed onto the stack, making those representations accessible to the implementation of append. For `data`, this means the variable follows a **call-by-value** policy, the policy for `dblist` is **call-by-reference**. What happens before and during the call is shown in Figure 4.5.

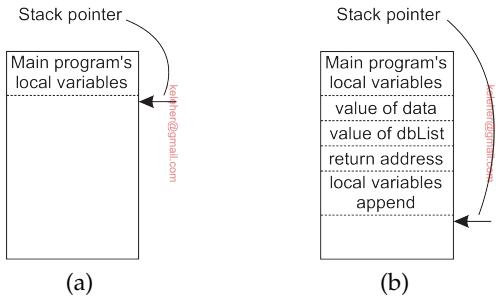


Figure 4.5: (a) Parameter passing in a local procedure call: the stack before the call to append. (b) The stack while the called procedure is active.

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Several things are worth noting. For one, a value parameter, such as `data`, is just an initialized local variable. The called procedure may modify it, but such changes do not affect the original value at the calling side.

When a parameter like `dbList` is actually a pointer to a variable rather than the value of the variable, something else happens. What is pushed onto the stack is the address of the list object as stored in main memory. When the value of `data` is appended to the list, a call to append *does* modify the list object. The difference between call-by-value and call-by-reference is quite important for RPC.

One other parameter passing mechanism also exists, although it is not used in most programming languages. It is called **call-by-copy/restore**. It consists of having the variable copied to the stack by the caller, as in call-by-value, and then copied back after the call, overwriting the caller's original value. Under most conditions, this achieves the same effect as call-by-reference, but in some situations, such as the same parameter being present multiple times in the parameter list, the semantics are different.

The decision of which parameter passing mechanism to use is normally made by the language designers and is a fixed property of the language. Every so often, it depends on the data type being passed. In C, for example, integers and other scalar types are always passed by value, whereas arrays are always passed by reference. Some Ada compilers use copy/restore for `inout` parameters, but others use call-by-reference. The language definition permits either choice, which makes the semantics a bit fuzzy. In Python, all variables are passed by reference, but some actually get copied to local variables, thus mimicking the behavior of copy-by-value.

RPC achieves its transparency analogously. When append is actually a remote procedure, a different version of append, called a **client stub**, is offered to the calling client. Like the original one, it, too, is called using a normal calling sequence. However, unlike the original one, it does not perform an append operation. Instead, it packs the parameters into a message and requests that message to be sent to the server, as illustrated in Figure 4.6. Following the call to send, the client stub calls `receive`, blocking itself until the reply comes back.

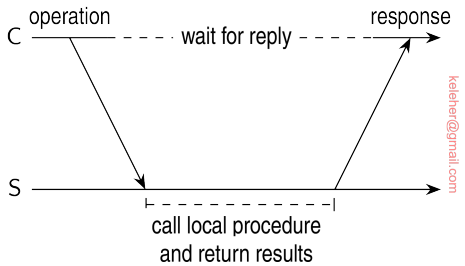


Figure 4.6: The principle of RPC between a client and server program.

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When the message arrives at the server, the server's operating system passes it to a **server stub**. A server stub is the server-side equivalent of a client stub: it is a piece of code that transforms requests coming in over the network into local procedure calls. Typically, the server stub will have called `receive` and be blocked waiting for incoming messages. The server stub unpacks the parameters from the message and then calls the server procedure in the usual way. From the server's perspective, it is as though it is being called directly by the client—the parameters and return address are all on the stack where they belong and nothing seems unusual. The server performs its work and then returns the result to the caller (in this case the server stub) in the usual way.

When the server stub gets control back after the call has completed, it packs the result in a message and calls `send` to return it to the client. Thereafter, the server stub usually does a call to `receive` again, to wait for the next incoming request.

When the result message arrives at the client's machine, the operating system passes it through the `receive` operation, which had been called previously, to the client stub, and the client process is subsequently unblocked. The client stub inspects the message, unpacks the result, copies it to its caller, and returns in the usual way. When the caller gets control following the call to `append`, all it knows is that it appended some data to a list. It has no idea that the work was done remotely at another machine.

This blissful ignorance for the client is the beauty of the whole scheme. As far as it is concerned, remote services are accessed by making ordinary (i.e., local) procedure calls, not by calling `send` and `receive`. All the details of the message passing are hidden away in the two library procedures, just as the details of actually making system calls are hidden away in traditional libraries. To summarize, a remote procedure call occurs in the following steps:

1. The client procedure calls the client stub in the normal way.
2. The client stub builds a message and calls the local operating system.
3. The client's OS sends the message to the remote OS.
4. The remote OS gives the message to the server stub.
5. The server stub unpacks the parameter(s) and calls the server.
6. The server does the work and returns the result to the stub.
7. The server stub packs the result in a message and calls its local OS.
8. The server's OS sends the message to the client's OS.
9. The client's OS gives the message to the client stub.
10. The stub unpacks the result and returns it to the client.

The first steps are shown in Figure 4.7 for an abstract two-parameter procedure `doit(a,b)`, where we assume that parameter `a` is of type `type1`, and `b` of type `type2`. The net effect of all these steps is to convert the local call by the client procedure to the client stub, to a local call to the server procedure,

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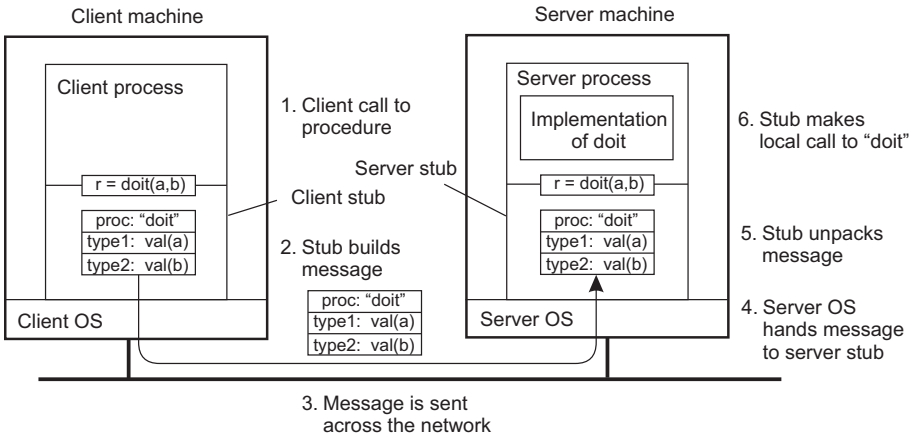


Figure 4.7: The steps involved in calling a remote procedure `doit(a,b)`. The return path for the result is not shown.

without either client or server being aware of the intermediate steps or the existence of the network.

Note 4.3 (More information: An example in Python)
To make matters concrete, let us consider how a remote procedure call could be implemented for the operation `append` discussed previously. Take a look at the Python code shown in Figure 4.8 (from which we omit nonessential code fragments).

The class `DBList` is a simple representation of a list object, mimicking what one would expect to see in a version that would be found in a database environment. The client stub, represented by the class `Client`, consists of an implementation of `append`. When called with parameters `data` and `dbList`, the following happens. The call is transformed into a tuple `(APPEND, data, dbList)` containing all the information the server would need to do its work. The client stub then sends the request off to the server, and subsequently waits for the response. In the `channel` package, a `recvFrom` operation always returns a (*sender*,*message*) pair, allowing the caller to identify the process who had sent the message. In our case, when the response comes in, the client stub finishes the call to `append` by simply passing the returned message from the server to the program that initially called the stub.

On the server side, we see that in the server stub, the server waits for any incoming message, and inspects which operation it is required to call. Again, the `channel` package returns the identifier of the original sender (line 24). Assuming the server received a request to call `append`, it then simply does a *local* call to its implementation of `append` with the appropriate parameters as also found in the request tuple. The result is then sent back to the identified client.

Note that an actual client would simply call `c.append(...)` where `c` is an instance of the class `Client`. Indeed, the call truly seems to take place locally.

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```
1  import channel, pickle
2
3  class DBList:
4    def append(self, data):
5      self.value.extend(data)
6      return self
7
8  class Client:
9    def append(self, data, dbList):
10     msglst = (APPEND, data, dbList)             # message payload
11     self.channel.sendTo(self.server, msglst)    # send message to server
12     msgrcv = self.channel.recvFrom(self.server) # wait for an incoming message
13
14     # A call to recvFrom returns a [senderID, message] pair
15     return msgrcv[1]                            # pass returned message to caller
16
17 class Server:
18   def append(self, data, dbList):
19     return dbList.append(data)
20
21   def run(self):
22     while True:
23       msgreq = self.channel.recvFromAny() # wait for any request
24       client = msgreq[0]                  # see who is the caller
25       msgrpc = msgreq[1]                  # fetch the actual request
26
27       # At this point, msgreq should have the form (operation, data, list)
28       if APPEND == msgrpc[0]:          # check what is being requested
29         result = self.append(msgrpc[1], msgrpc[2]) # do local call
30         self.channel.sendTo(client,result)         # return response
```

Figure 4.8: A simple RPC example for operation append.

4.2.2 Parameter passing

The function of the client stub is to take its parameters, pack them into a message, and send them to the server stub. While this sounds straightforward, it is not quite as simple as it at first appears.

Packing parameters into a message is called **parameter marshaling**. Returning to our append operation, we need to ensure that its two parameters (`data` and `dbList`) are sent over the network and correctly interpreted by the server. The thing to realize, is that the server will just be seeing a series of bytes coming in that constitute the original message sent by the client. However, no additional information on what those bytes mean is normally provided with the message. Also, we would be facing the same problem again: how should the meta-information be recognized as such by the server?

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Besides this interpretation problem, we also need to handle the case that the placement of bytes in memory may differ between machine architectures. In particular, we need to account for the fact that some machines, such as Intel processors, number their bytes from right to left, whereas many others, such as the older ARM processors, number them the other way (ARM now supports both). The Intel format is called **little endian** and the (older) ARM format is called **big endian**. Byte ordering is also important for networking: also here we can witness that machines may use a different ordering when transmitting (and thus receiving) bits and bytes. However, big endian is what is normally used for transferring bytes across a network.

The solution to this problem is to transform data that is to be sent to a machine- and network-independent format, next to making sure that both communicating parties expect the same *message data type* to be transmitted. The latter can typically be solved at the level of programming languages. The former is accomplished by using machine-*dependent* routines that transform data to and from machine- and network-independent formats.

Marshaling and unmarshaling is all about this transformation to neutral formats and forms an essential part of remote procedure calls.

Note 4.4 (More information: An example in Python revisited)

```
1  import channel, pickle
2
3  class Client:
4    def append(self, data, dbList):
5      msglst = (APPEND, data, dbList)             # message payload
6      msgsnd = pickle.dumps(msglst)               # wrap call
7      self.channel.sendTo(self.server, msgsnd)    # send request to server
8      msgrcv = self.channel.recvFrom(self.server) # wait for response
9      retval = pickle.loads(msgrcv[1])            # unwrap return value
10     return retval                               # pass it to caller
11
12 class Server:
13   def run(self):
14     while True:
15       msgreq = self.channel.recvFromAny() # wait for any request
16       client = msgreq[0]                  # see who is the caller
17       msgrpc = pickle.loads(msgreq[1])    # unwrap the call
18       if APPEND == msgrpc[0]:             # check what is being requested
19         result = self.append(msgrpc[1], msgrpc[2]) # do local call
20         msgres = pickle.dumps(result)              # wrap the result
21         self.channel.sendTo(client,msgres)         # send response
```

Figure 4.9: A simple RPC example for operation append, but now with proper marshaling.

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It is not difficult to see that the solution to remote procedure calling as shown in Figure 4.8 will not work in general. Only if the client and server are operating on machines that obey the same byte-ordering rules *and* have the same machine representations for data structures, will the exchange of messages as shown lead to correct interpretations. A robust solution is shown in Figure 4.9 (where we again have omitted code for brevity).

In this example, we use the Python `pickle` library for marshaling and unmarshaling data structures. Note that the code hardly changes in comparison to what we have shown in Figure 4.8. The only changes occur just before sending, and after receiving a message. Also note that both client and server are programmed to work on the same data structures, as we discussed above.

(We do note that, behind the scene, whenever a class instance is passed as a parameter, Python generally takes care of *pickling* and later *unpickling* the instance. In this sense, our explicit use of `pickle` is, strictly speaking, not necessary.)

We now come to a difficult problem: How are pointers, or in general, references passed? The answer is: only with the greatest of difficulty, if at all. A pointer is meaningful only within the address space of the process in which it is being used. Getting back to our `append` example, we stated that the second parameter, `dbList`, is implemented through a *reference* to a list stored in a database. If that reference is just a pointer to a local data structure somewhere in the caller's main memory, we cannot simply pass it to the server. The transferred pointer value will most likely be referring to something entirely different.

One solution is just to forbid pointers and reference parameters in general. However, these are so important that this solution is highly undesirable. In fact, it is often not necessary either. First, reference parameters are often used with fixed-sized data types, such as static arrays, or with dynamic data types for which it is easy to compute their size at runtime, such as strings or dynamic arrays. In such cases, we can simply copy the entire data structure to which the parameter is referring, effectively replacing the copy-by-reference mechanism by copy-by-value/restore. Although this is semantically not always identical, it frequently is good enough. An obvious optimization is that when the client stub knows the referred data will be only read, there is no need to copy it back when the call has finished. Copy-by-value is thus good enough.

More intricate data types can often be supported as well, and certainly if a programming language supports those data types. For example, a language such as Python or Java supports user-defined classes, allowing a language system to provide fully automated marshaling and unmarshaling of those data types. Note, however, that as soon as we are dealing with large, nested, or otherwise intricate dynamic data structures, automatic (un)marshaling may not be available, or even desirable.

The problem with pointers and references, as discussed so far, is that they make sense only locally: they refer to memory locations that have

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meaning only to the calling process. Problems can be alleviated by using *global* references: references that are meaningful to the calling and the called process. For example, if the client and the server have access to the same file system, passing a file handle instead of a pointer may do the trick. There is one important observation: both processes need to know exactly what to do when a global reference is passed. In other words, if we consider a global reference having an associated data type, the calling and called process should have the same picture of the operations that can be performed. Moreover, both processes should have agreement on exactly what to do when a file handle is passed. Again, these are typically issues that can be solved by proper programming-language support. We will return to this subject shortly.

Note 4.5 (Advanced: Parameter passing in object-based systems)
Object-based systems often use global references. Consider the situation that all objects in the system can be accessed from remote machines. In that case, we can consistently use **object references** as parameters in method invocations. References are passed by value, and thus copied from one machine to the other. When a process is given an object reference as the result of a method invocation, it can simply bind to the object referred to when needed later (see also Section 2.1.2).

Unfortunately, using only distributed objects can be highly inefficient, especially when objects are small, such as integers, or worse yet, Booleans. Each invocation by a client that is not co-located in the same server as the object, generates a request between different address spaces or, even worse, between different machines. Therefore, references to remote objects and those to local objects are often treated differently.

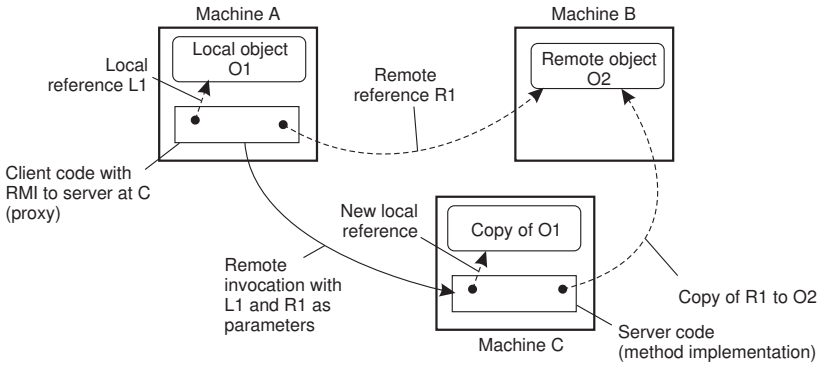


Figure 4.10: Passing an object by reference or by value.

When invoking a method with an object reference as parameter, that reference is copied and passed as a value parameter only when it refers to a remote object. In this case, the object is literally passed by reference. However, when the reference refers to a local object, that is an object in the same address space as the client, the referred object is copied as a whole and passed along with the invocation. In

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other words, the object is passed by value.

These two situations are illustrated in Figure 4.10 which shows a client program running on a machine A, and a server program on a machine C. The client has a reference to a local object O1 that it uses as a parameter when calling the server program on machine C. In addition, it holds a reference to a remote object O2 residing at machine B, which is also used as a parameter. When calling the server, a copy of O1 is passed to the server on machine C, along with only a copy of the reference to O2.

Note that whether we are dealing with a reference to a local object or a reference to a remote object can be highly transparent, such as in Java. In Java, the distinction is visible only because local objects are essentially of a different data type than remote objects. Otherwise, both types of references are treated very much the same (see also [Wollrath et al., 1996]). On the other hand, when using conventional programming languages such as C, a reference to a local object can be as simple as a pointer, which can never be used to refer to a remote object.

The side effect of invoking a method with an object reference as parameter is that we may be *copying* an object. Obviously, hiding this aspect is unacceptable so that we are consequently forced to make an explicit distinction between local and distributed objects. Clearly, this distinction not only violates distribution transparency, but also makes it harder to write distributed applications.

We can now also easily explain how global references can be implemented when using portable, interpreted languages such as Python or Java: use the entire client stub as a reference. The key observation is that a client stub is often just another data structure that is compiled into (portable) bytecode. That compiled code can actually be transferred across the network and executed at the receiver's side. In other words, there is no need for explicit binding anymore; simply copying the client stub to the recipient is enough to allow the latter to invoke the associated server-side object. Of course, it may be necessary to marshall that code before shipping it to the other machine, although strictly speaking, there is no reason why that other machine would work with different layouts.

4.2.3 RPC-based application support

From what we have explained so far, it is clear that hiding a remote procedure call requires that the caller and the callee agree on the format of the messages they exchange and that they follow the same steps when it comes to, for example, passing complex data structures. In other words, both sides in an RPC should follow the same protocol or the RPC will not work correctly. There are at least two ways in which RPC-based application development can be supported. The first one is to let a developer specify exactly what needs to be called remotely, from which complete client-side and server-side stubs can be generated. A second approach is to embed remote procedure calling as part of a programming-language environment.

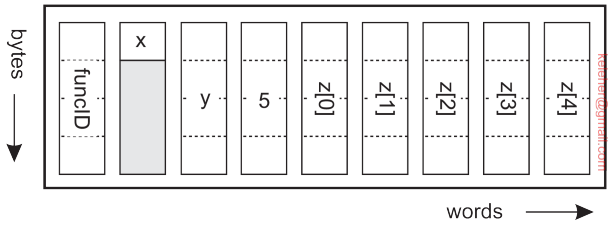
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Stub generation

Consider the function `someFunction` of Figure 4.11(a). It has three parameters, a character, a floating-point number, and an array of five integers. Assuming a word is four bytes, the RPC protocol might prescribe that we should transmit a character in the rightmost byte of a word (leaving the next three bytes empty), a float as a whole word, and an array as a group of words whose size is equal to the array length, preceded by a word giving the length, as shown in Figure 4.11(b). Thus given these rules, the client stub for `someFunction` knows that it must use the format of Figure 4.11(b), and the server stub knows that incoming messages for `someFunction` will have the format of Figure 4.11(b).

```
void someFunction(char x; float y; int z[5])
```

(a)



(b)

Figure 4.11: (a) A function. (b) The corresponding message, and the order in which bytes and words are sent across the network.

Defining the message format is one aspect of an RPC protocol, but it is not sufficient. What we also need is the client and the server to agree on the representation of simple data structures, such as integers, characters, Booleans, etc. For example, the protocol could prescribe that integers are represented in two's complement, characters in 16-bit Unicode, and floats in the IEEE standard #754 format, with everything stored in little endian. With this additional information, messages can be unambiguously interpreted.

With the encoding rules now pinned down to the last bit, the only thing that remains to be done is that the caller and callee agree on the actual exchange of messages. For example, it may be decided to use a connection-oriented transport service, such as TCP/IP. An alternative is to use an unreliable datagram service and let the client and server implement an error control scheme as part of the RPC protocol. In practice, several variants exist, and it is up to the developer to indicate the preferred underlying communication service.

Once the RPC protocol has been fully defined, the client and server stubs need to be implemented. Fortunately, stubs for the same protocol, but different

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procedures, normally differ only in their interface to the applications. An interface consists of a collection of procedures that can be called by a client, and which are implemented by a server. An interface is usually available in the same programming language as the one in which the client or server is written (although this is, strictly speaking, not necessary). To simplify matters, interfaces are often specified through an **Interface Definition Language** (**IDL**). An interface specified in such an IDL is then subsequently compiled into a client stub and a server stub, along with the appropriate compile-time or run-time interfaces. This process is sketched in Figure 4.12.

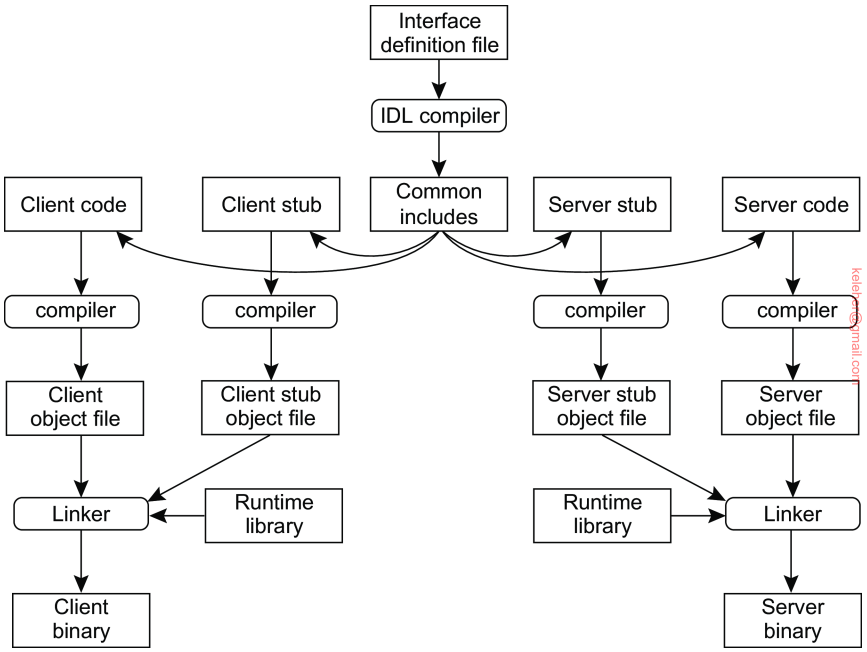


Figure 4.12: The principle of generating stubs from an interface definition file.

The figure shows the situation for compiled languages, yet it is not hard to imagine that a very similar situation holds for interpreted languages. Also note that it is possible that the client and server code are written in *different* languages. There is no principal reason why the client side cannot consist of components written in, for example, Python, while the server side is written in C. Of course, in that case, the box in Figure 4.12 representing common includes will then contain separate files for the client and server, respectively. Important in this scheme is the runtime library: it forms the interface to the actual runtime system, constituting the middleware.

Practice shows that using an interface definition language considerably simplifies client-server applications based on RPCs. Because it is easy to fully

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generate client and server stubs, all RPC-based middleware systems offer an IDL to support application development. In some cases, using the IDL is even mandatory.

Note 4.6 (More information: Language-based RPC in Python)
Let us see by an example of how remote procedure calling can be integrated in a language. We have been using the Python language for most of our examples, and will continue to do so now as well. In Figure 4.13 we show a simple server for our `DBList` data structure. In this case, it has two *exposed* operations: `exposed_append` for appending elements, and `exposed_value` to display what is currently in the list. We use the Python RPyC package for embedding RPCs.

```
1  import rpyc
2  from rpyc.utils.server import ForkingServer
3
4  class DBList(rpyc.Service):
5    value = [] # Used to build a list of strings
6
7    def exposed_append(self, data):
8      self.value.extend(str(data)) # Extend the list with the data
9      return self.value            # Return the current list
10
11 class Server:
12   # Create a forking server at inititalization time and immediately start it.
13   # For each incoming request, the server will spawn another process to handle
14   # that request. The process that started the (main) server can simply kill
15   # it when it's time to do so.
16   def __init__(self):
17     self.server = ForkingServer(DBList, hostname=SERVER, port=PORT)
18
19   def start(self):
20     self.server.start()
21
22 class Client:
23   def run(self):
24     conn = rpyc.connect(SERVER, PORT) # Connect to the server
25     conn.root.exposed_append(2)       # Call an exposed operation,
26     conn.root.exposed_append(4)       # and append two elements
27     print(conn.root.exposed_value())  # Print the result
```

Figure 4.13: Embedding RPCs in a language

The client is also shown in Figure 4.13. When a connection is made to the server, a new instance of `DBList` will be created and the client can immediately append values to the list. The exposed operations are called without further ado.

Note that as the client breaks the connection to the server, the list will be lost. It is a **transient object** and special measures will need to be taken to make it a **persistent object**.

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Language-based support

The approach described up until now is largely independent of a specific programming language. As an alternative, we can also embed remote procedure calling into a language itself. The main benefit is that application development often becomes much simpler. Furthermore, reaching a high degree of access transparency is often simpler, as many issues related to parameter passing can be circumvented altogether.

A well-known example in which remote procedure calling is fully embedded is Java, where an RPC is referred to as a **remote method invocation** (**RMI**). In essence, a client being executed by its own (Java) virtual machine can invoke a method of an object managed by another virtual machine. By simply reading an application's source code, it may be hard or even impossible to see whether a method invocation is to a local or to a remote object.

4.2.4 Variations on RPC

As in conventional procedure calls, when a client calls a remote procedure, the client will block until a reply is returned. This strict request-reply behavior is unnecessary when there is no result to return, or may hinder efficiency when multiple RPCs need to be performed. In the following, we look at two variations on the RPC scheme we have discussed so far.

Asynchronous RPC

To support situations in which there is simply no result to return to the client, RPC systems may provide facilities for what are called **asynchronous RPCs**. With asynchronous RPCs, the server, in principle, immediately sends a reply back to the client the moment the RPC request is received, after which it locally calls the requested procedure. The reply acts as an acknowledgment to the client that the server is going to process the RPC. The client will continue without further blocking as soon as it has received the server's acknowledgment. Figure 4.14(b) shows how client and server interact in the case of asynchronous RPCs. For comparison, Figure 4.14(a) shows the normal request-reply behavior.

Asynchronous RPCs can also be useful when a reply will be returned, but the client is not prepared to wait for it and do nothing in the meantime. A typical case is when a client needs to contact several servers independently. In that case, it can send the call requests one after the other, effectively establishing that the servers operate more or less in parallel. After all call requests have been sent, the client can start waiting for the various results to be returned. In cases such as these, it makes sense to organize the communication between the client and server through an asynchronous RPC combined with a **callback**, as shown in Figure 4.15. In this scheme, also referred to as **deferred synchronous RPC**, the client first calls the server, waits for the acceptance,

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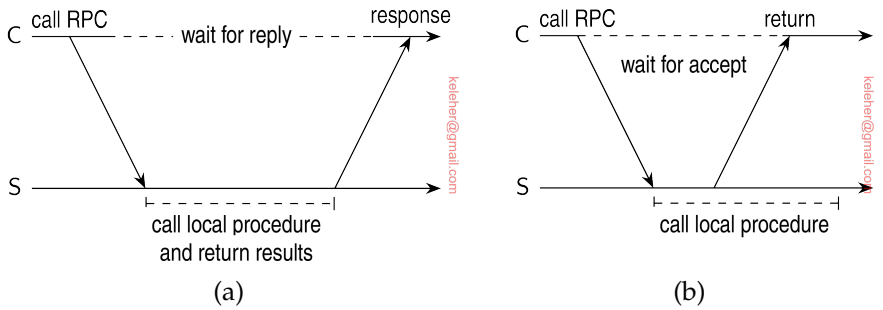


Figure 4.14: (a) The interaction between client and server in a traditional RPC. (b) The interaction using asynchronous RPC.

and continues. When the results become available, the server sends a response message that leads to a callback at the client's side. A callback is a user-defined function that is invoked when a special event happens, such as an incoming message. A straightforward implementation is to spawn a separate thread and let it block on the occurrence of the event while the main process continues. When the event occurs, the thread is unblocked and calls the function.

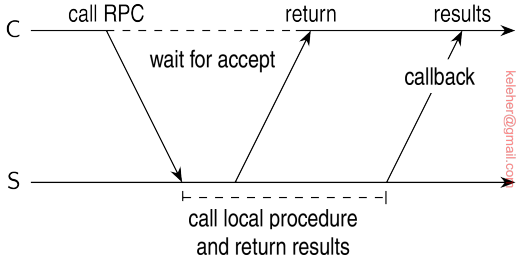


Figure 4.15: A client and server interacting through a deferred synchronous RPC.

It should be noted that variants of asynchronous RPCs exist in which the client continues executing immediately after sending the request to the server. In other words, the client does not wait for an acknowledgment of the server's acceptance of the request. We refer to such RPCs as **one-way RPCs**. The problem with this approach is that when reliability is not guaranteed, the client cannot know for sure whether its request will be processed. We return to these matters in Chapter 8. Likewise, in the case of deferred synchronous RPC, the client may poll the server to see whether the results are available yet, instead of letting the server calling back the client.

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Multicast RPC

Asynchronous and deferred synchronous RPCs facilitate another alternative to remote procedure calls, namely executing multiple RPCs at the same time. Adopting the one-way RPCs (i.e., when a server does not tell the client it has accepted its call request but immediately starts processing it, while the client continues just after issuing the RPC), a **multicast RPC** boils down to sending an RPC request to a group of servers. This principle is shown in Figure 4.16. In this example, the client sends a request to two servers, who subsequently process that request independently and in parallel. When done, the result is returned to the client where a callback takes place.

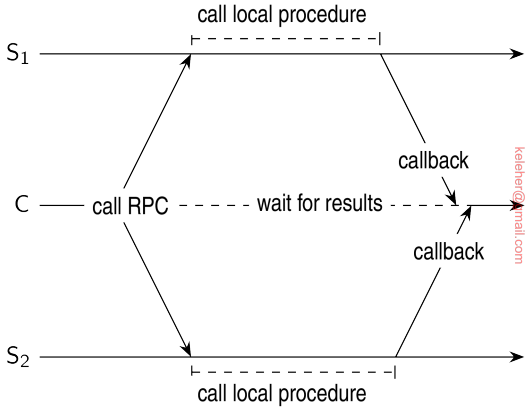


Figure 4.16: The principle of a multicast RPC.

There are several issues that we need to consider. First, as before, the client application may be unaware of the fact that an RPC is actually being forwarded to more than one server. For example, to increase fault tolerance, we may decide to have all operations executed by a backup server who can take over when the main server fails. That a server has been replicated can be completely hidden from a client application by an appropriate stub. Yet even the stub need not be aware that the server is replicated, for example because we are using a transport-level multicast address.

Second, we need to consider what to do with the responses. In particular, will the client proceed after all responses have been received, or wait just for one? It all depends. When the server has been replicated for fault tolerance, we may decide to wait for just the first response, or perhaps until a majority of the servers returns the same result. On the other hand, if the servers have been replicated to do the same work but on different parts of the input, their results may need to be merged before the client can continue. Again, such matters can be hidden in the client-side stub, yet the application developer will, at the very least, have to specify the purpose of the multicast RPC.

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4.3 Message-oriented communication

Remote procedure calls and remote object invocations contribute to hiding communication in distributed systems, that is, they enhance access transparency. Unfortunately, neither mechanism is always appropriate. In particular, when it cannot be assumed that the receiving side is executing at the time a request is issued, alternative communication services are needed. Likewise, the inherent synchronous nature of RPCs, by which a client is blocked until its request has been processed, may need to be replaced by something else.

That something else is messaging. In this section, we concentrate on message-oriented communication in distributed systems by first taking a closer look at what exactly synchronous behavior is and what its implications are. Then, we discuss messaging systems that assume that parties are executing at the time of communication. Finally, we will examine message-queuing systems that allow processes to exchange information, even if the other party is not executing at the time communication is initiated.

4.3.1 Simple transient messaging with sockets

Many distributed systems and applications are built directly on top of the simple message-oriented model offered by the transport layer. To better understand and appreciate the message-oriented systems as part of middleware solutions, we first discuss messaging through transport-level sockets.

Special attention has been paid to standardizing the interface of the transport layer to allow programmers to make use of its entire suite of (messaging) protocols through a simple set of operations. Furthermore, standard interfaces make it easier to port an application to a different machine. As an example, we briefly discuss the **socket interface** as introduced in the 1970s in Berkeley Unix, and which has been adopted as a POSIX standard (with only very few adaptations).

Conceptually, a **socket** is a communication end point to which an application can write data that are to be sent out over the underlying network, and from which incoming data can be read. A socket forms an abstraction over the actual port that is used by the local operating system for a specific transport protocol. In the following text, we concentrate on the socket operations for TCP, which are shown in Figure 4.17.

Servers generally execute the first four operations, normally in the order given. When calling the `socket` operation, the caller creates a new communication end point for a specific transport protocol. Internally, creating a communication end point means that the local operating system reserves resources for sending and receiving messages for the specified protocol.

The `bind` operation associates a local address with the newly created socket. For example, a server should bind the IP address of its machine together with a (possibly well-known) port number to a socket. Binding tells the operating

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Operation	Description
socket	Create a new communication end point
bind	Attach a local address to a socket
listen	Tell operating system what the maximum number of pending connection requests should be
accept	Block caller until a connection request arrives
connect	Actively attempt to establish a connection
send	Send some data over the connection
receive	Receive some data over the connection
close	Release the connection

Figure 4.17: The socket operations for TCP/IP.

system that the server wants to receive messages only on the specified address and port. In the case of connection-oriented communication, the address is used to receive incoming connection requests.

The `listen` operation is called only in the case of connection-oriented communication. It is a nonblocking call that allows the local operating system to reserve enough buffers for a specified maximum number of pending connection requests that the caller is willing to accept.

A call to `accept` blocks the caller until a connection request arrives. When a request arrives, the local operating system creates a new socket with the same properties as the original one, and returns it to the caller. This approach will allow the server to, for example, fork a process that will subsequently handle the actual communication through the new connection. The server can go back and wait for another connection request on the original socket.

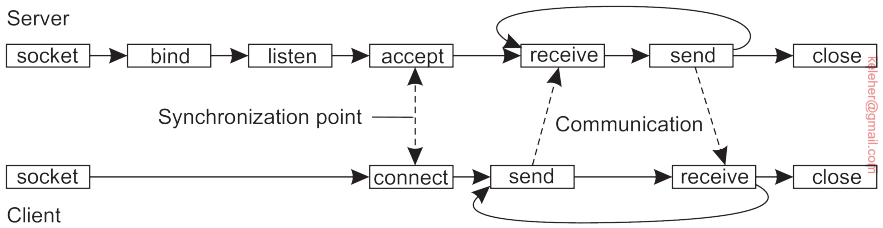


Figure 4.18: Connection-oriented communication pattern using sockets.

Let us now take a look at the client side. Here, too, a socket must first be created using the `socket` operation, but explicitly binding the socket to a local address is not necessary, since the operating system can dynamically allocate a port when the connection is set up. The `connect` operation requires that the caller specifies the transport-level address to which a connection request is to be sent. The client is blocked until a connection has been set up successfully, after which both sides can start exchanging information through the `send` and `receive` operations. Finally, closing a connection is symmetric when using

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sockets, and is established by having both the client and server call the `close` operation. Although there are many exceptions to the rule, the general pattern followed by a client and server for connection-oriented communication using sockets is as shown in Figure 4.18. Details on network programming using sockets and other interfaces in Unix can be found in [Stevens, 1998].

Note 4.7 (Example: A simple socket-based client-server system)
As an illustration of the recurring pattern in Figure 4.18, consider the simple socket-based client-server system shown in Figure 4.19 (see also Note 2.1). We see the server starting by creating a socket, and subsequently binding an address to that socket. It calls the `listen` operation, and waits for an incoming connection request. When the server accepts a connection, the socket library creates a separate connection, `conn`, which is used to receive data and send a response to the connected client. The server enters a loop receiving and sending messages, until no more data has been received. It then closes the connection.

```
1  from socket  import *
2
3  class Server:
4    def run(self):
5      s = socket(AF_INET, SOCK_STREAM)
6      s.bind((HOST, PORT))
7      s.listen(1)
8      (conn, addr) = s.accept()  # returns new socket and addr. client
9      while True:                # forever
10       data = conn.recv(1024)   # receive data from client
11       if not data: break       # stop if client stopped
12       conn.send(data+b"*")     # return sent data plus an "*"
13     conn.close()               # close the connection
```

(a)

```
1  class Client:
2    def run(self):
3      s = socket(AF_INET, SOCK_STREAM)
4      s.connect((HOST, PORT)) # connect to server (block until accepted)
5      s.send(b"Hello, world") # send same data
6      data = s.recv(1024)     # receive the response
7      print(data)             # print what you received
8      s.send(b"")             # tell the server to close
9      s.close()               # close the connection
```

(b)

Figure 4.19: A simple socket-based client-server system.

The client again follows the pattern from Figure 4.18. It creates a socket, and calls `connect` to request a connection with the server. Once the connection has been established, it sends a single message, waits for the response, and after printing the result, closes the connection.

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Note 4.8 (Advanced: Implementing stubs as global references revisited)
To provide a more in-depth insight in the working of sockets, let us look at a more elaborate example, namely the use of stubs as global references.

```
class DBClient:
  def __sendrecv(self, message):           # this is a private method
    sock = socket()                        # create a socket
    sock.connect((self.host, self.port))   # connect to server
    sock.send(pickle.dumps(message))       # send some data
    result = pickle.loads(sock.recv(1024)) # receive the response
    sock.close()                           # close the connection
    return result

  def create(self):
    self.listID = self.__sendrecv([CREATE])
    return self.listID

  def getValue(self):
    return self.__sendrecv([GETVALUE, self.listID])

  def appendData(self, data):
    return self.__sendrecv([APPEND, data, self.listID])
```

(a)

```
class Server:
    self.setOfLists = {}                 # init: no lists to manage

  def run(self):
    while True:
      (conn, addr) = self.sock.accept() # accept incoming call
      data = conn.recv(1024)            # fetch data from client
      request = pickle.loads(data)      # unwrap the request

      if request[0] == CREATE:               # create a list
        listID = len(self.setOfLists) + 1    # allocate listID
        self.setOfLists[listID] = []         # initialize to empty
        conn.send(pickle.dumps(listID))      # return ID

      elif request[0] == APPEND:             # append request
        listID = request[2]                  # fetch listID
        data   = request[1]                  # fetch data to append
        self.setOfLists[listID].append(data) # append it to the list
        conn.send(pickle.dumps(OK))          # return an OK

      elif request[0] == GETVALUE:           # read request
        listID = request[1]                  # fetch listID
        result = self.setOfLists[listID]     # get the elements
        conn.send(pickle.dumps(result))      # return the list
```

(b)

Figure 4.20: Implementing a list server in Python.

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We return to our example of implementing a shared list, which we now do using a list server, implemented in the form of the Python class shown in Figure 4.20(b). Figure 4.20(a) shows the stub implementation of a shared list. Again, we have omitted code for readability.

The `DBClient` class represents a client-side stub that, once marshaled, can be passed between processes. It provides three operations associated with a list: `create`, `getValue`, and `append`, with obvious semantics. A `DBClient` is assumed to be associated with one specific list as managed by the server. An identifier for that list is returned when the list is created. Note how the (internal) `sendrecv` operation follows the client-side pattern explained in Figure 4.18.

The server maintains lists, as shown in Figure 4.20(b). Its internal data structure is a `setOfLists` with each element being a previously created list. The server simply waits for incoming requests, unmarshals the request, and checks which operation is being requested. Results are sent back to the requesting client (which always issues the `sendrecv` operation implemented as part of `DBClient`). Again, we see that the server follows the pattern shown in Figure 4.18: it creates a socket, binds an address to it, informs the operating system to how many connections it should listen, and then waits to accept an incoming connection request. Once a connection has been established, the server receives data, sends a response, and closes the connection again.

```
1  class Client:
2    def __init__(self, port):
3      self.host = 'localhost'               # this machine
4      self.port = port                      # port it will listen to
5      self.sock = socket()                  # socket for incoming calls
6      self.sock.bind((self.host, self.port)) # bind socket to an address
7      self.sock.listen(2)                   # max num connections
8
9    def sendTo(self, host, port, data):
10     sock = socket()
11     sock.connect((host, port))    # connect to server (blocking call)
12     sock.send(pickle.dumps(data)) # send some data
13     sock.close()
14
15   def recvAny(self):
16     (conn, addr) = self.sock.accept()
17     return conn.recv(1024)
```

Figure 4.20: (c) Implementing a list server in Python: the client.

To use a stub as a global reference, we represent each client application by the class `Client` shown in Figure 4.20(c). The class is instantiated in the same process running the application (exemplified by the value of `self.host`), and will be listening on a specific port for messages from other applications, as well as the server. Otherwise, it merely sends and receives messages, coded through the operations `sendTo` and `recvAny`, respectively.

Now consider the code shown in Figure 4.20(d), which mimics two client applications. The first one creates a new list and appends data to it. Then note

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how dbClient1 is simply sent to the other client. Under the hood, we now know that it is marshaled in the operation sendTo (line 12) of class Client shown in Figure 4.20(c).

```
1  def client1():
2    c1   = Client(PORTC1)              # create client
3    dbC1 = DBClient(HOSTS,PORTS)       # create reference
4    dbC1.create()                      # create new list
5    dbC1.appendData('Client 1')        # append some data
6    c1.sendTo(HOSTC2,PORTC2,dbC1)      # send to other client
7
8  def client2():
9    c2   = Client(PORTC2)              # create a new client
10   data = c2.recvAny()                # block until data is sent
11   dbC2 = pickle.loads(data)          # receive reference
12   dbC2.appendData('Client 2')        # append data to same list
```

Figure 4.20: (d) Passing stubs as references.

The second client simply waits for an incoming message (line 12), unmarshals the result, knowing that it is a DBClient instance, and subsequently appends some more data to the same list as the one the first client appended data. Indeed, an instance of DBClient is seen to be passed as a global reference, seemingly along with all the operations that go with the associated class.

4.3.2 Advanced transient messaging

The standard socket-based approach toward transient messaging is very basic and, as such, rather brittle: a mistake is easily made. Furthermore, sockets essentially support only TCP or UDP, meaning that any extra facility for messaging needs to be implemented separately by an application programmer. In practice, we do often need more advanced approaches for message-oriented communication to make network programming easier, to expand beyond the functionality offered by existing networking protocols, to make better use of local resources, and so on.

Using messaging patterns: ZeroMQ

One approach toward making network programming easier is based on the observation that many messaging applications, or their components, can be effectively organized according to a few simple communication patterns. By subsequently providing enhancements to sockets for each of these patterns, it may become easier to develop a networked, distributed application. This approach has been followed in ZeroMQ and documented in [Hintjens, 2013; Akgul, 2013].

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Like in the Berkeley approach, ZeroMQ also provides sockets through which all communication takes place. Actual message transmission generally takes place over TCP connections, and like TCP, all communication is essentially connection-oriented, meaning that a connection will first be set up between a sender and receiver before message transmission can take place. However, setting up, and maintaining connections is kept mostly under the hood: an application programmer need not bother with those issues. To further simplify matters, a socket may be bound to multiple addresses, effectively allowing a server to handle messages from very different sources through a single interface. For example, a server can listen to multiple ports using a single blocking receive operation. ZeroMQ sockets can thus support *many-to-one* communication instead of just *one-to-one* communication, as is the case with standard Berkeley sockets. To complete the story: ZeroMQ sockets also support *one-to-many* communication, i.e., multicasting.

Essential to ZeroMQ is that communication is **asynchronous**: a sender will normally continue after having submitted a message to the underlying communication subsystem. An interesting side effect of combining asynchronous with connection-oriented communication, is that a process can request a connection setup, and subsequently send a message even if the recipient is not yet up-and-running and ready to accept incoming connection requests, let alone incoming messages. What happens, of course, is that a connection request and subsequent messages are queued at the sender's side, while a separate thread as part of ZeroMQ's library will take care that eventually the connection is set up and messages are transmitted to the recipient.

Simplifying matters, ZeroMQ establishes a higher level of abstraction in socket-based communication by *pairing* sockets: a specific type of socket used for sending messages is paired with a corresponding socket type for receiving messages. Each pair of socket types corresponds to a **communication pattern**. The three most important communication patterns supported by ZeroMQ are *request-reply*, *publish-subscribe*, and *pipeline*.

The **request-reply pattern** is used in traditional client-server communication, like the ones normally used for remote procedure calls. A client application uses a *request socket* (of type `REQ`) to send a request message to a server and expects the latter to respond with an appropriate response. The server is assumed to use a *reply socket* (of type `REP`). The request-reply pattern simplifies matters for developers by avoiding the need to call the `listen` operation, as well as the `accept` operation. Moreover, when a server receives a message, a subsequent call to `send` is automatically targeted toward the original sender. Likewise, when a client calls the `recv` operation (for receiving a message) after having sent a message, ZeroMQ assumes the client is waiting for a response from the original recipient. Note that this approach was effectively encoded in the local `sendrecv` operation of Figure 4.20(b), which we discussed in Note 4.8.

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Note 4.9 (Example: The request-reply pattern)
Let us look at a simple programming example to illustrate the request-reply pattern. Figure 4.21 shows a server that appends an asterisk to a received message. As before, it creates a socket, and binds it to a combination of a protocol (in this case TCP), and a host and port. In our example, the server is willing to accept incoming connection requests on two different ports. It then waits for incoming messages. The request-reply pattern effectively ties the receipt of a message to the subsequent response. In other words, when the server calls `send`, it will transmit a message to the same client from which it previously had received a message. Of course, this simplicity can be achieved only if the programmer indeed abides to the request-reply pattern.

```
1  import zmq
2
3  def server():
4    context = zmq.Context()
5    socket  = context.socket(zmq.REP)       # create reply socket
6    socket.bind("tcp://*:12345")            # bind socket to address
7
8    while True:
9      message = socket.recv()               # wait for incoming message
10     if not "STOP" in str(message):        # if not to stop...
11       reply = str(message.decode())+'*'   # append "*" to message
12       socket.send(reply.encode())         # send it away (encoded)
13     else:
14       break                               # break out of loop and end
15
16 def client():
17   context = zmq.Context()
18   socket  = context.socket(zmq.REQ)       # create request socket
19
20   socket.connect("tcp://localhost:12345") # block until connected
21   socket.send(b"Hello world")             # send message
22   message = socket.recv()                 # block until response
23   socket.send(b"STOP")                    # tell server to stop
24   print(message.decode())                 # print result
```

Figure 4.21: A ZeroMQ client-server system.

The client, also shown in Figure 4.21, creates a socket and connects to the associated server. When it sends a message, it can expect to receive, from that same server, a response. By sending the string "`STOP`", it tells the server it is done, after which the server will actually stop.

Interestingly, the asynchronous nature of ZeroMQ allows one to start the client *before* starting the server. An implication is that if, in this example, we would start the server, then a client, and after a while a second client, that the latter will be blocked until the server is restarted. Furthermore, note that ZeroMQ does not require the programmer to specify how many bytes are expected to be received. Unlike TCP, ZeroMQ uses messages instead of byte streams.

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In the case of a **publish-subscribe pattern**, clients *subscribe* to specific messages that are *published* by servers. We came across this pattern in Section 2.1.3 when discussing coordination. In effect, only the messages to which the client has subscribed will be transmitted. If a server is publishing messages to which no one has subscribed, these messages will be lost. In its simplest form, this pattern establishes multicasting messages from a server to several clients. The server is assumed to use a socket of type PUB, while each client must use SUB type sockets. Each client socket is connected to the socket of the server. By default, a client subscribes to no specific message. This means that as long as no explicit subscription is provided, a client will not receive a message published by the server.

Note 4.10 (Example: The publish-subscribe pattern)
Again, let us make this pattern more concrete through a simple example. Figure 4.22 shows an admittedly naive time server that publishes its current, local time, through a PUB socket. The local time is published every five seconds, for any interested client.

```
1  import multiprocessing
2  import zmq, time
3
4  def server():
5    context = zmq.Context()
6    socket = context.socket(zmq.PUB)          # create a publisher socket
7    socket.bind("tcp://*:12345")              # bind socket to the address
8    while True:
9      time.sleep(5)                           # wait every 5 seconds
10     t = "TIME " + time.asctime()
11     socket.send(t.encode())                 # publish the current time
12
13 def client():
14   context = zmq.Context()
15   socket = context.socket(zmq.SUB)          # create a subscriber socket
16   socket.connect("tcp://localhost:12345")   # connect to the server
17   socket.setsockopt(zmq.SUBSCRIBE, b"TIME") # subscribe to TIME messages
18
19   for i in range(5):      # Five iterations
20     time = socket.recv()  # receive a message related to subscription
21     print(time.decode())  # print the result
```

Figure 4.22: A multicasting socket-based setup.

A client is equally simple, as also shown in Figure 4.22. It first creates a SUB socket which it connects to the corresponding PUB socket of the server. To receive the appropriate messages, it needs to subscribe to messages that have TIME as their tag. In our example, a client will simply print the first five messages received from the server. Note that we can have as many clients as we want: the server's message will be multicasted to all subscribers. Most important in this example, is

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that it illustrates that the *only* messages received through the SUB socket, are the ones the client had subscribed to.

Finally, the **pipeline pattern** is characterized by the fact that a process wants to *push out* its results, assuming that there are other processes that want to *pull in* those results. The essence of the pipeline pattern is that a pushing process does not really care which other process pulls in its results: the first available one will do just fine. Likewise, any process pulling in results from multiple other processes will do so from the first pushing process making its results available. The intention of the pipeline pattern is thus seen to keep as many processes working as possible, pushing results through a pipeline of processes as quickly as possible.

Note 4.11 (Example: The pipeline pattern)
As our last example, consider the following template for keeping a collection of worker tasks busy. Figure 4.23 shows the code for a so-called **farmer task**: a process producing tasks to be picked up by others. In this example, we simulate the task by letting the producer pick a random number modeling the duration, or load, of the work to be done. This workload is then sent to the PUSH socket, effectively being queued until another process picks it up.

```
1  def producer():
2    context = zmq.Context()
3    socket  = context.socket(zmq.PUSH)      # create a push socket
4    socket.bind("tcp://127.0.0.1:12345")    # bind socket to address
5
6    while True:
7      workload = random.randint(1, 100)     # compute workload
8      socket.send(pickle.dumps(workload))   # send workload to worker
9      time.sleep(workload/NWORKERS)         # balance production by waiting
10
11 def worker(id):
12   context = zmq.Context()
13   socket  = context.socket(zmq.PULL)      # create a pull socket
14   socket.connect("tcp://localhost:12345") # connect to the producer
15
16   while True:
17     work = pickle.loads(socket.recv())     # receive work from a source
18     time.sleep(work)                       # pretend to work
```

Figure 4.23: A producer-worker pattern.

Such other processes are known as **worker tasks**, of which a sketch is also given in Figure 4.23. A worker task connects to a single PULL socket. Once it picks up some work, it simulates that it is actually doing something by sleeping for some time proportional to the received workload.

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The semantics of this push-pull pattern is such that the first available worker will pick up work from the producer, and likewise, if there are multiple workers ready to pick up work, each one of them will be provided with a task. How work distribution is actually done fairly requires some specific attention, which we will not discuss further here.

The Message-Passing Interface (MPI)

With the advent of high-performance multicomputers, developers have been looking for message-oriented operations that would allow them to easily write highly efficient applications. This means that the operations should be at a convenient level of abstraction (to ease application development), and that their implementation incurs only minimal overhead. Sockets were deemed insufficient for two reasons. First, they were at the wrong level of abstraction by supporting only simple send and receive operations. Second, sockets had been designed to communicate across networks using general-purpose protocol stacks such as TCP/IP. They were not considered suitable for the proprietary protocols developed for high-speed interconnection networks, such as those used in high-performance server clusters. Those protocols required an interface that could handle more advanced features, such as different forms of buffering and synchronization.

The result was that most interconnection networks and high-performance multicomputers were shipped with proprietary communication libraries. These libraries offered a wealth of high-level and generally efficient communication operations. Of course, all libraries were mutually incompatible, so that application developers now had a portability problem.

The need to be hardware and platform independent eventually lead to the definition of a standard for message passing, simply called the **Message-Passing Interface** or **MPI**. MPI is designed for parallel applications and as such is tailored to transient communication. It makes direct use of the underlying network. Furthermore, it assumes that serious failures such as process crashes or network partitions are fatal and do not require automatic recovery.

MPI assumes communication takes place within a known group of processes. Each group is assigned an identifier. Each process within a group is also assigned a (local) identifier. A (groupID, processID) pair therefore uniquely identifies the source or destination of a message, and is used instead of a transport-level address. There may be several, possibly overlapping, groups of processes involved in a computation and that are all executing at the same time.

At the core of MPI are messaging operations to support transient communication, of which the most intuitive ones are summarized in Figure 4.25. To understand their semantics, it helps to keep Figure 4.4 in mind, with the

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middleware formed entirely by an MPI implementation. In particular, the middleware maintains its own buffers and synchronization is often related to when the necessary data has been copied from the calling application to the middleware.

Operation	Description
MPI_BSEND	Append outgoing message to a local send buffer
MPI_SEND	Send a message and wait until copied to local or remote buffer
MPI_SSEND	Send a message and wait until transmission starts
MPI_SENDRECV	Send a message and wait for reply
MPI_ISEND	Pass reference to outgoing message, and continue
MPI_ISSEND	Pass reference to outgoing message, and wait until receipt starts
MPI_RECV	Receive a message; block if there is none
MPI_IRECV	Check if there is an incoming message, but do not block

Figure 4.24: Some of the more intuitive message-passing operations of MPI.

Transient asynchronous communication is supported by the MPI_BSEND operation. The sender submits a message for transmission, which is generally first copied to a local buffer in the MPI runtime system. When the message has been copied, the sender continues. The local MPI runtime system will remove the message from its local buffer and take care of transmission as soon as a receiver has called a receive operation.

There is also a blocking send operation, called MPI_SEND, of which the semantics are implementation dependent. The operation MPI_SEND may either block the caller until the specified message has been copied to the MPI runtime system at the sender's side, or until the receiver has initiated a receive operation. Synchronous communication by which the sender blocks until its request is accepted for further processing is available through the MPI_SSEND operation. Finally, the strongest form of synchronous communication is also supported: when a sender calls MPI_SENDRECV, it sends a request to the receiver and blocks until the latter returns a reply. Basically, this operation corresponds to a normal RPC.

Both MPI_SEND and MPI_SSEND have variants that avoid copying messages from user buffers to buffers internal to the local MPI runtime system. These variants essentially correspond to a form of asynchronous communication. With MPI_ISEND, a sender passes a pointer to the message, after which the MPI runtime system takes care of communication. The sender immediately continues. To prevent overwriting the message before communication completes, MPI offers operations to check for completion, or even to block if required. As with MPI_SEND, whether the message has actually been transferred to the receiver or that it has merely been copied by the local MPI runtime system to an internal buffer is left unspecified.

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Likewise, with MPI_ISSEND, a sender also passes only a pointer to the MPI runtime system. When the runtime system indicates it has processed the message, the sender is then guaranteed that the receiver has accepted the message and is now working on it.

The operation MPI_RECV is called to receive a message; it blocks the caller until a message arrives. There is also an asynchronous variant, called MPI_IRECV, by which a receiver indicates that it is prepared to accept a message. The receiver can check whether a message has indeed arrived, or block until one does.

The semantics of MPI communication operations are not always straightforward, and different operations can sometimes be interchanged without affecting the correctness of a program. The official reason why so many different forms of communication are supported is that it gives implementers of MPI systems enough possibilities for optimizing performance. Cynics might say the committee could not make up its collective mind, so it threw in everything. By now, MPI is in its fourth version with over 650 operations available. Being designed for high-performance parallel applications, it is perhaps easier to understand its diversity. More on MPI can be found in [Gropp et al., 2016]. The complete MPI-4 reference can be found in [Message Passing Interface Forum, 2021].

4.3.3 Message-oriented persistent communication

We now come to an important class of message-oriented middleware services, generally known as **message-queuing systems**, or just **message-oriented middleware** (**MOM**). Message-queuing systems provide extensive support for persistent asynchronous communication. The essence of these systems is that they offer intermediate-term storage capacity for messages, without requiring either the sender or receiver to be active during message transmission. An important difference with sockets and MPI is that message-queuing systems are typically targeted to support message transfers that are allowed to take minutes instead of seconds or milliseconds.

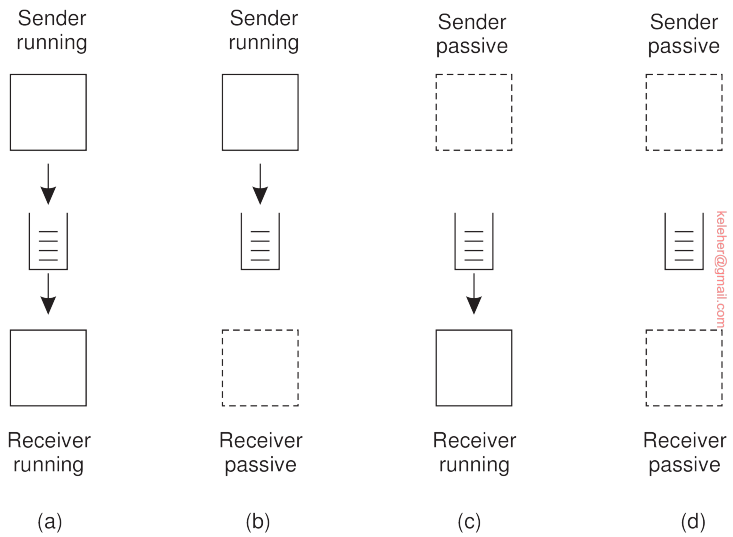
Message-queuing model

The basic idea behind a message-queuing system is that applications communicate by inserting messages in specific queues. These messages are forwarded over a series of communication servers and are eventually delivered to the destination, even if it was down when the message was sent. In practice, most communication servers are directly connected to each other. In other words, a message is generally transferred directly to a destination server. In principle, each application has its own private queue to which other applications can send messages. A queue can be read only by its associated application, but it is also possible for multiple applications to share a single queue.

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An important aspect of message-queuing systems is that a sender is generally given only the guarantees that its message will eventually be inserted in the recipient's queue. No guarantees are given about when, or even if the message will actually be read, which is completely determined by the behavior of the recipient.

These semantics permit communication to be decoupled in time, as also discussed in Section 2.1.3. There is thus no need for the receiver to be executing when a message is being sent to its queue. Likewise, there is no need for the sender to be executing the moment its message is picked up by the receiver. The sender and receiver can execute completely independently of each other. In fact, once a message has been deposited in a queue, it will remain there until it is removed, irrespective of whether its sender or receiver is executing. This gives us four combinations regarding the execution mode of the sender and receiver, as shown in Figure 4.25.



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Figure 4.25: Four combinations for loosely coupled communication using queues.

In Figure 4.25(a), both the sender and receiver are in execution during the entire transmission of a message. In Figure 4.25(b), only the sender is in execution, while the receiver is passive, that is, in a state in which message delivery is not possible. Nevertheless, the sender can still send messages. The combination of a passive sender and an active receiver is shown in Figure 4.25(c). In this case, the receiver can read messages that were sent to it, but it is not necessary that their respective senders are in execution as well. Finally, in Figure 4.25(d), we see the situation that the system is storing (and possibly transmitting) messages even while sender and receiver

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are passive. One may argue that only if this last configuration is supported, the message-queuing system truly provides persistent messaging.

Messages can, in principle, contain any data. The only important aspect from the perspective of middleware is that messages are properly addressed. In practice, addressing is done by providing a systemwide unique name of the destination queue. In some cases, message size may be limited, although it is also possible that the underlying system takes care of fragmenting and assembling large messages in a way that is completely transparent to applications. An effect of this approach is that the basic interface offered to applications can be simple, as shown in Figure 4.26.

Operation	Description
PUT	Append a message to a specified queue
GET	Block until the specified queue is nonempty, and remove the first message
POLL	Check a specified queue for messages, and remove the first. Never block
NOTIFY	Install a handler to be called when a message is put into the specified queue

Figure 4.26: Basic interface to a queue in a message-queuing system.

The PUT operation is called by a sender to pass a message to the underlying system that is to be appended to the specified queue. As we explained, this is a nonblocking call. The GET operation is a blocking call by which an authorized process can remove the longest pending message in the specified queue. The process is blocked only if the queue is empty. Variations on this call allow searching for a specific message in the queue, for example, using a priority, or a matching pattern. The nonblocking variant is given by the POLL operation. If the queue is empty, or if a specific message could not be found, the calling process simply continues.

Finally, most queuing systems also allow a process to install a handler as a *callback function*, which is automatically invoked whenever a message is put into the queue. Arranging this scheme is handled through a NOTIFY operation. Callbacks can also be used to automatically start a process that will fetch messages from the queue if no process is currently in execution. This approach is often implemented through a daemon on the receiver's side that continuously monitors the queue for incoming messages and handles accordingly.

General architecture of a message-queuing system

Let us now take a closer look at what a general message-queuing system looks like. First, queues are managed by **queue managers**. A queue manager is either a separate process, or is implemented through a library that is linked

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with an application. Secondly, as a rule of thumb, an application can put messages only into a *local* queue. Likewise, getting a message is possible by extracting it from a *local* queue only. As a consequence, if a queue manager QM_A handling the queues for an application A runs as a separate process, both processes QM_A and A will generally be placed on the same machine, or at worst on the same LAN. Also note that if *all* queue managers are linked into their respective applications, we can no longer speak of a persistent asynchronous messaging system.

If applications can put messages only into local queues, then clearly each message will have to carry information concerning its destination. It is the queue manager's task to make sure that a message reaches its destination. This brings us to several issues.

In the first place, we need to consider how the destination queue is addressed. Obviously, to enhance location transparency, it is preferable that queues have logical, location-independent names. Assuming that a queue manager is implemented as a separate process, using logical names implies that each name should be associated with a **contact address**, such as a (host,port)-pair, and that the name-to-address mapping is readily available to a queue manager, as shown in Figure 4.27. In practice, a contact address carries more information, notably the protocol to be used, such as TCP or UDP. We came across such contact addresses in our examples of advanced sockets in, for example, Note 4.9.

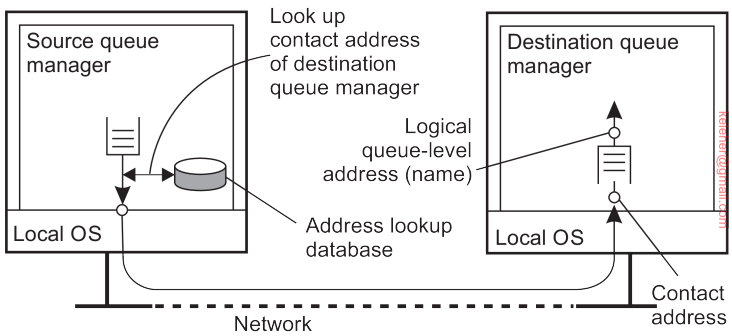


Figure 4.27: The relationship between queue-level naming and network-level addressing.

A second issue that we need to consider is how the name-to-address mapping is actually made available to a queue manager. A common approach is to simply implement the mapping as a lookup table and copy that table to all managers. Obviously, this leads to a maintenance problem, for every time that a new queue is added or named, many, if not all tables, need to be updated. There are various ways to alleviate such problems, which we will discuss in Chapter 6.

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This brings us to a third issue, related to the problems of efficiently maintaining name-to-address mappings. We have implicitly assumed that if a destination queue at manager QM_B is known to queue manager QM_A, then QM_A can directly contact QM_B to transfer messages. In effect, this means that (the contact address of) each queue manager should be known to all others. Obviously, when dealing with large message-queuing systems, we will have a scalability problem. In practice, there are often special queue managers that operate as **routers**: they forward incoming messages to other queue managers. In this way, a message-queuing system may gradually grow into a complete, application-level, **overlay network**.

If only a few routers need to know about the network topology, then a source queue manager need only to know to which adjacent router, say R, it should forward a message, given a destination queue. The router R, in turn, may need to keep track only of its adjacent routers to see where to forward the message to, and so on. Of course, we still need to have name-to-address mappings for all queue managers, including the routers, but it is not difficult to imagine that such tables can be much smaller and easier to maintain.

Message brokers

An important application area of message-queuing systems is integrating existing and new applications into a single, coherent distributed information system. If we assume that communication with an application takes place through messages, then integration requires that applications can understand the messages they receive. In practice, this requires the sender to have its outgoing messages in the same format as that of the receiver, but also that its messages adhere to the same semantics as those expected by the receiver. Sender and receiver essentially need to speak the same language, that is, adhere to the same messaging protocol.

The problem with this approach is that each time an application A is added to the system having its own messaging protocol, then for each other application B that is to communicate with A we will need to provide the means for converting their respective messages. In a system with N applications, we will thus need $N \times N$ messaging-protocol converters.

An alternative is to agree on a common messaging protocol, as is done with traditional network protocols. Unfortunately, this approach will generally not work for message-queuing systems. The problem is the level of abstraction at which these systems operate. A common messaging protocol makes sense only if the collection of processes that make use of that protocol indeed have enough in common. If the collection of applications that make up a distributed information system is highly diverse (which it often is), then inventing a one-size-fits-all solution is simply not going to work.

If we focus only on the format and meaning of messages, commonality can be achieved by lifting the level of abstraction, as is done with XML messages.

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In this case, messages carry information on their own organization, and what has been standardized is the way that they can describe their content. As a consequence, an application can provide information on the organization of its messages that can be automatically processed. Of course, this information is generally not enough: we also need to make sure that the *semantics* of messages are well understood.

Given these problems, the general approach is to learn to live with differences, and try to provide the means to make conversions as simple as possible. In message-queuing systems, conversions are handled by special nodes in a queuing network, known as **message brokers**. A message broker acts as an application-level gateway in a message-queuing system. Its main purpose is to convert incoming messages so that they can be understood by the destination application. Note that to a message-queuing system, a message broker is just another application, as shown in Figure 4.28. In other words, a message broker is generally not considered to be an integral part of the queuing system.

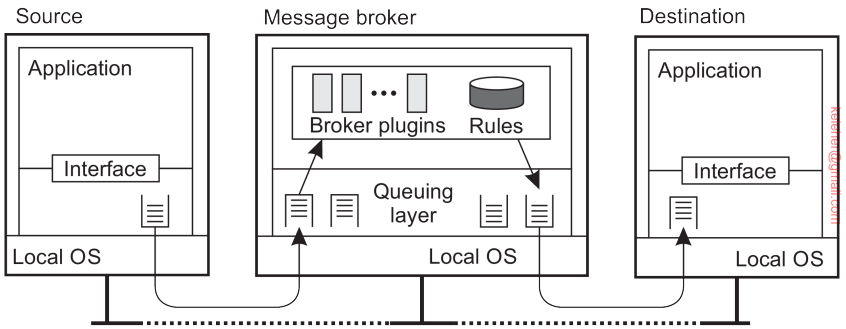


Figure 4.28: The general organization of a message broker in a message-queuing system.

A message broker can be as simple as a reformatter for messages. For example, assume an incoming message contains a table from a database in which records are separated by a special *end-of-record* delimiter and fields within a record have a known, fixed length. If the destination application expects a different delimiter between records, and also expects that fields have variable lengths, a message broker can be used to convert messages to the format expected by the destination.

In a more advanced setting, a message broker may act as an application-level gateway, in which information on the messaging protocol of several applications has been encoded. In general, for each pair of applications, we will have a separate subprogram capable of converting messages between the two applications. In Figure 4.28, this subprogram is drawn as a **plugin** to emphasize that such parts can be dynamically plugged in, or removed from a broker.

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Finally, note that often a message broker is used for advanced **enterprise application integration** (**EAI**), as we discussed in Section 1.3.2. In this case, rather than (only) converting messages, a broker is responsible for *matching* applications based on the messages that are being exchanged. In such a **publish-subscribe model**, applications send messages in the form of *publishing*. In particular, they may publish a message on topic X, which is then sent to the broker. Applications that have stated their interest in messages on topic X, that is, who have *subscribed* to those messages, will then receive these messages from the broker. More advanced forms of mediation are also possible.

At the heart of a message broker lies a repository of rules for transforming a message of one type to another. The problem is defining the rules and developing the plugins. Most message-broker products come with sophisticated development tools, but the bottom line is still that the repository needs to be filled by experts. Here we see a perfect example where commercial products are often misleadingly said to provide "intelligence," where, in fact, the intelligence is to be found only in the heads of those experts. It may be better to consistently talk about "advanced" systems (but even then, we often see that this stands for complex or complicated).

Note 4.12 (More information: A note on message-queuing systems)
Considering what we have said about message-queuing systems, it would appear that they have long existed in the form of implementations for e-mail services. E-mail systems are generally implemented through a collection of mail servers that store and forward messages on behalf of the users on hosts directly connected to the server. Routing is generally left out, as e-mail systems can make direct use of the underlying transport services. For example, in the mail protocol for the Internet, SMTP [Postel, 1982], a message is transferred by setting up a direct TCP connection to the destination mail server.

What makes e-mail systems special compared to message-queuing systems is that they are primarily aimed at providing direct support for end users. This explains, for example, why several groupware applications are based directly on an e-mail system [Khoshafian and Buckiewicz, 1995]. In addition, e-mail systems may have very specific requirements such as automatic message filtering, support for advanced messaging databases (e.g., to easily retrieve previously stored messages), and so on.

General message-queuing systems are not aimed at supporting only end users. An important issue is that they are set up to enable persistent communication between processes, regardless of whether a process is running a user application, handling access to a database, performing computations, and so on. This approach leads to a different set of requirements for message-queuing systems than pure e-mail systems. For example, e-mail systems generally need not provide guaranteed message delivery, message priorities, logging facilities, efficient multicasting, load balancing, fault tolerance, and so on for general usage.

General-purpose message-queuing systems, therefore, have a wide range

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of applications, including e-mail, workflow, groupware, and batch processing. However, as we have stated before, the most important application area is the integration of a (possibly widely dispersed) collection of databases and applications into a federated information system [Hohpe and Woolf, 2004]. For example, a query expanding several databases may need to be split into subqueries that are forwarded to individual databases. Message-queuing systems assist by providing the basic means to package each subquery into a message and routing it to the appropriate database. Other communication facilities we have discussed in this chapter are far less appropriate.

4.3.4 Example: Advanced Message Queuing Protocol (AMQP)

An interesting observation about message-queuing systems is that they have been developed in part to allow legacy applications to interoperate, yet at the same time we see that when it comes to operations between different message-queuing systems, we often hit a wall. As a consequence, once an organization chooses to use a message-queuing system from manufacturer X, they may have to settle for solutions that only X provides. Message-queuing solutions are thus in large part proprietary solutions. So much for openness.

In 2006, a working group was formed to change this situation, which resulted in the specification of the **Advanced Message-Queuing Protocol**, or simply **AMQP**. There are different versions of AMQP, with version 1.0 being the most recent one. There are also various implementations of AMQP, notably of versions before 1.0, which by the time version 1.0 was established, had gained considerable popularity. Because a pre-1.0 version is so different from the 1.0 version, yet has also a steady user base, we may see various pre-1.0 AMQP servers exist next to (their undeniably incompatible) 1.0 servers.

In this section, we will describe AMQP, but will more or less deliberately mix the pre-1.0 and 1.0 versions, sticking to the essentials and spirit of AMQP. Details can be found in the specifications [AMQP Working Group, 2008; OASIS, 2012]. Implementations of AMQP include RabbitMQ [Roy, 2018] and Apache's ActiveMQ.

Basics

AMQP revolves around applications, queue managers, and queues. Taking an approach that is common for many networking situations, we make a distinction between AMQP as a *messaging service*, the actual *messaging protocol*, and, finally, the *messaging interface* as offered to applications. To this end, it is easiest to consider having only a single queue manager, running as a single, separate server, forming the implementation of AMQP as a service. An application communicates with this queue manager through a local interface.

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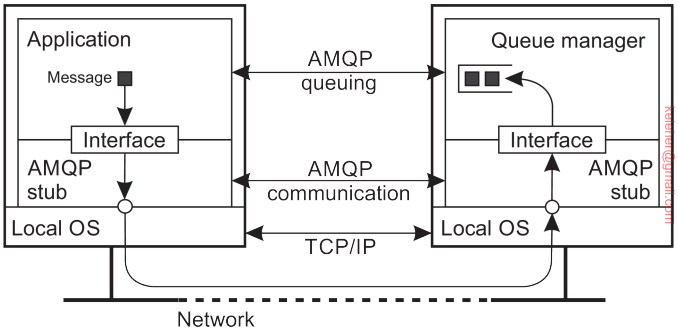


Figure 4.29: An overview of a single-server AMQP instance.

Between an application and the queue manager, communication proceeds according to the AMQP protocol.

This situation is shown in Figure 4.29 and should look familiar. The AMQP stub shields the application (as well as the queue manager) from the details concerning message transfer and communication in general. At the same time, it implements a message-queuing interface, allowing the application to make use of AMQP as a message-queuing service. Although the distinction between AMQP stub and queue manager is made explicit for queue managers, the strictness of the separation is left to an implementation. Nevertheless, if not strict, conceptually there is a distinction between handling queues and handling related communication as we shall make clear shortly.

AMQP communication

AMQP allows an application to set up a **connection** to a queue manager; a connection is a container for a number of one-way **channels**. Whereas the lifetime of a channel can be highly dynamic, connections are assumed to be relatively stable. This difference between connection and channel allows for efficient implementations, notably by using a single transport-layer TCP connection to multiplex many different channels between an application and a queue manager. In practice, AMQP assumes TCP is used for establishing AMQP connections.

Bidirectional communication is established through **sessions**: a logical grouping of two channels. A connection may have multiple sessions, but note that a channel need not necessarily be part of a session.

Finally, to actually transfer messages, a **link** is needed. Conceptually, a link, or rather its end points, keep track of the status of messages that are being transferred. It thus provides fine-grained flow control between an application and a queue manager, and, indeed, different control policies can be put simultaneously in place for different messages that are transferred

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through the same session or connection. Flow control is established through credits: a receiver can tell the sender how many messages it is allowed to send over a specific link.

When a message is to be transferred, the application passes it to its local AMQP stub. As mentioned, each message transfer is associated with one specific link. Message transfer normally proceeds in three steps.

1. At the sender's side, the message is assigned a unique identifier and is recorded *locally* to be in an **unsettled state**. The stub subsequently transfers the message to the server, where the AMQP stub also records it as being in an unsettled state. At that point, the server-side stub passes it to the queue manager.
2. The receiving application (in this case the queue manager), is assumed to handle the message and normally reports back to its stub that it is finished. The stub passes this information to the original sender, at which point the message at the original sender's AMQP stub enters a **settled state**.
3. The AMQP stub of the original sender now tells the stub of the original receiver that message transfer has been settled (meaning that the original sender will forget about the message from now on). The receiver's stub can now also discard anything about the message, formally recording it as being settled as well.

Note that because the receiving application can indicate to the underlying AMQP communication layer that it is done with a message, AMQP enables true end-to-end communication reliability. In particular, the application, be it a client application or an actual queue manager, can instruct the AMQP communication layer to keep hold of a message (i.e., a message stays in the unsettled state).

AMQP messaging

Messaging in AMQP logically takes place at the layer above the one handling communication. It is here that an application can indicate what needs to be done with a message, but can also see what has happened so far. Messaging formally takes place between two **nodes**, of which there are three types: a producer, a consumer, or a queue. Typically, producer and consumer nodes represent regular applications, whereas queues are used to store and forward messages. Indeed, a queue manager will typically consist of multiple queue nodes. In order for message transfer to take place, two nodes will have to establish a link between them.

The receiver can indicate to the sender whether its message was accepted (meaning that it was successfully processed), or rejected. Note that this means that a notification is returned to the original sender. AMQP also

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```python
import rabbitpy

def producer():
  connection = rabbitpy.Connection() # Connect to RabbitMQ server
  channel = connection.channel()     # Create new channel on the connection

  exchange = rabbitpy.Exchange(channel, 'exchange') # Create an exchange
  exchange.declare()

  queue1 = rabbitpy.Queue(channel, 'example1') # Create 1st queue
  queue1.declare()

  queue2 = rabbitpy.Queue(channel, 'example2') # Create 2nd queue
  queue2.declare()

  queue1.bind(exchange, 'example-key') # Bind queue1 to a single key
  queue2.bind(exchange, 'example-key') # Bind queue2 to the same key

  message = rabbitpy.Message(channel, 'Test message')
  message.publish(exchange, 'example-key') # Publish the message using the key
  exchange.delete()
```

(a)

```python
import rabbitpy

def consumer():
  connection = rabbitpy.Connection()
  channel = connection.channel()

  queue = rabbitpy.Queue(channel, 'example1')

  # While there are messages in the queue, fetch them using Basic.Get
  while len(queue) > 0:
    message = queue.get()
    print('Message Q1: %s' % message.body.decode())
    message.ack()

  queue = rabbitpy.Queue(channel, 'example2')

  while len(queue) > 0:
    message = queue.get()
    print('Message Q2: %s' % message.body.decode())
    message.ack()
```

(b)

Figure 4.30: A simple (a) producer and (b) consumer for RabbitMQ, adapted from [Roy, 2018].

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supports fragmentation and assembly of large messages, for which additional notifications are sent.

Of course, an important aspect of AMQP is its support for **persistent messaging**. Achieving persistence is handled through several mechanisms. First, a message can be marked as durable, indicating that the source expects any intermediate node, such as a queue, to be able to recover in the case of a failure. An intermediate node that cannot guarantee such durability will have to reject a message. Second, a source or target node can also indicate its durability: if durable, will it maintain its state, or will it also maintain the (unsettled) state of durable messages? Combining the latter with durable messages effectively establishes reliable message transfer and persistent messaging.

AMQP is truly a messaging protocol in the sense that it does not by itself support, for example, publish-subscribe primitives. It expects that such issues are handled by more advanced, proprietary queue managers, akin to the message brokers discussed in Section 4.3.3.

Finally, there is no reason why a queue manager cannot be connected to another queue manager. In fact, it is quite common to organize queue managers into an overlay network in which messages are routed from producers to their consumers. AMQP does not specify how the overlay network should be constructed and managed, and, indeed, different providers of AMQP-based systems offer different solutions. Of particular importance is specifying how messages should be routed through the network. The bottom line is that administrators will need to do a lot of this specification manually. Only in cases where overlays have regular structures, such as cycles or trees, it becomes easier to provide the necessary routing details.

AMQP in practice

One of the more popular implementations of message brokering is the RabbitMQ server, extensively described by Roy [2018]. The RabbitMQ server is based on the 0.9 version of AMQP, while also fully supporting version 1.0. An important difference between these two versions is the use of **exchanges**. Instead of directly manipulating queues, producers contact an exchange, which, in turn, places messages in one or several queues. In essence, an exchange allows a producer to use a synchronous or asynchronous RPC to a queue manager. An exchange never stores messages; it simply makes sure that a message is placed in an appropriate queue.

To illustrate this concept, consider Figure 4.30(a) showing a simple producer. As mentioned previously, AMQP embeds channels as part of more durable connections, and the first we do is create a means for communicating with the server (we are omitting various details; in this case we establish a default connection with the server). In the following lines, an exchange is created within the realm of the server, along with two queues. In Lines 16

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and 17, both queues are bound to the same exchange by a simple key named `example-key`. After creating a message, the producer tells the exchange that it should publish that message to any queue bound by the specified key.

As to be expected, the exchange will place the message in both queues, which is verified by the simple consumer shown in Figure 4.30(b). In this case, the consumer simply fetches messages from either queue and prints their content.

Exchanges and queues have many options, for which we refer the interested reader to Roy [2018]. It is important to note that using systems such as RabbitMQ, but also other messaging brokers, one can generally set up advanced overlay networks for application-level message routing. At the same time, maintenance of those networks can easily become a nightmare.

4.4 Multicast communication

An important topic in communication in distributed systems is the support for sending data to multiple receivers, also known as **multicast communication**. For many years, this topic has belonged to the domain of network protocols, where numerous proposals for network-level and transport-level solutions have been implemented and evaluated [Janic, 2005; Obraczka, 1998]. A major issue in all solutions was setting up the communication paths for information dissemination. In practice, this involved a huge management effort, often requiring human intervention. In addition, as long as there is no convergence of proposals, ISPs have shown to be reluctant to support multicasting [Diot et al., 2000].

With the advent of peer-to-peer technology, and notably structured overlay management, it became easier to set up communication paths. As peer-to-peer solutions are typically deployed at the application layer, various application-level multicasting techniques have been introduced. In this section, we will take a brief look at these techniques.

Multicast communication can also be accomplished in other ways than setting up explicit communication paths. As we also explore in this section, gossip-based information dissemination provides simple (yet often less efficient) ways for multicasting.

4.4.1 Application-level tree-based multicasting

The basic idea in **application-level multicasting** is that nodes are organized into an overlay network, which is then used to disseminate information to its members. An important observation is that network routers are not involved in group membership. As a consequence, the connections between nodes in the overlay network may cross several physical links, and as such, routing messages within the overlay may not be optimal in comparison to what could have been achieved by network-level routing.

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A crucial design issue is the construction of the overlay network. In essence, there are two approaches [El-Sayed et al., 2003; Hosseini et al., 2007; Allani et al., 2009]. First, nodes may organize themselves directly into a tree, meaning that there is a unique (overlay) path between every pair of nodes. An alternative approach is that nodes organize into a mesh network in which every node will have multiple neighbors and, in general, there exist multiple paths between every pair of nodes. The main difference between the two is that the latter generally provides higher robustness: if a connection breaks (e.g., because a node fails), there will still be an opportunity to disseminate information without having to immediately reorganize the entire network.

Note 4.13 (Advanced: Constructing a multicast tree in Chord)
To make matters concrete, let us consider a relatively simple scheme for constructing a multicast tree in Chord, which we described in Note 2.6. This scheme was originally proposed for Scribe [Castro et al., 2002b] which is an application-level multicasting scheme built on top of Pastry [Rowstron and Druschel, 2001]. The latter is also a DHT-based peer-to-peer system.

Assume a node wants to start a multicast session. To this end, it simply generates a multicast identifier, say, mid which is just a randomly chosen 160-bit key. It then looks up succ(mid), which is the node responsible for that key, and promotes it to become the root of the multicast tree that will be used for sending data to interested nodes. To join the tree, a node P simply executes the operation lookup(mid) having the effect that a lookup message with the request to join the multicast group mid will be routed from P to succ(mid). The routing algorithm itself will be explained in detail in Chapter 6.

On its way toward the root, the join request will pass several nodes. Assume it first reaches node Q. If Q had never seen a join request for mid before, it will become a **forwarder** for that group. At that point, P will become a child of Q whereas the latter will continue to forward the join request to the root. If the next node on the root, say, R is also not yet a forwarder, it will become one and record Q as its child and continue to send the join request.

On the other hand, if Q (or R) is already a forwarder for mid, it will also record the previous sender as its child (i.e., P or Q, respectively), but there will not be a need to send the join request to the root anymore, as Q (or R) will already be a member of the multicast tree.

Nodes such as P that have explicitly requested to join the multicast tree are, by definition, also forwarders. The result of this scheme is that we construct a multicast tree across the overlay network with two types of nodes: pure forwarders that act as helpers, and nodes that are also forwarders, but have explicitly requested to join the tree. Multicasting is now simple: a node merely sends a multicast message toward the root of the tree by again executing the lookup(mid) operation, after which that message can be sent along the tree.

We note that this high-level description of multicasting in Scribe does not do complete justice to its original design. The interested reader is encouraged to take a look at the details, which can be found in [Castro et al., 2002b].

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Performance issues in overlays

From the high-level description given above, it should be clear that although building a tree by itself is not that difficult once we have organized the nodes into an overlay, building an efficient tree may be a different story. Note that in our description so far, the selection of nodes that participate in the tree does not take into account any performance metrics: it is purely based on the (logical) routing of messages through the overlay.

To understand the problem at hand, take a look at Figure 4.31 which shows a small set of five nodes that are organized in a simple overlay network, with the node A forming the root of a multicast tree. The costs for traversing a physical link are also shown. Now, whenever A multicasts a message to the other nodes, the *logical* route, i.e., at the level of the overlay, is simply $A \rightarrow B \rightarrow E \rightarrow D \rightarrow C$. The actual route crosses the network-level links in the following order: $A \rightarrow Ra \rightarrow Rb \rightarrow B \rightarrow Rb \rightarrow Ra \rightarrow Re \rightarrow E \rightarrow Re \rightarrow Rc \rightarrow Rd \rightarrow D \rightarrow Rd \rightarrow Rc \rightarrow C$. It is thus seen that this message will traverse each of the links $\langle B, Rb\rangle$, $\langle Ra, Rb\rangle$, $\langle E, Re\rangle$, $\langle Rc, Rd\rangle$, and $\langle D, Rd\rangle$ twice. The overlay network would have been more efficient if we had not constructed *overlay links* $\langle B, E\rangle$, and $\langle D, E\rangle$, but instead $\langle A, E\rangle$ and $\langle C, E\rangle$. Such a configuration would have saved the double traversal across physical links $\langle Ra, Rb\rangle$ and $\langle Rc, Rd\rangle$.

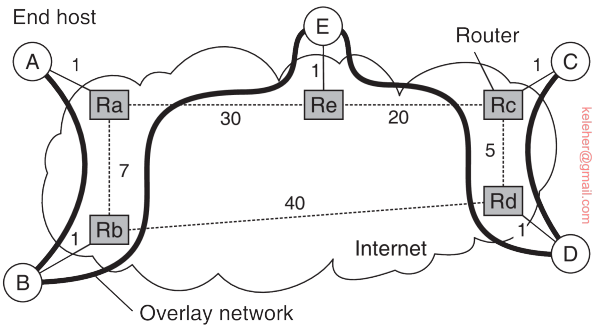


Figure 4.31: The relation between links in an overlay and network-level routes.

The quality of an application-level multicast tree is generally measured by three different metrics: link stress, stretch, and tree cost. **Link stress** is defined per link and counts how often a packet crosses the same link [Chu et al., 2002]. A link stress greater than 1 comes from the fact that although at a logical level a packet may be forwarded along two different connections, part of those connections may actually correspond to the same physical link, as we showed in Figure 4.31.

The **stretch** or **relative delay penalty** (**RDP**) measures the ratio of the delay between two nodes in the overlay, and the delay that those two nodes would experience in the underlying network. For example, messages from

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B to C follow the route B → Rb → Ra → Re → E → Re → Rc → Rd → D → Rd → Rc → C in the overlay network, having a total cost of 73 units. However, messages would have been routed in the underlying network along the path B → Rb → Rd → Rc → C, with a total cost of 47 units, leading to a stretch of 1.55. Obviously, when constructing an overlay network, the goal is to minimize the aggregated stretch, or similarly, the average RDP measured over all node pairs.

Finally, the **tree cost** is a global metric, generally related to minimizing the aggregated link costs. For example, if the cost of a link is taken to be the delay between its two end nodes, then optimizing the tree cost boils down to finding a minimal spanning tree in which the total time for disseminating information to all nodes is minimal.

To simplify matters somewhat, assume that a multicast group has an associated and well-known node that keeps track of the nodes that have joined the tree. When a new node issues a join request, it contacts this **rendezvous node** to obtain a (potentially partial) list of members. The goal is to select the best member that can operate as the new node's parent in the tree. Who should it select? There are many alternatives, and different proposals often follow very different solutions.

Consider, for example, a multicast group with only a single source. In this case, the selection of the best node is obvious: it should be the source (because in that case, we can be assured that the stretch will be equal to 1). However, in doing so, we would introduce a star topology with the source in the middle. Although simple, it is not difficult to imagine the source may easily become overloaded. In other words, selection of a node will generally be constrained in such a way that only those nodes may be chosen who have k or fewer neighbors, with k being a design parameter. This constraint severely complicates the tree-establishment algorithm, as a suitable solution may require that part of the existing tree is reconfigured. Tan et al. [2003] provide an extensive overview and evaluation of various solutions to this problem.

Note 4.14 (Advanced: Switch-trees)
As an illustration, let us take a closer look at one specific family, known as **switch-trees** [Helder and Jamin, 2002]. The basic idea is simple. Assume we already have a multicast tree with a single source as root. In this tree, a node P can switch parents by dropping the link to its current parent in favor of a link to another node. The only constraints imposed on switching links is that the new parent can never be a member of the subtree rooted at P (as this would partition the tree and create a loop), and that the new parent will not have too many immediate children. This last requirement is needed to limit the load of forwarding messages by any single node.

There are different criteria for deciding to switch parents. A simple one is to

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optimize the route to the source, effectively minimizing the delay when a message is to be multicasted. To this end, each node regularly receives information on other nodes (we will explain one specific way of doing this below). At that point, the node can evaluate whether another node would be a better parent in terms of delay along the route to the source, and if so, initiates a switch.

Another criterion could be whether the delay to the potential other parent is lower than to the current parent. If every node takes this as a criterion, then the aggregated delays of the resulting tree should ideally be minimal. In other words, this is an example of optimizing the cost of the tree, as we explained above. However, more information would be needed to construct such a tree, but as it turns out, this simple scheme is a reasonable heuristic leading to a good approximation of a minimal spanning tree.

As an example, consider the case where a node P receives information on the neighbors of its parent. Note that the neighbors consist of P's grandparent, along with the other siblings of P's parent. Node P can then evaluate the delays to each of these nodes and subsequently choose the one with the lowest delay, say Q, as its new parent. To that end, it sends a switch request to Q. To prevent loops from being formed due to concurrent switching requests, a node that has an outstanding switch request will simply refuse to process any incoming requests. In effect, this leads to a situation where only completely independent switches can be carried out simultaneously. Furthermore, P will provide Q with enough information to allow the latter to conclude that both nodes have the same parent, or that Q is the grandparent.

An important problem that we have not yet addressed is node failure. In the case of switch-trees, a simple solution is proposed: whenever a node notices that its parent has failed, it attaches itself to the root. At that point, the optimization protocol can proceed as usual and will eventually place the node at a good point in the multicast tree. Experiments described in [Helder and Jamin, 2002] show that the resulting tree is indeed close to a minimal spanning one.

4.4.2 Flooding-based multicasting

So far, we have assumed that when a message is to be multicasted, it is to be received by every node in the overlay network. Strictly speaking, this corresponds to **broadcasting**. In general, multicasting refers to sending a message to a subset of all the nodes, that is, a specific *group of nodes*. A key design issue when it comes to multicasting is to minimize the use of intermediate nodes for which the message is not intended. To make this clear, if the overlay is organized as a multi-level tree, yet only the leaf nodes are the ones who should receive a multicast message, then clearly there may be quite some nodes who need to store and subsequently forward a message that is not meant for them.

One simple way to avoid such inefficiency, is to construct an overlay network *per multicast group*. As a consequence, multicasting a message m to a

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group G is the same as broadcasting m to G. The drawback of this solution is that a node belonging to several groups, will, in principle, need to maintain a separate list of its neighbors for each group of which it is a member.

If we assume that an overlay corresponds to a multicast group, and thus that we need to broadcast a message, a naive way of doing so is to apply **flooding**. In this case, each node simply forwards a message m to each of its neighbors, except to the one from which it received m. Furthermore, if a node keeps track of the messages it received and forwarded, it can simply ignore duplicates.

To understand the performance of flooding, we model an overlay network as a connected undirected graph $G = (V, E)$ with nodes V and links E. Let v_0 be the node that initiates flooding. In principle, every node other node than v_0 will send out the message to its neighbors once, except to the neighbor from which it had received the message. We will thus see a total of $\delta(v_0) + \sum_{v \in V - v_0}(\delta(v) - 1)$ messages, with $\delta(v)$ the number of neighbors of the node v. For any undirected graph, $\sum_{v \in V} \delta(v) = 2 \cdot |E|$, with $|E|$ denoting the total number of edges. In other words, we will see $2|E| - |V| + 1$ messages being sent, making flooding quite inefficient. Only if G is a tree, will flooding be optimal, for in that case, $|E| = |V| - 1$. In the worst case, when G is fully connected, we have $|E| = \binom{|V|}{2}$ leading to an order of $|V|^2$ messages.

Suppose now that we have no information on the structure of the overlay network and that the best we can assume is that it can be represented as a **random graph**, which (to keep it simple) is a graph having a probability p_{edge} that two vertices are joined by an edge, also known as an **Erdös-Rényi graph** [Erdös and Rényi, 1959]. Note that we are actually considering our overlay network to be an unstructured peer-to-peer network, and that we do not have any information on how it is being constructed. With a probability p_{edge} that two nodes are joined, and a total of $\binom{|V|}{2}$ edges, it is not difficult to see that we can expect our overlay to have $|E| = \frac{1}{2} \cdot p_{edge} \cdot |V| \cdot (|V| - 1)$ edges. To give an impression of what we are dealing with, Figure 4.32 shows the relationship between the number of nodes and edges for different values of p_{edge}. As can be seen, the number of edges can easily become large, even for small values of p_{edge}.

To reduce the number of messages, we can also use **probabilistic flooding** as introduced by Banaei-Kashani and Shahab [2003] and formally analyzed by Oikonomou and Stavrakakis [2007]. The idea is simple: when a node is flooding a message m and needs to forward m to a specific neighbor, it will do so with a probability p_{flood}. The effect can be dramatic: the total number of messages sent will drop linearly in p_{flood}. However, there is also a risk: the lower p_{flood}, the higher the chance that not all nodes in the network will be reached. This risk is caused by the simple fact that all neighbors of a specific node Q may have decided *not* to forward m to Q. If Q has n neighbors, then this can happen roughly with a probability of $(1 - p_{flood})^n$. Clearly, the

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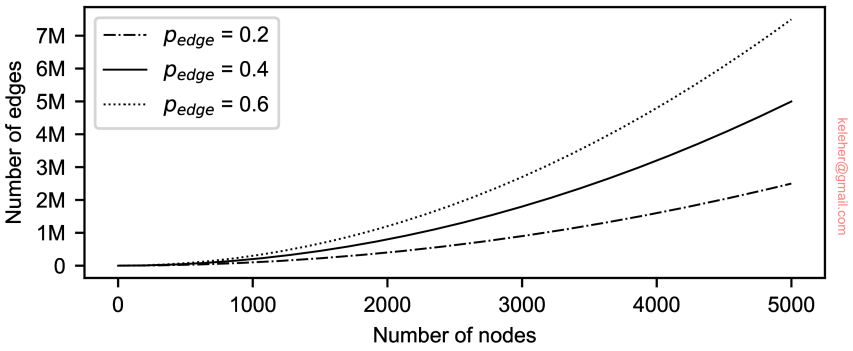


Figure 4.32: The size of a random overlay as function of the number of nodes.

number of neighbors plays an important role in deciding whether to forward a message, and, indeed, we can replace the static probability of forwarding with one that takes the degree of the neighbor into account. This has been further developed and analyzed by Sereno and Gaeta [2011]. To give an idea of the efficiency of probabilistic broadcasting: in a random network of 10,000 nodes and $p_{edge} = 0.1$, we need only set $p_{flood} = 0.01$ to establish a more than 50-fold reduction in the number of messages sent in comparison to full flooding.

When dealing with a structured overlay, that is, one having a more or less deterministic topology, designing efficient flooding schemes is simpler. As an example, consider an n-dimensional hypercube, shown in Figure 4.33 for the case $n = 4$, as also discussed in Section 2.4.1.

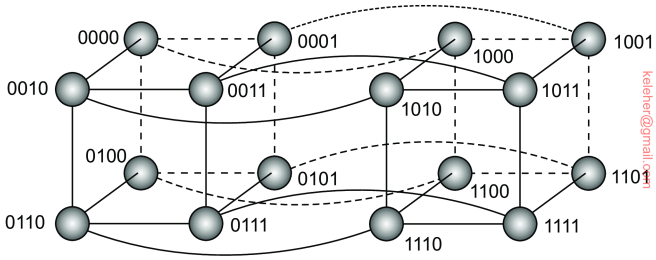


Figure 4.33: A simple peer-to-peer, four-dimensional hypercube.

A simple and efficient broadcast scheme has been designed by Schlosser et al. [2002] and relies on keeping track of neighbors *per dimension*. This is best explained by considering that every node in an n-dimensional hypercube is represented by a bit string of length n. Each edge in the overlay is labeled with its dimension. For the case $n = 4$, node 0000 will have as its neighbors the set $\{0001, 0010, 0100, 1000\}$. The edge between 0000 and 0001 is labeled

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"4" corresponding to changing the 4th bit when comparing 0000 to 0001 and *vice versa*. Likewise, the edge $\langle 0000, 0100 \rangle$ is labeled "2," and so forth. A node initially broadcasts a message m to all of its neighbors, and tags m with the label of the edge over which it sends the message. In our example, if node 1001 broadcasts a message, it will send the following:

- (m,1) to 0001
- (m,2) to 1101
- (m,3) to 1011
- (m,4) to 1000

When a node receives a broadcast message, it will forward it only along edges that have a higher dimension. In other words, in our example, node 1101 will forward m only to nodes 1111 (joined to 1101 by an edge labeled "3") and 1100 (joined by an edge with label "4"). Using this scheme, it can be shown that every broadcast requires precisely $N - 1$ messages, where $N = 2^n$, that is the number of nodes in a n-dimensional hypercube. This broadcasting scheme is therefore optimal in terms of the number of messages sent.

Note 4.15 (Advanced: Ring-based flooding)

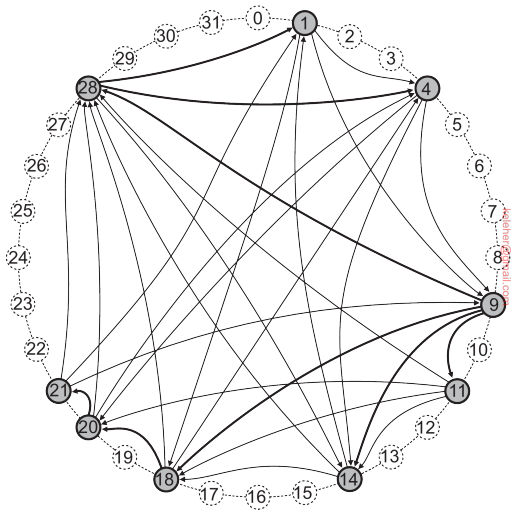


Figure 4.34: A Chord ring in which node 9 broadcasts a message.

A hypercube is a straightforward example of how we can effectively use knowledge of the structure of an overlay network to establish efficient flooding. In the case of Chord, we can follow an approach proposed by Ghodsi [2010]. Recall that in Chord each node is identified by a number p, and each resource (typically

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a file), is assigned a key k from the same space as used for node identifiers. The successor succ(k) of a key k is the node with the smallest identifier $p \geq k$. Consider the small Chord ring shown in Figure 4.34 and assume that node 9 wants to flood a message to all other nodes.

In our example, node 9 divides the identifier space into four segments (one for each of its neighbors). Node 28 is requested to make sure that the message reaches all nodes with identifiers $28 \leq k < 9$ (recall that we are applying modulo arithmetic); node 18 takes care of nodes with identifiers $18 \leq k < 28$; node 14 for $14 \leq k < 18$; and 11 for identifiers $11 \leq k < 14$.

Node 28 will subsequently divide the part of the identifier space it is requested to handle into two subsegments: one for its neighboring node 1 and another for 4. Likewise, node 18, responsible for segment $[18, 28)$ will "split" that segment into only one part: it has only one neighbor to delegate the flood to, and forwards the message to node 20, telling it that it should handle segment $[20, 28)$.

In the last step, only node 20 has work to do. It forwards the message to node 21, telling it to forward it to nodes known to it in the segment $[21, 28)$. As there are no such nodes anymore, the broadcast completes.

As in the case of our hypercube example, we see that flooding is done with $N - 1$ messages, with N being the number of nodes in the system.

4.4.3 Gossip-based data dissemination

An important technique for disseminating information is to rely on **epidemic behavior**, also referred to as **gossiping**. Observing how diseases spread among people, researchers have since long investigated whether simple techniques could be developed for spreading information in very large-scale distributed systems. The main goal of these **epidemic protocols** is to rapidly propagate information among a large collection of nodes using only local information. In other words, there is no central component by which information dissemination is coordinated.

To explain the general principles of these algorithms, we assume that all updates for a specific data item are initiated at a single node. In this way, we simply avoid write-write conflicts. The following presentation is based on the classical paper by Demers et al. [1987] on epidemic algorithms. An overview of epidemic information dissemination can be found in [Eugster et al., 2004].

Information dissemination models

As the name suggests, epidemic algorithms are based on the theory of epidemics, which studies the spreading of infectious diseases. In the case of large-scale distributed systems, instead of spreading diseases, they spread information. Research on epidemics for distributed systems also aims at an entirely different goal: whereas health organizations will do their best to prevent infectious diseases from spreading across large groups of people,

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designers of epidemic algorithms for distributed systems will try to "infect" all nodes with new information as fast as possible.

Using the terminology from epidemics, a node that is part of a distributed system is called **infected** if it holds data that it is willing to spread to other nodes. A node that has not yet seen this data is called **susceptible**. Finally, an updated node that is not willing or able to spread its data is said to have been **removed**. Note that we assume we can distinguish old from new data, for example because it has been timestamped or versioned. In this light, nodes are also said to spread updates.

A popular propagation model is that of **anti-entropy**. In this model, a node P picks another node Q at random, and subsequently exchanges updates with Q. There are three approaches to exchanging updates:

1. P only pulls in new updates from Q
2. P only pushes its own updates to Q
3. P and Q send updates to each other (i.e., a push-pull approach)

When it comes to rapidly spreading updates, only pushing updates turns out to be a bad choice. Intuitively, this can be understood as follows. First, note that in a pure push-based approach, updates can be propagated only by infected nodes. However, if many nodes are infected, the probability of each one selecting a susceptible node is relatively small. Consequently, chances are that a particular node remains susceptible for a long period simply because an infected node does not select it.

In contrast, the pull-based approach works much better when many nodes are infected. In that case, spreading updates is essentially triggered by susceptible nodes. Chances are big that such a node will contact an infected one to subsequently pull in the updates and become infected as well.

If only a single node is infected, updates will rapidly spread across all nodes using either form of anti-entropy, although push-pull remains the best strategy [Jelasity et al., 2007]. Define a **round** as spanning a period in which every node will have taken the initiative once to exchange updates with a randomly chosen other node. It can then be shown that the number of rounds to propagate a single update to all nodes is of the order $\mathcal{O}(log(N))$, where N is the number of nodes in the system. This indicates indeed that propagating updates is fast, but above all scalable.

Note 4.16 (Advanced: An analysis of anti-entropy)
A simple and straightforward analysis will give some idea on how well anti-entropy works. Consider a system with N nodes. One of these nodes initiates the spreading of a message m to all other nodes. Let p_i denote the probability that a node P has not yet received m after the ith round. We distinguish the following three cases:

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- With a pure pull-based approach, $p_{i+1} = (p_i)^2$: not only had P not yet been updated in the previous round, also P contacts a node that had not yet received m.
- With a pure push-based approach, $p_{i+1} = p_i \cdot (1 - \frac{1}{N-1})^{N(1-p_i)}$: again, P should not have been updated in the previous round, but also none of the updated nodes should contact P. The probability that not a single node contacts P is $1 - \frac{1}{N-1}$; we can expect that there are $N(1 - p_i)$ updated nodes in round i.
- In a push-pull approach, we can simply combine the two: P should not contact an updated node, and should not be contacted by one.

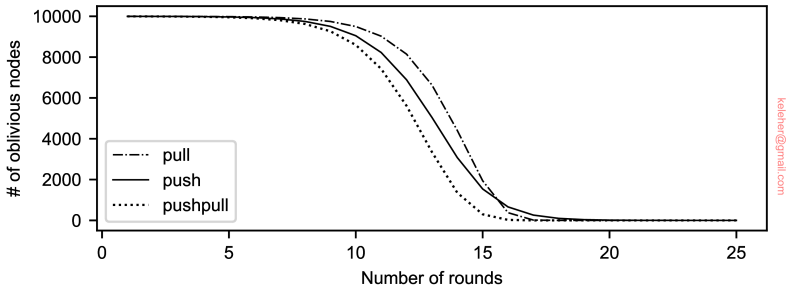


Figure 4.35: (a) The number of nodes that have not been updated as a function of the number of dissemination rounds.

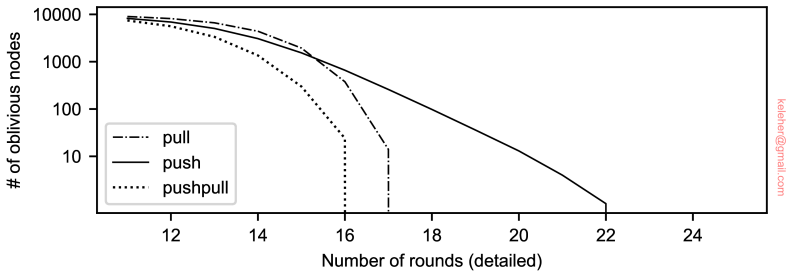


Figure 4.35: (b) Zooming in the differences between the three anti-entropy approaches when almost all nodes have been updated.

Figure 4.35(a) shows how quickly the probability of not yet being updated drops as a function of the number of rounds in a network of 10,000 nodes. After a slow start, the information is rapidly disseminated. In Figure 4.35(b) we have zoomed into what happens in the last number of rounds while also using a log scale for the y-axis. Indeed, assuming that nodes are up-and-running all the time,

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it turns out that an anti-entropy push-pull strategy can be an extremely effective dissemination protocol.

One specific variant of epidemic protocols is called **rumor spreading**. It works as follows. If node P has just been updated for a data item x, it contacts an arbitrary other node Q and tries to push the update to Q. However, it is possible that Q was already updated by another node. In that case, P may lose interest in spreading the update any further, say with probability p_{stop}. In other words, it then becomes removed.

Rumor spreading is gossiping as we mostly experience it in real life. When Bob has some hot news to spread around, he may phone his friend Alice, telling her all about it. Alice, like Bob, will be really excited to spread the rumor to her friends as well. However, she will become disappointed when phoning a friend, say Chuck, only to hear that the news has already reached him. Chances are that she will stop phoning other friends, for what good is it if they already know?

Rumor spreading turns out to be an excellent way of rapidly spreading news. However, it cannot guarantee that all nodes will actually be updated [Demers et al., 1987]. In fact, when there are many nodes that participate in the epidemics, the fraction s of nodes that will remain ignorant of an update, that is, remain susceptible, satisfies the equation:

$$s = e^{-(1/p_{stop}+1)(1-s)}$$

To get an idea of what this means, take a look at Figure 4.36, which shows s as a function of p_{stop}. Even for high values of p_{stop} we see that the fraction of nodes that remains ignorant is relatively low, and always less than approximately 0.2. For $p_{stop} = 0.20$ it can be shown that $s = 0.0025$. However, in those cases when p_{stop} is relatively high, additional measures will need to be taken to ensure that *all* nodes are updated.

Note 4.17 (Advanced: Analysis of rumor spreading)
To formally analyze the situation for rumor spreading, we let s denote the fraction of nodes that have not yet been updated, i.e., the fraction of susceptible nodes. Likewise, let i denote the fraction of infected nodes: the ones that have been updated and are still contacting other nodes to spread information. Finally, r is the fraction of nodes that have been updated, but have given up, i.e., they are passive and no longer play a role in disseminating information. Obviously, $s + i + r = 1$. Using theory from epidemics, it is not difficult to see the following:

$$\begin{array}{lllll} (1) & ds/dt & = & -s \cdot i \\ (2) & di/dt & = & s \cdot i - p_{stop} \cdot (1-s) \cdot i \\ \Rightarrow & di/ds & = & -(1+p_{stop}) + \frac{p_{stop}}{s} \\ \Rightarrow & i(s) & = & -(1+p_{stop}) \cdot s + p_{stop} \cdot \ln(s) + C \end{array}$$

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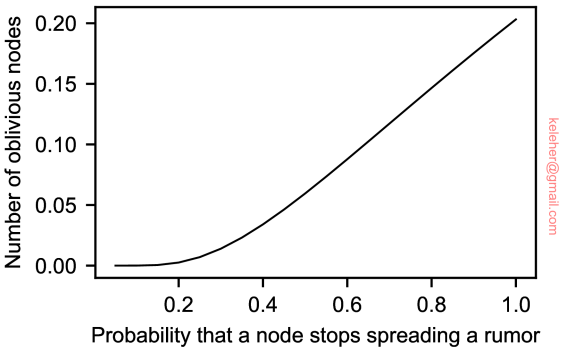


Figure 4.36: The relation between the fraction s of update-ignorant nodes and the probability p_{stop} that a node will stop gossiping once it contacts a node that has already been updated.

where we use the notation $i(s)$ to express i as a function of s. When $s = 1$, no nodes have yet been infected, meaning that $i(1) = 0$. This allows us to derive that $C = 1 + p_{stop}$, and thus

$$i(s) = (1 + p_{stop}) \cdot (1 - s) + p_{stop} \cdot \ln(s)$$

We are looking for the situation that there is no more rumor spreading, i.e., when $i(s) = 0$. Having a closed expression for $i(s)$ then leads to

$$s = e^{-(1/p_{stop}+1)(1-s)}$$

One of the main advantages of epidemic algorithms is their scalability since the number of synchronizations between processes is relatively small compared to other propagation methods. For wide-area systems, Lin and Marzullo [1999] have shown that it makes sense to take the actual network topology into account to achieve better results. In that case, nodes that are connected to only a few other nodes are contacted with a relatively high probability. The underlying assumption is that such nodes form a bridge to other remote parts of the network; therefore, they should be contacted as soon as possible. This approach is referred to as **directional gossiping** and comes in different variants.

This problem touches upon an important assumption that most epidemic solutions make, namely that a node can randomly select any other node to gossip with. This implies that, in principle, the complete set of nodes should be known to each member. In a large system, this assumption can never hold, and special measures will need to be taken to mimic such properties. We return to this issue in Section 5.5.2 when we discuss a **peer-sampling service**.

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Removing data

Epidemic algorithms are fantastic for spreading updates. However, they have a rather strange side effect: spreading the *deletion* of a data item is hard. The essence of the problem lies in the fact that deletion of a data item destroys all information on that item. Consequently, when a data item is simply removed from a node, that node will eventually receive old copies of the data item and interpret those as updates on something it did not have before.

The trick is to record the deletion of a data item as just another update, and keep a record of that deletion. In this way, old copies will not be interpreted as something new, but merely treated as versions that have been updated by a delete operation. The recording of a deletion is done by spreading **death certificates**.

Of course, the problem with death certificates is that they should eventually be cleaned up, or otherwise each node will gradually build a huge local database of historical information on deleted data items that is otherwise not used. Demers et al. [1987] propose to use what are called **dormant death certificates**. Each death certificate is timestamped when it is created. If it can be assumed that updates propagate to all nodes within a known finite time, then death certificates can be removed after this maximum propagation time has elapsed.

However, to provide hard guarantees that deletions are indeed spread to all nodes, only a very few nodes maintain dormant death certificates that are never thrown away. Assume node P has such a certificate for a data item x. If by any chance an obsolete update for x reaches P, P will react by simply spreading the death certificate for x again.

4.5 Summary

Having powerful and flexible facilities for communication between processes is essential for any distributed system. In traditional network applications, communication is often based on the low-level message-passing primitives offered by the transport layer. An important issue in middleware systems is to offer a higher level of abstraction that will make it easier to express communication between processes than the support offered by the interface to the transport layer.

One of the most widely used abstractions is the Remote Procedure Call (RPC). The essence of an RPC is that a service is implemented through a procedure, of which the body is executed at a server. The client is offered only the signature of the procedure, that is, the procedure's name along with its parameters. When the client calls the procedure, the client-side implementation, called a stub, takes care of wrapping the parameter values into a message and sending that to the server. The latter calls the actual procedure and returns the results, again in a message. The client's stub

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extracts the result values from the return message and passes it back to the calling client application.

RPCs offer synchronous communication facilities, by which a client is blocked until the server has sent a reply. Although variations of either mechanism exist by which this strict synchronous model is relaxed, it turns out that general-purpose, high-level message-oriented models are often more convenient.

In message-oriented models, the issues are whether communication is persistent, and whether communication is synchronous. The essence of persistent communication is that a message that is submitted for transmission, is stored by the communication system as long as it takes to deliver it. In other words, neither the sender nor the receiver needs to be up and running for message transmission to take place. In transient communication, no storage facilities are offered so that the receiver must be prepared to accept the message when it is sent.

In asynchronous communication, the sender is allowed to continue immediately after the message has been submitted for transmission, possibly before it has even been sent. In synchronous communication, the sender is blocked at least until a message has been received. Alternatively, the sender may be blocked until message delivery has taken place or even until the receiver has responded, as with RPCs.

Message-oriented middleware models generally offer persistent asynchronous communication, and are used where RPCs are not appropriate. They are often used to assist the integration of (widely dispersed) collections of databases into large-scale information systems.

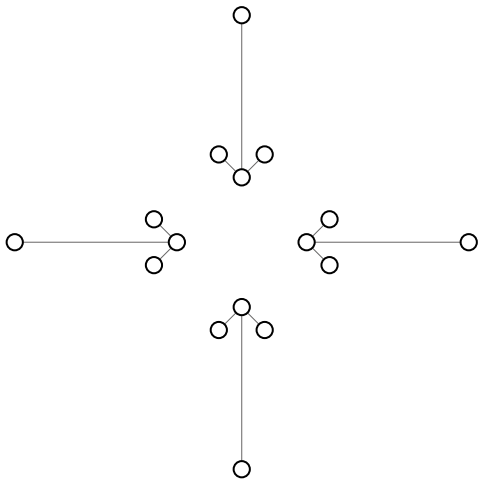
Finally, an important class of communication protocols in distributed systems is multicasting. The basic idea is to disseminate information from one sender to multiple receivers. We have discussed two different approaches. First, multicasting can be achieved by setting up a tree from the sender to the receivers. Considering that it is now well understood how nodes can self-organize into peer-to-peer system, solutions have also appeared to dynamically set up trees in a decentralized fashion. Second, flooding messages across the network is extremely robust, yet requires special attention if we want to avoid severe waste of resources as nodes may see messages multiple times. Probabilistic flooding, by which a node forwards a message with a certain probability, often proves to combine simplicity and efficiency, while being highly effective.

Another important class of dissemination solutions deploys epidemic protocols. These protocols have proven to be simple and extremely robust.

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05

COORDINATION



In the previous chapters, we have looked at processes and communication between processes. While communication is important, it is not the entire story. Closely related is how processes cooperate and synchronize with one another. Cooperation is partly supported by naming, which allows processes to at least share resources, or entities in general.

In this chapter, we mainly concentrate on how processes can synchronize and coordinate their actions. For example, it is important that multiple processes do not simultaneously access a shared resource, such as a file, but instead cooperate in granting each other temporary exclusive access. Another example is that multiple processes may sometimes need to agree on the ordering of events, such as whether message m_1 from process P was sent before or after message m_2 from process Q.

Synchronization and coordination are two closely related phenomena. In **process synchronization** we make sure that one process waits for another to complete its operation. When dealing with **data synchronization**, the problem is to ensure that two sets of data are the same. When it comes to **coordination**, the goal is to manage the interactions and dependencies between activities in a distributed system [Malone and Crowston, 1994]. From this perspective, one could state that coordination encapsulates synchronization.

As it turns out, coordination in distributed systems is often much more difficult compared to that in uniprocessor or multiprocessor systems. The problems and solutions that are discussed in this chapter are, by their nature, rather general, and occur in many situations in distributed systems.

We start with a discussion of the issue of synchronization based on actual time, followed by synchronization in which only relative ordering matters rather than ordering in absolute time.

Often, it is important that a group of processes can appoint one process as a coordinator, which can be done through election algorithms. We discuss various election algorithms in a separate section. Before that, we look into several algorithms for coordinating mutual exclusion to a shared resource. As a special class of coordination problems, we also dive into location systems, by which we place a process in a multidimensional plane. Such placements are useful when dealing with very large distributed systems.

We also consider three different gossip-based coordination problems: aggregation, peer sampling, and overlay construction.

Finally, we already came across publish-subscribe systems, but have not yet discussed in any detail how we actually match subscriptions to publications. There are many ways to do this, and we look at centralized as well as decentralized implementations.

Distributed algorithms come in all sorts and flavors and have been developed for different types of distributed systems. Many examples (and further references) can be found in Andrews [2000], Cachin et al. [2011], and Fokkink [2018]. More formal approaches to a wealth of algorithms can be found in

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textbooks from Attiya and Welch [2004], Lynch [1996], Santoro [2007], and Tel [2000].

5.1 Clock synchronization

In a centralized system, time is unambiguous. When a process wants to know the time, it simply makes a call to the operating system. If process A asks for the time, and then a little later process B asks for the time, the value that B gets will be higher than (or possibly equal to) the value A got. It will certainly not be lower. In a distributed system, achieving agreement on time is not trivial.

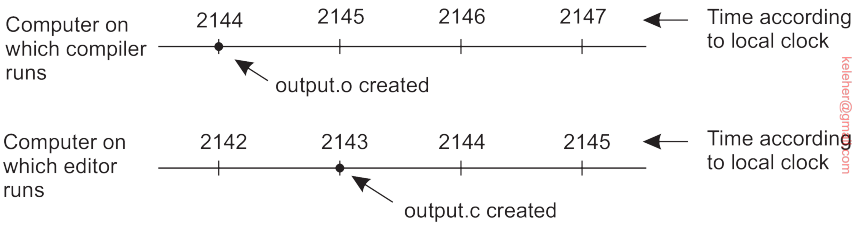
Just think, for a moment, about the implications of the lack of global time on the Unix `make` program, as a simple example. Normally, in Unix, large programs are split up into multiple source files, so that a change to one source file requires only one file to be recompiled, not all the files. If a program consists of 100 files, not having to recompile everything because one file has been changed greatly increases the speed at which programmers can work.

The way `make` normally works is simple. When the programmer has finished changing all the source files, she runs `make`, which examines the times at which all the source and object files were last modified. If the source file `input.c` has time 2151 and the corresponding object file `input.o` has time 2150, `make` knows that `input.c` has been changed since `input.o` was created, and thus `input.c` must be recompiled. On the other hand, if `output.c` has time 2144 and `output.o` has time 2145, no compilation is needed. Thus `make` goes through all the source files to find out which ones need to be recompiled and calls the compiler to recompile them.

Now imagine what could happen in a distributed system in which there was no global agreement on time. Suppose that `output.o` has time 2144 as above, and shortly thereafter `output.c` is modified but is assigned time 2143 because the clock on its machine is slightly behind, as shown in Figure 5.1. `Make` will not call the compiler. The resulting executable binary program will then contain a mixture of object files from the old sources and the new sources. It may crash, and the programmer will go crazy trying to understand what is wrong with the code.

There are many more examples where an accurate account of time is needed. The example above can easily be reformulated to file timestamps in general. In addition, think of application domains such as financial brokerage, security auditing, and collaborative sensing, and it will become clear that accurate timing is important. From a different perspective, Najafi et al. [2021] argue that accurate timing is essential for most systems research, if only to ensure that performance measurements can be properly compared. Since time is so basic to the way people think and the effect of not having all the clocks synchronized can be so dramatic, it is fitting that we begin our study of

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Figure 5.1: When each machine has its own clock, an event that occurred after another event may nevertheless be assigned an earlier time.

synchronization with the simple question: Would it be possible to synchronize all the clocks in a distributed system? The answer is surprisingly complicated.

5.1.1 Physical clocks

Nearly all computers often have several circuits for keeping track of time. Despite the widespread use of the word "clock" to refer to these devices, they are not actually clocks in the usual sense. **Timer** is perhaps a better word. A computer timer is usually a precisely machined quartz crystal. When kept under tension, quartz crystals oscillate at a well-defined frequency that depends on the kind of crystal, how it is cut, and the amount of tension. Associated with each crystal are two registers, a **counter** and a **holding register**. Each oscillation of the crystal decrements the counter by one. When the counter gets to zero, an interrupt is generated and the counter is reloaded from the holding register. In this way, it is possible to program a timer to generate an interrupt 60 times a second, or at any other desired frequency. Each interrupt is called one **clock tick**.

When the system is booted and initialized for the very first time, the user is asked for the current time zone, or even time, to convert time to the number of ticks after some known starting date and stored in memory. Most computers have a special battery-backed up CMOS RAM so that the date and time need not be entered on subsequent boots. At every clock tick, the interrupt service procedure adds one to the time stored in memory. In this way, the (software) clock is kept up to date.

With a single computer and a single clock, it does not matter much if this clock is off by a small amount. Since all processes on the machine use the same clock, they will still be internally consistent. For example, if the file `input.c` has time 2151 and file `input.o` has time 2150, `make` will recompile the source file, even if the clock is off by 2 and the true times are 2153 and 2152, respectively. All that really matters are the relative times.

As soon as multiple CPUs are introduced, each with its own clock, the situation changes radically. Although the frequency at which a crystal oscillator runs is usually fairly stable, it is impossible to guarantee that the crystals in

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different computers all run at the same frequency. In practice, when a system has n computers, all n crystals will run at slightly different rates, causing the (software) clocks gradually to get out of sync and give different values when read out. This difference in time values is called **clock skew**. As a consequence of this clock skew, programs that expect the time associated with a file, object, process, or message to be correct and independent of the machine on which it was generated (i.e., which clock it used) can fail, as we saw in the `make` example above.

In some systems (e.g., real-time systems), the actual clock time is important. Under these circumstances, external physical clocks are needed. For reasons of efficiency and redundancy, multiple physical clocks are generally considered desirable, which yields two problems: (1) how do we synchronize them with real-world clocks, and (2) how do we synchronize the clocks with each other?

Note 5.1 (More information: Determining real time)
Before answering these questions, let us digress slightly to see how time is actually measured. It is not nearly as easy as one might think, especially when high accuracy is required. Since the invention of mechanical clocks in the 17th century, time has been measured astronomically. Every day, the sun appears to rise on the eastern horizon, then climbs to a maximum height in the sky, and finally sinks in the west. The event of the sun's reaching its highest apparent point in the sky is called the **transit of the sun**. This event occurs at about noon each day. The interval between two consecutive transits of the sun is called the **solar day**. Since there are 24 hours in a day, each containing 3600 seconds, the **solar second** is defined as exactly $1/86400^{th}$ of a solar day. The geometry of the mean solar day calculation is shown in Figure 5.2.

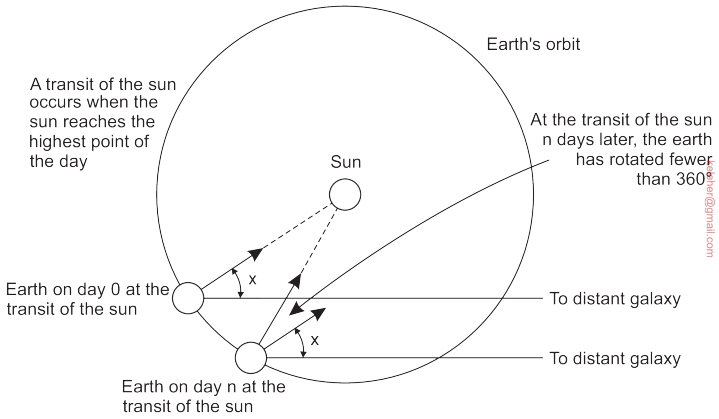


Figure 5.2: Computation of the mean solar day.

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In the 1940s, it was established that the period of the earth's rotation is not constant. The earth is slowing down due to tidal friction and atmospheric drag. Based on studies of growth patterns in ancient coral, geologists now believe that 300 million years ago there were about 400 days per year. The length of the year (the time for one trip around the sun) is not thought to have changed; the day has simply become longer. In addition to this long-term trend, short-term variations in the length of the day also occur, probably caused by turbulence deep in the earth's core of molten iron. These revelations lead astronomers to compute the length of the day by measuring a large number of days and taking the average before dividing by 86,400. The resulting quantity was called the **mean solar second**.

With the invention of the atomic clock in 1948, it became possible to measure time much more accurately, and independent of the wiggling and wobbling of the earth, by counting transitions of the cesium 133 atom. The physicists took over the job of timekeeping from the astronomers and defined the second to be the time it takes the cesium 133 atom to make exactly 9,192,631,770 transitions. The choice of 9,192,631,770 was made to make the atomic second equal to the mean solar second in the year of its introduction. Currently, several laboratories around the world have cesium 133 clocks. Periodically, each laboratory tells the Bureau International de l'Heure (BIH) in Paris how many times its clock has ticked. The BIH averages these to produce **International Atomic Time**, which is abbreviated to **TAI**. Thus TAI is just the mean number of ticks of the cesium 133 clocks since midnight on Jan. 1, 1958 (the beginning of time) divided by 9,192,631,770.

Although TAI is highly stable and available to anyone who wants to go to the trouble of buying a cesium clock, there is a serious problem with it; 86,400 TAI seconds is now about 3 msec less than a mean solar day (because the mean solar day is getting longer all the time). Using TAI for keeping time would mean that over the course of the years, noon would get earlier and earlier, until it would eventually occur in the wee hours of the morning. People might notice this and we could have the same kind of situation as occurred in 1582 when Pope Gregory XIII decreed that 10 days be omitted from the calendar. This event caused riots in the streets because landlords demanded a full month's rent and bankers a full month's interest, while employers refused to pay workers for the 10 days they did not work, to mention only a few of the conflicts. The Protestant countries, as a matter of principle, refused to have anything to do with papal decrees and did not accept the Gregorian calendar for 170 years.

BIH solves the problem by introducing **leap seconds** whenever the discrepancy between TAI and solar time grows to 800 msec. The use of leap seconds is illustrated in Figure 5.3. This correction gives rise to a time system based on constant TAI seconds but which stays in phase with the apparent motion of the sun. This time system is known as **Coordinated Universal Time** abbreviated to **UTC**.

Most electric power companies synchronize the timing of their 60-Hz or 50-Hz clocks to UTC, so when BIH announces a leap second, the power companies raise their frequency to 61 Hz or 51 Hz for 60 or 50 sec, to advance all the clocks in their distribution area. Since 1 sec is a noticeable interval for a computer, an operating system that needs to keep accurate time over a period of years must have special

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software to account for leap seconds as they are announced (unless they use the power line for time, which is usually too crude). The total number of leap seconds introduced into UTC so far is about 30.

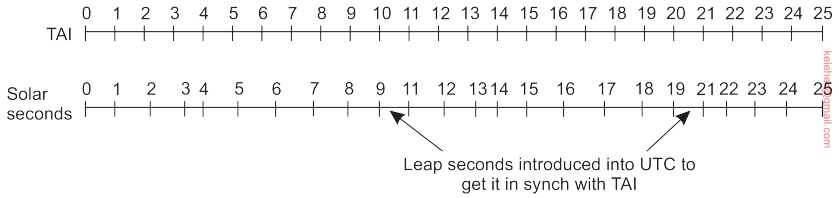


Figure 5.3: TAI seconds are of constant length, unlike solar seconds. Leap seconds are introduced when necessary to keep in phase with the sun.

The basis for keeping global time is a called **Coordinated Universal Time**, but is abbreviated as **UTC**. UTC is the basis of all modern civil timekeeping and is a worldwide standard. To provide UTC to people who need precise time, some 40 shortwave radio stations around the world broadcast a short pulse at the start of each UTC second. The accuracy of these stations is about $\pm$ 1 msec, but due to random atmospheric fluctuations that can affect the length of the signal path, in practice the accuracy is no better than $\pm$ 10 milliseconds.

Several earth satellites also offer a UTC service. The Geostationary Operational Environment Satellite can provide UTC accurately to 0.5 msec, and some other satellites do even better. By combining receptions from several satellites, ground timeservers can be built, offering an accuracy of 50 nanoseconds. UTC receivers are commercially available, and many computers are equipped with one.

5.1.2 Clock synchronization algorithms

If one machine has a UTC receiver, the goal becomes keeping all the other machines synchronized to it. If no machines have UTC receivers, each machine keeps track of its own time, and the goal is to keep all the machines together as well as possible. Many algorithms have been proposed for doing this synchronization. Surveys are provided by Ramanathan et al. [1990], Horauer [2004], Shin et al. [2011], and Levesque and Tipper [2016].

All clocks are based on some harmonic oscillator: an object that resonates at a certain frequency and from which we can subsequently derive time. Atomic clocks are based on the transitions of the cesium 133 atom, which is not only very high, but also very constant. Hardware clocks in most computers use a crystal oscillator based on quartz, which is also capable of producing

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a very high, stable frequency, although not as stable as that of atomic clocks. A software clock in a computer is derived from that computer's hardware clock. In particular, the hardware clock is assumed to cause an interrupt f times per second. When this timer goes off, the interrupt handler adds 1 to a counter that keeps track of the number of ticks (interrupts) since some agreed-upon time in the past. This counter acts as a software clock C, resonating at frequency F.

When the UTC time is t, denote by $C_p(t)$ the value of the software clock on machine p. The goal of **clock synchronization algorithms** is to keep the deviation between the respective clocks of any two machines in a distributed system, within a specified bound, known as the **precision** π:

$$\forall t, \forall \mathsf{p}, \mathsf{q} : |C_p(t) - C_q(t)| \leq \pi$$

Note that precision refers to the deviation of clocks only *between machines* that are part of a distributed system. When considering an external reference point, like UTC, we speak of **accuracy**, aiming to keep it bound to a value α:

$$\forall t, \forall \mathsf{p} : |C_p(t) - t| \leq \alpha$$

The whole idea of clock synchronization is that we keep clocks *precise*, referred to as **internal synchronization** or *accurate*, known as **external synchronization**. A set of clocks that are accurate within bound α, will be precise within bound $\pi = 2\alpha$. However, being precise does not allow us to conclude anything about the accuracy of clocks.

In a perfect world, we would have $C_p(t) = t$ for all p and all t, and thus $\alpha = \pi = 0$. Unfortunately, hardware clocks, and thus also software clocks, are subject to **clock drift**: because their frequency is not perfect and affected by external sources such as temperature, clocks on different machines will gradually start showing different values for time. This is known as the **clock drift rate**: the difference per unit of time from a perfect reference clock. A typical quartz-based hardware clock has a clock drift rate of some 10^{-6} seconds per second, or approximately 31.5 seconds per year. Computer hardware clocks exist that have much lower drift rates.

The specifications of a hardware clock include its maximum clock drift rate ρ. If $F(t)$ denotes the actual oscillator frequency of the hardware clock at time t and F its ideal (constant) frequency, then a hardware clock is living up to its specifications if

$$\forall t : (1 - \rho) \leq \frac{F(t)}{F} \leq (1 + \rho)$$

By using hardware interrupts we are directly coupling a software clock to the hardware clock, and thus also its clock drift rate. In particular, we have that

$$C_p(t) = \frac{1}{F} \int_0^t F(t)dt, \text{ and thus: } \frac{dC_p(t)}{dt} = \frac{F(t)}{F}$$

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which brings us to our goal, namely keeping the software clock drift rate also bounded to ρ:

$$\forall t : 1 - \rho \leq \frac{dC_p(t)}{dt} \leq 1 + \rho$$

Slow, perfect, and fast clocks are shown in Figure 5.4.

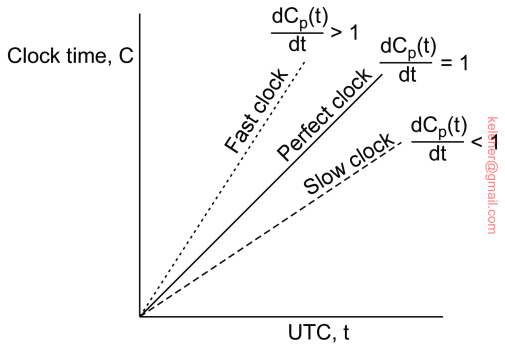


Figure 5.4: The relation between clock time and UTC when clocks tick at different rates.

If two clocks are drifting from UTC in the opposite direction, at a time Δt after they were synchronized, they may be as much as $2\rho \cdot \Delta t$ apart. If the system designers want to guarantee a precision π, that is, that no two clocks ever differ by more than π seconds, clocks must be resynchronized (in software) at least every $\pi/(2\rho)$ seconds. The various algorithms differ in precisely how this resynchronization is done.

Network Time Protocol

A common approach in many protocols, and originally proposed by Cristian [1989], is to let clients contact a timeserver. The latter can accurately provide the current time, for example because it is equipped with a UTC receiver or an accurate clock. The problem, of course, is that when contacting the server, message delays will have outdated the reported time. The trick is to find a good estimation for these delays. Consider the situation sketched in Figure 5.5.

In this case, A will send a request to B, timestamped with value T_1. B, in turn, will record the time of receipt T_2 (taken from its own local clock), and returns a response timestamped with value T_3, and piggybacking the previously recorded value T_2. Finally, A records the time of the response's arrival, T_4. Let us assume that the propagation delays from A to B is roughly the same as B to A, meaning that $\delta T_{req} = T_2 - T_1 \approx T_4 - T_3 = \delta T_{res}$. In that

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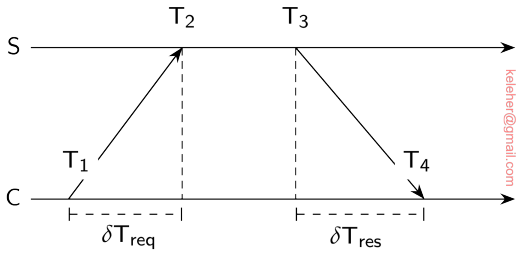


Figure 5.5: Getting the current time from a time server.

case, A can estimate its offset relative to B as

$$\theta = T_3 + \frac{(T_2 - T_1) + (T_4 - T_3)}{2} - T_4 = \frac{(T_2 - T_1) + (T_3 - T_4)}{2}$$

Of course, time is not allowed to run backward. If A's clock is fast, $\theta < 0$, meaning that A should, in principle, set its clock backward. This is not allowed as it could cause serious problems, such as an object file compiled just after the clock change having a time earlier than the source which was modified just before the clock change.

Such a change must be introduced gradually. One way is as follows. Suppose that the timer is set to generate 100 interrupts per second. Normally, each interrupt would add 10 msec to the time. When slowing down, the interrupt routine adds only 9 msec each time until the correction has been made. Similarly, the clock can be advanced gradually by adding 11 msec at each interrupt instead of jumping it forward all at once.

In the case of the **Network Time Protocol** (**NTP**), this protocol is set up pairwise between servers. In other words, B will also probe A for its current time. The offset θ is computed as given above, along with the estimation δ for the delay:

$$\delta = \frac{(T_4 - T_1) - (T_3 - T_2)}{2}$$

Eight pairs of (θ, δ) values are buffered, finally taking the minimal value found for δ as the best estimation for the delay between the two servers, and subsequently the associated value θ as the most reliable estimation of the offset.

Applying NTP symmetrically should, in principle, also let B adjust its clock to that of A. However, if B's clock is known to be more accurate, then such an adjustment would be foolish. To solve this problem, NTP divides servers into strata. A server with a **reference clock** such as a UTC receiver or an atomic clock, is known to be a **stratum-1 server** (the clock itself is said to operate at stratum 0). When A contacts B, it will adjust only its time if its own stratum level is higher than that of B. Moreover, after the synchronization, A's

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stratum level will become one higher than that of B. In other words, if B is a stratum-k server, then A will become a stratum-$(k+1)$ server if its original stratum level was already larger than k. Due to the symmetry of NTP, if A's stratum level was *lower* than that of B, B will adjust itself to A.

There are many important features about NTP, of which many relate to identifying and masking errors, but also security attacks (see, e.g., Malhotra et al. [2016]). Many security issues, such as also described by Levesque and Tipper [2016], can be dealt with by establishing a secure channel between a time server and its client. NTP was originally described in [Mills, 1992] and is known to achieve (worldwide) accuracy in the range of 1–50 msec. A detailed description of NTP can be found in [Mills, 2011].

Clock synchronization in wireless networks

An important advantage of more traditional distributed systems is that we can easily and efficiently deploy timeservers. Moreover, most machines can contact each other, allowing for a relatively simple dissemination of information. These assumptions are no longer valid in many wireless networks, notably sensor networks. Nodes are resource constrained, and multihop routing is expensive. In addition, it is often important to optimize algorithms for energy consumption. These and other observations have led to the design of very different clock synchronization algorithms for wireless networks. In the following, we consider one specific solution. Sivrikaya and Yener [2004] provide a brief overview of other solutions. An extensive survey can be found in [Sundararaman et al., 2005].

Reference broadcast synchronization (**RBS**) is a clock synchronization protocol that is quite different from other proposals [Elson et al., 2002]. First, the protocol does not assume that there is a single node with an accurate account of the actual time available. Instead of aiming to provide all nodes UTC time, it aims at merely internally synchronizing the clocks. Second, the solutions we have discussed so far are designed to bring the sender and receiver into sync, essentially following a two-way protocol. RBS deviates from this pattern by letting only the receivers synchronize, keeping the sender out of the loop.

In RBS, a sender broadcasts a reference message that allows its receivers to adjust their clocks. A key observation is that in a sensor network the time to propagate a signal to other nodes is roughly constant, provided no multi-hop routing is assumed. Propagation time in this case is measured from the moment that a message leaves the network interface of the sender. As a consequence, two important sources of variation in message transfer no longer play a role in estimating delays: the time spent to construct a message, and the time spent to access the network. This principle is shown in Figure 5.6.

Note that in protocols such as NTP, a timestamp is added to the message before it is passed on to the network interface. Furthermore, as wireless

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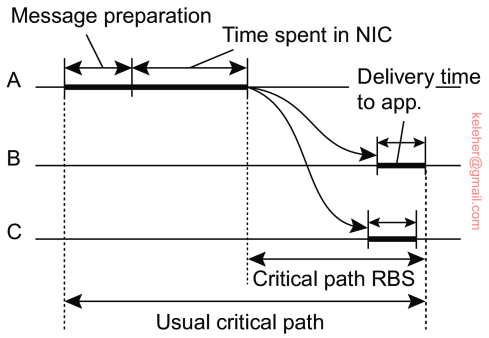


Figure 5.6: The usual critical path and the one used in RBS in determining network delays.

networks are based on a contention protocol, there is generally no saying how long it will take before a message can actually be transmitted. These factors of nondeterminism are eliminated in RBS. What remains is the delivery time at the receiver, but this time varies considerably less than the network-access time.

The idea underlying RBS is simple: when a node broadcasts a reference message m, each node p simply records the time $T_{p,m}$ that it received m. Note that $T_{p,m}$ is read from p's local clock. Ignoring clock skew, two nodes p and q can exchange each other's delivery times to estimate their mutual, relative offset:

$$Offset[p,q] = \frac{\sum_{k=1}^{M}(T_{p,k} - T_{q,k})}{M}$$

where M is the total number of reference messages sent. This information is important: node p will know the value of q's clock relative to its own value. Moreover, if it simply stores these offsets, there is no need to adjust its own clock, which saves energy.

Unfortunately, clocks can drift apart. The effect is that simply computing the average offset as done above will not work: the last values sent are simply less accurate than the first ones. Moreover, as time goes by, the offset will presumably increase. Elson et al. [2002] use a simple algorithm to compensate for this: instead of computing an average, they apply standard linear regression to compute the offset as a function:

$$Offset[p,q](t) = \alpha t + \beta$$

The constants α and β are computed from the pairs $(T_{p,k}, T_{q,k})$. This new form allows a much more accurate computation of q's current clock value by node p, and vice versa.

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Note 5.2 (More information: How important is an accurate account of time?)
So why is time such a big deal for distributed systems? As we shall discuss in the remainder of this chapter, reaching consensus on a global ordering of events is what we really want, and this can be achieved without any notion of global absolute time. However, as will become clear, alternative methods for distributed coordination do not come easy.

Life would be much simpler if processes in a distributed system could time-stamp their events with infinite precision. Although infinite precision is asking too much, we can come practically close. Researchers at Google were confronted with the fact that their customers would really like to make use of a globally distributed database that supported transactions. Such a database would need to serve massive numbers of clients, rendering the use of, for example, a central transaction processing monitor as we discussed in Section 1.3.2, infeasible. Instead, for their Spanner system, Google decided to implement a true-time service, called TrueTime [Corbett et al., 2013]. This service provides three operations:

Operation	Result
`TT.now()`	A time interval $[T_{lwb}, T_{upb}]$ with $T_{lwb} < T_{upb}$
`TT.after(t)`	True if timestamp `t` has definitely passed
`TT.before(t)`	True if timestamp `t` has definitely not arrived

The most important aspect is that T_{lwb} and T_{upb} are *guaranteed* bounds. Of course, if $\epsilon = T_{upb} - T_{lwb}$ is large, say 1 hour, then implementing the service is relatively easy. Impressively enough, $\epsilon = 6ms$. To achieve this accuracy, the TrueTime service makes use of *time-master* machines of which there are several per data center. Time-slave daemons run on every machine in a data center and query multiple time masters, including ones from other data centers, very similar to what we described for NTP. Many time masters are equipped with accurate GPS receivers, while many others are independently equipped with atomic clocks. The result is a collection of time sources with a high degree of mutual independence (which is important for reasons of fault tolerance). Using a version of an algorithm developed by Marzullo and Owicki [1983], outliers are kept out of the computations. Meanwhile, the performance of TrueTime is continuously monitored and "bad" time machines are (manually) removed to give at least very high guarantees for the accuracy of the TrueTime service.

With a guaranteed accuracy of 6 milliseconds, building a transactional system becomes much easier: transactions can actually be timestamped, even by different servers, with the restriction that timestamping may need to be delayed for ϵ time units. More precisely, to know for sure that a transaction has committed, reading the resulting data may impose a wait for ϵ units. This is achieved by pessimistically assigning a timestamp to a transaction that writes data to the global database and making sure that clients never see any changes before the assigned timestamp (which is relatively easy to implement).

There are many details to this approach, which can be found in [Corbett et al., 2013]. As we are still dealing with a time interval, taking more traditional

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ordering mechanisms into account it is possible to improve results, as explained by Demirbas and Kulkarni [2013].

5.2 Logical clocks

Clock synchronization is naturally related to time, although it may not be necessary to have an accurate account of the real time: it may be sufficient that every node in a distributed system agrees on *a* current time. We can go one step further. For running `make` it is adequate that two nodes agree that `input.o` is outdated by a new version of `input.c`, for example. In this case, keeping track of each other's events (such as a producing a new version of `input.c`) is what matters. For these algorithms, it is conventional to speak of the clocks as **logical clocks**.

In a seminal paper, Lamport [1978] showed that although clock synchronization is possible, it need not be absolute. If two processes do not interact, it is not necessary that their clocks be synchronized because the lack of synchronization would not be observable and thus could not cause problems. Furthermore, he pointed out that what usually matters is not that all processes agree on exactly what time it is, but rather that they agree on the *order in which events occur*. In the `make` example, what counts is whether `input.c` is older or newer than `input.o`, not their respective absolute creation times.

5.2.1 Lamport's logical clocks

To synchronize logical clocks, Lamport defined a relation called **happens-before**. The expression $a \rightarrow b$ is read "event a happens before event b" and means that all processes agree that first event a occurs, then afterward, event b occurs. The happens-before relation can be observed directly in two situations:

1. If a and b are events in the same process, and a occurs before b, then $a \rightarrow b$ is true.
2. If a is the event of a message being sent by one process, and b is the event of the message being received by another process, then $a \rightarrow b$ is also true. A message cannot be received before it is sent, or even at the same time it is sent, since it takes a finite, nonzero amount of time to arrive.

Happens-before is a transitive relation, so if $a \rightarrow b$ and $b \rightarrow c$, then $a \rightarrow c$. If two events, x and y, happen in different processes that do not exchange messages (not even indirectly via third parties), then $x \rightarrow y$ is not true, but neither is $y \rightarrow x$. These events are said to be **concurrent**, which simply means

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that nothing can be said (or need be said) about when the events happened or which event happened first.

What we need is a way of measuring a notion of time such that for every event, a, we can assign it a time value $C(a)$ on which all processes agree. These time values must have the property that if $a \rightarrow b$, then $C(a) < C(b)$. To rephrase the conditions we stated earlier, if a and b are two events within the same process and a occurs before b, then $C(a) < C(b)$. Similarly, if a is the sending of a message by one process and b is the reception of that message by another process, then $C(a)$ and $C(b)$ must be assigned in such a way that everyone agrees on the values of $C(a)$ and $C(b)$ with $C(a) < C(b)$. In addition, the clock time, C, must always go forward (increasing), never backward (decreasing). Corrections to time can be made by adding a positive value, never by subtracting one.

Now let us look at the algorithm Lamport proposed for assigning times to events. Consider the three processes depicted in Figure 5.7. The processes run on different machines, each with its own clock. For the sake of argument, we assume that a clock is implemented as a software counter: the counter is incremented by a specific value every T time units. However, the value by which a clock is incremented differs per process. The clock in process P_1 is incremented by 6 units, 8 units in process P_2, and 10 units in process P_3, respectively. (Below, we explain that Lamport clocks are, in fact, **event counters**, which explains why their value may differ between processes.)

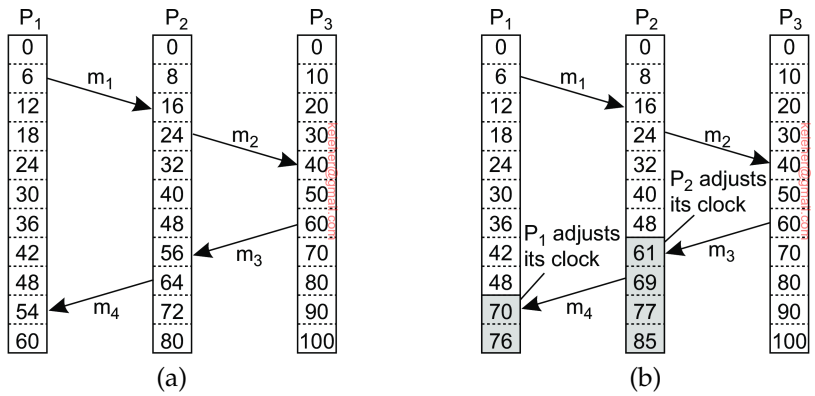


Figure 5.7: (a) Three processes, each with its own (logical) clock. The clocks run at different rates. (b) Lamport's algorithm corrects their values.

At time 6, process P_1 sends message m_1 to process P_2. How long this message takes to arrive depends on whose clock you believe. In any event, the clock in process P_2 reads 16 when it arrives. If the message carries the starting time, 6, in it, process P_2 will conclude that it took 10 ticks to make the journey. This value is certainly possible. According to this reasoning, message m_2 from P_2 to P_3 takes 16 ticks, again a plausible value.

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Now consider message m_3. It leaves process P_3 at 60 and arrives at P_2 at 56. Similarly, message m_4 from P_2 to P_1 leaves at 64 and arrives at 54. These values are clearly impossible. It is this situation that must be prevented.

Lamport's solution follows directly from the happens-before relation. Since m_3 left at 60, it must arrive at 61 or later. Therefore, each message carries the sending time according to the sender's clock. When a message arrives and the receiver's clock shows a value before the time the message was sent, the receiver fast forwards its clock to be one more than the sending time. In Figure 5.7, we see that m_3 now arrives at 61. Similarly, m_4 arrives at 70.

Let us formulate this procedure more precisely. At this point, it is important to distinguish three different layers of software, as we already encountered in Section 2.2: the network, a middleware layer, and an application layer, as shown in Figure 5.8. What follows is typically part of the middleware layer.

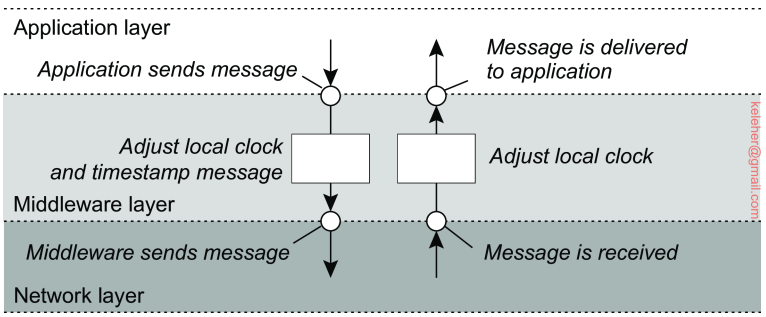


Figure 5.8: The positioning of Lamport's logical clocks in distributed systems.

To implement Lamport's logical clocks, each process P_i maintains a *local* counter C_i. These counters are updated according to the following steps [Raynal and Singhal, 1996]:

1. Before executing an event (i.e., sending a message over the network, delivering a message to an application, or some other internal event), P_i increments C_i: $C_i \leftarrow C_i + 1$.
2. When process P_i sends a message m to process P_j, it sets m's timestamp $ts(m)$ equal to C_i after having executed the previous step.
3. Upon the receipt of a message m, process P_j adjusts its own local counter as $C_j \leftarrow \max\{C_j, ts(m)\}$ after which it then executes the first step and delivers the message to the application.

In some situations, an additional requirement is desirable: no two events ever occur at the same time. To achieve this goal, we also use the unique process identifier to break ties and use tuples instead of only the counter's values. For example, an event at time 40 at process P_i will be timestamped as $\langle 40, i\rangle$. If we also have an event $\langle 40, j\rangle$ and $i < j$, then $\langle 40, i\rangle < \langle 40, j\rangle$.

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Note that by assigning the event time $C(a) \leftarrow C_i(a)$ if a happened at process P_i at time $C_i(a)$, we have a distributed implementation of the global time value we were initially seeking for; we have thus constructed a **logical clock**.

Example: Totally ordered multicasting

As an application of Lamport's logical clocks, consider the situation in which a database has been replicated across several sites. For example, to improve query performance, a bank may place copies of an account database in two different cities, say New York and San Francisco. A query is always forwarded to the nearest copy. The price for a fast response to a query is partly paid in higher update costs because each update operation must be carried out at each replica.

In fact, there is a more stringent requirement regarding updates. Assume a customer in San Francisco wants to add $100 to her account, which currently contains $1,000. At the same time, a bank employee in New York initiates an update by which the customer's account is to be increased with 1 percent interest. Both updates should be carried out at both copies of the database. However, due to communication delays in the underlying network, the updates may arrive in the order as shown in Figure 5.9.

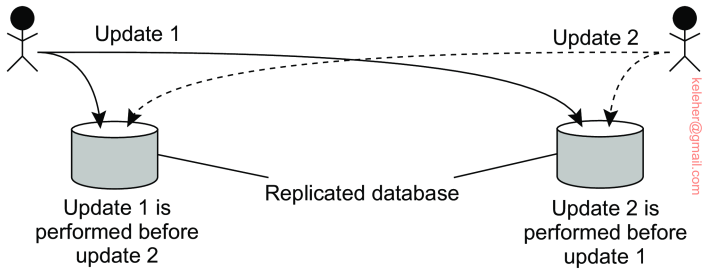


Figure 5.9: Updating a replicated database and leaving it in an inconsistent state.

The customer's update operation is performed in San Francisco before the interest update. In contrast, the copy of the account in the New York replica is first updated with the 1 percent interest, and after that with the $100 deposit. Consequently, the San Francisco database will record a total amount of $1,111, whereas the New York database records $1,110.

The problem that we are faced with is that the two update operations should have been performed in the same order at each copy. Although it makes a difference whether the deposit is processed before the interest update or the other way around, which order is followed is not essential from a consistency perspective. The important issue is that both copies should be the

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same after the updates have taken place. In general, situations such as these require a **totally ordered multicast**, that is, a multicast operation by which all messages are delivered in the same order to each receiver. Lamport's logical clocks can be used to implement totally ordered multicasts in a completely distributed fashion.

Consider a group of processes multicasting messages to each other. Each message is always timestamped with the current (logical) time of its sender. When a message is multicasted, it is conceptually also sent to the sender. In addition, we assume that messages from the same sender are received in the order they were sent, and that no messages are lost.

When a process receives a message, it is put into a local queue, ordered according to its timestamp. The receiver multicasts an acknowledgment to the other processes. Note that if we follow Lamport's algorithm for adjusting local clocks, the timestamp of the received message is lower than the timestamp of the acknowledgment. The interesting aspect of this approach is that all processes will eventually have the same copy of the local queue (provided no messages are removed).

A process can deliver a queued message to the application it is running only when that message is at the head of the queue and has been acknowledged by each other process. At that point, the message is removed from the queue and handed over to the application; the associated acknowledgments can simply be removed. Because each process has the same copy of the queue, all messages are delivered in the same order everywhere. In other words, we have established totally ordered multicasting. We leave it as an exercise to the reader to figure out that it is not strictly necessary that each multicast message has been explicitly acknowledged. It is sufficient that a process *reacts* to an incoming message, either by returning an acknowledgment or sending its own multicast message.

Totally ordered multicasting is an important vehicle for replicated services, where the replicas are kept consistent by letting them execute the same operations in the same order everywhere. As the replicas essentially follow the same transitions in the same finite state machine, it is also known as **state machine replication** [Schneider, 1990].

Note 5.3 (Advanced: Using Lamport clocks to achieve mutual exclusion)
To further illustrate the usage of Lamport's clocks, let us see how we can use the previous algorithm for totally ordered multicasting to establish access to what is commonly known as a **critical region**: a section of code that can be executed by at most one process at a time. This algorithm is very similar to the one for multicasting, as essentially all processes need to agree on the order by which processes are allowed to enter their critical region.

Figure 5.10(a) shows the code that each process executes when requesting, releasing, or allowing access to the critical region (again, omitting details). Each

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process maintains a request queue as well as a logical clock. To enter the critical region, a call to `requestToEnter` is made, which results in inserting an `ENTER` message with timestamp `(clock,procID)` into the local queue and sending that message to the other processes. The operation `cleanupQ` essentially sorts the queue. We return to it shortly.

```
1  class Process:
2    def __init__(self, chanID, procID, procIDSet):
3      self.chan.join(procID)
4      self.procID     = int(procID)
5      self.otherProcs.remove(self.procID)
6      self.queue      = []                    # The request queue
7      self.clock      = 0                     # The current logical clock
8
9    def requestToEnter(self):
10     self.clock = self.clock + 1                          # Increment clock value
11     self.queue.append((self.clock, self.procID, ENTER)) # Append request to q
12     self.cleanupQ()                                      # Sort the queue
13     self.chan.sendTo(self.otherProcs, (self.clock, self.procID, ENTER)) # Send request
14
15   def ackToEnter(self, requester):
16     self.clock = self.clock + 1                          # Increment clock value
17     self.chan.sendTo(requester, (self.clock, self.procID, ACK)) # Permit other
18
19   def release(self):
20     tmp = [r for r in self.queue[1:] if r[2] == ENTER]  # Remove all ACKs
21     self.queue = tmp                                     # and copy to new queue
22     self.clock = self.clock + 1                          # Increment clock value
23     self.chan.sendTo(self.otherProcs, (self.clock, self.procID, RELEASE)) # Release
24
25   def allowedToEnter(self):
26     commProcs = set([req[1] for req in self.queue[1:]]) # See who has sent a message
27     return (self.queue[0][1] == self.procID and len(self.otherProcs) == len(commProcs))
```

Figure 5.10: (a) Using Lamport's logical clocks for mutual exclusion.

When a process P receives an `ENTER` message from process Q, it can simply acknowledge that Q can enter its critical region, even if P wants to do so as well. In the latter case, P's `ENTER` request will have a lower logical timestamp than the `ACK` message sent by P to Q, meaning that P's request will have been inserted into Q's queue *before* P's `ACK` message.

Finally, when a process leaves its critical region, it calls `release`. It cleans up its local queue by removing all received `ACK` messages, leaving only the `ENTER` requests from other processes. It then multicasts a `RELEASE` message.

To actually enter a critical region, a process will have to repeatedly call `allowedToEnter` and when returned `False`, will have to block on a next incoming message. The operation `allowedToEnter` does what is to be expected: it checks if the calling process's `ENTER` message is at the head of the queue, and sees if all other processes have sent a message as well. The latter is encoded through the set

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commProcs, which contains the procIDs of all processes having sent a message by inspecting all messages in the local queue from the second position and onwards.

```
1   def receive(self):
2     msg = self.chan.recvFrom(self.otherProcs)[1]        # Pick up any message
3     self.clock = max(self.clock, msg[0])                # Adjust clock value...
4     self.clock = self.clock + 1                         # ...and increment
5     if msg[2] == ENTER:
6       self.queue.append(msg)                            # Append an ENTER request
7       self.ackToEnter(msg[1])                           # and unconditionally allow
8     elif msg[2] == ACK:
9       self.queue.append(msg)                            # Append a received ACK
10    elif msg[2] == RELEASE:
11      del(self.queue[0])                                # Just remove first message
12    self.cleanupQ()                                     # And sort and cleanup
```

Figure 5.10: (b) Using Lamport's logical clocks for mutual exclusion: handling incoming requests.

What to do when a message is received is shown in Figure 5.10(b). First, the local clock is adjusted according to the rules for Lamport's logical clocks explained above. When receiving an ENTER or ALLOW message, that message is simply inserted into the queue. An entry request is always acknowledged, as we just explained. When a RELEASE message is received, the original ENTER request is removed. Note that this request is at the head of the queue. Thereafter, the queue is cleaned up again.

At this point, note that if we would clean up the queue by only sorting it, we may get into trouble. Suppose that processes P and Q want to enter their respective critical regions at roughly the same time, but that P is allowed to go first based on logical-clock values. P may find Q's request in its queue, along with ENTER or ALLOW messages from other processes. If its own request is at the head of its queue, P will proceed and enter its critical region. However, Q will also send an ALLOW message to P as well, in addition to its original ENTER message. That ALLOW message may arrive *after* P had already entered its critical region, but *before* ENTER messages from other processes. When Q eventually enters, and leaves its critical region, Q's RELEASE message would result in removing Q's original ENTER message, but not the ALLOW message it had previously sent to P. By now, that message is at the head of P's queue, effectively blocking the entrance to the critical region of other processes in P's queue. Cleaning up the queue thus also involves removing old ALLOW messages.

5.2.2 Vector clocks

Lamport's logical clocks lead to a situation where all events in a distributed system are totally ordered with the property that if event a happened before event b, then a will also be positioned in that ordering before b, that is, $C(a) < C(b)$.

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However, with Lamport clocks, nothing can be said about the relationship between two events a and b by merely comparing their time values $C(a)$ and $C(b)$, respectively. In other words, if $C(a) < C(b)$, then this does not necessarily imply that a indeed happened before b. Something more is needed for that.

To explain, consider the messages as sent by the three processes shown in Figure 5.11. Denote by $T_{snd}(\mathsf{m_i})$ the logical time at which message $\mathsf{m_i}$ was sent, and likewise, by $T_{rcv}(\mathsf{m_i})$ the time of its receipt. By construction, we know that for each message $T_{snd}(\mathsf{m_i}) < T_{rcv}(\mathsf{m_i})$. But what can we conclude in general from $T_{rcv}(\mathsf{m_i}) < T_{snd}(\mathsf{m_j})$ for different messages $\mathsf{m_i}$ and $\mathsf{m_j}$?

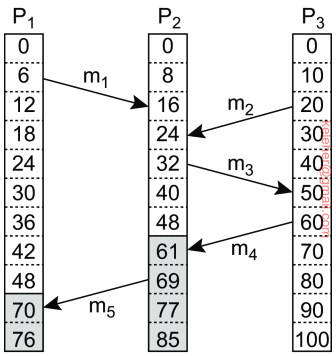


Figure 5.11: Concurrent message transmission using logical clocks.

In the case for which $\mathsf{m_i} = \mathsf{m_1}$ and $\mathsf{m_j} = \mathsf{m_3}$, we know that these values correspond to events that took place at process $\mathsf{P_2}$, meaning that $\mathsf{m_3}$ was indeed sent after the receipt of message $\mathsf{m_1}$. This may indicate that the sending of message $\mathsf{m_3}$ depended on what was received through message $\mathsf{m_1}$. At the same time, we also know that $T_{rcv}(\mathsf{m_1}) < T_{snd}(\mathsf{m_2})$. However, as far as we can tell from Figure 5.11, the sending of $\mathsf{m_2}$ has nothing to do with the receipt of $\mathsf{m_1}$.

The problem is that Lamport clocks do not capture **causality**. In practice, causality is captured by means of **vector clocks**. To better understand where these come from, we follow the explanation as given by Baquero and Preguica [2016]. In fact, tracking causality is simple if we assign each event a unique name such as the combination of a process ID and a locally incrementing counter: $\mathsf{p_k}$ is the kth event that happened at process P. The problem then boils down to keeping track of **causal histories**. For example, if two local events happened at process P, then the causal history $H(\mathsf{p_2})$ of event $\mathsf{p_2}$ is $\{\mathsf{p_1}, \mathsf{p_2}\}$.

Now assume that process P sends a message to process Q (which is an event at P and thus recorded as $\mathsf{p_k}$ for some k), and that at the time of arrival (an event for Q), the most recent causal history of Q was $\{\mathsf{q_1}\}$. To track

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causality, P also sends its most recent causal history (assume it was $\{p_1, p_2\}$, extended with p_3 expressing the sending of the message). Upon arrival, Q records the event (q_2), and merges the two causal histories into a new one: $\{p_1, p_2, p_3, q_1, q_2\}$.

Checking whether an event p causally precedes an event q can be done by checking whether $H(p) \subset H(q)$ (i.e., it should be a *proper* subset). In fact, with our notation, it even suffices to check whether $p \in H(q)$, assuming that q is always the last local event in $H(q)$.

The problem with causal histories, is that their representation is not very efficient. However, there is no need to keep track of all successive events from the same process: the last one will do. If we subsequently assign an index to each process, we can represent a causal history as a vector, in which the jth entry represents the number of events that happened at process P_j. Causality can then be captured by means of **vector clocks**, which are constructed by letting each process P_i maintain a vector VC_i with the following two properties:

1. $VC_i[i]$ is the number of events that have occurred so far at P_i. In other words, $VC_i[i]$ is the local logical clock at process P_i.
2. If $VC_i[j] = k$ then P_i knows that k events have occurred at P_j. It is thus P_i's knowledge of the local time at P_j.

The first property is maintained by incrementing $VC_i[i]$ at the occurrence of each new event that happens at process P_i. The second property is maintained by piggybacking vectors along with messages that are sent. In particular, the following steps are performed:

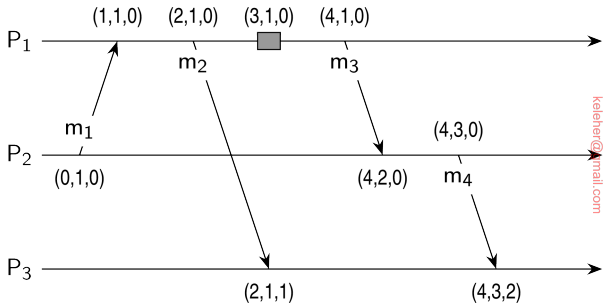
1. Before executing an event (i.e., sending a message over the network, delivering a message to an application, or some other internal event), P_i executes $VC_i[i] \leftarrow VC_i[i] + 1$. This is equivalent to recording a new event that happened at P_i.
2. When process P_i sends a message m to P_j, it sets m's (vector) timestamp $ts(m)$ equal to VC_i after having executed the previous step (i.e., it also records the sending of the message as an event that takes place at P_i).
3. Upon the receipt of a message m, process P_j adjusts its own vector by setting $VC_j[k] \leftarrow \max\{VC_j[k], ts(m)[k]\}$ for each k (which is equivalent to merging causal histories), after which it executes the first step (recording the receipt of the message) and then delivers the message to the application.

Note that if an event a has timestamp $ts(a)$, then $ts(a)[i] - 1$ denotes the number of events processed at P_i that causally precede a. As a consequence, when P_j receives a message m from P_i with timestamp $ts(m)$, it knows about the number of events that have occurred at P_i that causally preceded the

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sending of m. More important, however, is that P_j is also told how many events at *other* processes have taken place, known to P_i, before P_i sent message m. In other words, timestamp $ts(m)$ tells the receiver how many events in other processes have preceded the sending of m, and on which m may causally depend.

To see what this means, consider Figure 5.12 which shows three processes. In Figure 5.12(a), P_2 sends a message m_1 at logical time $VC_2 = (0,1,0)$ to process P_1. Message m_1 thus receives timestamp $ts(m_1) = (0,1,0)$. Upon its receipt, P_1 adjusts its logical time to $VC_1 \leftarrow (1,1,0)$ and delivers it. Message m_2 is sent by P_1 to P_3 with timestamp $ts(m_2) = (2,1,0)$. Before P_1 sends another message, m_3, an event happens at P_1, eventually leading to timestamping m_3 with value $(4,1,0)$. After receiving m_3, process P_2 sends message m_4 to P_3, with timestamp $ts(m_4) = (4,3,0)$.



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Figure 5.12: Capturing potential causality when exchanging messages.

Now consider the situation shown in Figure 5.12(b). Here, we have delayed sending message m_2 until after message m_3 has been sent, and after the event had taken place. It is not difficult to see that $ts(m_2) = (4,1,0)$, while $ts(m_4) = (2,3,0)$. Compared to Figure 5.12(a), we have the following situation:

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Situation	$ts(m_2)$	$ts(m_4)$	$ts(m_2) < ts(m_4)$	$ts(m_2) > ts(m_4)$	Conclusion
Figure 5.12(a)	$(2,1,0)$	$(4,3,0)$	Yes	No	m_2 may causally precede m_4
Figure 5.12(b)	$(4,1,0)$	$(2,3,0)$	No	No	m_2 and m_4 may conflict

We use the notation $ts(a) < ts(b)$ if and only if for all k, $ts(a)[k] \leq ts(b)[k]$ and there is at least one index k' for which $ts(a)[k'] < ts(b)[k']$. Thus, by using vector clocks, process P_3 can detect whether m_4 may be causally dependent on m_2, or whether there may be a potential conflict. Note, by the way, that without knowing the actual information contained in messages, it is not possible to state with certainty that there is indeed a causal relationship, or perhaps a conflict.

Note 5.4 (Advanced: Enforcing causal communication)
Using vector clocks, it is now possible to ensure that a message is delivered only if all messages that may have causally preceded it have been received as well. To enable such a scheme, we will assume that messages are multicast within a group of processes. Note that this **causally ordered multicasting** is weaker than totally ordered multicasting. Specifically, if two messages are not in any way related to each other, we do not care in which order they are delivered to applications. They may even be delivered in different order at different locations.

For enforcing causal message delivery, we assume that clocks are adjusted only when sending and *delivering* messages (note, again, that messages are not adjusted when they are *received* by a process, but only when they are delivered to an application). In particular, only upon sending a message, will process P_i increment $VC_i[i]$ by 1. Only when it delivers a message m with timestamp $ts(m)$, will it adjust $VC_i[k]$ to $\max\{VC_i[k], ts(m)[k]\}$ for each k.

Now suppose that P_j receives a message m from P_i with (vector) timestamp $ts(m)$. The delivery of the message to the application layer will then be delayed until the following two conditions are met:

1. $ts(m)[i] = VC_j[i] + 1$
2. $ts(m)[k] \leq VC_j[k]$ for all $k \neq i$

The first condition states that m is the next message that P_j was expecting from process P_i. The second condition states that P_j has delivered all the messages that have been delivered by P_i when it sent message m. Note that there is no need for process P_j to delay the delivery of its own messages.

As an example, consider three processes P_1, P_2, and P_3 as shown in Figure 5.13. At local time $(1,0,0)$, P_1 multicasts message m to the other two processes. Note that $ts(m) = (1,0,0)$. Its receipt and subsequent delivery by P_2, will bring the logical clock at P_2 to $(1,0,0)$, effectively indicating that it has received one message from P_1, has itself sent no message so far, and has not yet delivered a message from P_3. P_2 then decides to multicast m^*, at updated time $(1,1,0)$, which arrives at P_3 *sooner* than m.

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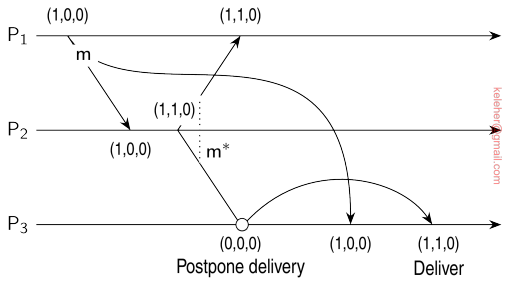


Figure 5.13: Enforcing causal communication.

When comparing the timestamp of m with its current time, which is $(0,0,0)$, P_3 concludes that it is still missing a message from P_1 which P_2 apparently had delivered before sending m^*. P_3 therefore, decides to postpone the delivery of m^* (and will also not adjust its local, logical clock). Later, after m has been received and delivered by P_3, which brings its local clock to $(1,0,0)$, P_3 can deliver message m^* and also update its clock.

A note on ordered message delivery Some middleware systems, notably ISIS and its successor Horus [Birman and van Renesse, 1994], provide support for totally ordered and causally ordered (reliable) multicasting. There has been some controversy whether such support should be provided as part of the message-communication layer, or whether applications should handle ordering (see, e.g., Cheriton and Skeen [1993]; Birman [1994]). Matters have not been settled, but more important is that the arguments still hold today.

There are two main problems with letting the middleware deal with message ordering. First, because the middleware cannot tell what a message actually contains, only *potential* causality is captured. For example, two messages from the same sender that are completely independent will always be marked as causally related by the middleware layer. This approach is overly restrictive and may lead to efficiency problems.

A second problem is that not all causality may be captured. Consider some digital chat room. Suppose Alice posts a message. If she then phones Bob telling about what she just wrote, Bob may post another message as a reaction without having seen Alice's posting. In other words, there is a causality between Bob's posting and that of Alice due to *external* communication. This causality is not captured by the chat room system.

In essence, ordering issues, like many other application-specific communication issues, can be adequately solved by looking at the application for which communication is taking place. This is also known as the **end-to-end principle** in systems design [Saltzer et al., 1984]. A drawback of having only application-level solutions is that a developer is forced to concentrate on issues that do not immediately relate to the core functionality of the application. For example, ordering may

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not be the most important problem when developing a messaging system such as the one for a chat room. In that case, having an underlying communication layer handle ordering may turn out to be convenient. We will come across the end-to-end argument several times.

5.3 Mutual exclusion

Fundamental to distributed systems is the concurrency and collaboration among multiple processes. In many cases, this also means that processes will need to simultaneously access the same resources. To prevent that such concurrent accesses corrupt the resource, or make it inconsistent, solutions are needed to grant mutual exclusive access by processes. In this section, we take a look at some important and representative distributed algorithms that have been proposed. Surveys of distributed algorithms for mutual exclusion are provided by Saxena and Rai [2003] and Velazquez [1993]. Various algorithms are also presented in [Kshemkalyani and Singhal, 2008].

5.3.1 Overview

Distributed mutual exclusion algorithms can be classified into two different categories. In **token-based solutions** mutual exclusion is achieved by passing a special message between the processes, known as a **token**. There is only one token available, and who ever has that token is allowed to access the shared resource. When finished, the token is passed on to a next process. If a process having the token is not interested in accessing the resource, it passes it on.

Token-based solutions have a few important properties. First, depending on how the processes are organized, they can fairly easily ensure that every process will get a chance at accessing the resource. In other words, they provide guarantees for **safety** by avoiding what is known as **starvation**. Second, **deadlocks** by which several processes are indefinitely waiting for each other to proceed, can easily be avoided, contributing to their simplicity. The main drawback of token-based solutions is a rather serious one: when the token is lost (e.g. because the process holding it crashed), an intricate distributed procedure needs to be started to ensure that a new token is created, but above all, that it is also the only token.

As an alternative, many distributed mutual exclusion algorithms follow a **permission-based approach**. In this case, a process wanting to access the resource first requires the permission from other processes. There are many ways toward granting such permission and in the sections that follow we will consider a few of them.

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5.3.2 A centralized algorithm

A straightforward way to achieve mutual exclusion in a distributed system is to simulate how it is done in a one-processor system. One process is elected as the coordinator. Whenever a process wants to access a shared resource, it sends a request message to the coordinator stating which resource it wants to access and asking for permission. If no other process is currently accessing that resource, the coordinator sends back a reply granting permission, as shown in Figure 5.14(a). When the reply arrives, the requester can go ahead.

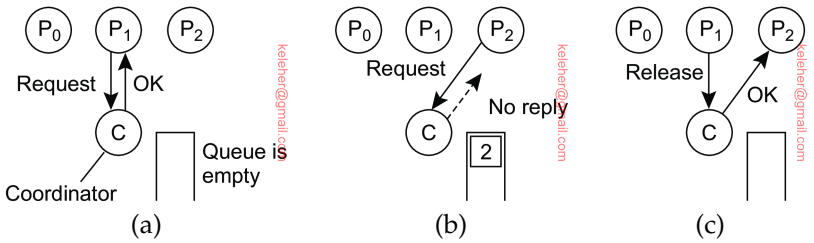


Figure 5.14: (a) Process P_1 asks for permission to access a shared resource. Permission is granted. (b) Process P_2 asks permission to access the same resource, but receives no reply. (c) When P_1 releases the resource, the coordinator replies to P_2.

Now suppose that another process, P_2 in Figure 5.14(b) asks for permission to access the resource. The coordinator knows that a different process is already at the resource, so it cannot grant permission. The exact method used to deny permission is system dependent. In Figure 5.14(b) the coordinator just refrains from replying, thus blocking process P_2, which is waiting for a reply. Alternatively, it could send a reply saying "permission denied." Either way, it queues the request from P_2 for the time being and waits for more messages.

When process P_1 is finished with the resource, it sends a message to the coordinator releasing its exclusive access, as shown in Figure 5.14(c). The coordinator takes the first item off the queue of deferred requests and sends that process a grant message. If the process was still blocked (i.e., this is the first message to it), it unblocks and accesses the resource. If an explicit message has already been sent denying permission, the process will have to poll for incoming traffic or block later. Either way, when it sees the grant, it can go ahead as well.

It is easy to see that the algorithm guarantees mutual exclusion: the coordinator lets only one process at a time access the resource. It is also fair, since requests are granted in the order in which they are received. No process ever waits forever (no starvation). The scheme is easy to implement, too, and requires only three messages per use of resource (request, grant, release). Its simplicity makes it an attractive solution for many practical situations.

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The centralized approach also has shortcomings. The coordinator is a single point of failure, so if it crashes, the entire system may go down. If processes normally block after making a request, they cannot distinguish a dead coordinator from "permission denied" since in both cases no message comes back. In addition, in a large system, a single coordinator can become a performance bottleneck. Nevertheless, the benefits coming from its simplicity often outweigh the potential drawbacks. Moreover, distributed solutions are not necessarily better, as we illustrate next.

5.3.3 A distributed algorithm

Using Lamport's logical clocks, and inspired by Lamport's original solution for distributed mutual exclusion (which we discussed in Note 5.3), Ricart and Agrawala [1981] provided the following algorithm. Their solution requires a total ordering of all events in the system. That is, for any pair of events, such as messages, it must be unambiguous which one actually happened first.

The algorithm works as follows. When a process wants to access a shared resource, it builds a message containing the name of the resource, its process number, and the current (logical) time. It then sends the message to all other processes, conceptually including itself. The sending of messages is assumed to be reliable; that is, no message is lost.

When a process receives a request message from another process, the action it takes depends on its own state regarding the resource named in the message. Three different cases have to be clearly distinguished:

- If the receiver is not accessing the resource and does not want to access it, it sends back an OK message to the sender.
- If the receiver already has access to the resource, it simply does not reply. Instead, it queues the request.
- If the receiver wants to access the resource as well but has not yet done so, it compares the timestamp of the incoming message with the one contained in the message that it has sent everyone. The lowest one wins. If the incoming message has a lower timestamp, the receiver sends back an OK message. If its own message has a lower timestamp, the receiver queues the incoming request and sends nothing.

After sending out requests asking permission, a process sits back and waits until everyone else has given permission. As soon as all the permissions are in, it may go ahead. When it is finished, it sends OK messages to all processes in its queue and deletes them all from the queue. If there is no conflict, it clearly works. However, suppose that two processes try to access the resource simultaneously, as shown in Figure 5.15(a).

Process P_0 sends everyone a request with timestamp 8, while at the same time, process P_2 sends everyone a request with timestamp 12. P_1 is not

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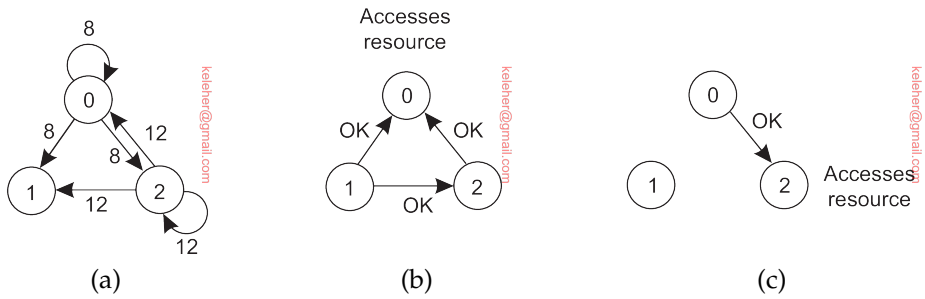


Figure 5.15: (a) Two processes want to access a shared resource at the same moment. (b) P_0 has the lowest timestamp, so it wins. (c) When process P_0 is done, it sends an OK also, so P_2 can now go ahead.

interested in the resource, so it sends OK to both senders. Processes P_0 and P_2 both see the conflict and compare timestamps. P_2 sees that it has lost, so it grants permission to P_0 by sending OK. Process P_0 now queues the request from P_2 for later processing and accesses the resource, as shown in Figure 5.15(b). When it is finished, it removes the request from P_2 from its queue and sends an OK message to P_2, allowing the latter to go ahead, as shown in Figure 5.15(c). The algorithm works because in the case of a conflict, the lowest timestamp wins and everyone agrees on the ordering of the timestamps.

With this algorithm, mutual exclusion is guaranteed without deadlock or starvation. If the total number of processes is N, then the number of messages that a process needs to send and receive before it can enter its **critical region** is $2 \cdot (N-1)$: $N-1$ request messages to all other processes, and subsequently $N-1$ OK messages, one from each other process. A critical region is a series of instructions to be executed by a process, which requires mutually exclusive access.

Unfortunately, this algorithm has N points of failure. If any process crashes, it will fail to respond to requests. This silence will be interpreted (incorrectly) as denial of permission, thus blocking all subsequent attempts by all processes to enter any of their respective critical regions. The algorithm can be patched up as follows. When a request comes in, the receiver always sends a reply, either granting or denying permission. Whenever either a request or a reply is lost, the sender times out and keeps trying until either a reply comes back or the sender concludes that the destination is dead. After a request is denied, the sender should block, waiting for a subsequent OK message.

Another problem with this algorithm is that either a multicast communication primitive must be used, or each process must maintain the group membership list itself, including processes entering the group, leaving the group, and crashing. The method works best with small groups of processes

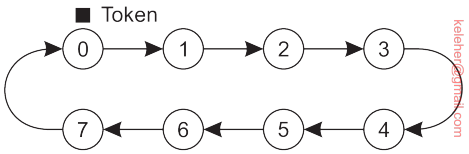
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that never change their group memberships. Finally, note that *all* processes are involved in *all* decisions concerning accessing the shared resource, which may impose a burden on processes running on resource-constrained machines.

Various minor improvements are possible to this algorithm. For example, getting permission from everyone is overkill. All that is needed is a method to prevent two processes from accessing the resource at the same time. The algorithm can be modified to grant permission when it has collected permission from a simple majority of the other processes, rather than from all of them.

5.3.4 A token-ring algorithm

An entirely different approach to deterministically achieving mutual exclusion in a distributed system is illustrated in Figure 5.16. In software, we construct an overlay network in the form of a logical ring in which each process is assigned a position in the ring. All that matters is that each process knows who is next in line after itself.



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Figure 5.16: An overlay network constructed as a logical ring with a token circulating between its members.

When the ring is initialized, process P_0 is given a **token**. The token circulates around the ring. Assuming there are N processes, the token is passed from process P_k to process $P_{(k+1) \bmod N}$ in point-to-point messages. When a process acquires the token from its neighbor, it checks to see if it needs to access the shared resource. If so, the process goes ahead, does all the work it needs to, and releases the resources. After it has finished, it passes the token along the ring. It is not permitted to enter the resource immediately again using the same token.

If a process is handed the token by its neighbor and is not interested in the resource, it just passes the token along. As a consequence, when no processes need the resource, the token just circulates around the ring.

The correctness of this algorithm is easy to see. Only one process has the token at any instant, so only one process can actually get to the resource. Since the token circulates among the processes in a well-defined order, starvation cannot occur. Once a process decides it wants to have access to the resource, at worst it will have to wait for every other process to use the resource.

This algorithm has its own problems. If the token is ever lost, for example because its holder crashes or due to a lost message containing the token, it must be regenerated. In fact, detecting that it is lost may be difficult, since the

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amount of time between successive appearances of the token on the network is unbounded. The fact that the token has not been spotted for an hour does not mean that it has been lost; somebody may still be using it.

The algorithm also runs into trouble if a process crashes, but recovery is relatively easy. If we require a process receiving the token to acknowledge receipt, a dead process will be detected when its neighbor tries to give it the token and fails. At that point, the dead process can be removed from the group, and the token holder can throw the token over the head of the dead process to the next member down the line, or the one after that, if necessary. Of course, doing so requires that everyone maintains the current ring configuration.

5.3.5 A decentralized algorithm

Let us take a look at a fully decentralized solution. Lin et al. [2004] propose to use a voting algorithm. Each resource is assumed to be replicated N times. Every replica has its own coordinator for controlling the access by concurrent processes.

However, whenever a process wants to access the resource, it will simply need to get a majority vote from $m > N/2$ coordinators. We assume that when a coordinator does not give permission to access a resource (which it will do when it had granted permission to another process), it will tell the requester.

The assumption is that when a coordinator crashes, it recovers quickly but will have forgotten any vote it gave before it crashed. Another way of viewing this is that a coordinator resets itself at arbitrary moments. The risk that we are taking is that a reset will make the coordinator forget that it had previously granted permission to some process to access the resource. As a consequence, it may incorrectly grant this permission again to another process after its recovery.

Let $p = \Delta t/T$ be the probability that a coordinator resets during a time interval Δt, while having a lifetime of T. The probability $\mathbb{P}[k]$ that k out of m coordinators reset during the same interval is then

$$\mathbb{P}[k] = \binom{m}{k} p^k (1-p)^{m-k}$$

If f coordinators reset, then the correctness of the voting mechanism will be violated when we have more than m other coordinators think it is okay to allocate the resource, that is, when $N - (m - f) \geq m$, or, in other words, when $f \geq 2m - N$. The probability that such a violation occurs is $\sum_{k=2m-N}^{m} \mathbb{P}[k]$. To give an impression of what this could mean, Figure 5.17 shows the probability of violating correctness for different values of N, m, and p. Note that we compute p by considering the number of seconds per hour that a coordinator resets, and also taking this value to be the average time needed to access

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a resource. Our values for p are considered to be (very) conservative. The conclusion is that, in general, the probability of violating correctness can be so low that it can be neglected in comparison to other types of failure.

N	m	p	Violation
8	5	3 sec/hour	$< 10^{-5}$
8	6	3 sec/hour	$< 10^{-11}$
16	9	3 sec/hour	$< 10^{-4}$
16	12	3 sec/hour	$< 10^{-21}$
32	17	3 sec/hour	$< 10^{-4}$
32	24	3 sec/hour	$< 10^{-43}$

N	m	p	Violation
8	5	30 sec/hour	$< 10^{-3}$
8	6	30 sec/hour	$< 10^{-7}$
16	9	30 sec/hour	$< 10^{-2}$
16	12	30 sec/hour	$< 10^{-13}$
32	17	30 sec/hour	$< 10^{-2}$
32	24	30 sec/hour	$< 10^{-27}$

Figure 5.17: Violation probabilities for various parameter values of decentralized mutual exclusion.

To implement this scheme, we can use a system in which a resource is replicated N times. Assume that the resource is known under its unique name rname. We can then assume that the ith replica is named rname_i which is then used to compute a unique key using a known hash function. As a consequence, every process can generate the N keys given a resource's name, and subsequently look up each node responsible for a replica (and controlling access to that replica) using some commonly used naming system.

If permission to access the resource is denied (i.e., a process gets less than m votes), it is assumed that it will back off for some randomly chosen time, and make a next attempt later. The problem with this scheme is that if many nodes want to access the same resource, it turns out that the utilization rapidly drops. In that case, there are so many nodes competing to get access that eventually no one can get enough votes, leaving the resource unused. A solution to solve this problem can be found in [Lin et al., 2004].

Note 5.5 (More information: A comparison of the mutual-exclusion algorithms)
A brief comparison of the mutual exclusion algorithms we have looked at is instructive. In Figure 5.18 we have listed the algorithms and two performance properties: the number of messages required for a process to access and release a shared resource, and the delay before access can occur (assuming messages are passed sequentially over a network).

In the following, we assume only point-to-point messages (or, equivalently, count a multicast to N processes as N messages).

- The centralized algorithm is simplest and also most efficient. It requires only three messages to enter and leave a critical region: a request, a grant to enter, and a release to exit.

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- The distributed algorithm requires $N-1$ request messages, one to each of the other processes, and an additional $N-1$ grant messages, for a total of $2(N-1)$.
- With the token ring algorithm, the number is variable. If every process constantly wants to enter a critical region, then each token pass will result in one entry and exit, for an average of one message per critical region entered. At the other extreme, the token may sometimes circulate for hours without anyone being interested in it. In this case, the number of messages per entry into a critical region is unbounded.
- The decentralized case requires sending N messages to coordinators, and another N responses. If it does not get a majority, it will have to release (at most) $N/2$ votes. If it did get enough votes, it will have to send an additional N release messages later on. A process may need to go through $k \geq 1$ attempts.

Algorithm	Messages per entry/exit	Delay before entry (in message times)
Centralized	3	2
Distributed	$2(N-1)$	$2(N-1)$
Token ring	$1, \ldots, \infty$	$0, \ldots, N-1$
Decentralized	$2kN + (k-1)N/2 + N, k = 1, 2, \ldots$	$2kN + (k-1)N/2$

Figure 5.18: A comparison of four mutual exclusion algorithms.

The delay from the moment a process needs to enter a critical region until its actual entry also varies. For a worst-case analysis, we assume that messages are sent one after the other (i.e., there are never two or more messages in transit at the same time), and that message transfer time is roughly the same everywhere. Delay can then be expressed in **message transfer time units**, or simply **MTTU**. Under these assumptions, when the time using a resource is short, the dominant factor in the delay is determined by the total number of messages sent through the system before access can be granted. When resources are used for a long period of time, the dominant factor is waiting for everyone else to take their turn. In Figure 5.18 we show the former case.

- It takes only two MTTUs to enter a critical region in the centralized case, caused by a request message and the subsequent grant message sent by the coordinator.
- The distributed algorithm requires sending $N-1$ request messages, and receiving another $N-1$ grant messages, adding up to $2(N-1)$ MTTUs.
- For the token ring, the delay varies from 0 MTTU (in case the token had just arrived) to $N-1$ (for when the token had just departed).
- In the case of decentralized, the delay is dependent on the number of times a process needed to return the (minority of) votes. With having to go through $k \geq 1$ attempts, a process may see $2kN + (k-1)N/2$ MTTUs.

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> Virtually all algorithms suffer badly in the event of crashes. Special measures and additional complexity must be introduced to avoid having a crash bring down the entire system. It is somewhat ironic that distributed algorithms are generally more sensitive to crashes than centralized ones. In this sense, it should not come as a surprise that, indeed, centralized mutual exclusion is widely applied: it is simple to understand the behavior, and relatively easy to increase the fault tolerance of the centralized server.

5.3.6 Example: Simple locking with ZooKeeper

For many practical reasons, mutual exclusion in distributed systems is often done with a centralized coordinator, not in the least because the behavior of these solutions is much easier to understand than many other noncentralized versions. Let us briefly look at a system that is designed for coordination tasks in distributed systems, and which is by now also widely deployed. **ZooKeeper** was developed to provide facilities for supporting various coordination tasks, including locking, leader election, monitoring, to name but just a few. It has been designed for scalability and fault tolerance. For our purposes now, we concentrate only on a relatively simple setup, namely to support locking through a single server. ZooKeeper's fault tolerance aspects are discussed in Chapter 8. The system was first described in [Hunt et al., 2010], to be later followed up with a practical description by Junqueira and Reed [2014].

ZooKeeper basics

As mentioned, ZooKeeper is designed to facilitate different coordination functions. An important design principle is its lack of blocking primitives: clients send messages to the ZooKeeper service and, in principle, are always immediately returned a response. In the case of locking, this means that a client will, for example, be informed whether it was able to grab a lock. If the lock could not be acquired, a new attempt will be necessary, as we discuss shortly.

To facilitate a range of coordination functions, ZooKeeper maintains a namespace, organized as a tree. Operations on the tree are simple: creating and deleting nodes, as well as reading and updating the data contained in a node (if any). A partial update of a node is not possible: all the node's data will be overwritten in the case of an update. In addition, a client can check with the server whether a node exists. At this point, it should be become clear how one could implement a locking service in ZooKeeper: to acquire a lock, simply let a process create a special node, say `lock`, but have that operation fail if the node already exists. Releasing a lock is done by merely deleting the node `lock`. We return to some important details below.

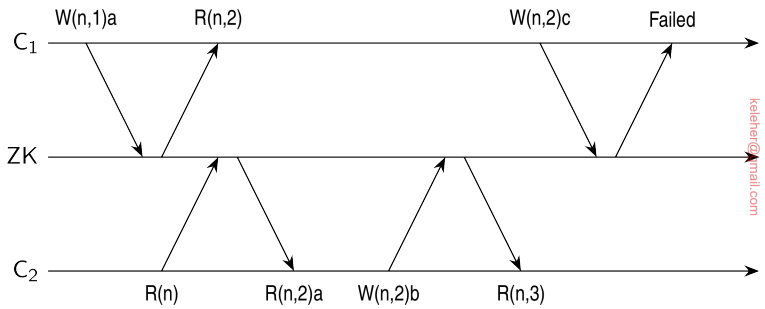
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There are two types of nodes in ZooKeeper. Persistent nodes need to be created and deleted explicitly. In contrast, ephemeral nodes are created explicitly, but are automatically removed if the connection with the creating client closes (perhaps because of a failure), or expires. Of course, an ephemeral node can also be explicitly deleted.

An important consequence of not supporting blocking operations, is that a client will have to regularly check the status of the namespace: have nodes been added or removed, have nodes in the namespace been changed? However, polling is generally not considered to be very efficient. To this end, ZooKeeper supports a notification mechanism by which a client can subscribe to a change in a node or a branch in the tree. When a change happens, the client receives a message.

There are a few things that need to be considered to make this all work. First, suppose a client subscribes to changes at a specific node after having read the state of that node. If the node is updated twice in a row, we want to prevent that the client sees the second update before being notified, as the notification dealt with changes to the *previous* state, i.e., the state after the first update took place.

Second, it should be clear that there can be a lot of concurrency among competing clients. As we shall discuss in detail in Chapter 7, problematic are, in particular, the situations in which one client C_1 reads data to subsequently decide to update that data. If, between the read and intended update another client C_2 performs an update, client C_1 may now request an update based on out-of-date information. Such intended updates can be prevented by using version numbers, as shown in Figure 5.19.



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Figure 5.19: Preventing updates based on out-of-date information through versions (based on [Junqueira and Reed, 2014].

We use the notation $W(n,k)a$ to denote the request to write the value a to node n, under the assumption that its current version is k. $R(n,k)$ denotes that the current version of node n is k. The operation $R(n)$ tells that a client wants to read the current value of node n, and $R(n,k)a$ means that the value a from

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node n is returned with its current version k. In Figure 5.19 we see that client C_1 writes value a to node n, assuming its version is 1. If no write has ever taken place, this assumption is correct. The ZooKeeper service (ZK) returns the new version number, namely 2.

Meanwhile, client C_2 issues a read request for node n and is returned the previously written value a having version number 2. C_2 decides to update n, assuming its current version is still 2, by means of the write operation W(n, 2)b. ZooKeeper accepts the write, and returns the new version number 3. This concurrent update places C_1 in the position that its intended write operation W(n, 2)c will fail: ZooKeeper states that its assumption about the version number is incorrect. The best what C_1 could do is read n again to see if it still wants to update the node.

Clearly, each client may need to try several times before an update actually takes place. This may not always seem so efficient, yet this approach ensures that data maintained by ZooKeeper is at least consistent with what clients expect it to be. As mentioned, we will return to consistency in much more detail in Chapter 7.

A ZooKeeper locking protocol

Implementing a lock using ZooKeeper is now fairly straightforward. ZooKeeper uses the same path notation for nodes as in Unix file systems. That means that we can create a node /lock at the root of the tree. The existence of that node means that a client has successfully acquired the lock. Releasing the lock is simply done by deleting node /lock.

Any client wanting to create the same node will receive the message that the node already exists and thus that the operation failed. Capturing such exceptions is at the core of blocking a client until /lock is removed. When the node already exists, the client will request ZooKeeper to send a notification when the node is deleted. Until that moment, the client simply waits (i.e., locally blocks) until notified, after which it tries to create the node again.

There are many subtleties to deal with. For example, a decent locking mechanism should allow a client to bail out after several attempts. Likewise, if a client who has created /lock crashes before deleting it again, we need to make sure that ZooKeeper will delete the node. Furthermore, we will find ourselves in an unfortunate race when the following happens:

1. A client C_1 creates a node /lock.
2. A client C_2 wants to acquire the lock but is notified that the associated node already exists.
3. Before C_2 subscribes to a notification, C_1 releases the lock, i.e., deletes /lock.
4. Client C_2 subscribes to changes to /lock and blocks locally.

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This is analogous to the situation we described before: C_2 should be able to subscribe to changes based on the *existence* of /lock. In other words, when performing the fourth step, C_2 should be immediately notified that the situation has already changed since it last visited the tree. These subtleties are handled by ZooKeeper.

5.4 Election algorithms

Many distributed algorithms require one process to act as coordinator, initiator, or otherwise perform some special role. In general, it does not matter which process takes on this special responsibility, but one of them has to do it. In this section, we will look at algorithms for **electing a coordinator**. We will often also speak of a leader election.

If all processes are the same, with no distinguishing characteristics, there is no way to select one of them to be special. Consequently, we will assume that each process P has a unique identifier $id(P)$. In general, election algorithms attempt to locate the process with the highest identifier and designate it as coordinator. The algorithms differ in the way they locate the coordinator.

Furthermore, we also assume that every process knows the identifier of every other process. In other words, each process has complete knowledge of the process group in which a coordinator must be elected. What the processes do not know is which ones are currently up and which ones are currently down. The goal of an election algorithm is to ensure that when an election starts, it concludes with all processes agreeing on whom the new coordinator is to be. There are many algorithms and variations, of which several important ones are discussed in the textbooks by Tel [2000] and Lynch [1996].

5.4.1 The bully algorithm

A well-known solution for electing a coordinator is the **bully algorithm** devised by Garcia-Molina [1982]. In the following, we consider N processes $\{P_0, \ldots, P_{N-1}\}$ and let $id(P_k) = k$. When any process notices that the coordinator is no longer responding to requests, it initiates an election. A process, P_k, holds an election as follows:

1. P_k sends an ELECTION message to all processes with higher identifiers: $P_{k+1}, P_{k+2}, \ldots, P_{N-1}$.
2. If no one responds, P_k wins the election and becomes coordinator.
3. If one of the higher-ups answers, it takes over and P_k's job is done.

At any moment, a process can get an ELECTION message from one of its lower-numbered colleagues. When such a message arrives, the receiver sends an OK message back to the sender to indicate that it is alive and will take

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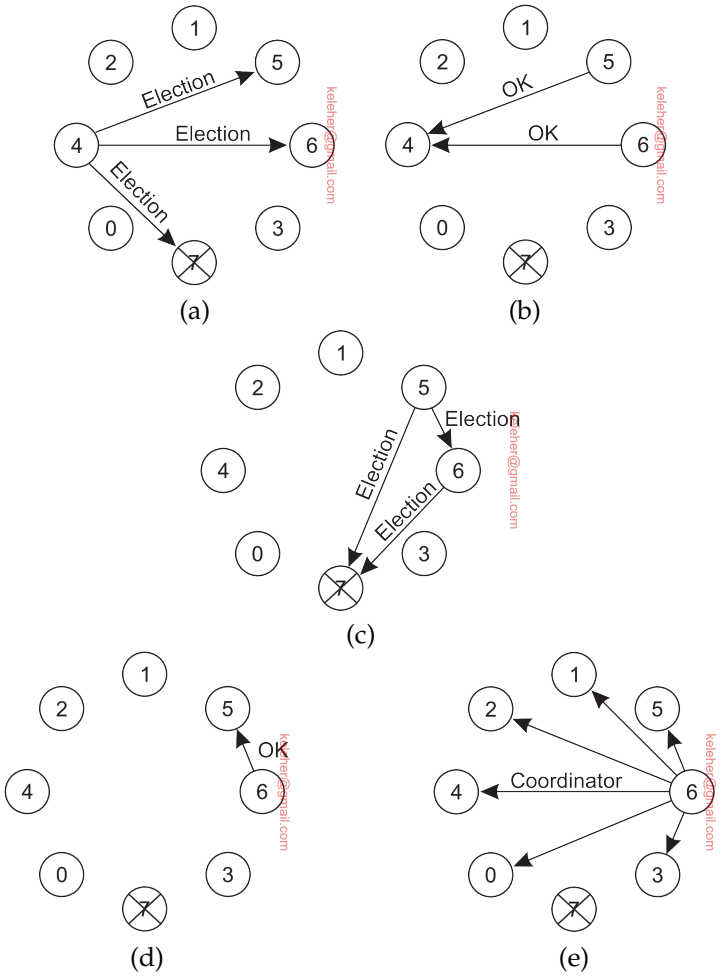


Figure 5.20: The bully election algorithm. (a) Process 4 holds an election. (b) Processes 5 and 6 respond, telling 4 to stop. (c) Now 5 and 6 each hold an election. (d) Process 6 tells 5 to stop. (e) Process 6 wins and tells everyone.

over. The receiver then holds an election, unless it is already holding one. Eventually, all processes give up but one, and that one is the new coordinator. It announces its victory by sending all processes a message telling them that starting immediately it is the new coordinator.

If a process that was previously down comes back up, it holds an election. If it happens to be the highest-numbered process currently running, it will win the election and take over the coordinator's job. Thus the biggest guy in town always wins, hence the name "bully algorithm."

In Figure 5.20 we see an example of how the bully algorithm works. The group consists of eight processes, with identifiers numbered from 0 to 7.

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Previously, process P_7 was the coordinator, but it has just crashed. Process P_4 is the first one to notice this, so it sends ELECTION messages to all the processes higher than it, namely P_5, P_6, and P_7, as shown in Figure 5.20(a). Processes P_5 and P_6 both respond with OK, as shown in Figure 5.20(b). Upon getting the first of these responses, P_4 knows that its job is over, knowing that either one of P_5 or P_6 will take over and become coordinator. Process P_4 just sits back and waits to see who the winner will be (although at this point it can make a pretty good guess).

In Figure 5.20(c) both P_5 and P_6 hold elections, each one sending messages only to those processes with identifiers higher than itself. In Figure 5.20(d), P_6 tells P_5 that it will take over. At this point, P_6 knows that P_7 is dead and that it (P_6) is the winner. If there is state information to be collected from disk or elsewhere to pick up where the old coordinator left off, P_6 must now do what is needed. When it is ready to take over, it announces the takeover by sending a COORDINATOR message to all running processes. When P_4 gets this message, it can now continue with the operation it was trying to do when it discovered that P_7 was dead, but using P_6 as the coordinator this time. In this way, the failure of P_7 is handled, and the work can continue.

If process P_7 is ever restarted, it will send all the others a COORDINATOR message and bully them into submission.

5.4.2 A ring algorithm

Consider the following election algorithm that is based on the use of a **(logical) ring**. Unlike some ring algorithms, this one does not use a token. We assume that each process knows who its successor is. When any process notices that the coordinator is not functioning, it builds an ELECTION message containing its own process identifier and sends the message to its successor. If the successor is down, the sender skips over the successor and goes to the next member along the ring, or the one after that, until a running process is located. At each step along the way, the sender adds its own identifier to the list in the message, effectively making itself a candidate to be elected as coordinator.

Eventually, the message gets back to the process that started it all. That process recognizes this event when it receives an incoming message containing its own identifier. At that point, the message type is changed to COORDINATOR and circulated once again, this time to inform everyone else who the coordinator is (the list member with the highest identifier) and who the members of the new ring are. When this message has circulated once, it is removed and everyone goes back to work.

In Figure 5.21 we see what happens if two processes, P_3 and P_6, discover simultaneously that the previous coordinator, process P_7, has crashed. Each of these builds an ELECTION message and each of them starts circulating its message, independent of the other one. Eventually, both messages will go all the way around, and both P_3 and P_6 will convert them into COORDINATOR

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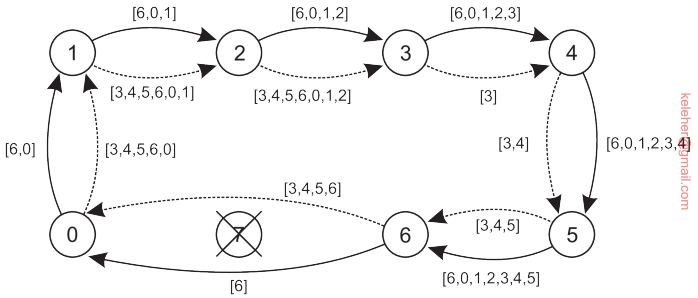


Figure 5.21: Election algorithm using a ring. The solid line shows the election messages initiated by P_6; the dashed one those by P_3.

messages, with exactly the same members and in the same order. When both have gone around again, both will be removed. It does no harm to have extra messages circulating; at worst it consumes a little bandwidth, but this is not considered wasteful.

5.4.3 Example: Leader election in ZooKeeper

Let us look at a practical example of how leader election takes place in a distributed system. As mentioned, ZooKeeper is logically centralized coordination service. In Section 5.3.6 we considered a simple setup with just a single ZooKeeper server. In practice, we see that ZooKeeper maintains a (relatively small) set of servers, forming what is called an **ensemble**. For a client, an ensemble appears as just a single server. For ZooKeeper, this ensemble is also coordinated by a single server, called the leader. The other servers are called followers and essentially act as up-to-date standbys for whenever the leader malfunctions. In that case, one of the followers will be elected as the new leader. From the client's perspective, a malfunctioning leader is mostly invisible: as soon as one of the standbys has taken over the role of leader, everything proceeds as if nothing happened. We return to many details of such fault-tolerant behavior in Chapter 8, but for now, let us dive a bit deeper into how the server in a ZooKeeper ensemble choose their leader.

It is important to note that a leader-election algorithm *within* a ZooKeeper ensemble is fundamentally different from devising a leader-election algorithm *by means of* ZooKeeper. The first deals with the *implementation of ZooKeeper* as a coordination service; the second with *using ZooKeeper* as a coordination service, perhaps as part of the implementation of a larger distributed system.

Which leader-election algorithm is used within a ZooKeeper ensemble is not that important, as long as, in the end, a single leader is chosen and enough followers are running as up-to-date standbys. Enough in this case means that a majority of the servers that form an ensemble are operating properly. For now, we will simply assume this to be the case.

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The default election algorithm in an implementation of ZooKeeper is a simple version of bullying and works as follows [Junqueira and Reed, 2014]. Each server s in the ensemble has an identifier *id*(s), as well as a monotonically increasing counter *tx*(s) of the latest transaction it handled. Think of a transaction as a series of operations the server has executed, such as the combination of adding a node and registering that it should send a client a notification when anything changes. Typically, the leader of the ensemble will have performed the transaction on its own, locally stored, namespace and instructed its followers to do so as well on their respective local copy of that namespace.

When a follower s believes something is wrong with the leader (e.g., it suspects that the leader crashed), it sends out an ELECTION message to all other servers, along with the pair (voteID,voteTX). For this first message, it sets voteID to *id*(s) and voteTX to *tx*(s). During an election, each server s maintains two variables. The first, *leader*(s), records the identifier of the server that s believes may turn out to be the final leader, and is initialized to *id*(s). The second, *lastTX*(s) is what s has learned to be the most recent transaction, initially being its own value, namely *tx*(s).

When a server s^* receives (voteID,voteTX), it proceeds as follows:

- If $lastTX(s^*) <$ voteTX, then s^* just received more up-to-date information on the most recent transaction. In that case, it sets
  - $leader(s^*) \leftarrow$ voteID
  - $lastTX(s^*) \leftarrow$ voteTX
- If $lastTX(s^*) =$ voteTX *and* $leader(s^*) <$ voteID, then s^* knows as much about the most recent transaction as what it was just sent, but its perspective on which server will be the next leader needs to be updated:
  - $leader(s^*) \leftarrow$ voteID

Each time a server s^* receives a (voteID,voteTX) message, it may update its own information on whom it suspects to be the next leader. If s^* believes it should be the next leader, and has not sent out a message stating this, it will broadcast the pair ($id(s^*)$,$tx(s^*)$). Under the assumption that communication is reliable, this broadcast alone should do the job. Typically, s^* will send out a message stating that it is the leader, when either initiating an election, or when receiving a message with a lower voteTX than its own $tx(s^*)$, or receiving a message with an up-to-date value for voteTX, but with a smaller voteID than $id(s^*)$. Once a leader has been elected, all servers will make sure that they synchronize on the latest transaction.

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Note 5.6 (Advanced: ZooKeeper leader election in Python)

```
class Process:
  def __init__(self, chanID, procID, procIDSet, initTX):
    self.txID       = initTX       # Your own most recent transaction
    self.leader     = self.procID  # Who you believe may become leader
    self.lastTX     = self.txID    # What is the most recent transaction
    self.noleader   = False        # Are you still in the race for leader?

  def receive(self):
    while True:
      msg             = self.chan.recvFrom(self.otherProcs)
      sender, payload = msg[0], msg[1]
      if payload[0] == ELECTION: # A process started an election
        voteID, voteTX = payload[1], payload[2]

        if self.lastTX < voteTX: # You're not up to date on most recent transaction
          self.leader = voteID   # Record the suspected leader
          self.lastTX = voteTX   # As well as the likely most recent transaction

        elif (self.lastTX == voteTX) and (self.leader < voteID): # Wrong leader
          self.leader = voteID   # Update your suspected leader

        elif (self.procID > voteID) and (self.txID >= voteTX) and (not self.noleader):
          # At this point, you may very well be the new leader (having a sufficiently
          # high process identifier as well as perhaps the most recent transaction).
          # No one has told you so far that you could not be leader. Tell the others.
          self.chan.sendTo(self.otherProcs, (LEADER, self.procID, self.txID))

      if payload[0] == LEADER:
        # Check if the sender should indeed be leader
        if ((self.lastTX < payload[2]) or
            ((self.lastTX == payload[2]) and (self.leader <= payload[1]))):
          # The sender is more up-to-date than you, or is equally up-to-date but
          # has a higher process identifier. Declare yourself follower.
          self.chan.sendTo(sender, (FOLLOWER, self.procID))
        else:
          # Sender is wrong: you have information that the sender based its decision
          # on outdated information
          self.chan.sendTo(sender, (NOLEADER))
```

Figure 5.22: A simplified version of ZooKeeper's leader election for an ensemble of servers.

Coming to the conclusion that one of the servers in a ZooKeeper ensemble is now indeed the new leader, can be a bit tricky. One way is to let a server come to the conclusion that it may never become a leader in the current round, in which case it tells the alleged leader that it will become a follower. This means that as soon as a server has collected enough followers, it can promote itself to leader. In ZooKeeper, this happens when a server has a majority of the other servers in the ensemble as followers. If we take an oversimplified approach by assuming

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that messages are never lost, nor do servers crash during an election, a leader is elected when all other servers are its followers. The code is shown in Figure 5.22 (again, omitting many details).

5.4.4 Example: Leader election in Raft

As another example of a leader-election algorithm that is in practical use, let us take a look at **Raft** [Ongaro and Ousterhout, 2014]. We will return to Raft in Chapter 8 when discussing consensus algorithms and focus here only on how it selects a leader.

Raft operates in a setting in which a handful of known replicated servers (typically five) collaborate by ensuring that each server executes the same set of operations, and all in the same order. To this end, one of the servers is elected as leader to tell the others what the exact order is of those operations. The protocol assumes that messages may be lost and that servers may crash.

Each server can be in one of three states: follower, candidate, or leader. Furthermore, the protocol operates in **terms**, where during each term there is exactly one leader, although it could have perhaps crashed. Each term is numbered, starting with 0. The leader is assumed to regularly send out a message, either containing information on an operation that should be carried out, or otherwise a heartbeat message to tell the other servers that their leader is still up and running.

Each server initially starts in the follower state (that is, there is initially no leader). Adopting the terminology from Howard et al. [2015], after a *follower timeout*, a following server concludes that the leader may have crashed (which, by the way, may be a false conclusion). As a result, it enters the candidate state and starts an election, volunteering to be the new leader. An election starts with a broadcast to all other servers, along with increasing the term number by 1. At that point, three situations may happen.

1. A candidate may receive a message from an alleged leader. If that server indicates it is operating in the same term as the candidate server, the latter will become follower again for the current term.
2. When a following server receives an election message for the first time (in a new term), it simply votes for the candidate and ignores any other election messages (for that new term). Therefore, a candidate server can also receive a vote. If it has a majority of votes (i.e., more than half of the servers, including itself), it promotes itself as leader for the new term.
3. As long as there is no alleged leader, or not enough votes have been received, the candidate server waits until a *candidate timeout* happens. At that point, the candidate server will simply start a new election (and again, for a next term).

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Of course, if all servers enter the candidate state, we bear the risk of indefinitely not being able to cast enough votes for a single server. A simple solution is to slightly vary the *follower timeout* and *candidate timeout* on a per-server basis. The result is that, generally, there is only a single candidate for the new term, allowing each other server to vote for that candidate and become a follower. As soon as a server becomes leader, it will send out a heartbeat message to the rest.

5.4.5 Elections in large-scale systems

Many leader-election algorithms apply to only relatively small distributed systems. In such cases, the various nodes exchange information among each other, often through several rounds, to eventually collectively decide on the new leader. In Chapter 8, we shall see that this collective decision-making can take place even when a fraction of the participating nodes exhibits nonconforming behavior, as in the case of security attacks or faults. The problem of electing a leader becomes rather nasty when there are potentially many processes to choose from. This is the case, for example, in **permissionless blockchains**.

Proof of work

A relatively simple, yet by now heavily criticized solution, is to have the candidates run a computational race, referred to as **proof of work**. Running such a race, which is done by solving a computational puzzle, should be known to be possible, yet difficult. The first process solving the puzzle, wins, and may proceed as leader to append a block of validated transactions to the existing blockchain.

The type of problem used for blockchains is based on what is known as **hashing**. As introduced in Section 1.2.5, a hash function takes as input a (possibly large) data set and produces a fixed-length string of bits, typically of length 1024 or 2048, called a *hash*. A cryptographically well-defined hash function has several important properties [Ferguson and Schneier, 2003]:

1. Computing the hash of a given data set is relatively easy, i.e., it does not require significant computational efforts.
2. However, given a specific hash, it is computationally very difficult to find a corresponding data set with the same associated hash.
3. With very high probability, any two different input data sets will lead to two seemingly unrelated, different hashes. Even if the two data sets are minimally different, their associated hashes will most likely be very different.

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The third property is used to securely protect blocks of validated transactions against adversarial modifications: even the change of a single bit will not go unnoticed. Combined with the second property, it becomes virtually impossible to modify a block meaningfully, such that the modified block's hash is the same as the original block's hash.

To set up a race between validators, each validator computes the hash of its block of validated transactions. Note again that a hash is technically just a fixed-length string of bits. We denote the computed hash over a block a **digest**. The validator is then required to find an *additional* bit string, called a **nonce**, such that the hash computed over the digest and the nonce when taken together produces a bit string with a predetermined number of leading zeroes. Given property 2 of hash functions, we know that it is computationally very difficult to find a data set that matches a given hash. In our case, we have as input the hash of the block of validated transactions (i.e., the digest), and are now required to find a bit string (i.e., nonce) that, taken together with the digest, has an associated hash that starts with a given number of zeroes. This is computationally very difficult. In essence, each validator simply needs to go through a most likely lengthy trial-and-error process of generating a nonce and checking whether that nonce, combined with the digest, will lead to the required result.

By controlling how many leading zeroes that the outcome should have, we essentially control the difficulty of the computations. For example, with just 1 leading zero, there is a 50% chance that a generated nonce will lead to the desired result. Demanding 2 leading zeroes reduces this probability to 25%, 3 leading zeroes to 12.5%, and so forth. With 64 leading zeroes, which is common practice, the chance that an arbitrarily chosen nonce will do the trick is a mere 0.0000 0000 0000 0000 05% (that is, 17 zeroes after the decimal dot). Put differently, a validator will on average need to check about 18 billion billion nonces to find one that leads to the desired result. Using dedicated supercomputers, this takes about 10 minutes. It would take an average laptop about 100 years.

With increasing hardware capacity, the time it takes to find a nonce will drop. For this reason, the difficulty of finding a nonce is deliberately controlled in such a way that a race will have an expected duration. For Bitcoin systems, a popular application of blockchains, the duration is approximately 10 minutes. If races tend to become shorter, then the difficulty for finding a nonce is increased. If races turn out to be too lengthy, the difficulty is lowered. Adjusting the difficulty is done by regularly computing the average time it took to run the race over the last 2000 blocks of transactions, or so. The adjustment of the difficulty is done by all validators using the same globally known method, and is therefore done in a completely decentralized manner.

There is a reason for properly setting the expected duration of a race. Suppose we have very lengthy races. This means that validators would have

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to do a lot of work to find a proper nonce. The good part of the story is that the chance that two validators find a nonce at more or less the same time is relatively small (although this depends, of course, on how many validators participate in the race). As a consequence, the chance that there will be only one winner who will have enough time to successfully broadcast its victory to all others is relatively high.

The downside of the story, however, is that with lengthy races, the transaction processing capacity may be too low. Suppose that we have a race duration of 10 minutes and an average number of 2500 transactions per block (which are common values for Bitcoin). This means that, effectively, the system can handle about 4 transactions per second. Increasing the number of transactions per block is an obvious path to follow, yet even such improvements still lead to low rates in comparison to many centralized solutions with a fixed, trusted third party validating transactions.

On the other hand, assume we would have very short-lasting races. In that case, two validators may more easily concurrently find a proper nonce, both will declare victory, append their block to the blockchain (which was already massively replicated), and before other validators had the chance to stop their work, we would find ourselves with different copies of the same blockchain. There are different ways to correct for these concurrent versions of what is supposed to be a single blockchain, but obviously, it comes at the price of invalidating potentially many transactions, and rolling back to a valid state. Of course, the advantage of a smaller race duration is that, in principle, we can handle more transactions per time unit.

Proof of stake

As a reaction to the waste of computational resources found in proof-of-work systems, much effort has been spent on alternative leader-election algorithms for permissionless blockchains. An important class is formed by so-called **proof-of-stake** systems. Leaving out many details, as well as the many variants of such systems, the basic principle is as follows. An easy-to-read overview of the principles of proof-of-stake leader election is provided by Nguyen et al. [2019].

First, we need to make the assumption that each transaction as recorded in a blockchain has one or more associated tokens. Moreover, at each moment, each token has exactly one owner. This is typically the case with monetary transactions, where a token is associated with a digital coin. Because blockchains are fully readable by any participant, we may also assume that tokens may be *passed* between owners, and that, indeed, *copying* a token and associating it with multiple owners cannot go undetected. How this can be realized is discussed in Section 9.4.3. Each token will, therefore, have only a single owner. Second, we make also make the (realistic) assumption that for

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a given blockchain, there are a total of N tokens, which may vary over time depending on the application for which a blockchain is being used.

A simple leader-election process for blockchains then consists of a function that generates a random number k between 1 and N. That number is used as an index of the kth token, of which the owner becomes the next leader. The function used to generate a random number is public, meaning that any participant can execute it to determine k. Obviously, the more tokens a process P owns, the higher the probability that P will be elected as leader.

The simplicity of this algorithm and the fact that there are still so many race-based blockchains, indicates that there are many issues to deal with before proof-of-stake solutions can actually work. To understand, we need to realize that with proof of work, there is a price to pay to become leader, namely using considerable computational resources. If that price is compensated by a reward (namely a transaction fee), we have some sort of balance. In many proof-of-stake systems, there may also be a reward, but there is no price to be paid in advance for becoming a leader. As a result, these systems are generally more vulnerable to security attacks, for example, by letting participants generate invalid transactions that will disrupt the work of honest leaders.

Leader election in blockchains is part of reaching consensus on the status of a blockchain. We return to these matters in Section 8.2.6.

Note 5.7 (Advanced: Selecting superpeers)
Many leader-election algorithms often concentrate on the selection of only a single node. There are situations when several nodes should actually be selected, such as in the case of **super peers** in peer-to-peer networks, which we discussed in Section 2.4.3.

The following requirements need to be met for super-peer selection (see also [Lo et al., 2005]):

1. Normal nodes should have low-latency access to super peers.
2. Super peers should be evenly distributed across the overlay network.
3. There should be a predefined portion of super peers relative to the total number of nodes in the overlay network.
4. Each super peer should not need to serve more than a fixed number of normal nodes.

Fortunately, these requirements are relatively easy to meet in most peer-to-peer systems, given the fact that the overlay network is either structured (as in DHT-based systems), or randomly unstructured (as, for example, can be realized with gossip-based solutions). Let us take a look at solutions proposed by Lo et al. [2005].

In the case of DHT-based systems, the basic idea is to reserve a fraction of the identifier space for super peers. In a DHT-based system, each node receives a random and uniformly assigned m-bit identifier. Now suppose we reserve the

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first (i.e., leftmost) k bits to identify super peers. For example, if we need N superpeers, then the first $\lceil log_2(N) \rceil$ bits of any *key* can be used to identify these nodes.

To explain, assume we have a (small) Chord system with $m = 8$ and $k = 3$. When looking up the node responsible for a specific key K, we can first decide to route the lookup request to the node responsible for the pattern K $\wedge$ 11100000 which is then treated as the superpeer. (We use the binary operator "$\wedge$" to denote a bitwise and.) Note that each node with identifier ID can check whether it is a super peer by looking up ID $\wedge$ 11100000 to see if this request is routed to itself. Provided node identifiers are uniformly assigned to nodes, it can be seen that with a total of N nodes the number of super peers is, on average, equal to $2^{k-m}N$.

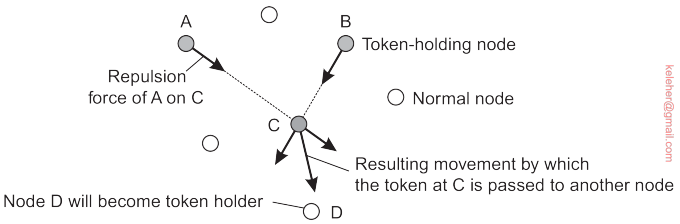


Figure 5.23: Moving tokens in a two-dimensional space using repulsion forces.

A different approach is based on positioning nodes in an m-dimensional geometric space. In this case, assume we need to place N super peers *evenly* throughout the overlay. The basic idea is simple: a total of N tokens are spread across N randomly chosen nodes. No node can hold more than one token. Each token represents a repelling force by which another token is inclined to move away. The net effect is that if all tokens exert the same repulsion force, they will move away from each other and spread themselves evenly in the geometric space.

This approach requires that nodes holding a token learn about other tokens. To this end, we can use a gossiping protocol, by which a token's force is disseminated throughout the network. If a node discovers that the total forces that are acting on it exceed a threshold, it will move the token in the direction of the combined forces, as shown in Figure 5.23. When a token is held by a node for a given amount of time, that node will promote itself to superpeer.

5.4.6 Elections in wireless environments

Traditional election algorithms are generally based on assumptions that are not realistic in wireless environments. For example, they assume that message passing is reliable and that the topology of the network does not change. These assumptions are false in most wireless environments, especially when dealing with mobile devices.

Only few protocols for elections have been developed that work in mobile environments. Vasudevan et al. [2004] propose a solution that can handle

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failing nodes and partitioning networks. An important property of their solution is that the *best* leader can be elected rather than just a random one as was more or less the case in the previously discussed solutions. Their protocol works as follows. To simplify our discussion, we concentrate only on ad hoc networks and ignore that nodes can actually move. This may be the case, for example, in sensor networks.

Consider such a wireless ad hoc network. To elect a leader, any node in the network, called the source, can initiate an election by sending an ELECTION message to its immediate neighbors (i.e., the nodes in its range). When a node receives an ELECTION for the first time, it designates the sender as its parent, and subsequently sends out an ELECTION message to all its immediate neighbors, except for the parent. When a node receives an ELECTION message from a node besides its parent, it merely acknowledges the receipt.

When node R has designated node Q as its parent, it forwards the ELECTION message to its immediate neighbors (excluding Q) and waits for acknowledgments to come in before acknowledging the ELECTION message from Q. This waiting has an important consequence. First, note that neighbors who have already selected a parent will immediately respond to R. More specifically, if all neighbors already have a parent, R is a leaf node and will be able to report back to Q quickly. In doing so, it will also report information such as its battery lifetime and other resource capacities.

This information will later allow Q to compare R's capacities to that of other downstream nodes, and select the best eligible node for leadership. Of course, Q had sent an ELECTION message only because its own parent P had done so as well. In turn, when Q eventually acknowledges the ELECTION message previously sent by P, it will pass the most eligible node to P as well. In this way, the source will eventually get to know which node is best to be selected as leader, after which it will broadcast this information to the rest.

This process is illustrated in Figure 5.24. Nodes have been labeled a to j, along with their capacity. Node a initiates an election by broadcasting an ELECTION message to nodes b and j, as shown in Figure 5.24(b) After that step, ELECTION messages are propagated to all nodes, ending with the situation shown in Figure 5.24(e), where we have omitted the last broadcast by nodes f and i. From there on, each node reports to its parent the node with the best capacity, as shown in Figure 5.24(f). For example, when node g receives the acknowledgments from its children e and h, it will notice that h is the best node, propagating $[h, 8]$ to its own parent, node b. In the end, the source will note that h is the best leader and will broadcast this information to all other nodes.

When multiple elections are initiated, each node will decide to join only one election. To this end, each source tags its ELECTION message with a unique identifier. Nodes will participate only in the election with the highest identifier, stopping any running participation in other elections.

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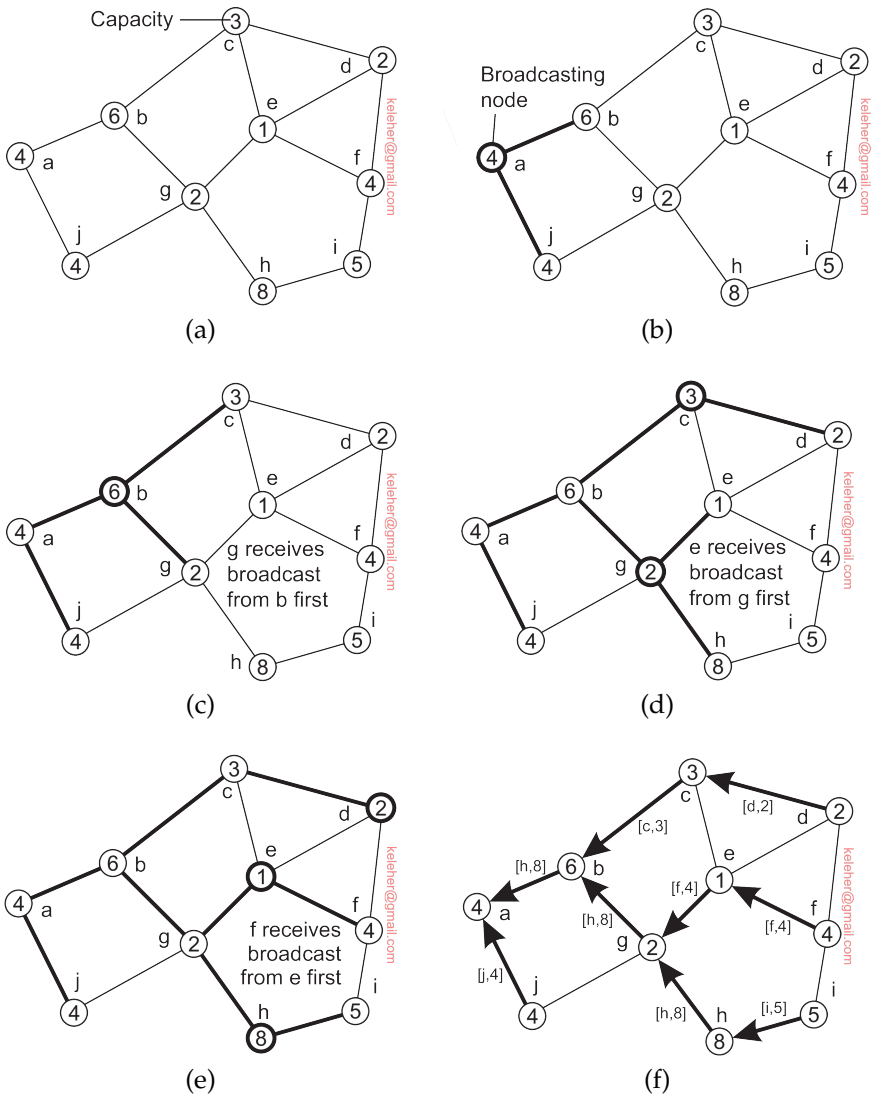


Figure 5.24: Election algorithm in a wireless network, with node a as the source. (a) Initial network. (b)–(e) The build-tree phase (last broadcast step by nodes f and i not shown). (f) Reporting of best node to source.

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With some minor adjustments, the protocol can be shown to operate also when the network partitions, and when nodes join and leave. The details can be found in Vasudevan et al. [2004].

5.5 Gossip-based coordination

As a next topic in coordination, we take a look at a few important examples in which gossiping is deployed. In the following, we look at aggregation, large-scale peer sampling, and overlay construction, respectively.

5.5.1 Aggregation

Let us take a look at some interesting applications of **epidemic protocols**. We already mentioned spreading updates, which is perhaps the most widely deployed application. In the same light, gossiping can be used to discover nodes that have a few outgoing wide-area links, to subsequently apply directional gossiping.

Another interesting application area is simply collecting, or actually aggregating information [Jelasity et al., 2005]. Consider the following information exchange. Every node P_i initially chooses an arbitrary number, say v_i. When node P_i contacts node P_j, they each update their value as:

$$v_i, v_j \leftarrow (v_i + v_j)/2$$

Obviously, after this exchange, both P_i and P_j will have the same value. In fact, it is not difficult to see that eventually all nodes will have the same value, namely the average of all initial values. Propagation speed is again exponential.

What use does computing the average have? Consider the situation that all nodes P_i have set v_i to zero, except for P_1 who has set v_1 to 1:

$$v_i \leftarrow \begin{cases} 1 & \text{if } i = 1 \\ 0 & \textit{otherwise} \end{cases}$$

If there are N nodes, then eventually each node will compute the average, which is $1/N$. As a consequence, every node P_i can estimate the size of the system as being $1/v_i$.

Computing the average may be difficult when nodes regularly join and leave the system. One practical solution to this problem is to introduce epochs. Assuming that node P_1 is stable, it simply starts a new epoch now and then. When node P_i sees a new epoch for the first time, it resets its own variable v_i to zero and starts computing the average again.

Of course, other results can also be computed. For example, instead of having a fixed node such as P_1 start the computation of the average, we

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can easily pick a random node as follows. Every node P_i initially sets v_i to a random number from the same interval, say $(0, 1]$, and also stores it permanently as m_i. Upon an exchange between nodes P_i and P_j, each change their value to:

$$v_i, v_j \leftarrow \max\{v_i, v_j\}$$

Each node P_i for which $m_i < v_i$ will lose the competition for being the initiator in starting the computation of the average. In the end, there will be a single winner. Of course, although it is easy to conclude that a node has lost, it is much more difficult to decide that it has won, as it remains uncertain whether all results have come in. The solution to this problem is to be optimistic: a node always assumes it is the winner until proven otherwise. At that point, it simply resets the variable it is using for computing the average to zero. Note that by now, several computations (in our example, computing a maximum and computing an average) may be executing simultaneously.

5.5.2 A peer-sampling service

An important aspect in epidemic protocols is the ability of a node P to choose another node Q *at random* from all available nodes in the network. When giving the matter some thought, we may actually have a serious problem: if the network consists of thousands of nodes, how can P ever pick one of these nodes at random without having a complete overview of the network? For smaller networks, one could often resort to a central service that had registered every participating node. Obviously, this approach can never scale to large networks.

A solution is to construct a fully decentralized **peer-sampling service**, or **PSS** for short. As it turns out, and somewhat counter-intuitive, a PSS can be built using an epidemic protocol. As explored by Jelasity et al. [2007], each node maintains a list of c neighbors, where, ideally, each of these neighbors represents a randomly chosen *live* node from the current set of nodes. This list of neighbors is also referred to as a **partial view**. There are many ways to construct such a partial view. In the solution of Jelasity et al., it is assumed that nodes regularly exchange entries from their partial view. Each entry identifies another node in the network, and has an associated age that indicates how old the reference to that node is. Two threads are used, as shown in Figure 5.25.

The different selection operations are specified as follows:

- `selectPeer`: Randomly select a neighbor from the local partial view
- `selectToSend`: Select some other entries from the partial view, and add to the list intended for the selected neighbor.
- `selectToKeep`: Add received entries to the partial view, remove repeated items, and shrink the view to c items.

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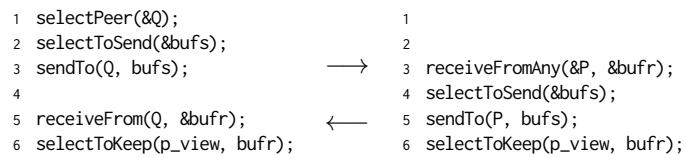


Figure 5.25: Communication between the (a) active and (b) passive thread in a peer-sampling service.

The active thread takes the initiative to communicate with another node. It selects that node from its current partial view. It continues by constructing a list containing $c/2 + 1$ entries, including an entry identifying itself. The other entries are taken from the current partial view. After sending the list to the selected neighbor, it waits for a response.

That neighbor, meanwhile, will also have constructed a list through the passive thread shown in Figure 5.25(b) whose activities strongly resemble that of the active thread.

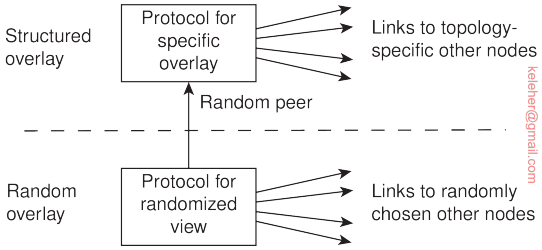
The crucial point is the construction of a new partial view. This view, for contacting as well as for the contacted peer, will contain exactly c entries, part of which will come from the received list. In essence, there are two ways to construct the new view. First, the two nodes may decide to discard the entries that they had sent to each other. Effectively, this means that they will *swap* part of their original views. The second approach is to discard as many *old* entries as possible (meaning, in practice, that after every gossiping round, the age of each entry in every partial view is incremented by one).

As it turns out, as long as peers regularly run the exchange algorithm just described, selecting a random peer from a thus dynamically changing partial view, is statistically indistinguishable from randomly selecting a peer from the entire network. Of course, selecting a peer from a partial view should occur at approximately the same frequency as the refreshing of partial views. We have thus constructed a fully decentralized gossip-based peer-sampling service. A simple and often-used implementation of a peer-sampling service is Cyclon [Voulgaris et al., 2005].

5.5.3 Gossip-based overlay construction

Although it would seem that structured and unstructured peer-to-peer systems form strict independent classes, this need actually not be the case (see also Castro et al. [2005]). One key observation is that by carefully exchanging and selecting entries from partial views, it is possible to construct and maintain specific topologies of overlay networks. This topology management is achieved by adopting a two-layered approach, as shown in Figure 5.26.

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Figure 5.26: A two-layered approach for constructing and maintaining specific overlay topologies using techniques from unstructured peer-to-peer systems.

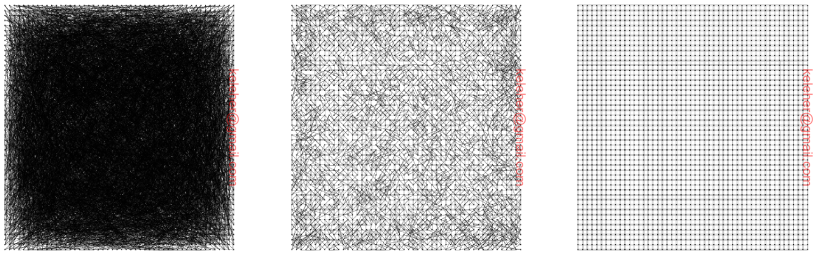
The lowest layer constitutes an unstructured peer-to-peer system in which nodes periodically exchange entries of their partial views with the aim to provide a peer-sampling service. Accuracy in this case refers to the fact that the partial view should be filled with entries referring to randomly selected *live* nodes.

The lowest layer passes its partial view to the higher layer, where an additional selection of entries takes place. This then leads to a second list of neighbors corresponding to the desired topology. Jelasity and Kermarrec [2006] propose to use a *ranking function* by which nodes are ordered according to some criterion relative to a given node. A simple ranking function is to order a set of nodes by increasing distance from a given node P. In that case, node P will gradually build up a list of its nearest neighbors, provided the lowest layer continues to pass randomly selected nodes.

As an illustration, consider a logical grid of size $N \times N$ with a node placed on each point of the grid. Every node is required to maintain a list of c nearest neighbors, where the distance between a node at (a_1, a_2) and (b_1, b_2) is defined as $d_1 + d_2$, with $d_i = \min(N - |a_i - b_i|, |a_i - b_i|)$. If the lowest layer periodically executes the protocol as outlined in Figure 5.25, the topology that will evolve is a two-dimensional torus, shown in Figure 5.27. In this example, $N = 50$ and we show the results after first initializing each list with random entries. Figure 5.27(b) shows the results after five rounds of exchanges, while the final result, shown in Figure 5.27(c) is achieved after 20 rounds.

Note 5.8 (Advanced: Gossip-based overlay construction in Python)
To better appreciate and understand the construction of overlay networks through gossiping, let us see what a concrete implementation in Python would look like. As we have mentioned, the principle is simple: we have two separate layers and each node is responsible for exchanging links with selected neighbors. A straightforward implementation would seem to be to use a separate thread per layer, as suggested in Figure 5.26. However, such a choice will instantly lead to potential deadlock problems. Suppose node A decides to exchange links with

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(a) (b) (c)

Figure 5.27: Generating a torus overlay network using a two-layered unstructured peer-to-peer system.

node B, but B has chosen another node, say C for exchanging links. If C chooses A for exchanging links, we may have a deadlock if these three nodes first collectively choose their respective neighbor for exchanging links, and then collectively wait for that neighbor to respond.

The general solution to this problem is to essentially avoid **synchronous request-response behavior**, but instead to use **asynchronous, persistent communication**. In other words, we let the middleware store a message before it is delivered, after which the sender can continue doing other things. This type of communication is precisely supported by the channel package used for many of our Python examples. The outline of the code is shown in Figure 5.28. (The complete code is readily available as accompanying online material for the book.)

Note that for each layer, the approach is essentially always the same: select a peer, decide on the links to exchange, and send that information asynchronously to the selected peer. Thereafter, a node that initiated an exchange will have to wait for a response. When that comes in, it merely needs to update its own view. (There are many subtleties that need to be considered, which we will discuss shortly.) In principle, there will be a regular call to `maintainViews()`, if only to ensure that a node will also react to incoming exchange requests. Note that in our solution, we simply let a node ignore an incoming exchange request when it has an outstanding initiated exchange itself. Ignoring a request is done by telling the initiating node to give up its own attempt. Although this approach will prevent deadlocks, it bears the risk that no exchange will take place at all.

As mentioned, the actual code for handling gossip-based overlay construction involves more than what we have shown in Figure 5.28. Crucial are the criteria for selecting a neighboring peer, as well as the selection of links to exchange. As explained by Voulgaris and van Steen [2013], there are several things that can be done to speed up convergence of the overlay in comparison to random choices. At the same time, including random choices is essential to achieve convergence to the requested overlay.

In the first place, at the top layer for constructing the overlay, links should be exchanged that make most sense for the receiving peer. In our example, this

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means that when a node A sends links to a node B, it should select the ones to neighbors that are closest to B. Furthermore, B should not return links that A had just sent: it is much better that A learns about new nodes first.

```
def maintainViews():
  for viewType in [viewOverlay, viewPSS]: # For each view, do the same
    peer[viewType] = None
    if time to maintain viewType: # This viewType needs to be updated
      peer[viewType] = selectPeer(viewType)            # Select a peer
      links = selectLinks(viewType, peer[viewType])    # Select links
      sendTo(peer[viewType], Request[viewType], links) # Send links asynchronously

  while True:
    block = (peer[viewOverlay] != None) or (peer[viewPSS] != None)
    sender, msgType, msgData = recvFromAny(block) # Block if expecting something

    if msg == None: # All work has been done, simply return from the call
      return

    for viewType in [viewOverlay, viewPSS]: # For each view, do the same
      if msgType == Response[viewType]:  # Response to previously sent links
        updateOwnView(viewType, msgData) # Just update the own view

      elif msgType == Request[viewType]: # Request for exchanging links
        if peer[viewType] == None:       # No outstanding exchange request
          links = selectLinks(viewType, sender)     # Select links
          sendTo(sender, Response[viewType], links) # Send them asynchronously
          updateOwnView(viewType,msgData)           # Update own view
        else: # This node already has a pending exchange request, ignore this one
          sendTo(sender, IgnoreRequest[viewType])

      elif msgType == IgnoreRequest[viewType]: # Request has been denied, give up
        peer[viewType] = None
```

Figure 5.28: Pseudocode in Python for implementing a gossip-based approach toward overlay construction.

Moreover, when A selects a peer, we should try to prevent selecting one with which a recent exchange has taken place. In other words, A should try to maximize diversity in selecting a peer. This diversity can be obtained by selecting peers in a round-robin fashion. This approach also has the benefit that in a dynamic network, nodes that have left are guaranteed to be discovered quickly and can thus be removed from a view.

If we were to use only the top layer, the greediness of exchanging only best links may lead to a situation that a collection of nodes will jointly form a closed set within the network: jointly, they are referring to each other and as such, may be missing out on better nodes. This is the reason introducing randomness is essential, and why we need to incorporate information from the lower random overlay in the selection process of the higher structured overlay.

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This aspect is completely analogous to getting stuck in a local optimum when optimizing through hill-climbing procedures. What is needed is to introduce some randomness to ensure that indeed a global optimum will eventually be reached.

5.5.4 Secure gossiping

Gossiping is an attractive means for coordination. However, the speed by which a large collection of nodes manages to synchronize, is also an inherent vulnerability. The time it takes to exchange and disseminate proper information is also the time it takes to spread false information. This is perhaps best illustrated by means of an example. Consider a peer-sampling service for some large network. Assume that each node maintains a partial view of size c, and that, on average, $c/2$ references to neighboring nodes are exchanged. Let us also assume that there are c colluding attackers. Their behavior is simple: each attacker also maintains a partial view, but when returning $c/2$ references, only references to other colluding attackers are returned. The effect is that, gradually, the partial view of each benign node in the network is polluted by references to the attackers, to the extent that each partial view contains *only* references to attackers.

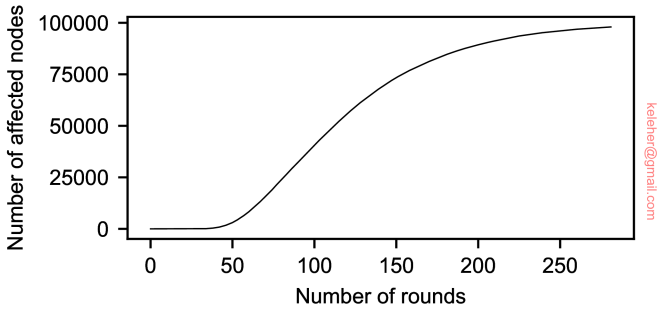


Figure 5.29: The speed by which a hub attack can take effect.

The effect can be dramatic. Figure 5.29 shows the speed by which partial views are fully polluted, that is, each contains references only to attackers. In this example, we consider a network of 100,000 nodes, a partial view size of 30, and only 30 attackers. After less than 300 rounds, all benign nodes have fully polluted partial views. Also note that, completely consistent with the push-pull behavior of epidemic protocols, it may take a while before any serious effect is seen. However, as soon as a few hundred nodes are completely

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polluted, they increasingly contribute to contaminating the rest of the nodes, and so on.

It can be argued that there is no standard security measure that can combat these kinds of attacks. For example, even with using secure channels by which both parties have genuinely authenticated each other, there is no reason to assume that both of them will also behave as they should. In other words, authentication provides no guarantees for trust. In the case of gossiping systems, security often comes in the form of trying to detect and prevent malicious behavior. For our example, the question is how can a benign node can detect that it is under attack, and which countermeasures can it take to mitigate the effects of an attack?

A solution in the case of peer sampling, and developed by Jesi et al. [2010], works as follows. The essence is to discover if there are nodes that behave badly. The attack we just described is characterized by the fact that, eventually, many nodes will be referencing a very small of set of nodes. In terms of graphs, this means that the *indegree* of some nodes is extraordinarily high, while for most nodes, their indegree will be low, and actually 0. An indegree of 0 means that a node is unknown to any other node in the network.

When using Cyclon [Voulgaris et al., 2005] as the peer-sampling service, we know that the indegrees follow a normal distribution, as shown in Figure 5.30(a). In this example, the y-axis shows the fraction of nodes with a specific indegree value. Even with a mere 30 colluding attackers, we already see that after 10 rounds, the indegree distribution has started to becoming wider and shifting to the left, as shown in Figure 5.30(b). This continues, and after 40 rounds (Figure 5.30(e)), we already have that about 15% of all nodes have an indegree of 0. (Note that for Figure 5.29, we were counting the number of nodes that were referencing *only* attacking nodes, which is different from measuring indegrees. An attacked node can still be referenced by others.) Finally, after less than 300 rounds, virtually all nodes have an indegree of 0; the attackers have now full control over the entire network.

Figure 5.30 also gives us a hint toward a solution. As explained by Jesi et al. [2010], each node can gather statistics on which nodes are being referenced. If certain nodes are gradually being referenced unusually more often than others, then something fishy may be going on. There are a few issues we need to deal with. First, spreading of "bad" links can be so efficient, that we need to consider that once a benign node has come to the conclusion that it has identified an attacker, it may already be too late. Second, we also want to prevent that every benign node has to build up complete knowledge of the entire network, partly also because attackers have to be identified quickly. Third, attackers do not want to be caught, i.e., be recognized as acting maliciously, as they collectively do need some time to be successful.

Notably the last aspect, not wanting to be caught, provides a solution to mitigating an attack. In essence, a benign node can force an attacker to behave

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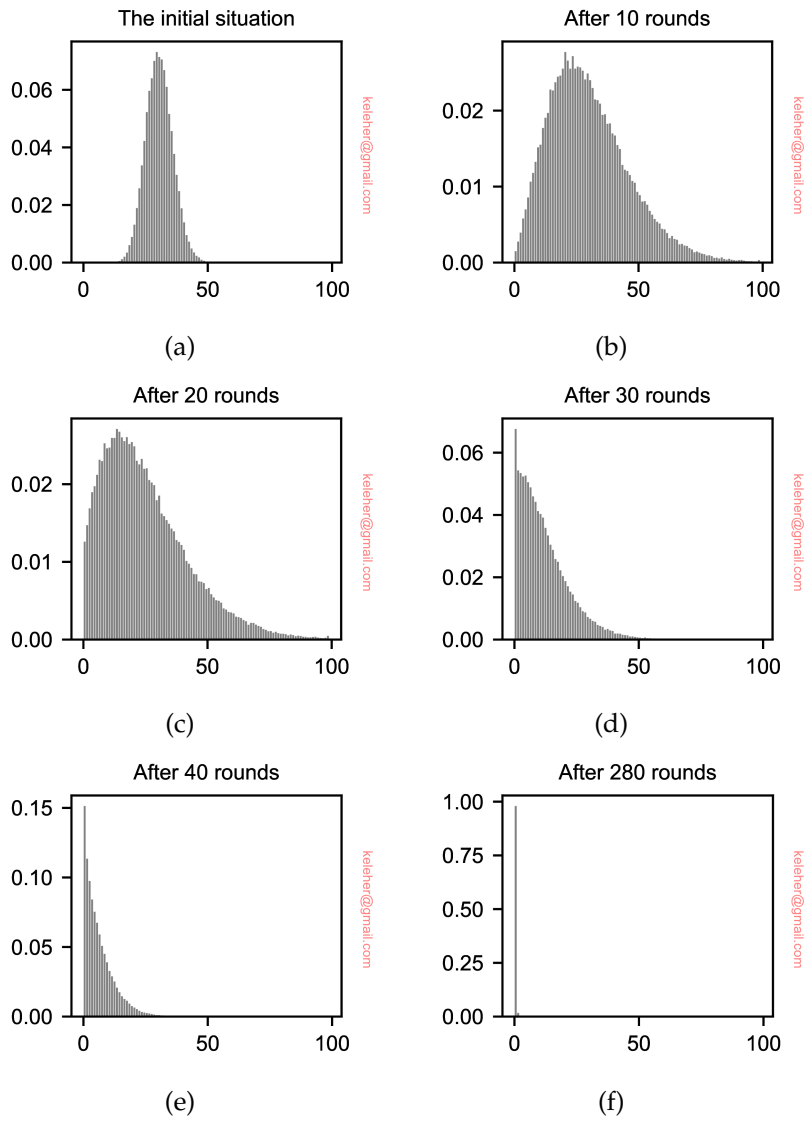


Figure 5.30: The development of the indegree distribution at several stages of a peer-sampling service under attack. Each x-axis shows the indegree, and the y-axis the fraction of nodes with that degree.

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according to the rules. What happens is that when a benign node initiates an exchange of links with a selected node, it will process the returned links either for updating its own partial view, or for gathering statistics (it may do both, but this may not be on par with efficient resource usage). Moreover, to accelerate the process of discovering attackers, a node will initiate several link exchanges simultaneously (and, in the end, process only one of the replies). A malicious node is now faced with a dilemma: always returning references to its colluders will quickly reveal the attackers. The only thing that it can do, is occasionally play by the rules and return links to other, benign nodes. As it turns out, this dilemma is enough to force the attackers to behave decently enough, effectively rendering an attack useless. Note that this procedure will also be applied when an *attacker* initiates an exchange: the recipient can decide to use the links passed to it for updating its statistics on indegrees, as well as use it for running the exchange protocol. The attacker will, however, never know whether the recipient is going to collect statistics. To fly under the radar, it will have to behave according to the rules of the game enough in order not to be caught. The details can be found in [Jesi et al., 2010].

There are many other types of attacks possible. For example, attackers may decide to initiate information exchange at a much higher frequency than specified by the protocol. With a push-based approach, this may result in attackers rapidly polluting values of benign nodes, which are then further disseminated. Mousazadeh and Ladani [2015] come to the conclusion that for such attacks, in principle, only pulling should be allowed (as the benign nodes will then collectively dictate the pace at which information is disseminated).

Likewise, in the case of aggregation, attackers may deliberately return incorrect values. To illustrate, suppose that when collectively computing the average value as we described in Section 5.5.1, an attacker A always returns its own value v_A. Every node v_i communicating with A will therefore compute

$$v_i \leftarrow (v_i + v_A)/2$$

which gradually converges to the value v_A. In all these cases, attacks can be mitigated only by trying to detect if a node is not behaving according to the protocol. That detection mechanism, of course, should be known to the attacker, while at the same time will be effective if it forces an attacker to try to go ahead unnoticed. In turn, if well designed, the mechanism will force the attacker to actually behave well, or at least to the extent that any possible damage remains limited. More on how these **data-injection attacks** work and can be mitigated, are described by Gentz et al. [2016].

5.6 Distributed event matching

Let us now draw our attention to distributed event matching. Event matching, or more precisely, **notification filtering**, is at the heart of **publish-subscribe systems**. The problem boils down to the following:

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- A process specifies through a subscription S in which events it is interested.
- When a process publishes a notification N on the occurrence of an event, the system needs to see if S *matches* N.
- In the case of a match, the system should send the notification N, possibly including the data associated with the event that took place, to the subscriber.

As a consequence, we need to facilitate at least two things: (1) matching subscriptions against events, and (2) notifying a subscriber in the case of a match. The two can be separated, but this need not always be the case. In the following, we assume the existence of a function *match*(S, N) which returns true when subscription S matches the notification N, and false otherwise. As we argue below, publish-subscribe systems turn out to face difficult scalability problems when the matching function cannot be easily distributed across multiple servers. This is often the case when subscriptions can be expressive, as in content-based publish-subscribe systems. We discuss scalability of distributed event matching separately below.

Another, rather nasty problem, is combining security and privacy while keeping publishers and subscribers mutually unaware of each other, but also ensuring that servers have only enough information available to perform their matching function. In this sense, it may seem that the goals of publish-subscribe systems, and those of secure distributed systems conflict. We return to this issue below as well.

5.6.1 Centralized implementations

A simple, naive implementation of event matching is to have a fully centralized server that handles all subscriptions and notifications. In such a scheme, a subscriber simply submits a subscription, which is subsequently stored. When a publisher submits a notification, that notification is checked against each and every subscription, and when a match is found, the notification is copied and forwarded to the associated subscriber.

Obviously, this is not a very scalable solution. Nevertheless, provided the matching can be done efficiently and the server itself has enough processing power, the solution is feasible for many cases. For example, using a centralized server is the canonical solution for implementing **Linda tuple spaces**. Likewise, many publish-subscribe systems that run within a single department or organization can be implemented through a central server. Important, in these cases, is that the matching function can be implemented efficiently. In practice, this is often the case when dealing with topic-based filtering: matching then resorts to checking for equality of attribute values.

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Note that a simple way to scale up a centralized implementation, is to deterministically divide the work across multiple servers. A standard approach is to make use of two functions, as explained by Baldoni et al. [2009]:

- a function *sub2node*(S), which takes a subscription S and maps it to a nonempty subset of servers
- a function *not2node*(N), which takes a notification N and maps it to a nonempty subset of servers.

The servers to which *sub2node*(S) is mapped are called the **rendezvous nodes** for S. Likewise, *sub2node*(N) are the rendezvous nodes for N. The only constraint that needs to be satisfied, is that for any subscription S and matching notification N, $sub2node(S) \cap not2node(N) \neq \emptyset$. In other words, there must be at least one server that can handle the subscription when there is a matching notification. In practice, this constraint is satisfied by topic-based publish-subscribe systems by using a hashing function on the names of the topics.

The idea of having a central server can be extended by distributing the matching across multiple servers and dividing the work. The servers, generally referred to as **brokers**, are organized into an overlay network. The issue then becomes how to route notifications to the appropriate set of subscribers. A straightforward way to make sure that notifications reach their subscribers, is to deploy flooding. There are essentially two approaches. First, we store each subscription at every broker, while publishing notifications only at a single broker. The latter will handle identifying the matching subscriptions and subsequently copy and forward the notification. The alternative is to store a subscription only at one broker, while broadcasting notifications to all brokers. In that case, matching is distributed across the brokers, which may lead to a more balanced workload among the brokers.

Note 5.9 (Example: TIB/Rendezvous)
Flooding notifications is used in **TIB/Rendezvous**, of which the basic architecture is shown in Figure 5.31 [TIBCO]. In this approach, a notification is a message tagged with a compound keyword describing its content, such as news.comp.os.books. A subscriber provides (parts of) a keyword, or indicating the messages it wants to receive, such as news.comp.$*$.books. These keywords are said to indicate the **subject** of a message.

Fundamental to its implementation is the use of broadcasting common in local-area networks, although it also uses more efficient communication facilities when possible. For example, if it is known exactly where a subscriber resides, point-to-point messages will generally be used. Each host on such a network will run a **rendezvous daemon**, which takes care that messages are sent and delivered according to their subject. Whenever a message is published, it is multicasted to each host on the network running a rendezvous daemon. Typically, multicasting

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is implemented using the facilities offered by the underlying network, such as IP-multicasting or hardware broadcasting.

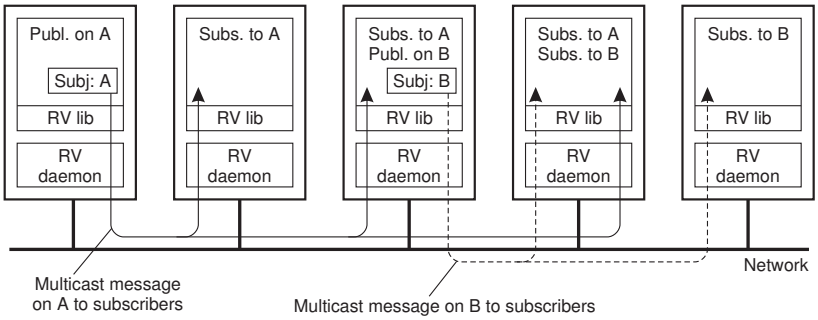


Figure 5.31: The principle of a publish/subscribe system as implemented in TIB/Rendezvous.

Processes that subscribe to a subject pass their subscription to their local daemon. The daemon constructs a table of (*process, subject*), entries and whenever a message on subject S arrives, the daemon simply checks in its table for local subscribers, and forwards the message to each one. If there are no subscribers for S, the message is discarded immediately.

When using multicasting as is done in TIB/Rendezvous, there is no reason why subscriptions cannot be elaborate and be more than string comparison, as is currently the case. The crucial observation here is that because messages are forwarded to every node anyway, the potentially complex matching of published data against subscriptions can be done entirely locally without further network communication needed.

When subscriptions are allowed to be more expressive, scalability may easily become a problem. In practice, expressive subscriptions take the form of (*attribute*,*value*) pairs in which a *value* returns as an expression on the possible values for the specified attribute. Examples include range values ("$1 \leq x < 10$"), containment ("$x \in \{\mathsf{red}, \mathsf{blue}\}$"), prefix and suffix expressions ("`url.startswith("https")`"), etc. More complicated expressions including combinations of conjunctions and disjunctions are, in theory, also possible, thus gradually resembling what one would find in SQL queries for databases. It is not hard to imagine that expressive subscriptions can be handled by centralized brokers, but when scalability is at stake, we may need to limit expressiveness altogether to the point that we find ourselves looking at topic-based event matching.

One way out that has been explored is combining peer-to-peer networks and publish-subscribe systems, as described in Kermarrec and Triantafillou [2013]. As the authors show, there is no simple general solution, and most solutions often have very specific limitations.

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Selective routing

If we consider that publish-subscribe is all about routing messages between publishers and subscribers, one thought is to install specific filters in an overlay network that effectively ignore paths toward nodes that are not interested in what is being published. In other words, instead of flooding a network with either subscriptions or publications, we apply **selective routing**, as proposed by Carzaniga et al. [2004]. Consider a publish-subscribe system consisting of N brokers to which clients (i.e., applications) can send subscriptions and retrieve notifications. Carzaniga et al. propose a two-layered routing scheme, in which the lowest layer consists of a shared broadcast tree connecting the N brokers. There are various ways for setting up such a tree, ranging from network-level multicast support to application-level multicast trees, as we discussed in Chapter 4. Here, we also assume that such a tree has been set up with the N brokers as end nodes, along with a collection of intermediate nodes forming routers. Note that the distinction between a server and a router is only a logical one: a single machine may host both kinds of processes.

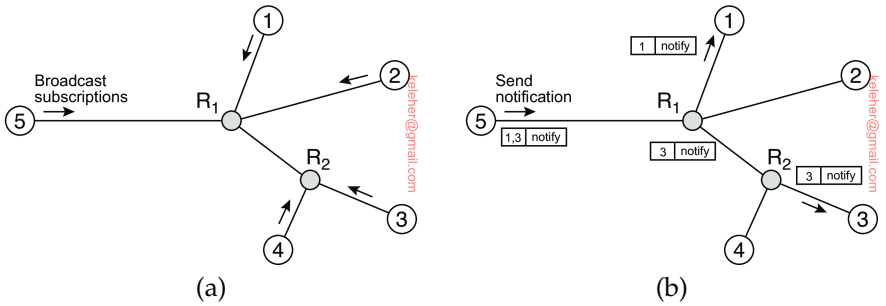


Figure 5.32: A first approach toward content-based routing in which brokers (a) first broadcast subscriptions to later (b) forward notifications only to relevant brokers.

Every broker broadcasts its subscriptions to all other brokers, as shown in Figure 5.32(a). As a result, every broker will be able to compile a list of (*subject*, *destination*) pairs. Then, whenever a process publishes a notification N, its associated broker prepends the destination brokers to that message and forwards the notification to the others, shown in Figure 5.32(b). When the message reaches a router, the latter can use the list to decide on the paths that the message should follow, as shown in Figure 5.32.

We can now refine the capabilities of routers for deciding where to forward notifications. To that end, each broker broadcasts its subscription across the network so that routers can compose **routing filters**. For example, assume that node 3 in Figure 5.32 subscribes to notifications for which an attribute a lies in the range $[0, 3]$, but that node 4 wants messages with $a \in [2, 5]$. In this

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case, router R_2 will create a routing filter as a table with an entry for each of its outgoing links (in this case three: one to node 3, one to node 4, and one toward router R_1), as shown in Figure 5.33.

Interface	Filter
To node 3	$a \in [0,3]$
To node 4	$a \in [2,5]$
Toward router R_1	(unspecified)

Figure 5.33: A partially filled routing table.

More interesting is what happens at router R_1. In this example, the subscriptions from nodes 3 and 4 dictate that any notification with a lying in the interval $[0,3] \cup [2,5] = [0,5]$ should be forwarded along the path to router R_2, and this is precisely the information that R_1 will store in its table. It is not difficult to imagine that more intricate subscription compositions can be supported.

This simple example also illustrates that whenever a node leaves the system, or when it is no longer interested in specific notifications, it should cancel its subscription and essentially broadcast this information to all routers. This cancellation, in turn, may lead to adjusting various routing filters. Late adjustments will at worst lead to unnecessary traffic, as notifications may be forwarded along paths for which there are no longer subscribers. Nevertheless, timely adjustments are needed to keep performance at an acceptable level.

Gossiping

Selective routing may help to avoid broadcasting notifications to all routers, but ultimately, it may still not be enough to reach acceptable scalable solutions. Another approach is based on gossiping. The basic idea is that subscribers interested in the same notifications form their own overlay network (which is constructed through gossiping as we describe further below), so that once a notification is published, it merely needs to be routed to the appropriate overlay. For the latter, a **random walk** can be deployed. This approach is following in TERA [Baldoni et al., 2007].

A more sophisticated approach toward combining gossiping and event matching is followed in Sub-2-Sub [Voulgaris et al., 2006]. It is still one of the very few content-based approaches that combines scalability with expressiveness of subscriptions. Consider a publish-subscribe system in which data items can be described by N attributes $a_1, \ldots, a_N$ whose value can be directly mapped to a floating-point number. Such values include, for example, floats, integers, enumerations, Booleans, and strings. A subscription S takes the form of a tuple of (*attribute, value/range*) pairs, such as

$$S = \langle a_1 \rightarrow 3.0, a_4 \rightarrow [0.0,\ 0.5) \rangle$$

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In this example, S specifies that a_1 should be equal to 3.0, and a_4 should lie in the interval $[0.0, 0.5)$. Other attributes are allowed to take on any value. For clarity, assume that every node i enters only one subscription S_i.

Note that each subscription S_i actually specifies a subset $\mathbf{S_i}$ in the N-dimensional space of floating-point numbers. Such a subset is also called a **hyperspace**. For the system as a whole, notifications that fall in the union $\overline{\mathbf{S}} = \cup \mathbf{S_i}$ of these hyperspaces are the only ones of interest. The whole idea is to automatically partition $\overline{\mathbf{S}}$ into M disjoint hyperspaces $\overline{\mathbf{S}}_\mathbf{1}, \ldots, \overline{\mathbf{S}}_\mathbf{M}$ such that each falls completely in one of the subscription hyperspaces $\mathbf{S_i}$, and together they cover all subscriptions. More formally, we have that:

$$(\overline{\mathbf{S}}_\mathbf{m} \cap \mathbf{S_i} \neq \emptyset) \Rightarrow (\overline{\mathbf{S}}_\mathbf{m} \subseteq \mathbf{S_i})$$

Sub-2-Sub keeps M minimal in the sense that there is no partitioning with fewer parts $\overline{\mathbf{S}}_\mathbf{m}$. To this end, for each hyperspace $\overline{\mathbf{S}}_\mathbf{m}$, it registers exactly those nodes i for which $\mathbf{S_m} \subseteq \mathbf{S_i}$. In that case, when a notification is published, the system need merely find the $\overline{\mathbf{S}}_\mathbf{m}$ to which the associated event belongs, from which point it can forward the notification to the appropriate nodes.

To this end, nodes regularly exchange subscriptions through gossiping. If two nodes i and j notice that their respective subscriptions intersect, that is, $\mathbf{S_{ij}} \equiv \mathbf{S_i} \cap \mathbf{S_j} \neq \emptyset$ they will record this fact and keep references to each other. If they discover a third node k with $\mathbf{S_{ijk}} \equiv \mathbf{S_{ij}} \cap \mathbf{S_k} \neq \emptyset$, the three of them will connect to each other so that a notification N from $\mathbf{S_{ijk}}$ can be efficiently disseminated. Note that if $\mathbf{S_{ij}} - \mathbf{S_{ijk}} \neq \emptyset$, nodes i and j will maintain their mutual references, but now associate it strictly with $\mathbf{S_{ij}} - \mathbf{S_{ijk}}$.

In essence, what we are seeking is a means to cluster nodes into M different groups, such that nodes i and j belong to the same group if and only if their subscriptions S_i and S_j intersect. Moreover, nodes in the same group should be organized into an overlay network that allows efficient dissemination of a data item in the hyperspace associated with that group. This situation for a single attribute is sketched in Figure 5.34.

Here, we see a total of seven nodes, in which the horizontal line for node i indicates its range of interest for the value of the single attribute. Also shown is the grouping of nodes into disjoint ranges of interests in values of the attribute. For example, nodes 3, 4, 7, and 10 will be grouped together representing the interval [16.5, 21.0]. Any data item with a value in this range should be disseminated to only these four nodes.

To construct these groups, the nodes are organized into a gossip-based unstructured network. Each node maintains a list of references to other neighbors (i.e., a **partial view**), which it periodically exchanges with one of its neighbors. Such an exchange allows a node to learn about random other nodes in the system. Every node keeps track of the nodes it discovers with overlapping interests (i.e., with an intersecting subscription).

At a certain moment, every node i will generally have references to other nodes with overlapping interests. As part of exchanging information with a

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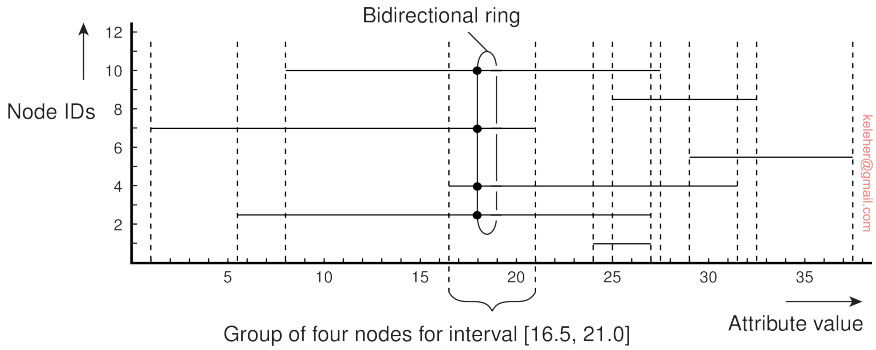


Figure 5.34: Grouping nodes for supporting range queries in a gossip-based publish-subscribe system.

node j, node i orders these nodes by their identifiers and selects the one with the lowest identifier $i_1 > j$, such that its subscription overlaps with that of node j, that is, $\mathbf{S}_{j,i_1} \equiv \mathbf{S}_{i_1} \cap \mathbf{S}_j \neq \emptyset$.

The next one to be selected is $i_2 > i_1$ such that its subscription also overlaps with that of j, but only if it contains elements not yet covered by node i_1. In other words, we should have that $\mathbf{S}_{j,i_1,i_2} \equiv (\mathbf{S}_{i_2} - \mathbf{S}_{j,i_1}) \cap \mathbf{S}_j \neq \emptyset$. This process is repeated until all nodes that have an overlapping interest with node i have been inspected, leading to an ordered list $i_1 < i_2 < \cdots < i_n$. Note that a node i_k is in this list because it covers a region $\mathbf{R}$ of common interest to node i and j not yet jointly covered by nodes with a lower identifier than i_k. In effect, node i_k is the *first* node that node j should forward a notification to that falls in this unique region $\mathbf{R}$. This procedure can be expanded to let node i construct a bidirectional ring. Such a ring is also shown in Figure 5.34.

Whenever a notification N is published, it is disseminated as quickly as possible to *any* node that is interested in it. As it turns out, with the information available at every node finding a node i interested in N is simple. From there on, node i need simply forward N along the ring of subscribers for the particular range that N falls into. To speed up dissemination, shortcuts are maintained for each ring as well.

5.6.2 Secure publish-subscribe solutions

A characteristic feature of publish-subscribe systems is that publishers and subscribers are **referentially decoupled**, and possibly also **temporally decoupled**. Notably, referential decoupling means that in a secure system messages should be able to flow from a publisher to subscribers while guaranteeing mutual anonymity. To guarantee confidentiality and integrity of messages, communicating parties usually set up a secure channel, but such an approach would break the party's anonymity.

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A simple solution is to let a (possibly distributed) broker handle messages. The broker would handle all the matching and storage, if necessary. Obviously, such a broker would need to be trusted. Yet, it is not obvious at all that such trust is justified. For example, many publish-subscribe services are offered by third parties operating in a cloud setting. It is often unclear to which extent they can actually provide guarantees when it comes to protecting data. In addition, messages may often contain information that should definitely not, often even by law, be revealed to third parties, such as medical records.

We now seem to have several clashing requirements. First, for any publish-subscribe system, we need to ensure that the communicating parties remain referentially decoupled, that is, we need to guarantee mutual anonymity. Not knowing where messages come from imposes integrity issues: how can we be sure the messages have not been tampered with? Second, we cannot simply assume that brokers can be trusted. Facing confidentiality issues, we then cannot allow them to see (all of) the content of messages, yet this may impose serious routing problems. But there is more: if we cannot trust brokers, how can we ensure availability in the sense that subscribers, entitled to specific messages, actually receive those messages? Equally important is to ensure that messages are not delivered to unauthorized subscribers. Cui et al. [2021] also address an additional requirement: assuming there are colluders in place, brokers should not learn about the interests of innocent subscribers.

These problems are not unique to publish-subscribe systems, and have become more prevalent with the advent of cloud-based services. The key question is how to let a third party search through data to make decisions, yet without revealing that data. This is also known as **searchable encryption** [Bosch et al., 2014]. For publish-subscribe systems, an important technique is to conduct secure keyword search. Boneh et al. [2004] introduced **Public Key Encryption with Keyword Search**, generally known as **PEKS**. We will explain only the principle of PEKS without going into the cryptographic details.

In PEKS, a message is accompanied by a collection of keywords. To that end, using a public key PK, a message m and its n keywords $\mathsf{KW}_1, \ldots, \mathsf{KW}_n$ are stored at a server as the message m^*:

$$\mathsf{m}^* = [PK(\mathsf{m})|PEKS(\mathsf{PK}, \mathsf{KW}_1)|PEKS(\mathsf{PK}, \mathsf{KW}_2)|\cdots|PEKS(\mathsf{PK}, \mathsf{KW}_n)]$$

For each keyword KW_i a trapdoor $\mathsf{T}_{\mathsf{KW}_i}$ is generated: $\mathsf{T}_{\mathsf{W}}(\mathsf{m}^*)$ will return true if $\mathsf{W} \in \{\mathsf{KW}_1, \ldots, \mathsf{KW}_n\}$ and false otherwise.

Using PEKS, we can now set up a cloud-based publish-subscribe system with (conceptually) a single broker that resides in the cloud. Such a scheme is described in [Yang et al., 2017] and sketched in Figure 5.35. We note we have deliberately simplified matters significantly to focus on the essentials.

An important role is played by the key authority: a trusted third party that generates keys for publishers and subscribers. In our example, it provides the

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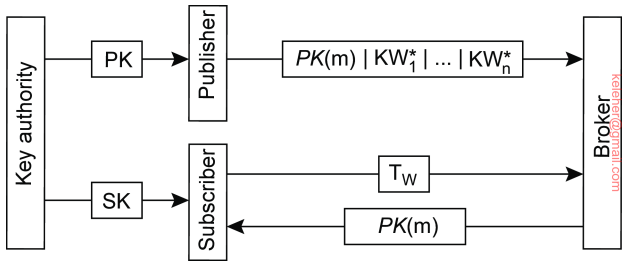


Figure 5.35: The principle of privacy-preserving publish-subscribe.

publisher with a key PK that can be used for encrypting the publication m. In addition, the key is used for constructing the encrypted tags $KW_i^* = PEKS(PK, KW_i)$. Likewise, each subscriber is given a secret key SK that it can use to generate a trapdoor T_W to be used by the broker to see whether the tag W is associated with the publication m. If there is a match, the broker returns $PK(m)$ which can then be decrypted using SK.

Our simplified scheme misses many important details, for which we refer the interested reader to Yang et al. [2017]. The authors assume that the broker is **honest-but-curious** and does not collude with publishers or subscribers. Honest-but-curious means that the server will behave according to the specifications, but may keep track of everything it does and use that information. Cui et al. [2021] describe a system that drops the assumption on collusion, and is one of the few known publish-subscribe systems that can handle colluding brokers. An overview of the security issues in publish-subscribe systems is given by Uzunov [2016]. For a good overview on preserving confidentiality, we refer to Onica et al. [2016]. The authors also come to the conclusion that security becomes easier if brokers can be trusted. Otherwise, good support for operating on encrypted data is needed.

5.7 Location systems

When looking at large distributed systems that are dispersed across a wide-area network, it is often necessary to take proximity into account. Just imagine a distributed system organized as an overlay network in which two processes are neighbors in the overlay network, but are actually placed far apart in the underlying network. If these two processes communicate a lot, it may have been better to ensure that they are also physically placed in each other's proximity. In this section, we take a look at location-based techniques to coordinate the placement of processes and their communication.

5.7.1 GPS: Global Positioning System

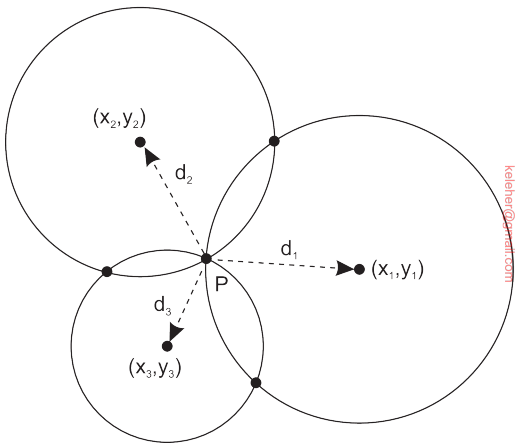
Let us start by considering how to determine your geographical position anywhere on Earth. This positioning problem is by itself solved through a highly specific, dedicated distributed system, namely **GPS**, which is an

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acronym for **Global Positioning System**. GPS is a satellite-based distributed system that was launched in 1978. Although it initially was used mainly for military applications, it by now has found its way to many civilian applications, notably for traffic navigation. However, many more application domains exist. For example, modern smartphones now allow owners to track each other's position. This principle can easily be applied to tracking other things as well, including pets, children, cars, boats, and so on.

GPS uses up to 72 satellites, each circulating in an orbit at a height of approximately 20,000 km. Each satellite has up to four atomic clocks, which are regularly calibrated from special stations on Earth. A satellite continuously broadcasts its position, and time stamps each message with its local time. This broadcasting allows every receiver on Earth to accurately compute its own position using, in principle, only four satellites. To explain, let us first assume that all clocks, including the receiver's, are synchronized.

To compute a position, consider first the two-dimensional case, as shown in Figure 5.36, in which three satellites are drawn, along with the circles representing points at the same distance from each respective satellite. We see that the intersection of the three circles is a unique point.



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Figure 5.36: Computing a node's position in a two-dimensional space.

This principle of intersecting circles can be expanded to three dimensions, meaning that we need to know the distance to four satellites to determine the longitude, latitude, and altitude of a receiver on Earth. This positioning is all fairly straightforward, but determining the distance to a satellite becomes complicated when we move from theory to practice. There are at least two important real-world facts that we need to take into account:

1. It takes a while before data on a satellite's position reaches the receiver.
2. The receiver's clock is generally not in sync with that of a satellite.

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Assume that the timestamp from a satellite is completely accurate. Let Δ_r denote the deviation of the receiver's clock from the actual time. When a message is received from satellite S_i with timestamp T_i, then the measured delay Δ_i by the receiver consists of two components: the actual delay, along with its own deviation:

$$\Delta_i = (T_{now} - T_i) + \Delta_r$$

where T_{now} is the actual current time. As signals travel with the speed of light, c, the *measured* distance $\tilde{d}_i$ to satellite S_i is equal to $c \cdot \Delta_i$. With

$$d_i = c \cdot (T_{now} - T_i)$$

being the real distance between the receiver and satellite S_i, the measured distance can be rewritten to $\tilde{d}_i = d_i + c \cdot \Delta_r$. The real distance is now computed as:

$$\tilde{d}_i - c \cdot \Delta_r = \sqrt{(x_i - x_r)^2 + (y_i - y_r)^2 + (z_i - z_r)^2}$$

where x_i, y_i, and z_i denote the coordinates of satellite S_i. What we see now is a system of quadratic equations with four unknowns (x_r, y_r, z_r, and also Δ_r). We thus need four reference points (i.e., satellites) to find a unique solution that will also give us Δ_r. A GPS measurement will thus also give an account of the actual time.

So far, we have assumed that measurements are perfectly accurate. Of course, they are not. There are many sources of errors, starting with the fact that the atomic clocks in the satellites are not always in perfect sync, the position of a satellite is not known precisely, the receiver's clock has a finite accuracy, the signal propagation speed is not constant (as signals appear to slow down when entering, e.g., the ionosphere), and so on. On average, this leads to an error of some 5–10 meters. Special modulation techniques, as well as special receivers, are needed to improve accuracy. Using so-called **differential GPS**, by which corrective information is sent through wide-area links, accuracy can be further improved. More information can be found in [LaMarca and de Lara, 2008], as well as an excellent overview by Zogg [2002].

5.7.2 When GPS is not an option

A major drawback of GPS is that it generally cannot be used indoors. For that purpose, other techniques are necessary. An increasingly popular technique is to make use of the numerous WiFi access points available. The basic idea is simple: if we have a database of known access points along with their coordinates, and we can estimate our distance to an access point, then with only three detected access points, we should be able to compute our position. Of course, it really is not that simple at all.

A major problem is determining the coordinates of an access point. A popular approach is to do this through **war driving**: using a WiFi-enabled

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device along with a GPS receiver, someone drives or walks through an area and records observed access points. An access point can be identified through its SSID or its MAC-level network address. An access point AP should be detected at several locations before its coordinates can be estimated. A simple method is to compute the centroid: assume we have detected AP at N different locations $\{\vec{x_1}, \vec{x_2}, \ldots, \vec{x_N}\}$, where each location $\vec{x_i}$ consists of a (*latitude, longitude*)-pair as provided by the GPS receiver. We then simply estimate AP's location $\vec{x}_{AP}$ as

$$\vec{x}_{AP} = \frac{\sum_{i=1}^{N} \vec{x_i}}{N}.$$

Accuracy can be improved by taking the observed signal strength into account, and giving more weight to a location with relatively high observed signal strength than to a location where only a weak signal was detected. In the end, we obtain an estimation of the coordinates of the access point. The accuracy of this estimation is strongly influenced by:

- the accuracy of each GPS detection point $\vec{x_i}$
- the fact that an access point has a nonuniform transmission range
- the number of sampled detection points N.

Studies show that estimates of the coordinates of an access point may be tens of meters off from the actual location (see, e.g., Kim et al. [2006] or Tsui et al. [2010]). Moreover, access points come and go at a relatively high rate. Nevertheless, locating and positioning access points is widely popular, exemplified by the open-access Wigle database which is populated through crowdsourcing.[1]

5.7.3 Logical positioning of nodes

Instead of trying to find the absolute location of a node in a distributed system, an alternative is to use a logical, proximity-based location. In **geometric overlay networks** each node is given a position in an m-dimensional geometric space, such that the distance between two nodes in that space reflects a real-world performance metric. Computing such a position is the core business of a **Network Coordinates System**, or simply **NCS**, which are surveyed by Donnet et al. [2010]. The simplest, and most applied example, is where distance corresponds to internode latency. In other words, given two nodes P and Q, then the distance $\hat{d}(\mathsf{P}, \mathsf{Q})$ reflects how long it would take for a message to travel from P to Q and *vice versa*. We use the notation $\hat{d}$ to denote distance in a system where nodes have been assigned coordinates.

There are many applications of geometric overlay networks. Consider the situation where a Website at server O has been replicated to multiple servers

[1] See wigle.net.

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$S_1, \ldots, S_N$ on the Internet, as typically happens with a **Content Delivery Network** (CDN). When a client C requests a page from O, the CDN may decide to redirect that request to the server closest to C, that is, the one that will give the best response time. If the geometric location of C is known, as well as those of each replica server, the CDN can then simply pick that server S_i for which $\hat{d}(C, S_i)$ is minimal.

Another example is optimal replica placement. Consider again a CDN that has gathered the positions of clients for one of its major customers. If the CDN were to replicate content to N servers, it can compute the N best positions where to place replicas such that the average client-to-replica response time is minimal. Performing such computations is almost trivially feasible if clients and servers have geometric positions that reflect internode latencies.

As a last example, consider **position-based routing** (see, e.g., [Popescu et al., 2012] or [Bilal et al., 2013]). In such schemes, a message is forwarded to its destination using only positioning information. For example, a naive routing algorithm to let each node forward a message to the neighbor closest to the destination. Although it can be easily shown that this specific algorithm need not converge, it illustrates that only local information is used to decide. There is no need to propagate link information or such to all nodes in the network, as is the case with conventional routing algorithms.

Centralized positioning

Positioning a node in an m-dimensional geometric space requires $m + 1$ distance measures to nodes with known positions. Assuming that node P wants to compute its own position, it contacts three other nodes with known positions and measures its distance to each of them. Contacting only one node would tell P about the circle it is located on; contacting only two nodes would tell it about the position of the intersection of two circles (which generally consists of two points); a third node would subsequently allow P to compute its actual location.

Node P can compute its own coordinates (x_P, y_P) by solving the three *quadratic* equations with the two unknowns x_P and y_P:

$$\tilde{d}_i = \sqrt{(x_i - x_P)^2 + (y_i - y_P)^2} \qquad (i = 1, 2, 3)$$

Here, we use $\tilde{d}$ to denote measured, or estimated distance. As said, $\tilde{d}_i$ generally corresponds to measuring the latency between P and the node at (x_i, y_i). This latency can be estimated as being half the round-trip delay, but it should be clear that its value will be different over time. The effect is a different positioning whenever P would want to recompute its position. Moreover, if other nodes would use P's current position to compute their own coordinates, then it should be clear that the error in positioning P will affect the accuracy of the positioning of other nodes.

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It should also be clear that measured distances between a set of nodes will generally not even be consistent. For example, assume we are computing distances in a one-dimensional space, as shown in Figure 5.37. In this example, we see that although R measures its distance to Q as 2.0, and $\tilde{d}(P,Q)$ has been measured to be 1.0, when R measures $\tilde{d}(P,R)$ it finds 2.8, which is clearly inconsistent with the other two measurements.

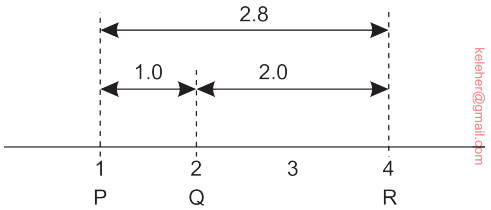


Figure 5.37: Inconsistent distance measurements in a one-dimensional space.

Figure 5.37 also suggests how this situation can be improved. In our simple example, we could solve the inconsistencies by merely computing positions in a two-dimensional space. This by itself, however, is not a general solution when dealing with many measurements. In fact, considering that Internet latency measurements may violate the **triangle inequality**, it is generally impossible to resolve inconsistencies completely. The triangle inequality states that in a geometric space, for any arbitrary three nodes P, Q, and R it must always be true that

$$d(P,R) \leq d(P,Q) + d(Q,R).$$

There are various ways to approach these issues. One common approach, proposed by Ng and Zhang [2002], is to use N special nodes $L_1, \ldots, L_N$, known as **landmarks**. Landmarks measure their pairwise latencies $\tilde{d}(L_i, L_j)$ and subsequently let a central node compute the coordinates for each landmark. To this end, the central node seeks to minimize the following aggregated error function:

$$\sum_{i=1}^{N} \sum_{j=i+1}^{N} \left(\frac{\tilde{d}(L_i,L_j) - \hat{d}(L_i,L_j)}{\tilde{d}(L_i,L_j)} \right)^2$$

where, again, $\hat{d}(L_i, L_j)$ corresponds to the distance after nodes L_i and L_j have been positioned.

The hidden parameter in minimizing the aggregated error function is the dimension m. Obviously, we have that $N > m$, but nothing prevents us from choosing a value for m that is much smaller than N. In that case, a node P measures its distance to each of the N landmarks and computes its coordinates by minimizing

$$\sum_{i=1}^{N} \left(\frac{\tilde{d}(L_i,P) - \hat{d}(L_i,P)}{\tilde{d}(L_i,P)} \right)^2$$

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As it turns out, with well-chosen landmarks, m can be as small as 6 or 7, with $\hat{d}(\mathsf{P},\mathsf{Q})$ being no more than a factor 2 different from the actual latency $d(\mathsf{P},\mathsf{Q})$ for arbitrary nodes P and Q [Szymaniak et al., 2004; 2008].

Decentralized positioning

Another way to tackle this problem is to view the collection of nodes as a huge system in which nodes are attached to each other through springs. In this case, $|\tilde{d}(\mathsf{P},\mathsf{Q}) - \hat{d}(\mathsf{P},\mathsf{Q})|$ indicates to what extent nodes P and Q are displaced relative to the situation in which the system of springs would be at rest. By letting each node (slightly) change its position, it can be shown that the system will eventually converge to an optimal organization in which the aggregated error is minimal. This approach is followed in Vivaldi, described in [Dabek et al., 2004a].

In a system with N nodes $\mathsf{P}_1,\ldots,\mathsf{P}_N$, Vivaldi aims at minimizing the following aggregated error:

$$\sum_{i=1}^{N}\sum_{j=1}^{N}|\tilde{d}(\mathsf{P}_i,\mathsf{P}_j) - \hat{d}(\mathsf{P}_i,\mathsf{P}_j)|^2$$

where $\tilde{d}(\mathsf{P}_i,\mathsf{P}_j)$ is the *measured* distance (i.e., latency) between nodes P_i and P_j, and $\hat{d}(\mathsf{P}_i,\mathsf{P}_j)$ the distance computed from the network coordinates of each node. Let $\vec{x}_i$ denote the coordinates of node P_i. In a situation that each node is placed in a geometric space, the *force* that node P_i exerts on node P_j is computed as:

$$\vec{F}_{ij} = \big(\tilde{d}(\mathsf{P}_i,\mathsf{P}_j) - \hat{d}(\mathsf{P}_i,\mathsf{P}_j)\big) \times u(\vec{x}_i - \vec{x}_j)$$

with $u(\vec{x}_i - \vec{x}_j)$ denoting the unit vector in the direction of $\vec{x}_i - \vec{x}_j$. In other words, if $F_{ij} > 0$, node P_i will push P_j away from itself, and will otherwise pull it toward itself. Node P_i now repeatedly executes the following steps:

1. Measure the latency $\tilde{d}_{ij}$ to node P_j, and also receive P_j's coordinates $\vec{x}_j$.
2. Compute the error $e = \tilde{d}(\mathsf{P}_i,\mathsf{P}_j) - \hat{d}(\mathsf{P}_i,\mathsf{P}_j)$
3. Compute the direction $\vec{u} = u(\vec{x}_i - \vec{x}_j)$.
4. Compute the force vector $F_{ij} = e \cdot \vec{u}$
5. Adjust own position by moving along the force vector: $\vec{x}_i \leftarrow \vec{x}_i + \delta \cdot \vec{u}$.

A crucial element is the choice of δ: too large and the system will oscillate; too small and convergence to a stable situation will take a long time. The trick is to have an adaptive value, which is large when the error is large as well, but small when only small adjustments are needed. Details can be found in [Dabek et al., 2004a].

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5.8 Summary

Strongly related to communication between processes is the issue of how processes in distributed systems synchronize. Synchronization is all about doing the right thing at the right time. A problem in distributed systems, and computer networks in general, is that there is no notion of a globally shared clock. In other words, processes on different machines have their own idea of what time it is.

There are various ways to synchronize clocks in a distributed system, but all methods are essentially based on exchanging clock values, while considering the time it takes to send and receive messages. Variations in communication delays, and the way those variations are dealt with, largely determine the accuracy of clock synchronization algorithms.

Knowing the absolute time is often not necessary. What counts is that related events at different processes happen in the correct order. Lamport showed that by introducing a notion of logical clocks, it is possible for a collection of processes to reach global agreement on the correct ordering of events. In essence, each event e, such as sending or receiving a message, is assigned a globally unique logical timestamp $C(e)$ such that when event a happened before b, $C(a) < C(b)$. Lamport timestamps can be extended to vector timestamps: if $C(a) < C(b)$, we even know that event a causally preceded b.

An important class of synchronization algorithms is that of distributed mutual exclusion. These algorithms ensure that in a distributed collection of processes, at most one process at a time has access to a shared resource. Distributed mutual exclusion can easily be achieved if we make use of a coordinator that keeps track of whose turn it is. Fully distributed algorithms also exist, but have the drawback that they are generally more susceptible to communication and process failures.

Synchronization between processes often requires that one process acts as a coordinator. In those cases where the coordinator is not fixed, it is necessary that processes in a distributed computation decide on who is going to be that coordinator. Such a decision is taken by election algorithms. Election algorithms are primarily used in cases where the coordinator can crash. However, they can also be applied for the selection of superpeers in peer-to-peer systems.

The most important aspect in gossip-based coordination is being able to select another peer randomly from an entire overlay. As it turns out, we can implement such a peer-sampling service using gossiping, by ensuring that the partial view is refreshed regularly and randomly. Combining peer sampling with a selective replacement of entries in a partial view allows us to efficiently construct structured overlay networks.

Particularly challenging when it comes to coordination is distributed event matching, which sits at the core of publish-subscribe systems. Relatively

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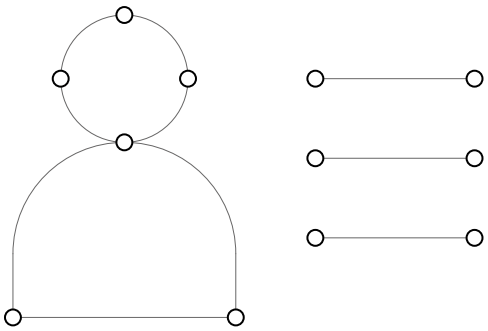
simple is the case when we have a central implementation where matching subscriptions against notifications can be done by essentially doing one-to-one comparisons. However, as soon as we aim at distributing the load, we are faced with the problem of deciding beforehand which node is responsible for which part of the subscriptions, without knowing what kind of notifications to expect. This is particularly problematic for content-based matching, which, in the end, requires advanced filtering techniques to route notifications to the proper subscribers.

Related to these synchronization problems is positioning nodes in a geometric overlay. The basic idea is to assign each node coordinates from an m-dimensional space such that the geometric distance can be used as an accurate measure for the latency between two nodes. The method of assigning coordinates strongly resembles the one applied in determining the location and time in GPS.

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06

NAMING



Names play an important role in all computer systems. They are used to sharing resources, to uniquely identify entities, to refer to locations, and more. An important issue with naming is that a name can be resolved to the entity it refers to. Name resolution thus allows a process to access the named entity. To resolve names, it is necessary to implement a naming system. The difference between naming in distributed systems and nondistributed systems lies in the way naming systems are implemented.

In a distributed system, the implementation of a naming system is itself often distributed across multiple machines. How this distribution is done plays a key role in the efficiency and scalability of the naming system. In this chapter, we concentrate on three different, important ways that names are used in distributed systems.

First, we consider so-called flat-naming systems. In such systems, entities are referred to by an identifier that, in principle, has no meaning at all. In addition, flat names bear no structure, implying that we need special mechanisms to trace the location of such entities. We discuss various approaches, ranging from distributed hash tables to hierarchical location services.

In practice, humans prefer to use readable names. Such names are often structured, as is well known from the way Web pages are referred to. Structured names allow for a highly systematic way of finding the server responsible for the named entity, as exemplified by the Domain Name System. We discuss the general principles, as well as scalability issues.

Finally, humans often prefer to describe entities through various characteristics, leading to a situation in which we need to resolve a description using the attributes assigned to an entity. As we shall see, this type of name resolution is notoriously difficult, especially with searching.

Name resolution and routing are closely related to each other. Normally, we look up an address given a name, to subsequently access the named entity through its address. As an alternative to looking up an address, researchers have been exploring how a name can be directly used to route toward the entity, to subsequently return a copy of its associated data. This so-called named-based routing is gradually maturing as an alternative for a future Internet. We discuss it briefly in this chapter.

6.1 Names, identifiers, and addresses

A name in a distributed system is a string of bits or characters that is used to refer to an entity. An entity in a distributed system, can be practically anything. Typical examples include resources such as hosts, printers, disks, and files. Other well-known examples of entities that are often explicitly named are processes, users, mailboxes, Web pages, graphical windows, messages, network connections, and so on.

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